

THE CAMBRIDGE HANDBOOK OF
PARENTING

Interdisciplinary Research and Application

*Edited by Amanda Sheffield Morris
and Julia Mendez Smith*



The Cambridge Handbook of Parenting

Parenting is a critical influence on the development of children across the globe. This handbook brings together scholars with expertise on parenting science and interventions for a comprehensive review of current research. It begins with foundational theories and research topics, followed by sections on parenting children at different ages, factors that affect parenting such as parental mental health or socioeconomic status, and parenting children with different characteristics such as depressed and anxious children or youth who identify as LGBTQ. It concludes with a section on policy implications, as well as prevention and intervention programs that target parenting as a mechanism of change. Global perspectives and the cultural diversity of families are highlighted throughout. Offering in-depth analysis of key topics such as risky adolescent behavior, immigration policy, father engagement, family involvement in education, and balancing childcare and work, this is a vital resource for understanding the most effective policies to support parents in raising healthy children.

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The Cambridge Handbook of Parenting

Interdisciplinary Research and Application

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This book is dedicated broadly to our family and friends who encourage us to be the best parents we can be. We especially honor the years we have spent with our parents, and we celebrate the blessings of our amazing children, Caleb, Mollie, RJ, Alex, and Danny. We are so grateful for our spouses, Mike and Rob, who share in all our parenting joys and challenges. We are also thankful for our many friends, students, and allies in academia who work with us in forging new, more inclusive directions in our parenting scholarship.

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Introduction

Working as co-editors and with our colleagues on this Handbook during the context of the global pandemic of 2019–2020 was certainly unanticipated and unlike any challenges we have ever known. As parents, we gained new appreciation for our research on how parenting shapes child development even as we struggled to use this knowledge within our own homes and lives. Future generations will have to uncover the role of the Covid-19 pandemic on parenting approaches and impacts on children around the globe. For now, we can honestly say we are both relieved and thrilled to share this work with our community of scholars to advance the goals of science and practice.

We also know that new possibilities are emerging for topics that we have not yet sufficiently addressed, such as how the features of the pandemic including social distancing, separation and isolation, remote learning, compromised health care and access to vaccines, and social connectedness through technology, create a unique and important context through which to view parenting and child development during this time. Just as the years after the Great Depression and the Great Recession provided a window to study impacts on cohorts of children, the pandemic too will likely become one of those historical turning points where shifts in thinking about parental influences will occur and influence our field for the better.

In this Handbook, we have captured the appealing range of topics related to parenting

while encouraging our authors to use a culturally and developmentally grounded approach with a pointed consideration of how their work may link to practice or policy implications. In Part I of the Handbook, *Foundations of Parenting*, we begin with chapters that provide theories and research that cut across age groups and topics, including a history of parenting science, the biology of parenting and attachment, parenting and brain development, and parenting and culture. In this section we also include chapters on parenting behaviors and strategies that influence development across childhood including discipline and punishment, emotion socialization, and parenting and children's cognitive and language development. These chapters set the stage for Part II, which is organized by developmental age/stage, *Parenting across Development: Social, Emotional, and Cognitive Influences*. In this section, we begin with a chapter on prenatal bonding and the transition to parenting, and end with a chapter on emerging adulthood. These two chapters bookend chapters on infancy and early childhood, middle childhood, and adolescence. Importantly, these chapters bring to light how parenting differs across children's development, emphasizing how parents and children change and grow together over time.

The chapters within Part III, on *Parental Factors that Impact Parenting*, provide a deeper examination of the variation within families and contexts that helps to shape the experiences and relationships within a family.

Authors explore a range of topics including parents with mental health challenges, the specific contributions of fathers as parents, and the important context of race and ethnicity. In keeping with our goals for identifying gaps, we see there is much work to be done in considering how parents (including fathers) make positive contributions to children's development, and how programs and policies can facilitate their success in child-rearing. We learn that parents' experiences with discrimination and barriers to accessing parenting supports remain an important challenge for researchers to address with their work.

The chapters within Part IV, on *Child Factors that Impact Parenting*, are a look at specific circumstances impacting children that can require parents to develop particular skills or accommodations to match the unique circumstances of their children. Here, we see an applied lens in examining the issues involved in raising children with disabilities, children with psychological disorders, including attention deficit hyperactivity disorder (ADHD), and anxiety and depression. We also address more general situations, including children with histories of adversity and children who identify as LGBTQ+. These chapters offer a deeper understanding of parenting in context and how children and parental transactions over time are a powerful influence on child development.

Finally, Part V, *Parent Education, Intervention, and Policy*, is a collection of chapters authored by researchers who are working vigorously on issues of intervention, prevention, and policy to create needed supports for parents and families. Examples include a vast array of topics from family engagement in education, immigration policy, employment for parents and access to child care supports, technology supports for

parents, and parenting interventions for preventing adolescent risky behavior. Given the number of challenges faced by parents across the development of a child (or more than one), such interventions tested with rigorous scientific methods and approaches that are brought to scale are another deep challenge for the field to continue to work on in earnest.

Upon completion of this project, it was not hard for us to see where we might head next. We applaud our authors' work, for they have pointed us to many future directions and gaps that must be at the center of our parenting research agenda. These include a true embrace of the biopsychosocial model and how our methods can better capture the lasting influences of parenting practices, while also capturing the moment-to-moment exchanges that build these pathways. We encouraged our authors to draw out possible implications for practice and policy, taking into account the state of the science on the optimal solutions or promising strategies. We see ample room for research on how and when parents serve as advocates and protectors for their children's rights and healthy development, particularly in the context of discrimination and bias within systems that intend to serve children's needs. Finally, many areas of the parenting research remain constrained by lack of diversity within the populations of study, and we need more conversations about how to move more rapidly beyond WEIRD (White, Educated, Industrialized, Rich, and Democratic) samples into global, interconnected, and cross-cultural work to develop a complete picture of parenting. Only this will permit us to fully shape practices and policies that can be more responsive and supportive of the needs of our most underserved families, such that we can achieve greater equity in child outcomes.

Part I

Foundations of Parenting

1 Foundational Theories and the Establishment of Parenting Science Research

George W. Holden

One of the most fundamental, important, and enduring questions in psychology concerns how the dual elements of genes and the environment influence the development of an individual, commonly referred to as nature vs. nurture. Psychology has now moved beyond pitting those two influences against each other and instead is asking the question, How do the two fundamental components of life work together (Anreiter, Sokolowski, & Sokolowski, 2018)? For example, the concept of epigenesis addresses how the environment modifies the expression of the genotype.

Parents are at the hub of the nature/nurture question – due to bestowing the child’s genetic nature (at least for biological parents) and for providing the environment. Although nurturing a child comes from many sources, no manifestation is more important than the child rearing provided by parents. However, it is not well understood what aspects of parental nurture provide the active ingredient for influencing a child’s development. Is it a parental style, one or more particular behaviors, a dimension of parenting, guidance along developmental pathways, or perhaps the experiences that the parent may or may not provide?

This chapter addresses two topics concerning the development of the research into questions about parents and their role in children’s development: the foundational theories and the establishment of the empirical research. In only a little over 100 years – five generations – research into parent–child relationships has developed from a cottage industry

practiced by only a handful of methodologically naïve individuals to a major industry engaged in by thousands of PhDs, MDs, and other professionals and researchers investigating a wide range of questions. Every year a prodigious number of research articles are published by this army of investigators in a wide variety of journals, chapters, and books. The goal of this chapter is to provide some structure for understanding this increasingly vast body of research. One framework concerns the theoretical foundations of contemporary research and the second framework concerns the development of discrete approaches undertaken to study parenting.

Foundational Theories

Psychology is replete with theories, and every content area of psychology, whether it be developmental, social, clinical, health or abnormal psychology, is populated with multiple theories. Parenting research has benefited from the fertile land of psychological theories. For example, social psychology theories, such as attribution theory (e.g., Kelley, 1967) and the theory of reasoned action (Ajzen & Fishbein, 1980) have provided fruitful conceptual tools for understanding parenting behavior. More recently, self-determination theory (e.g., Deci & Ryan, 2012), has offered insights into child rearing, parent–child relationships, and the ways parents influence their children’s development of internalization (e.g., Grolnick, Deci & Ryan, 1997). However, none of these

theories form what can be considered to be foundational theories of parenting.

Theorizing about the role of parental behavior on children's development began with Sigmund Freud. His stage theory (oral, anal, phallic, latency, and genital stages) focused on child fixations and parents' role in helping children successfully navigate through each stage (White, 1960). Despite recognizing the important role that parents play in their children's mental health, he devoted relatively few words to explicating adaptive and maladaptive child-rearing practices. More problematic was that when Freudian theory was put to empirical tests, it did not fare well (e.g., Chodoff, 1966). Consequently, his model of development fell out of favor.

In contrast, six distinct foundational theoretical perspectives gradually emerged that have been supported by multiple empirical studies and currently form the conceptual underpinning of contemporary research. Not all of the theoretical perspectives reflect or adhere to a single theory; some can be better characterized as theoretical orientations. Nevertheless, distinct conceptions about parents or parent-child relations can be identified. The six foundational conceptualizations are evolutionary theory, attachment theory, socialization theories, behavioral genetics theory, social cognition theories, and systems theories.

Evolutionary Theory

Chronologically speaking, the first foundational theory about parent-child relationships is evolutionary theory. Darwin's articulation of the concept of natural selection (1859) and sexual selection (1876), created the elements of evolutionary theory and set the stage for thinking about parents and children from this perspective. This model of development views parenting from a distal vantage point. Darwin's concepts of survival, fitness, and

natural selection have led to a distinctive perspective for thinking about parents and children. Specifically, the question at the core of evolutionary theory with regard to parents is how has child rearing evolved over the past 35,000 years, when anatomically modern humans emerged and lived in hunter-gatherer communities. What behaviors optimize the survival of their offspring? Given that 99% of human generations have lived in hunter-gatherer societies, evolutionary theory attempts to explicate how a parent's behavioral propensities reflects and is affected by selection processes.

One striking example of this theory comes from Konrad Lorenz (1943) and other ethologists, who proposed that the endearing facial features of infants, what is commonly called "cuteness" – a reaction to observing large heads and eyes but small noses and chins – led to increased survival of offspring (see Kringelbach et al., 2016). Those distinctive characteristics result in eliciting caregiver positive affect and attention.

Evolutionary theory has resulted in asking unique questions like: What is an individual's mating preferences?; Why do parents nurture some children but abuse others?; and What parental roles and behaviors are evolutionarily selected for? Evolutionary theory has led to speculation about parental investment (Trivers, 1974) and mate selection. Parents (especially mothers in traditional societies) devote a great deal of time, energy, money, and thought to rearing their children.

Why do some parents spend so much time with some children while other parents may be largely uninvolved? Taking an evolutionary perspective, Geary (2006) argued that the answer lies in the amount of shared genetic material, the offspring's likelihood of survival, and the future likelihood that the child will have children. Differential parental investment offers one explanation of why some parents physically abuse their children. According to

at least some evidence (Daly & Wilson, 1996), parents are less likely to maltreat biological children than non-biological children. With regard to mate selection, from an evolutionary perspective, females value the “three Gs” in males: good genes, good providers, and good fathers (Chang, Lu, & Zhu, 2017). In contrast, males seek females with physical features indicating fertility, women who will be faithful, a good provider, and a good mother.

It is important to recognize that contemporary evolutionary views of child rearing (e.g., Bjorklund & Jordan, 2013; Narvaez et al., 2016) are not deterministic; rather they reflect speculation about predispositions that have been selected for over hundreds of thousands of years. The evolutionary perspective focuses on identifying what those genetically inbred inclinations are. Such tendencies are not inevitable and can be overcome by awareness and resistance.

Attachment Theory

Evolutionary theory provided an important theoretical influence on the development of attachment theory as developed by John Bowlby (1988). Bowlby, a psychiatrist with ample training in Freudian and neo-Freudian theory, developed his evolution-informed theory that focused on the development of emotional bonds between the parent and child. The core concept is the need of children to develop secure bonds with their parents. This bond is based on caregivers’ responsivity. Children then create internal working models of whether other people are trustworthy and whether they themselves are worthy of love. Bowlby teamed up with Mary Ainsworth (e.g., 1967) who provided initial data to support the theory in Uganda and then Baltimore.

Attachment theory has continued to advance beyond simply the development of affectional ties between a parent and child,

and the repercussions of being securely or insecurely attached. The theory has expanded into such topics as the internalization of values and predicting the quality of intimate relations (Cassidy & Shaver, 2016).

The theory has also spawned other attachment-related theories. One prominent theory that is narrower in scope than Bowlby’s is Emotional Security Theory (Davies & Cummings, 1994). This theory was designed to account for children’s emotional reactions to, and the impact of, marital discord. In its latest formulation, the theory goes beyond the parent–child relationship in positing how children react to a variety of familial and extrafamilial social threats (Davies & Martin, 2013). A second attachment-related theory is Interpersonal Acceptance-Rejection Theory (Rohner, 2016). The theory is centered around the idea that parental love results in positive outcomes and rejection negatively affects a child’s psychological and behavioral functioning. It has now been expanded to consider the impact of perceptions of acceptance and rejection in other relationships across the lifespan.

Socialization Theories

Socialization can be simply defined as “the process whereby the child becomes a social being” (Grusec & Lytton, 1988, p. 161). But it is far from a simple process. It consists of all the ways that children are taught – intentionally or not – or assisted, and become mature, competent, and contributing members of a society (Grusec & Davidov, 2010). Theories that address the ways that parents and other social agents affect their children’s development began with Freud. But it was the provocative work by John B. Watson (1928), who based his views about development on classical conditioning theory, that helped to prompt theorists to account for how children learn from their social environment. Among

the many social learning theories, the two most influential have been those articulated by B. F. Skinner (1976) and Albert Bandura (2018).

Skinnerian theory, also known as operant conditioning theory, focused on analyzing the behaviors that functioned to increase or decrease the re-occurrence of actions. His theory concerned the stimuli that served as rewards or punishers and the behavioral consequences when stimuli were applied or withdrawn. His meticulous empirical animal research underlying his theory led to a rich understanding of the properties of effective actions that serve as reinforcers and punishers.

Skinner's theoretical explication of operant principles continues to profoundly influence socialization theories – particularly in the area of discipline – and the origins and maintenance of many of children's behavior problems. His theory helps to explicate the common errors parents make by reinforcing undesired behaviors and failing to reward desired actions (e.g., O'Leary, 1995). His theory has also led to the development of narrower theories, such as Gerald Patterson's coercion theory, developed to account for coercive cycles in parent–child relationships (Patterson, Reid, & Dishion, 1992).

Learning principles also played a key role in Albert Bandura's socialization theory. His legendary modeling studies with Bobo dolls (Bandura, Ross, & Ross, 1963) set Bandura on a course of understanding the fundamental roles that direct learning and observational learning play in the establishment of new behavior. What is observed may or may not be mimicked, depending on a series of cognitive components (e.g., attention, memory, imagery, and motivation). Once a behavior is established, it can then be readily maintained through reinforcement. Bandura also showed that children are more likely to imitate those models they perceive as powerful and nurturant – both attributes common to parents.

Although his theory can be regarded as a socialization theory, as it developed, it has grown to be progressively more cognitive; its name has changed to *social cognitive theory* (Bandura, 2018) due to his emphasis on motivation, expectancies, and self-efficacy. It now emphasizes the role of human agency in driving actions and an individual's feelings of self-efficacy. This theory helped to focus attention on parental self-efficacy as a fundamental component of effective parenting (Bugental, Blue, & Cruzcosa, 1989; Jones & Prinz, 2005).

The newest socialization theory is the Social Relational Theory (Kuczynski & De Mol, 2014). This theory is designed to capture the dynamic nature of transactional processes inherent in socialization; both the parent and child are recognized as social agents. The process of child rearing is conceptualized as a series of interchanges based on expectations (broken as well as met), conflicts (between parents' and children's needs), solving these conflicts, ambivalence (in the face of competing goals), and ambiguity (e.g., uncertainty about how the child will behave in the future). The inherently dynamic nature of parenting is nicely captured – although such a theoretical model is extremely difficult to study.

Behavioral Genetics Theory

A fourth foundational theory comes from behavioral genetics, which focuses on understanding the role that genes play in individual differences. A foreshadowing of this conceptualization of parenting appeared in 1895, with a paper authored by the mathematician and biostatistician Karl Pearson. Since then, researchers have explored the dual influences of genes and the environment using various methods. The traditional method for investigating human behavioral genetics consisted of twin studies, adoption, and family studies (Sherman et al., 1997). By assessing the

similarities and differences between monozygotic and dizygotic twins, adopted and biological children, and family members sharing different amounts of genes, an estimate of heritability could be generated. Those methods have now been largely supplanted with more sophisticated approaches, including quantitative trait loci analysis, biometric model fitting, and animal models.

Early behavioral genetic conceptions appeared as deterministic and unidirectional views about the effects of genotypes and genetic inheritance. The theory has seen dramatic change, partly due to an increasingly sophisticated understanding of how genes work in relation to the environment. A second factor prompting change was the low magnitude of the correlation coefficients between genes and outcomes, such as the impact of child rearing and genotype–environment interactions (Rutter, Moffitt, & Caspi, 2005). Nongenetic factors, including the environment and measurement error, must account for the rest of the differences. Those views evolved into a parallel focus on the role of the environment (e.g., McGuire, Segal, & Hershberger, 2012). For example, the concept of shared and nonshared environment emerged in order to better explain phenotypic differences between twins (Plomin & Daniels, 1987; Turkheimer & Waldron, 2000). Understanding gene–environment interactions has become a prominent focus of the theory. As Scarr and McCartney (1983) explicated, genes can play one or more roles – a *passive* role in the environment, an *active* role in directing a child to seek out certain environments, or an *evocative* role when parents react to a child’s phenotype in a particular way unique to that child.

Behavioral genetics theory has been useful for understanding normative and abnormal individual differences in children, as well as individual differences in parenting (Klahr & Burt, 2014). More recent work has focused

on understanding the role of particular genes and gene X environment interactions (McCormack et al., 2009). Investigations now address how genetic variants are related to neurotransmitters and hormones such as dopamine, serotonin, oxytocin, and vasopressin, all of which have been shown to influence parental behavior. Researchers are also examining the role of specific genes in development and how they may interact with the environment, including dopamine D4 receptor (DRD4) and monoamine oxidase (MAO-A), a gene that codes for the enzyme that breaks down serotonin (Grusec et al., 2013).

Social Cognition Theories

A very different set of conceptualizations concerning child rearing emerged with social cognition theories, designed to explicate how cognitions about oneself and others impact behavior. The theories have not been specific to parents, but rather reflect concepts about adult social cognition. The early research into parental attitudes foreshadowed the foundational role that these theories would play.

There are a number of social cognitive theories developed by social psychologists. An early and prominent one in parenting research was attribution theory (Kelley, 1967). There are now a large number of investigations into parental attributions about children’s behavior and how those assessments drive both behavior and emotions in parents (e.g., Dix et al., 1986; Lansford et al., 2011). Another prominent social cognition theory is the Theory of Planned Behavior (Ajzen & Fishbein, 1980). This theory recognizes that behavior stems from behavioral intentions that may or may not be expressed. Those intentions are influenced by an individual’s background, the social environment, behavioral preferences, and perceptions about how normative the behavior is. This particular theory has been

useful for guiding efforts to change parenting behavior by modifying child-rearing attitudes (e.g., Durrant et al., 2014) and helping parents to prevent obesity in their children (e.g., Andrews, Silk, & Eneli, 2010).

A third example of a prominent social cognition theory focuses on social information processing (e.g., Crick & Dodge, 1994). This theoretical approach centers on how individuals think about and then respond to problems. Behavior is recognized as developing from a sequential series of steps, starting with an individual's identification of a problem. Subsequent steps include the assessment of the causes of the problem, generation of potential solutions, evaluation of those solutions, and then testing and evaluating a potential solution. Social information processing theory has been applied to parental problem-solving (e.g., Holden, 1988) and to parents who maltreat their children (Azar et al., 2017; Milner, 2003).

Systems Theories

A final foundational perspective for parenting research comes from systems theories. These theories conceptualize parents as individuals embedded in and potentially influenced by larger systems. There are two prominent systems theories that relate to parents. The single most comprehensive system theory is the Bio-Ecological Systems Theory (Bronfenbrenner, 1979; Bronfenbrenner & Morris, 2006). Urie Bronfenbrenner conceptualized the developing child as being rooted in a series of contexts that interact directly or indirectly with one another and with the person. Bronfenbrenner's well-known theory explicates how children's characteristics network with multiple, hierarchically organized levels of the environment. He identified four levels that range from the immediate setting to the broad, cultural characteristics that reflect how a country's social policies or customs can

affect the context of a child's development. The theory highlights the connections and interactions between the contextual levels as well as how a child's development can modify the system of relationships.

The Bio-Ecological Systems Theory helped to focus attention on the role that context plays in the lives of children and their parents. The theory provided a structure to integrate diverse research results, such as the influence of different types of external environments (e.g., work, social networks, and neighborhoods) on the adaptive and maladaptive functioning of families (Gibson et al., 2010). It also provides researchers with a useful framework for studying the different contextual influences on individuals, the interconnections between variables, and how the child-rearing environment helps to shape children's development.

The second leading systems conceptualization of parent-child relationships is the Family Systems Theory. This theoretical orientation addresses the nature of family dynamics, different family structures, and such concepts as roles, communication patterns, boundaries, triangulation, and power relations (Bowen, 1978; Minuchin, 1985). It focuses on understanding connections between the parents, the parent and child, and the child's behavior. One of the first developmental psychologists to recognize the importance of family systems as it relates to parenting was Ross Parke (1978), who wrote about how the family constellation and contextual variables influenced the mother-infant dyad.

A key family system concept for parents is co-parenting – how mothers and fathers function together in their roles as parents and, in particular, whether the parents are mutually supportive and involved. For example, if one parent takes over the tasks of an ill parent, that would demonstrate support and cooperation. Alternatively, if one parent disparages and undermines the efforts of the other parent, that

would be an example of negative co-parenting. Investigators have identified a number of separate components of co-parenting, including conflict, disparagement, cooperation, and triangulation, all of which help to better explicate how parents interact together with regard to child rearing (McHale & Lindahl, 2011).

Summary of the Foundational Theories

Each of the six theories reflects a distinctive theoretical foundation for investigating and understanding parent–child relationships. With each theoretical orientation comes a set of lenses with which to view and understand parenting. These lenses provide a distinctive angle of viewing the phenomena, with some theories, notably evolutionary or systems theories, providing a wider angle while others, such as social cognitive theories focus on dyadic relations. Each foundational theory has unique concepts and particular aspects of the parent–child relationship that it focuses on.

These six foundational theories have helped to guide and evaluate parenting research for almost 100 years. However, few researchers stick to one theoretical approach. More common is a researcher using different theoretical perspectives over time or incorporating multiple theories in one study. Two examples of investigations that merge multiple theoretical approaches into one study are the integration of attachment theory with family system theory (Rothbaum, Rosen, Ujiie, & Uchida, 2002) and the combination of attachment theory with a socialization theory (Juffer, Bakersman-Kranenburg & van IJzendoorn, 2017).

The History and Establishment of Parenting Science Research

In his 1998 authoritative account of the history of developmental psychology, Robert Cairns identified three ages of research: the emergent

period (1890–1919), the middle period (1920–1946), and the modern period (1947–present). Parenting research proved to be a late-comer in developmental research; virtually no studies about parents appeared during the emergent period. In fact, parents, child rearing, and parent–child relationships were rarely mentioned in early research publications. Much of the work in the emergent period of research was atheoretical and simply descriptive of children’s behavior, often ignoring the parents.

Nevertheless, at least one foundational theory served to initiate or motivate early studies into child development relationships. This was the case with “baby diaries” that were motivated by evolutionary theory (Darwin, 1877; Preyer, 1882). Those diaries ignored parents and even when a researcher carefully observed newborn social behavior, parents went unmentioned (e.g., Blanton, 1917). Parents continued to be mostly neglected through the 1920s, despite adolescent delinquency becoming a topic of inquiry (Harris, 1953). John B. Watson, the father of behaviorism, disregarded the role of parents in many of his journal articles. In their joint paper on studies in infant psychology, Watson and his wife included the word “parent” just five times (Watson & Watson, 1921). In two of the cases, “feeble-minded” parents were described as producing similarly impaired children.

One investigation conducted during the emergent period of developmental research was a questionnaire study about a basic child-rearing practice: discipline techniques. In collaboration with his mentor, G. Stanley Hall, Charles Sears (1899) developed a series of questionnaires to study the attitudes of 486 adults toward the punishment of children. Sears failed to mention the role of parents but he found that adults regarded misbehaviors as most punishable when perceived as intentional, persistent, or due to repeated

carelessness; findings that foreshadowed a more methodologically rigorous study almost 90 years later (e.g., Catron & Masters, 1993). With the exception of that study, empirical studies into child-rearing topics or parents were missing during the initial period of developmental psychology.

The Emergence of Research into Parenting

The neglected parent began to receive attention during the middle period, when socialization emerged as a dominant research concern (Harris, 1953). One study on the effects of parental communications provides a good illustration. Hurlock (1924) investigated the role of praise and reproof as incentives for children, presaging the past 20 years of research on the effects of praise by Carol Dweck and many others (e.g., Mueller & Dweck, 1998).

A second topic that began to receive attention was that of parental love and affection. Along the lines of Watson (1928), who advised parents in his book against kissing and coddling their children, the psychiatrist David Levy began publishing a series of papers and books on the problem of excess mothering – a problem he labeled “maternal overprotection” (Levy, 1931). Levy warned that mothers who engaged in infantilization of the child failed to allow their children to experience autonomy, persisted in providing excess contact, or engaged in too much or too little discipline, which could damage a child’s development (Levy, 1943). His work led the way to studying other parental “excesses” in parenting, such as perfectionism, indulgence, permissiveness, and strictness (see review by Bakwin & Bakwin, 1942) as well as underscoring the challenging nature of parenting, with its changing demands for providing children with both guidance and freedom.

The bulk of research in the late 1920s and 1930s reflected investigations into a century-long exploration of parental discipline and its impact on children (Becker, 1964). What was probably the first systematic attempt to measure child-rearing attitudes appeared in 1927 with the publication of Gertrude Laws’ Columbia University dissertation. She recruited fifty mothers enrolled in a parent education class and had them fill out 346 questions across four types of surveys. The surveys consisted of four different ways of assessing attitudes, including crossing out annoying words and making free associations. The central surveys assessed the mothers’ attitudes about child-rearing practices (e.g., spanking, nagging, isolating, breastfeeding), and perceptions of their children’s current positive and negative behaviors in light of their expectations (e.g., obeys, gets up cheerfully, bullies playmates, displays social prejudice). She found that mothers displayed what most parents could today consider as appropriate child-rearing attitudes: most mothers thought they should not ridicule or humiliate (90% and 82%, respectively) their children, nor “throw cold water on their enthusiasms” (80%). Laws recognized a number of methodological limitations with her study, including the problem we now label as “social desirability” and the need to develop “relatively objective” means of measuring parental attitudes and practices, a need that continues today.

By the early 1930s, parental attitudes had become the research rage. Some of the impetus for that work came from Sigmund Freud and his disciples. Freud devoted little attention to parents or child-rearing techniques but, as indicated previously, did position parents in a critical role for their children’s psychological development. It was left to his disciples, namely Edward Jones (1923), Karen Horney (1933), and Margaret Ribble (1943) to consider the impact of parental attitudes on

children. Another parental attitude researcher was Ralph Stogdill, who investigated such topics as attitudes toward undesired child behavior, parental control, and “mental hygiene” (e.g., 1931, 1936). Other investigators documented attitudes about children’s independence (Koch et al., 1934), self-reliance (Ojemann, 1935), and sex education (Ackerley, 1935). There were enough published studies into child-rearing attitudes to allow Stogdill to publish the first review of research into parental attitudes in 1936.

During this time more and more child-rearing practices began to be linked to developmental outcomes. For example, Goodwin Watson (1934) recruited 210 graduate students to respond to a 200-item survey. He discovered that students from strict families reported many more negative attributes than the other students with lax parents: they disliked their parents, had “quarrelsome” attitudes, were more socially maladjusted, and suffered more from feelings of guilt, worry, and anxiety.

A similar type of investigation was published by Percival Symonds in 1938 on the topic of parental acceptance and rejection. He collected reports from thirty-two pairs of former students and divided them into those who reported being rejected or accepted by their parents. His impressionistic account indicated that youth who felt accepted by their parents were cooperative, friendly, stable emotionally, and cheerful. In contrast, youth who reported experiencing parental rejection tended to be restless, engage in attention-getting behavior, and be known as troublemakers in school. That study served as a forerunner to contemporary international research on parental acceptance/rejection (e.g., Rhoner & Khaleque, 2010).

Researchers were not the only drivers of investigations into child rearing. Cora Bussey Hillis recognized the need for research into the best ways to rear children and petitioned the

Iowa legislature to establish a child study laboratory. Her efforts eventually came to fruition in 1917. Simultaneously, in the early 1900s, the child development branch of the home economics movement gained steam with the impetus of developing educational material for parents (Smuts, 2008). Shortly thereafter the child guidance movement began and provided another practical impetus for research into parent–child relationships (Horn, 1989). In the 1930s, interest in parent education was galvanized, largely due to the 1930 White House Conference on Child Health and Protection (Croake & Glover, 1977)

Early Landmark Parenting Studies

What Cairns (1998) dubbed as the “middle period” of developmental science brought the first set of landmark parenting investigations. The researchers involved in this trilogy of studies dramatically changed the nature and scope of parenting studies by increasing the types of methodologies used, expanding the number of variables studied, and, in general, enhancing the rigor of the research. As a consequence, these changes served to promote improvements into parenting research efforts.

The first prominent longitudinal study began in 1929. Researchers at the Fels Research Institute developed scales for observers to rate 125 families on 30 variables including global adjustment, rapport with child, and severity of punishments (Champney, 1941). Given factor analysis had not been developed, Albert Baldwin and his colleagues (Baldwin, Kalhorn, & Breese, 1945) used a procedure called “syndrome analysis” to identify three fundamental patterns (warmth, democracy, indulgence) along with several secondary ones. Their comprehensive methodology included maternal interviews, home observations, and child observations in nursery school. They

found that warm-democratic parenting was linked to several positive child outcomes, including greater spontaneity, more advanced cognitive development, and less anxiety.

The second landmark study came from Stanford University researchers. In 1957 Robert Sears, Eleanor Maccoby, and Richard Levin published a book describing their study of 379 mothers of young children. Mothers participated in extensive interviews addressing dozens of child-rearing practices including feeding, aggression, dependency, toilet training, sexual development, and disciplinary practices as well as their children's behavior and indicators of conscience development. Maternal reports were then rated by the researchers and those ratings were correlated with children's behavior. Despite efforts to find a consistent pattern of child rearing linked to the children's sex or birth order, or the mothers' age, education, or socio-economic level, few significant patterns could be discerned. Instead, the researchers were left with the hollow conclusion that parental behavior "varied widely."

A final landmark study also emerged from California. In 1971, Diana Baumrind published her highly influential monograph. She used interviews, questionnaires, and observations of 146 white mothers of preschool-aged children to arrive at a typology of authoritarian, authoritative, and permissive parenting styles. Child behavior patterns were associated with each style. As found in virtually all developmental psychology textbooks, the parents with authoritative styles, characterized by warmth and responsiveness but firm control, who provided explanations with high behavioral expectations, had children who were likely to be independent, socially skilled, confident, and perform well in school. When the children became adolescents they were least likely to develop mental health or behavioral problems (Baumrind, 1991). In contrast,

authoritarian parents were strict, demanded obedience, and were unlikely to listen to their children's points of view. Their children tended to be less competent, have lower self-esteem, and experienced more problems as they developed. Baumrind's third group, consisting of permissive parents, were warm and responsive but did not have high expectations, had few rules, and were indulgent. Their children were most likely to be impulsive, egocentric, and exhibit poor relationship skills.

Textbooks commonly tout her findings, but a close reading reveals the results were not clear cut. The associations between parental style and child behavior were affected by the sex of the child as well as the parent's personality. In addition, critics recognized methodological limitations with all three of those studies, such as problems associated with data derived from a single reporter, a neglect of fathers, and the use of homogeneous, non-representative samples of parents. Another problem was that Baumrind's typology was incomplete. As pointed out by Maccoby and Martin (1983), neglectful parents who were low on both warmth and control had been omitted from her typology.

One other important limitation to those landmark studies is now well recognized: the unidirectional view of influence and the failure to recognize a child's impact on parents (Bell, 1968; Lewis, 1981). In a review, Bell (1968) discussed how children's behavior could readily elicit particular behavioral responses from parents. For example, parents do not need to exert control if their children are compliant or cooperative. Children of authoritative parents are likely to elicit warm, open parental communication simply by exhibiting more mature and compliant behavior. That observation about the role of child characteristics in prompting specific parental behaviors led to a body of research detailing how children's characteristics can impact parental behavior.

Despite the limitations of and qualifications to her findings, Baumrind's conceptualization of parenting styles continues to be influential and guide many contemporary parenting studies (Larzelere, Morris, & Harrist, 2013). The three landmark studies helped to create an enduring appetite for studies into parenting, parental roles, and how child rearing is associated with children's behavior.

By the 1950s, investigations into parenting were appearing with regularity in journals such as *Child Development* (since 1930), *Merrill Palmer Quarterly* (since 1954), and *Developmental Psychology* (since 1969). Research into parenting has expanded rapidly over the last 40 years or so. Researchers now explore a wide range of parenting topics that go beyond the traditional topics of parenting style, child-rearing discipline, or parent-child affectionate relationships. In the 1970s, there was an explosion of parenting topics that began to be studied, including fathers (Lamb, 1975), Black and "Chicano" parents (Bartz & Levine, 1978), and grandparents (e.g., Kahana & Kahana, 1970). The bias in some of the clinical research that continued through the early 1980s to blame mothers for child problems has now been recognized (Caplan & Hall-McCorquodale, 1985).

Currently hundreds of diverse topics concerning parent-child relations are being studied. Topics include examining the impact of family structure or characteristics (e.g., single-parent-headed households, sexual orientation of parents), relations between variables, parents, and children (e.g., sensitive caregiving, child aggression, use of screen time), and parenting processes, such as the transition to parenting or the aftermath of a divorce. There is now one journal (*Parenting: Science and Practice*) exclusively devoted to research on parents and child rearing, along with hundreds of other journals that regularly publish empirical parenting articles. In an effort to organize the vast research

literature on parenting, different approaches can be identified and will be described next.

Contemporary Research Approaches

At least eight distinct approaches to parenting research can be identified in research studies (Holden, 2021). Some approaches derived from the landmark studies, while other ways of studying parents emerged from newer conceptualizations or methodological advances. The eight approaches are defined and described with examples in Table 1.1; they will be briefly summarized below.

Studying *parenting traits* is the oldest of the eight approaches. With this approach, child rearing is conceptualized as reflecting one particular parenting global style. Instead of considering parenting dimensions, which are variable-centered (e.g., warmth, control), a parenting style encompasses multiple dimensions and is person-centered. These styles of interacting with children are assumed to be constant and stable over time – assumptions without sufficient empirical support (Holden & Miller, 1999). More than a dozen different parenting styles have been studied. Examples include accepting vs. rejecting, child-centered vs. parent-centered, involved vs. autonomous, facilitator vs. regulator, along with Baumrind's parenting styles discussed previously.

This approach is commonly used to link parenting styles with child outcomes. For example, using Baumrind's typology, researchers found that authoritative and autonomy-granting parents typically have youth who demonstrate greater competence and well-being (Lamborn et al., 1991; Ryan, Deci, & Grolnick, 1995). Many other styles have been examined, including sensitive, warm, overprotective, and, more recently, helicopter parents (e.g., Fingerman et al., 2012).

In reaction to the trait approach is the *child effects and transactions* approach. The

Table 1.1. *Eight major approaches to studying parenting*

Approach	Focus	Example
Trait	The relation between stable parenting characteristics and child outcome	Characterizing parents as authoritative, authoritarian, permissive, or neglectful
Child effects and transactions	How child characteristics or behavior affect parenting	Demonstrating how a child's challenging behavior can affect parenting
Social learning	The effects of parental use of social learning principles on children	Illustrating how positive reinforcement or punishment affects child behavior
Social address	How place of residence impacts parental behavior	Showing how parenting differs or is similar across racial/ethnic groups
Parent cognition	The ways parents think about their children and/or child rearing	Documenting how parents' attribution of intentionality can affect their emotion
Ecological momentary	Examination of parenting can change from moment to moment	Revealing how parental behavior can change across situations
Behavioral genetics	The dual roles played by genes and the environment on parenting and children	Explicating cross-generational phenotypic similarities and differences in child rearing
Large sample longitudinal datasets	The impact of antecedent, moderating, or mediating parenting variables on child outcomes	Testing causal relations between use of corporal punishment and child behavior

underlying model reflects the fact that parent–child relationships and influence are bidirectional and transactional (Bell, 1968). Studies classified under this label demonstrate how parenting can be a response to particular child behaviors or characteristics, or that parents and children modifying their behavior in reaction to the behavior of the other, fall under this category.

A large body of studies has accrued, demonstrating how general child characteristics (e.g., age, sex), physical attributes (e.g., height and weight, attractiveness), and specific behaviors (e.g., cooperativeness, type of misbehavior, activity level) elicit specific reactions or differential parental behavior. That approach, which began as illustrations of bidirectional influence in relationships, has progressed to

include transactional models – reflecting how children influence their parents and then parents subsequently influence their children. The concept of differential susceptibility fits under this approach. This concept captures the idea that not all children respond in the same way to the parenting and the environment (e.g., Belsky, 1997; Van Zeijl et al., 2007). Some children, due to their genotype, may be more malleable or susceptible to negative or positive environmental influences – such as parenting practices (Belsky & Pluess, 2009).

A third category of studies can be categorized under the *social learning approach* because parent–child relationships are conceptualized from a social learning framework. This approach, a dominant one since the 1950s,

views the parent as the dispenser of learning principles. Terms like modeling, attention, reinforcers, and punishment are tell-tale indicators of this approach. Gerald Patterson's analysis of coercive cycles in parent-child interactions (e.g., Granic & Patterson, 2006) is an exemplar. Other examples can readily be found when there is a focus on stimulus and response relations, as is often found in analyses of parental discipline (e.g., Dowling et al., 2009).

Like all approaches, the social learning approach is not static but has evolved. The approach has increasingly become more cognitively oriented over the past several decades. Bandura led the way for this development when he changed the name of his social learning theory to Social Cognitive Theory (as was previously discussed).

A fourth approach to parenting research concerns how child-rearing practices and cognitions vary across locations. There is now a large body of research that reflects the *social address* approach and documents how locations can influence child rearing. Anthropologists, such as Margaret Mead (1928), published early illustrations of this. Sociologists soon adopted the approach, for example in a study that explicated the relationships between social class, race, and child rearing (Davis & Havighurst, 1946). Today the social address approach is often reflected in studies that examine such variables as nationality and culture, region of the country, socio-economic status, or race/ethnicity.

The social address approach set the stage for current work into cross-cultural and international research (e.g., Bornstein, 2012; Minturn & Lambert, 1964). It also ushered in a wide variety of research, including neighborhood effects on parents' disciplinary practices (Ma, Grogan-Kaylor & Klein, 2018), how Latinx immigrant parents adapt to life in the United States (Perreira, Chapman, & Stein,

2006), and Canadian aboriginal parents' child rearing (Muir & Bohr, 2014). Increasingly, social address studies into child rearing focus on the reasons why a particular location may impact parenting.

The dawn of the fifth approach to studying parents, the *parent cognition* approach, began with the parent attitude research of the 1930s. But credit for the first comprehensive empirical study into parental cognition goes to Lois Stolz (1967), who published an empirically-based monograph focusing on the cognitive influences that drive parental behavior. Walter Emmerich (1969) was another early pioneer of this approach, with his examination of parental goals and some of parents' "means-ends" beliefs. Ross Parke (1978) ignited interest in the approach when he pointed out that the study of parental thinking was necessary in order to understand how objective and subjective cognitions mediate child-rearing behavior. Historically, like other parenting topics, most of the research into parental social cognition has focused on mothers. However paternal attitudes are increasingly being studied (e.g., Lansford et al., 2011).

The parent cognition approach has grown to include many different aspects of social cognition. Researchers now study attributions, attitudes, expectations, beliefs, problem-solving, self- and other-perceptions, and various types of reflective thought, such as reflective parenting (Holden & Smith, 2019). Although linking mothers' and fathers' thinking to their behavior needs more attention, considerable strides have been made in understanding the numerous facets of parental cognition.

Ecological momentary approach is a sixth method researchers have used to investigate child rearing. It is uniquely suited to capture dynamic and complex parent-child interactions. This approach focuses on how parent or child behavior may change moment to

moment or minute to minute, both within and between families. In contrast to survey methods, the observations collected under this approach, whether they are daily diaries, videos, or audio recordings, provide large amounts of rich ecologically valid, naturally occurring data. Early examples of this approach, as they applied to parenting research, appeared in studies with infants. For instance, Howard Moss (1967) examined parent–infant interactions at 60-second intervals over an 8-hour period, in order to document subtle changes in moment-to-moment interactions. This and other infancy studies became uniquely useful for revealing how parents promote the development of communication, reciprocity, and turn-taking in their infants (e.g., Brazelton, Koslowski, & Main, 1974).

The ecological momentary process approach is highly time intensive and challenging to utilize. However, it provides distinctive data, from latency information to subtle changes, that could not otherwise be studied. Holden, Williamson, and Holland (2014) used digital recording devices to record home interactions between mothers and their young children. Using an event-sampling approach, they discovered temporal latencies related to mothers' use of corporal punishment that had never been documented before. The approach is being used to address a wide variety of questions regarding parents and family functioning (Smyth & Heron, 2013).

In 1895, the famous statistician Karl Pearson, a founder of the field of statistics, authored a paper on inheritance from parents. This paper signaled the launch of a seventh approach to parenting research – that of *behavioral genetics*. This approach focuses on trying to tease apart the role of genes in parental behavior as well as the role of the environment. Much of the early work used twin and adoption studies to estimate genetic

inheritance (Plomin & Bergeman, 1991). Since then many investigations have attempted to assess the relative influence of genes and the environment as well as their interaction in parenting and a variety of children's outcomes (e.g., Plomin, DeFries, & Loehlin, 1977).

Investigators taking the behavioral genetics approach identified the important conceptual distinction concerning a child's environment – that parts of a child's experience are shared with other siblings, while other parts of a child's experience may take place in non-shared environments (Rutter & Silberg, 2002). For example, using this distinction, one can better tease apart genetic vs. parenting influences on antisocial behavior (e.g., Deater-Deckard & Dodge, 1997). More recent work utilizes the genome and identifies specific genes, from the 25,000 that are found on our chromosomes, that may be associated with aggression (e.g., dopamine D4 receptor and monoamine oxidase). Behavioral genetics research into parenting currently focuses on gene–environment interactions (e.g., Belsky & van IJzendoorn, 2017) and epigenetics (e.g., Scorza, et al., 2019).

The eighth common approach to parenting studies involves using *large sample, longitudinal datasets* to assess the relations between parenting characteristics or behavior, and children's development or outcomes. Among the first of these studies was the Panel Study of Income Dynamics. Begun in 1968, it is a nationally representative sample of over 18,000 individuals from 5,000 families in the United States. Findings from that study, along with a supplemental research component, have resulted in more than 4,500 journal articles addressing a range of topics that include the child, family structure, poverty, racial/ethnic differences and child, youth, and young adults' outcomes. Two other exemplars of this approach are the NICHD Study of Early Child Care, which commenced in 1991 with

1,364 families, and the National Survey of Children's Health, with more than 100,000 children and youth, which launched in 2003. There are now many more examples of these longitudinal studies. Although expensive to conduct, this approach is best suited among the eight to address causal relations regarding children's outcomes. Enough large sample longitudinal studies are now available to allow for reviews of such topics as fathers' involvement and children's outcomes (Sarkadi et al., 2008) and the role of child rearing in the development of childhood obesity (Luppino et al., 2010).

The hallmark of these large datasets is the use of advanced statistical methods, such as structural equation modeling, hierarchical linear models, and multivariate linear models, to reveal how parenting variables are related to children's development. With the longitudinal and large sample design, antecedent and consequent relations can be identified, as well as moderators and mediators.

Trends in Empirical Research

Given the robust ongoing research into parenting via multiple approaches, it can be hazardous to generalize about trends. That said, five general trends in empirical research into parenting can be identified. The first trend is that investigations into parent-child relationships are no longer the province of just a few disciplines, such as developmental psychology or sociology. Researchers from a large number of disciplines are now publishing investigations into parent-child relations in a wide variety of journals. It is not unusual to find parenting studies authored by anthropologists, criminologists, evolutionary biologists, family life educators, human ecologists, nursing professionals, pediatricians and other physicians, school psychologists, or social workers, to name just a few.

A second trend concerns work to fill in research gaps. The single most glaring hole in the parenting area has been the neglect of fathers. Although some research into the role of fathers appeared in the 1950s (e.g., Aberle & Naegele, 1952; Stolz, 1954), fathers were largely missing as a focus of parenting research (Biller, 1974; Lamb, 1975). Since then, a considerable amount of attention has been devoted to fathers and fathering (e.g., Lamb, 2000), although both continue to be under-represented in research.

Likewise, other types of parents have been neglected. Minority parents and parents from non-North American or European countries are under-represented in parenting research. Historically, white, college-educated and middle-class parents have predominantly been studied. Similarly, parents from different cultures and ethnicities were largely overlooked in research studies until the past few decades. Increasingly, researchers include diverse samples in their studies, including economically challenged (Middlemiss, 2003) or homeless families (Herbers et al., 2014), as well as affluent parents (e.g., Luthar, & Latedresse, 2006). Investigations into gay and lesbian parents appear with some frequency now (e.g., Patterson, 2009; Reczek, 2020). Child-rearing contributions from grandparents have also been neglected, despite the early effort to investigate that topic by Staples and Smith (1954). Efforts to correct that omission began more than 20 years ago, as evidenced by the publication of a review of the research into grandparents (Hayslip & Kaminski, 2005).

Recognition of the need to collect diverse samples has also been extended to parents from different countries. Like other areas of psychological research, parenting studies have largely consisted of WEIRD (Western European Industrialized Rich Democracies) samples (Stewart & Bond, 2002). Efforts to correct that deficit continue, and investigators

using the social address approach have extended their reach to most parts of the globe as cross-culture studies become increasingly common (e.g., Bornstein, 2012).

A third trend is the expansion of approaches. Some trait approach researchers are working to better understand child-rearing traits by returning to a dimensional view of parenting and the recognition that parenting cognition and behavior may differ across social domains, such as moral, conventional, and prudential domains (Smetana, 2017). Parent cognition studies are now being enriched through the study of implicit attitudes (e.g., Sturge-Apple et al., 2015) or unconscious influences on parenting (Holden, 2020). Studies into socialization have expanded areas of inquiry by including such topics as psychological control (Barber, 1996) and racial socialization (Lesane-Brown, 2006). Over the past decade or so, some of the more recent novel areas of research are parenting and sleep (McQuillan et al., 2019); positive parenting (Smith & Holden, 2020); screen use and social media (Lauricella, Wartella, & Rideout, 2015); parenting young adults (Cui et al., 2019), and parental burnout (Mikolajczak et al., 2018). Perhaps the most novel studies, and certainly the most methodologically sophisticated, are investigations into the neurobiology of parents. It has now been documented, with fMRI studies and other methods, that parents' brains change once a child is born (Swain et al., 2012).

A final trend is the increasingly sophisticated and nuanced understanding of parents. These advances have been achieved through expansion of areas of study in theoretical perspectives, the integration of multiple approaches, recognition of the impact of different social domains, and the inclusion of moderators and mediators in analyses. Researchers' understanding of parents has shifted from a static model to a view of

parents as thinking and dynamic, who modify their behavior based on a number of factors in the child, situation, and context. A number of researchers, such as Kuczynski, Pitman, and Mitchell (2009), have recognized the cognitively challenging task for parents of achieving an appropriate balance between parent- and child-initiated interactions – a balance that responsive parents recognize changes in synchrony with the child's development.

One aspect of research that has not shown as much advancement or sophistication is research methodology. Most notable is the continued reliance on parental self-reports. Although the validity of self-reports as well as global ratings has been a concern since the 1960s, and problems in them have frequently been recognized (Holden & Edwards, 1989), the vast majority of parenting research continues to rely on questionnaires. This is particularly problematic in areas where sensitive questions are asked, such as reporting on the use of harsh punishment or conflictual interactions. Self-presentation or demand characteristics are a likely source of bias. Observational studies are too rarely used, in part because of the expense and time they require as well as concerns about replicability from small sample sizes. Parenting researchers need to develop alternatives to self-report measures, such as is being done by Christina Rodriguez, who has created several analog methods to supplement or supplant self-report data. For example, in a series of laboratory studies Rodriguez, Russa, and Kircher (2015) had adults work on solving an unsolvable puzzle while hearing recordings of a crying baby or a tantruming toddler. For individuals with low frustration tolerance, the aversive analog procedure elicited physiological changes and intentions to use punitive discipline, akin to what can happen in homes.

Conclusion

As this chapter has described, today's science of parenting is based on multiple foundational theories and a long research history. Both the relevant theories and approaches to empirical inquiries reflect impressive developments over the past 100 years. In particular, the methodology used to study parenting has evolved from simple self-report surveys to the use of multiple methods, longitudinal designs, behavioral observations, and even brain imaging. If parenting was once thought to be simple and mechanistic, theoretical models and countless studies now recognize that it is complex, dynamic, and multi-determined. Clearly, a theoretically informed and empirically-based science of parenting is now upon us.

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2 Parenting, Challenges, Brain Development, and Attachment Strategies

Anne Rifkin-Graboi and Gwendolyn Ngoh

Parvati and Shiva's sons were arguing again, this time over a mango said to contain the truth of the universe. "Alright boys, enough. Whichever one of you is the first to circle the globe three times gets the fruit." Murugan, the faster of the two, leaped onto his peacock, and off he went. But when he returned, Ganesha was holding the fruit. "Dear brother, how is this possible when I am so much quicker than you?" asked Murugan. To which Ganesha replied, "It is because, Dear Brother, I circled Mother and Father three times for, to me, they are the world."

(Paraphrasing of the Hindu story, "The Golden Mango")

The basic themes of this chapter derive from three premises. The first is that when we are young, our parents are, in many ways, our world. At the very least, they are needed to manage most of life's challenges. The second is that the way challenges are managed has implications for brain development. The third is an offshoot of the others: we bring our histories into our new encounters. In keeping with ideas concerning differential susceptibility to context (Ellis et al., 2011), for some people, and in some circumstances, our biopsychosocial make-ups, forged through experience and/or genetic factors, make us more receptive to continued alterations in further functioning in response to new encounters throughout the lifespan.

We Need Our Caregivers: Attachment as a "Universally Given" Construct amongst Human Infants

Relatively speaking, humanity invests heavily in rearing young over long periods of time. Prolonged development has advantages, and in fact may allow for more time to adapt to environmental changes (Del

Giudice, Ellis, & Shirtcliff, 2011). However, prolonged development also involves prolonged dependency, especially during infancy when neurodevelopment is extremely protracted (Miller et al., 2012). Without caregivers, human infants would not, for example, be able to eat, thermoregulate, or avoid predation. Even after becoming able to independently locomote and navigate, human infants, have not, historically, had access to dens with narrow openings enabling retreat; likewise, we cannot easily escape in trees or the sky. Instead, our best chance of survival in times of alarm may be to escape to a person – our attachment figure, a tendency activated in the presence of true danger, and when we encounter stimuli correlated with future danger and uncertainty (Bowlby, 1969).

That is, young infants often treat "natural cues" – things such as darkness and stranger approach, as alarming even though they are not always dangerous in and of themselves. Likewise, separation from the mother may be threatening to infants and children because like darkness, it may lead to enhanced vulnerability (Bowlby, 1973), and compromised group

membership (also see Rifkin-Graboi, Borelli, & Bosquest, 2009). Group membership is so important to human survival that even indirect laboratory threats to it elicit physiological stress responses (Dickerson & Kemeny, 2004). As further evidence of the importance of group membership, the simple presence of another person is enough to alter our perception of physical features related to physical expenditure (Schnall et al., 2008).

Yet despite this helplessness, human infants are still able to signal needs. Infants can do this through gaze, cries, facial expressions, or, as they get older, motoric approach, and language (e.g., Bowlby, 1958; Soltis, 2004). Such communicatory signals may be used to indicate needs stemming from the complementary “exploratory” and “attachment” behavioral systems. These two systems reflect basic priorities in the human infant (and perhaps most animal) life – the need to learn about new things and the need for safety and security. Importantly, in early life when we are most helpless, we cannot independently achieve either of these, leading us to form a special type of affiliative bond, or “attachment relationship” with special, unique, stronger, and wiser caregivers, termed “attachment figures” (Ainsworth, 1989).

In particular, infants are likely to form preferences for and attachments to caregivers over time and repeated encounters (Benoit, 2004). Bidirectional engagement of the infant and caregiver through response-contingent stimulation such as game-like interactions, where infant cooing and smiling generates positive caregiver responses, may promote the formation of infants’ preferences for a specific caregiver (Watson, 1972). Therefore, infants and their caregivers must spend some amount of time together for an attachment relationship to develop. Moreover, whether the dyad engages in social behavior outside of instrumental

routine care may also determine whether an attachment relationship forms (Ainsworth, 1967; but see Kermorian & Leiderman, 1986).

The non-interchangeable nature of attachment relationships is considered key to distinguishing such relationships from other affiliative bonds (Suomi, 2016). Indeed, monotropy, the idea that an infant’s attachment relationship is focused on a single individual or a select group of individuals, was initially thought to be central to attachment theory (Bowlby, 1958). Monotropy may make it easier for infants to construct and organize themselves with one caregiver than with multiple caregivers, which may be useful in times of danger when infants need to quickly decide whom to signal (Sroufe & McIntosh, 2011). However, if the primary attachment figure is unavailable, alternative attachment figures are likely to be sought (Hazan & Shaver, 1994). Thus, while many may seek out their mothers in times of alarm (Carr & Landau, 2012), they may also become attached to multiple caregivers in their lives, including fathers and grandparents.

Amongst species universally recognized as forming “true attachment” relationships (i.e., monkeys stemming from Asia and Africa, as well as apes), and those where it is less clear whether “attachment relationships” have formed, the absence of an adult caregiver can have profound detrimental effects. When young rhesus macaques are, unnaturally, reared in peer groups without an adult caregiver, they are comparatively behaviorally inhibited, anxious, have difficulties forming subsequent mating and parenting relationships, and/or are more impulsive and aggressive (Suomi, 2016). Because these animals have not been raised in isolation, the sequelae are likely specifically due to the lack of an attachment figure.

Likewise, many children raised in orphanages exhibit such symptoms, and are at higher

risk for reactive attachment and disinhibited social engagement disorder – suggesting that the ability to enter unique enduring relationships has been compromised by the lack of early one-on-one experience (Fox, Nelson, & Zeanah, 2017). Work demonstrating that when children are subsequently placed into foster-care, factors such as foster-parent responsiveness to distress (Humphreys et al., 2015) and the quality of the foster-parent–child relationship (McLaughlin, Zeanah, Fox, & Nelson, 2012) lead to a reduction in many of these effects, strongly suggests that some of the aforementioned correlates of institutionalization may be consequences of the lack of attachment relationships, *per se*.

Furthermore, work performed by Sagi-Schwartz and colleagues (reviewed in Aviezer et al., 1994) examining a (outdated) child-rearing practice amongst some community-living groups, or *kibbutzim*, highlights the importance of one-on-one experience. Unlike children in institutions, *kibbutz* children have access to their families during the day and are not exposed to environmental deprivation. However, at night, some *kibbutzim* used to have children sleep in a central location, separately from their families, while under adult supervision. As a “natural cue to danger,” night-time is likely to evoke feelings of vulnerability. Nevertheless, such communally-sleeping children generally formed multiple attachment relationships, including attachment relationships to their parents. However, not having a caregiver who was uniquely devoted to particular children during the night had negative consequences for mother–child relationships, observed even in comparison to infants in other *kibbutzim* who were raised communally, but slept with their families. Hence, this work demonstrates the importance of having a devoted caregiver, specifically during challenging times.

Cost of Responses to Challenge Depends on the Experience of Threat, Production of Support Signals, and the Regulation of Behavior

Responses to challenging events can be considered to have a potential physical cost (Lupien et al., 2006). The lowest-cost solution to a challenging event is behavioral in nature (see Lupien et al., 2006; Rifkin-Graboi et al., 2009). However, low-cost solutions are unlikely when a situation is experienced as exceedingly threatening or frightening either at the onset, or over time due to failures in *regulation*. In addition, for an infant or young child, a low-cost behavioral solution generally requires parental assistance. When offspring are very young, every situation represents a challenge, which may or may not be mitigated by the parent. With age, children may develop skills needed to independently manage, but still often require parental support. While parental physical support may not be necessary, parental social support may determine whether the challenge is experienced as surmountable.

Although there are certain situations universally likely to be perceived as cause for concern, the extent to which they are experienced as manageable, threatening, or overwhelming is influenced by experience. For example, Mary Ainsworth’s strange situation (Ainsworth, 1967; Ainsworth et al., 1978), a series of standardized separation and reunion episodes between mother and child, with the introduction of a stranger (novel adult) in a novel but enticing environment, is expected to universally challenge infants and young children because it exposes them to three natural cues to danger: separation from the caregiver (and even being alone), placement in a novel room with novel toys, and interaction with a stranger. Nevertheless, several differences have been

observed with regards to how infants experience, signal, and regulate their distress in response to this confluence of natural cues to danger. Below, using the strange situation as an example, we discuss ways that experience may uniquely impact each of these processes. However, in the real world, effects are likely not mutually exclusive.

Experiencing the Situation

The extent to which infants experience the strange situation as “moderately stressful” may be influenced by temperamental factors, originating from genetic variation and/or exposure to other negative experiences, many of which co-occur with poverty (e.g., Bernier & Meins, 2008). Additionally, in keeping with Bowlby’s ideas that natural cues to danger are always more threatening when they are combined (Bowlby, 1973), children who are upset for other reasons or unwell are hypothesized to find the strange situation more threatening than those who are not (Granqvist et al., 2016). In this vein, it may not be surprising that Israeli children whose everyday lives are filled with day care that is of low quality or with low caregiver-to-child ratios, appear to find the strange situation more distressing than do other infants (Sagi et al., 2002).

Furthermore, the extent to which children experience the situation as only “moderately distressing” may be influenced by exposure to past similar experiences. Children from divorced families who experience uneasy overnight transitions between parents may find the strange situation more challenging than other children (Solomon & George, 1999). Inconsistent parenting, perhaps signaling that the parent is not always “present” for the child, associates with more child wariness at the beginning of the experiment (Cassidy & Berlin, 1994). Likewise, a history of abandonment

(Dozier et al., 2008), or recent past participation in the strange situation itself may influence infants’ reactions to subsequent strange situations (Granqvist et al., 2016).

Signaling Needs

Second, past exposure to sensitive and responsive care may influence the degree to which children signal feelings of distress, and, accordingly, are able to use the most straightforward behavioral solution to distress – seeking comfort from the caregiver to regulate their feelings. Meta-analytic findings (Verhage et al., 2016) demonstrate an association between caregiver responsiveness and child behavior during the strange situation. When a child, who has consistently been exposed to sensitive care, enters the strange situation’s novel toy-filled room, the child is able to explore the new environment while the parent remains present. During the standardized separation portions of the strange situation, such children show decreases in exploration. When the parent returns, such children outwardly signal their distress and seek out the parent, thus using behavioral signals to indicate need. In turn they receive comfort and so recognize that the threat has ended and can return to exploration. This type of distress management strategy is termed “security” (Ainsworth et al., 1978).

In contrast, children who have received inconsistent care signal needs in inconsistent ways. As discussed above, such children may overestimate their appraisals of inherent danger. They may also combine signals of distress, and signals for comfort, with displays of impassivity or anger and a desire to explore before they are truly ready to re-engage with the environment. These children are said to use a “resistant” or “ambivalent” strategy (Ainsworth et al., 1978). In addition, an “avoidant” strategy has also been identified. Specifically, when children have been rejected

by their caregivers, they are less likely to overtly signal their needs. Likewise, they do not receive comfort, nor do they return to full exploration (Ainsworth et al., 1978). It is possible that this rejection need not be extreme; even returning from low-quality care may instigate a heightened need for *exceptionally* sensitive care, which may remain unfulfilled by *normative* sensitivity, leaving children with the perceived (if not actual) experience of rejection (Aviezer, Sagi-Schwartz, & Koren-Karie, 2003). Indeed, higher rates of attachment insecurity have been reported amongst children who experience insensitivity alongside other day-care associated risks (reviewed in Belsky, 2009).

Thus, experience prior to challenging situations likely influences whether and how children signal their needs. As children age, additional factors may also influence what behavioral displays are considered “appropriate” and, ultimately, how children signal distress and whether they subsequently receive support. For example, in a study of emotional expressiveness in response to disappointing situations with four- and seven-year-old Chinese American and European American children, Chinese American children were reported to be less emotionally expressive than European American children, though this pattern depended on children’s age, with European American children showing greater decreases in negative emotion over time (Garrett-Peters & Fox, 2007). This may represent cultural variations in the adoption of acceptable behavioral expressions to disappointing situations across age. Garrett-Peters and Fox (2007) posit that the expression of positive emotions in response to disappointment may be a highly specific cultural norm for European American children. An important question, then, is whether within a (sub) culture, children signal in a manner that can be correctly interpreted by important caregivers.

Some evidence suggests that, indeed, culture not only moderates how signals are produced but also how they are interpreted. For example, children who signal distress during challenging situations are likely to be perceived by Japanese mothers as signaling needs for security, and by American mothers as exhibiting externalizing problem behavior (Rothbaum et al., 2007). Japanese mothers are also more likely to anticipate their children’s needs before children explicitly signal, while American mothers are more likely to respond after explicit behavioral displays. Still, multiple studies have reported a correspondence between secure-versus-insecure attachment amongst Japanese mothers and offspring (e.g., Kondo-Ikemura et al., 2018), perhaps suggesting that there are either limitations on the degree to which culture impacts the signaling of basic needs *or* that important aspects of sensitivity and/or security involve understanding the nuances in and intention of the partner’s behavior.

Regulating Responses

Not all children who try to signal attachment and exploration needs can fully execute these signals and regulate their behavior. Regardless of whether children use a secure, avoidant, or resistant/ambivalent strategy, some children (e.g., roughly 15 percent of middle-class non-clinical samples (van IJzendoorn, Schuengel, & Bakermans-Kranenburg, 1999)) display behavioral disorganization and/or dysregulation (hereafter disorganization) during the strange situation. Frequently, children exhibiting disorganization have had past frightening experiences with the caregiver with whom they enter the strange situation. In most cases, the fear likely has its origins in covert cues, with parents having behaved in subtly frightening, threatening, dissociated, or otherwise exceedingly anomalous ways (Madigan

et al., 2006). Less frequently, the prior fear may be due to having directly experienced abuse by the caregiver (Carlson et al., 1989).

As alluded to above, in some cases such experience may prime the child to assess the strange situation as more threatening. In addition, and as proposed by Main and Hesse (1990), children who attempt to manage the strange situation in the presence of a frightening caregiver are likely to experience a conflict during times of distress. On the one hand, as with other children, they are inclined to either signal their needs for comfort (security and resistance) or emit misleading signals regarding exploratory needs (avoidance and resistance). However, their fear of the caregiver makes it too difficult to maintain organized signals and so they show odd conflict behaviors including stilling, freezing, trance-like gazes, and mistimed or asynchronous movements (Main & Solomon 1990). In other words, children who have experienced frightening care may be least equipped to achieve a behavioral solution to challenges.

Moreover, achievement of a behavioral solution may also be influenced by baseline differences in regulatory capacity. For example, though this work requires replication, some evidence suggests that differences in a brain region (i.e., the hippocampus) important to fear learning, distress regulation, and behavioral conflict management, interacts with past caregiving to influence disorganized behavior during the strange situation (Rifkin-Graboi, Tan, et al., 2019). Likewise, Bosmans et al. (2020) have recently proposed that infants' responses to distress may be influenced by the extent to which they consistently appraise the attachment figure with rewarding feelings of safety, a learning outcome that may be genetically influenced. Indeed, variation in genes, including those thought important to reward and fear circuitry, have been investigated with attachment disorganization.

Interestingly, however, findings from large-scale observational studies do not consistently indicate that the majority of such genetic variation is relevant to disorganization (Luijk et al., 2011; Roisman et al., 2013).

When Low-Cost Solutions are Not Possible, Physiological Responses Ensurue

Above, we discuss how children's perception and experience of situations, their signaling of needs and receipt of support, and their regulatory skills and capacity may influence whether a challenge is met with a behavioral solution. What happens when it seems unlikely that a low-cost behavioral response will mitigate threat? Higher-cost physiological responses occur, such as enhanced sympathetic nervous system (SNS) arousal and decreased parasympathetic nervous system (PNS) control, as well as hypothalamic-pituitary-adrenal (HPA) activity (Lupien et al., 2006). Stress hormones, important to activity and/or energy release, are essential in circumstances requiring fleeing or fighting. SNS responses include an increase in noradrenaline (norepinephrine) and adrenaline (epinephrine) and so enhance heart rate and blood pressure. These peptides may also influence neural activity, indirectly contributing to the activation of the slower-acting, HPA system. In response to input from sources such as SNS activation, the HPA increases levels of corticotrophin releasing factor/hormone (CRF/CRH), adrenocorticotrophic hormone (ACTH), and ultimately cortisol, which releases sugar (energy) into the bloodstream for muscular use, increasing blood pressure (Ulrich-Lai & Herman, 2009).

Such higher-cost physiological responses are sometimes observed amongst insecurely attached children in the face of challenges that securely attached children may more fully mitigate via behavior. For example, when children were brought to a new preschool with

their mothers, the insecurely attached children exhibited comparatively more cortisol secretion than did their securely attached counterparts (Ahnert et al., 2004). Furthermore, as reviewed by Hane and Fox (2016), many, but not all, studies indicate that attachment security influences whether a physiological response occurs during the strange situation and how prolonged it is. Likewise, in contrast to infants exposed to higher levels of maternal responsiveness, following a parent–child challenge, infants exposed to lower maternal responsiveness relied less on a behavioral means of distress management (i.e., gazing towards the mother), and exhibited increased cardiac activity (Haley & Stansbury, 2003). Additionally, in a longitudinal follow-up, nine-year-olds, whose caregivers had participated in the control group of a randomized control trial (RCT) aimed at improving sensitive caregiving in infancy, exhibited markers of less ANS control (i.e., Resting Sinus Arrhythmia [RSA]) at the beginning of a laboratory session with their parents, than did children of parents in the intervention group (Tabachnick et al., 2019). However, in contrast to the idea that sensitive care predicts appraising all challenges as less threatening, other work examining sensitivity and infant responses to maternally evoked frustration versus short separation finds that the association between sensitivity and the magnitude of cortisol increase (as well as normative decrease) may be influenced by the type of challenge encountered (Atkinson et al., 2013). Likewise, one large-scale study investigating mothers and children in their homes demonstrated that infants of more sensitive mothers exhibited more extreme cortisol increases to challenges than did infants of less sensitive mothers (Blair et al., 2006); however, Blair et al. (2006) also found an association between higher levels of sensitive caregiving and lower baseline cortisol levels at the beginning of the study visit, as well as enhanced

regulation following challenge. This latter point is important because regulation of the cortisol response is influenced by feedback loops involving neurochemicals like cortisol. Thus, there may be interactive effects of basal levels and stress responsivity on cortisol levels over time (Ulrich-Lai & Herman, 2009).

When cortisol is released into the body, it travels back to bind to receptors along the HPA axis, as well as in various brain regions that contribute to HPA activity. Though the relations are complex, in general, when cortisol acts through receptors in brain regions associated with emotions such as anger and fear (i.e., the amygdala) the stress response is continued, but when it acts through receptors in regions associated with control (i.e., the hippocampus and prefrontal cortex) the stress response may be curtailed (Ulrich-Lai & Herman, 2009). Hence, whether feedback loops enhance or decrease activity is partially dependent upon the structures involved, an especially important point concerning longer-term effects of stress neurochemicals on neuroanatomy. Also, other neurochemicals (e.g., oxytocin, serotonin, and dopamine) may play a role (see Stockhorst & Antov, 2016), including ones which are implicated in genetic and epigenetic studies of caregiving, attachment, and development (e.g., Bakermans-Kranenburg et al., 2008; Bosmans, Young, & Hankin, 2018).

Impact of High-Cost Responses on Neurophysiology: Adaptation, Accelerated Development and Basal Functioning

Over time, whether because of increased exposure to stressful experiences, or prolonged physiological stress responses, more trait-like neural changes may occur. In animals, decreases in hippocampal properties important to stress regulation and increases in amygdala

properties likely involved in stress enhancement have been observed (McEwen, Nasca, & Gray, 2016). Likewise, in humans, lower levels of parental sensitivity and related constructs associate with smaller total brain volume (Kok et al., 2015), smaller hippocampal volume (Luby et al., 2016), and, in one small-scale study, larger left amygdala volume into adulthood (Lyons-Ruth et al., 2016). This pattern of results is consistent with ideas that stress tends to increase the effectiveness of areas that promote reactivity and decrease that of regions that exert control.

However, whether adversity predicts decreased or enhanced volume may depend upon the timing of exposure and assessment. For example, exposure to early life stress hormones may speed up brain aging. For example, if stress impacts brain aging, then in cases when a structure (or parameter) is expected to first show growth, then plateau, and then decline, early life stress exposure will associate with larger volumes in childhood, no differences in midlife, and smaller volumes during aging. It can also mean that adult-like patterns of connectivity between brain regions may be observed at earlier stages. Though much work remains to be done, complementing animal research (e.g., Sullivan & Perry, 2015) an increasing number of human studies suggests accelerated fronto-limbic development in response to early childhood caregiving adversity. For example, given that hippocampal volume is expected to increase with age, results indicating that less sensitive and/or nurturing care in early life associates with larger hippocampal volumes at six months (Rifkin-Graboi et al., 2015), ten years (Bernier et al., 2019), and in early-to-mid adolescence (Rao et al., 2010) may support ideas related to “accelerated development” (but see Kok et al., 2015; Luby et al., 2016). Likewise, associations between greater early life caregiving adversity and more mature

hippocampal microstructure values at 4.5 and 6 years (Lee et al., 2019), and early life caregiving adversity and more advanced memory capabilities (Dunn et al., 2016; Rifkin-Graboi et al., 2018) have been observed. Furthermore, resting-state (baseline) connections between brain regions important to fear regulation (i.e., amygdala-prefrontal connectivity) expected to become solidified in late childhood and adolescence, may be observed earlier in those exposed to insensitive early care (Thijssen et al., 2017). Such accelerated development may also be seen amongst children and teens who were institutionalized in early life. When faced with a challenge (i.e., viewing fear faces) formerly institutionalized children demonstrated more mature patterns of amygdala-prefrontal connectivity than did never-institutionalized children (Gee et al., 2013). Likewise, formerly institutionalized 7–16-year-olds taking part in an aversive fear learning paradigm demonstrate a comparatively advanced pattern of neural coactivation (i.e., involving the prefrontal cortex, amygdala, and hippocampus, Silvers et al., 2016).

Regardless of whether changes reflect accelerated development, trait-like changes in neurocircuitry regulating stress responses may also impact normative levels of hormones. For example, children whose parents had previously participated in an intervention to promote sensitive caregiving exhibited lower levels of daily cortisol than those who had not. Still, similar to a study examining behavioral outcomes (Bakermans-Kranenburg et al., 2008a), the effect upon cortisol was observed in interaction with a genetic factor associated with less efficient dopamine receptors and reward sensitivity (Bakermans-Kranenburg et al., 2008b).

The above work demonstrates the complex interplay between systems responsible for stress regulation, as well as the potential

influence of individual factors related to the reinforcement of learned behaviors. In addition, past and concurrent experience may interact to impact function. In many ways, the processes involved in experiencing, signaling, and responding to challenges are themselves impacted by cognitive functions (e.g., memory, emotional lability, and executive functioning including planning, cognitive flexibility, and inhibitory control) which rely, even in an acute sense, on many of the neurotransmitters, hormones, and brain regions discussed above (e.g., Shields et al., 2016).

Enrichment and Exploration

Caregiving may also impact neurodevelopment via effects on exploration. When offspring receive information that the world is either inherently dangerous, or dangerous because the parent may not be there to support them when danger does indeed arise, they may be more likely to meet new encounters with withdrawal rather than approach, and to decrease their exploratory behavior. At a physiological level, insecure attachment in childhood associates with greater right frontal asymmetry, often linked to emotional and behavioral withdrawal, even at baseline (Dawson et al., 2001).

Likewise, there is evidence that early life caregiving practices associate with exploratory behavior. Non-human primate infants born to low-ranking mothers unable to protect them from conspecifics (Suomi, 2016), and rodents born to mothers that have experienced unpredictable resources (e.g., Perry et al., 2018) show less exploratory behavior than do their conspecifics. Likewise, Schieche and Spangler (2005) reported that infant–mother attachment security assessed at 12 months of infant age significantly predicted greater exploratory behavior in toddlers aged 22 months. Similarly, Belsky and colleagues (1984) found

that higher scores on a measure to assess the home environment, including constructs such as maternal responsivity and involvement, predicted higher-quality infant play. In a second experiment, Belsky (1997) also assessed attachment with mothers and fathers alongside both the quality of play that infants independently emitted and the highest quality of play they could produce when elicited by an experimenter. In this experiment they found that secure attachment predicted independent play that was comparatively more likely to reflect the highest quality of elicited play, supporting the idea that when needs for security and comfort are met, children are better able to fully engage with their environment. Such work may also point to connections between secure attachment and mastery-motivation, as infants' play quality has also been linked to persistence behavior (Jennings et al., 1979). Indeed, sensitive, early life caregiving, and secure attachment, relate to more mastery-motivation and/or confidence when encountering difficult, but solvable, new problems (e.g., Baker, Sonnenschein, & Gilat, 1996; Moss & St-Laurent, 2001; also see Frodi, Bridges, & Grolnick, 1985 for mixed findings).

When young children explore less, they may be less likely to encounter and learn from new experiences. As indicated in “enrichment” studies, opportunities to engage with a stimulating environment associate with changes such as neurogenesis (cell birth), and dendritic growth (increasing the chances of connections) (Simpson & Kelly, 2011). As such, it may not be surprising that children who have experienced sensitive, warm, or responsive care and/or secure attachment display school success and academic competence (Williford, Carter, & Pianta, 2016), and enhanced executive functioning or aspects of cognition such as flexibility and inhibition (e.g., Matte-Gagne et al., 2018; Merz et al., 2016).

Lifespan Changes and Intergenerational Transmission

Given the multitude of ways early experience may shape brain development, it is sensible to consider mechanisms for change across the lifespan and into the next generation. Behavioral research suggests at least a partial role for “enduring effects” – that is, effects of infant experience in later functioning that cannot be fully accounted for by intervening life events (e.g., Fraley, Roisman, & Haltigan, 2013; Kovan, Chung, & Sroufe, 2009; Raby et al., 2015). Such “enduring effects” are also consistent with research suggesting that early life experience influences working models or schemas (Fraley et al., 2013; Raby et al., 2015), which impact expectations for support during challenging times. As one example, preschool girls who, as infants, experienced insensitive care, are better able than their counterparts to remember socioemotional relations. However, as they are worse at remembering non-socioemotional relations, this suggests that their early life experience has influenced what types of information are prioritized for future consideration (Rifkin-Graboi et al., 2021). As another example, when infants view a cartoon depicting an “adult” shape responding to a separation from a “child” shape, experience with sensitive care predicts greater looking towards the “adult” shape. However, this relation is only observed when the child figure appears to be upset, suggesting that those exposed to more sensitive care specifically expect adults to respond to child distress (Biro et al., 2015). Moreover, studies linking early life experience to adolescent and adult secure base scripts (e.g., Steele et al., 2014) and Adult Attachment Interview-assessed attachment representations (e.g., Roisman et al., 2014) also suggest that the quality of early life care influences our mental models surrounding support and caregiving.

What are the neurocognitive processes underlying the development of these mental models?

Processes Underpinning Associations between Early Life Care and Adult Functioning: Scripts, Schemas, and Neurocircuitry

Similar to Bowlby (1969), Main and Hesse (2000; Main, 2000) note that infants build working models based on their experience. They also suggest that these models can direct attentional processes under moderate distress and so maintain the child’s internal organization, but falter under more extreme distress leading to physical and/or mental disorganization (e.g., Hesse & Main, 2000; Main, 1981; 2000). This line of thinking leads to a “Top-Down” model of schema development (see Figure 2.1a) suggesting that early experience guides the development of executive and attentional control that limit the impact of distress; when executive control fails, disorganization ensues, and a physiological stress response is likely. Over time, such physiological responses may further affect executive (prefrontal) control, increase allostatic load, and ultimately also disrupt the limbic system and systemic HPA activity (Lupien et al., 2006). This is an intriguing possibility and, indeed, some evidence suggests that insecure-dismissing adults attempt to exclude negative information from processing (Leyh et al., 2016; but see Zeijlmans van Emmichoven et al., 2003 for mixed findings), and when they are prevented from doing so, HPA activity increases (Rifkin-Graboi, 2008). Moreover, as noted above there is a rapidly increasing amount of research linking exposure to early life sensitive care and cognitive flexibility and other aspects of executive functioning (e.g., Matte-Gagne et al., 2018).

However, as the prefrontal cortex develops comparatively late (e.g., Gogtay et al., 2004), a

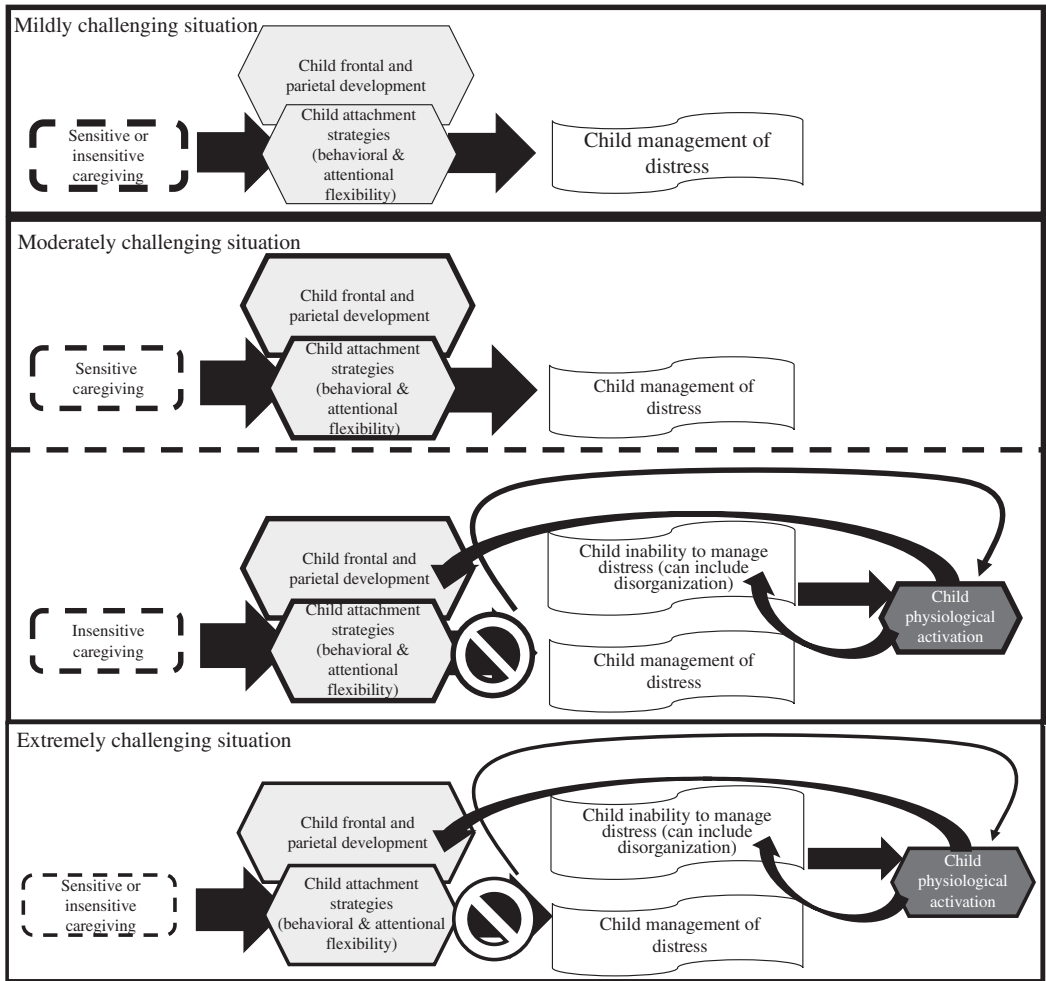
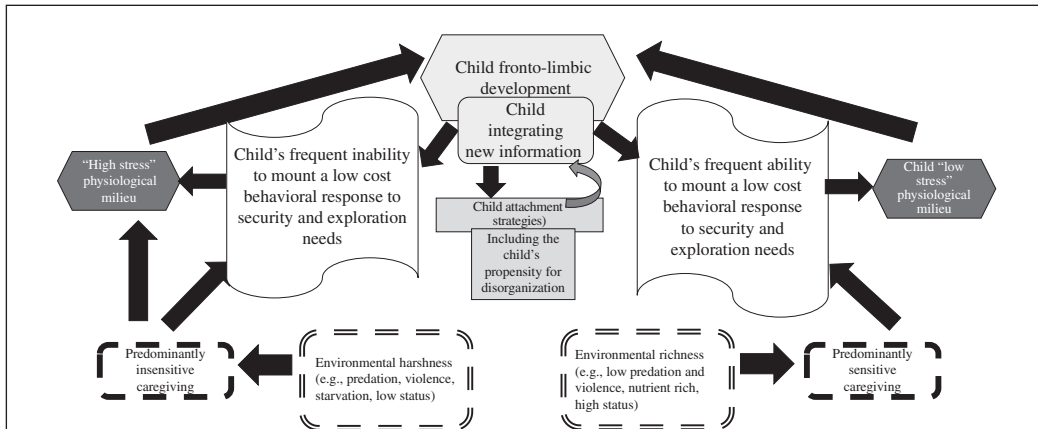


Figure 2.1a Parental sensitivity and the neurodevelopment of offspring attachment strategies and disorganization: “Top-Down” Model

second possibility, which we will call the “Bottom-Up” model of schema development (see Figure 2.1b), more centrally focuses on the limbic system in explaining the links between early life experience and attachment-related schemas (Rifkin-Graboi, Qiang, Qiu et al., 2019). Recalling the earlier discussion of experiencing, signaling, and regulation, when infants do not consistently receive sensitive caregiving, future encounters may be experienced as insurmountable threats, rather than manageable challenges. In turn, their day-to-day

lives may be filled with greater exposure to stress hormones, impacting early limbic (and to a certain extent, prefrontal) development, and influencing schema shaping (how new information is integrated and new memories are formed). These schemas may then further influence challenge appraisal and subsequent HPA response. Furthermore, in this model, insecure attachment schema development may be considered a by-product of neural development occurring in a conditionally adaptive way to manage a generally



Note. Both the “Top-Down” (Figure 2.1a) and “Bottom-Up” (Figure 2.1b) models describe ways that parenting may influence stress responsivity, frontal function, cognitive strategies, and attachment formation. However, the primacy of the associations and their presumed adaptive significance differs across the two models. In the “Top Down” model, caregiving’s effect on attachment schemas (and so attentional processing and frontal-parietal development) is primary. In the “Bottom-Up” model, differences in attachment strategies are secondary, while the extent of caregiver support in the face of frequent challenges plays a greater role, leading to downstream effects on frontal development, and differences in attachment strategies occurring as a byproduct. To illustrate the differences in these models, imagine two children, Betty and Cindy-Ann. Betty’s family life is relatively good and her mother experiences relatively low stress. In contrast, Cindy-Ann’s family lives in harsher circumstances and her mother experiences far more adversity. Betty’s mother is frequently sensitive and responsive, whereas Cindy-Ann’s mother is not.

In the “Top-Down” model, because Betty receives sensitive care, she learns she can count on her mother’s support in almost all circumstances. In other words, she develops a secure attachment schema; and so in normal circumstances, Betty does not need to restrict her attention, can access both positive and negative feelings, and display her thoughts and emotions openly. Such cognitive flexibility and attentional shifting, a hallmark of frontal (frontal-parietal) activity, may facilitate regulation in subsequent challenges, where she can select from various possible behavioral solutions to achieve her goals. Only when Betty is confronted with an exceptionally frightening challenge, in which she cannot find a behavioral solution, do more physiological solutions to distress become activated. Conversely, because Cindy-Ann is accustomed to receiving insensitive care, she has limited attentional capacity, reflected as an insecure attachment schema. Perhaps Cindy-Ann has experienced inconsistent care, and keeps her attentional focus on the mother during all circumstances. Or, maybe she has been frequently rejected and so purposefully defends against negative feelings to appear strong and in need of little support. Either way, she has learned to narrow the range of possibilities available to her when a challenge arises which helps her to maintain organization across much of her life. But, these same attentional constraints limit her ability to flexibly search for and find an appropriate behavioral solution when confronted with even moderate challenges.

Alternatively, consider these children’s development from the “Bottom-Up” perspective. Because Cindy-Ann and her mother live in a more adverse environment, Cindy-Ann is likely to face more than her fair-share of challenges growing up. Still, challenges arise in all environments, and so, from a young age, both girls do encounter situations where they need support. However, while Betty’s mother frequently helps Betty to explore and manage distressing circumstances, Cindy-Ann’s mother is less supportive. Perhaps she frustrates Cindy-Ann’s attempts at autonomy, or perhaps she does not help her when she is distressed. In some cases she may even display behavior that frightens Cindy-Ann. Over time, such differences pile up: Betty has experienced a variety of situations that were met with behavioral responses, whereas Cindy-Ann has experienced repeated physiological stress responses. These different physiological milieus alter brain development across the lifespan, perhaps starting with limbic regions and ultimately prefrontal circuitry. Such downstream changes in neurocircuitry are accompanied by changes in memory (autobiographical and schema formation), emotion perception and regulation, and attentional flexibility and inhibitory processing. These neurocognitive differences may be adaptive in varying environments (rich versus adverse) and further determine how the girls experience, signal, and regulate during subsequent challenges. Because, in the “Bottom-Up” Model, attachment schemas are constructed via the building blocks of memory, emotion, and attentional organization, in some ways the development of attachment models is a by-product of such change. Still, these schemas may also impact future social challenges and tendencies to search out social support.

Figure 2.1b Parental sensitivity and the neurodevelopment of offspring attachment strategies and disorganization: “Bottom-Up” model

“dangerous” world, rather than to specifically manage concurrent (Main, 1981) or predicted interpersonal relationships important to growth and development (e.g., Belsky, Steinberg, & Draper, 1991).

The idea that the hippocampus, and other limbic regions, may be important in early life to shaping subsequent working models is in many ways consistent with Bosmans et al. (2020) “Learning Theory of Attachment”.

This theory suggests that children form associations between stressful situations and feelings of anxiety, but that these associations can be extinguished when the situation fails to elicit such emotional responses. Moreover, when children learn that the presence of a caregiver is a reliable predictor of a disruption to the threat–anxiety association, and relatedly experience a decrease in cortisol, and an increase in oxytocin and dopamine, they begin

to think of the caregiver as a “safety cue.” The theory further suggests that individual variation in the degree of sensitivity needed to promote security may depend in part on the child’s ability to learn that the caregiver associates with a decrease in fear. This supposition is supported by initial findings suggesting that links between insensitive care and disorganization are more pronounced amongst those with larger left hippocampi at birth (i.e., a time before exposure to variation in care; Rifkin-Graboi, Tan, et al., 2019). It is also supported by work indicating that exposure to early life sensitive caregiving may shape hippocampal networks (Wang et al., 2019), including those involved in fear extinction (Rifkin-Graboi et al., 2015), with the latter findings further corroborated by work linking early life sensitivity to relational memory (Rifkin-Graboi et al., 2018) and fear extinction processes (Tsotsi et al., 2019).

Adolescent and Adult Neurocircuitry and Well-Being: An Example of Equifinality Dependent on Childhood History?

Regardless of whether “Top-Down” or “Bottom-Up” processes are primary in linking experience and schemas, several processes including emotion (amygdala), memory (hippocampus), and executive functioning (prefrontal cortex) are likely to interactively impact functioning. Moreover, how such activity influences every-day adult life may differ by early experience and attachment security. One of the most intriguing aspects of Bosman and colleagues’ (2020) “Learning Theory of Attachment” is the idea that while conditioned connections may be weakened or strengthened, they are not erased. This idea then suggests that the brain processes involved in concurrent well-being may differ amongst those with notably varied childhood experiences.

In fact, increasing evidence suggests that experience may influence what types of processes are used when encountering potential

challenges. For example, adolescents’ current interest in social relationships is more important in predicting peer interactions amongst those who were insecurely attached in infancy than for those who were in secure relationships (Weinfield, Ogawa, & Sroufe, 1997). As such, while early life experience may initially influence neurocircuitry involved in emotion and memory, the impact of such bottom-up processes may be moderated by cognitive control. Similar to what is observed amongst infants of depressed mothers and infants in insecure attachment relationships (Dawson et al., 1992, 2001), late teen and young adult participants who are Insecure-Dismissing on the Adult Attachment Interview show a neurophysiological pattern (i.e., right frontal EEG lateralization) during baseline conditions that suggests withdrawal tendencies (Kunl, Leyh, & Spangler, 2016). However, when similarly aged young adults are asked to perform emotional tasks, insecurity, and perhaps especially *avoidance*, in infancy related to increased prefrontal activation (Moutsiana et al., 2014). Furthermore, in the same study, formerly insecure infants showed the least connectivity between regions associated with reward and executive functioning, but were able to increase subject positive feelings via increases in other prefrontal regions. Likewise, when mothers are asked to view sad images of their babies, in comparison to secure mothers, dismissing mothers show decreased reward circuitry activity (i.e., within the ventral striatum) and enhanced activity in a region associated with disgust (i.e., the insula), as well as enhanced activity in a region important to executive functioning (i.e., the dorsolateral prefrontal cortex activity; Strathearn et al., 2009).

Clinicians and policy makers may wish to pay special attention to whether future research also suggests that early past experience influences which type of processes are used to achieve well-being and positive social relationships. Such findings could suggest that

knowledge about early life experience may help in determining whether a person will most benefit from programs aimed to reduce stress or boost security versus those associated with enhancing prefrontal function and executive control (e.g., monitoring, perspective taking, and regulation).

Caveats and Conclusions

Several exciting questions concerning the impact of parenting on neurodevelopment remain. As briefly discussed, genetic variation may influence how people are influenced by changes in their environments, including those encountered through intervention programs (e.g., Bakermans-Kranenburg et al., 2008a; 2008b). However, extant literature on the interplay between genetic and environmental factors is limited and requires further examination. Research has also only begun to examine the role of epigenetics in associations between parenting, attachment, and development. Epigenetic mechanisms, including methylation, can be important in determining DNA expression and can be influenced by both environmental factors and heritable nucleotide sequences (Bagot & Meaney, 2010). Some work suggests that sensitivity and/or attachment may relate to methylation important to stress responsiveness (e.g., Bosmans et al., 2018; Jones-Mason et al., 2016; van Ijzendoorn et al., 2010). Second, despite predictions from early life care to mental health (Groh et al., 2017), this chapter has not explored ways in which maternal mental health and postnatal experience may combine to impact brain development. Moreover, we only briefly touched upon the complex ways culture and parenting interact.

Still, despite these caveats, we have described two pathways through which parental behavior likely impacts children's neurodevelopment – (1) parents can influence whether and how children respond to many of life's challenges; and (2) parents can support

children's exploratory behavior to enrich growth. Though the brain remains susceptible to change across development, initial caregiving experiences may shape the trajectories of systems and regions that interactively impact the way future experience is experienced and managed. In turn, such circuitry may further solidify initial patterns and shape the extent to which additional brain regions are recruited during adult life.

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3 Parenting and Brain Development

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Experience shapes brain development, and the quality of the caregiving environment is one of the most impactful elements of experience for the developing child. A longstanding body of research underscores the central role of parents in child development, with maladaptive parenting practices –characterized by parental insensitivity and harshness – consistently linked with poorer social, emotional, and cognitive outcomes (e.g., Amato & Fowler, 2002; Valiente et al., 2007). While associations between parenting and child development are inextricably linked by shared genetic factors, evidence shows that the quality of the caregiving environment exerts effects over and above shared genetics (Plomin et al., 2008), underscoring parenting as a key leverage point for enhancing healthy development.

Given its effects on development, there has been a rising interest in how the caregiving environment is embedded at the neural level, especially in view of the notion that changes in brain function and structure may explain, in part, individual differences in trajectories of adjustment. The most prominent body of knowledge characterizing the effects of parenting on brain development comes from studies focusing on the most extreme forms of maladaptive parenting – child abuse, neglect, or maltreatment. While research of this kind, often focusing on clinical populations or highly victimized youth, is undoubtedly important, models of developmental psychopathology emphasize the importance of studying individual differences characteristic of both

abnormal and normative development, as this knowledge is mutually informative and may allow for greater understanding of mechanistic processes involved in a full continuum of development (Cicchetti & Cohen, 1995). Researchers within the last decade have reinforced this perspective, calling for research not just among clinical populations of maltreated children but also among children who have experienced normative ranges of parenting behavior (Belsky & de Haan, 2011).

To this end, this chapter begins with a brief primer on methods used to study neural development, and then summarizes research on the effects of child maltreatment on human brain development. We focus more comprehensively on recent studies of the effects of normative variations in parenting. With respect to the latter, we first discuss research modeling parenting's direct influence on youth brain development, and follow this with a review of a small but promising area of research modeling youths' differential susceptibility to the caregiving environment. We conclude with a discussion of future directions.

Methods for Measuring the Developing Brain

Technological advancements over the past two decades have allowed us to understand how the brain changes across childhood, adolescence, and into adulthood. The most common neuroimaging methods to study the developing brain in the context of parenting include

Table 3.1. *Neuroimaging modalities commonly used in the study of parenting and brain development*

Modality	Activity Measured/Description	Advantages (+)/Disadvantages (–)
Structural magnetic resonance imaging (sMRI)	Anatomical volume and cortical thickness of soft tissue: grey and white matter	• High spatial resolution (+) • No temporal information regarding cognitive processes (–) • Nonportable (–)
Task-based functional magnetic resonance imaging (fMRI)	Metabolic activity (blood oxygen level dependent [BOLD] signal) during task engagement to assess specific cognitive processes	• High spatial resolution (+) • Low temporal resolution (–) • Nonportable (–)
Resting state functional magnetic resonance imaging (rsfMRI)	Metabolic activity (BOLD signal) during rest to identify intrinsic networks	• High spatial resolution (+) • Low temporal resolution (–) • Nonportable (–)
Electroencephalogram (EEG)	Electric potentials from neuronal activity during tasks or rest	• High temporal resolution (+) • Low spatial resolution (–) • Less susceptible to movement, thus used with younger samples (e.g., infants) (+)
Functional near-infrared spectroscopy (fNIRS)	Metabolic changes in hemoglobin concentrations	• High temporal resolution (+) • Low spatial resolution (–) • Portable, thus often used in parent–child shared task activity (i.e., studies of neural synchrony) (+)

structural MRI (sMRI), which measures the anatomy of the brain, as well as task-based and resting-state functional MRI (fMRI), which measure the function of the brain. See Table 3.1 for a description of these modalities.

Structural MRI. sMRI seeks to measure changes in the anatomy of the brain in both grey and white matter. The most common metric is cortical volume, which is measured in terms of thickness (i.e., the distance between the white matter/grey matter cortical boundary and the grey matter/CSF cortical boundary) and surface area (i.e., the area of the white matter/grey matter cortical boundary and the grey matter/CSF cortical boundary; Vijayakumar, Mills, Alexander-Bloch, Tamnes, & Whittle, 2018). One of the most consistent findings

from longitudinal sMRI research is that grey matter volume in the frontal and parietal lobes increases during childhood followed by a decline (i.e., thinning) in adolescence, before stabilizing in young adulthood (Mills, Lalonde, Clasen, Giedd, & Blakemore, 2014). In contrast to cortical grey matter volume, cerebral white matter volume increases from birth and across the second decade of life before stabilizing in young adulthood (Mills, Goddings, Herting, Meuwese, Blakemore, et al., 2016). The regions exhibiting the most robust decreases in grey matter volume during adolescence are in regions implicated in higher-order cognition (e.g., the dorsal pre-frontal cortex), and are the same regions showing the most robust increases in white matter

volume (Mills et al., 2014). Grey matter changes are thought to reflect synaptic reorganization, including synaptic proliferation followed by synaptic pruning, whereas increases in white matter are thought to reflect continued axonal myelination that promotes efficient information flow in the brain (Blakemore, 2012). Other structural imaging modalities include diffusion MRI, which seeks to measure microstructural properties of the connecting white matter fiber bundles in the brain; however, this method has less commonly been used in the study of parenting and brain development.

Task-Based Functional MRI. fMRI seeks to understand how the brain functions during different psychological processes. fMRI measures the blood oxygenation level dependent (or BOLD) signal, which is an indirect measure of neural activity. When neural activity increases, there is an increase in blood oxygenation associated with blood flow to different regions of the brain. Because blood oxygenation varies according to the level of neural activity, these differences can be used to detect brain activity. While most research to date has focused on functional brain activation in isolated brain regions, there has been a growing emphasis on understanding functional connectivity, or how neural regions are co-activated. A hierarchical cascade of changes in functional connectivity patterns is proposed to occur, whereby development of subcortico-subcortical connectivity (e.g., amygdala-ventral striatum connectivity) occurs before that of cortico-subcortical connectivity (e.g., amygdala-prefrontal cortex), which serves as a necessary precursor to more complex neural interactions (Casey, Heller, Gee, & Cohen, 2019). This shift in connectivity patterns is thought to underlie emotional development across adolescence. Other methods utilized less commonly in the literature of parenting and brain development include event-related potentials (ERP), which measures the timing of electrical activity in the brain in response to discrete

events (Burani et al., 2019) and functional near-infrared spectroscopy (fNIRS), which is an optical imaging technique that measures changes in hemoglobin concentrations within the brain (Cai, Dong, & Niu, 2018).

Resting State MRI. Other research has employed resting state fMRI to examine organization within and between functional networks during task-independent activity or at rest. A consistent finding emerging across this research is enhanced connectivity *within* large-scale functional brain networks, such as the executive control network, coupled with reduced connectivity *between* networks during the adolescence period (Dosenbach et al., 2010). This shift to greater within-network connectivity is thought to reflect better specialization and increasing efficiency in neural processing (Blakemore, 2012).

Evidence from Studies of Childhood Maltreatment

Seminal work on the neural consequences of maltreatment includes studies of physical, sexual, and emotional abuse, and neglect, including research with youth raised in extreme circumstances of social-emotional deprivation characteristic of institutional rearing. Given the high levels of co-occurrence among maltreatment experiences and other forms of trauma and adversity (e.g., exposure to violence, poverty, abuse by extra-familial members), many studies in this domain have more broadly examined the effects of early life stress or adversity on the developing brain. While a comprehensive review of this subject can be found elsewhere (e.g., see Teicher, Samson, Anderson, & Ohashi, 2016 for further review), we briefly summarize evidence on childhood maltreatment and alterations in brain structure and function.

The majority of research in this area has focused on the effects of childhood

maltreatment involved in stress or threat systems. In particular, disruptions in the hypothalamic pituitary adrenal (HPA) axis and its effects on brain development have been a predominant focus. In this pathway, excessive release of glucocorticoids due to repeated or prolonged stress is thought to result in altered morphology (i.e., via neuronal loss, disruptions in neurogenesis) and function in neural regions densely populated with glucocorticoid receptors. These regions include key structures of the limbic system involved in emotional processing, learning, and memory, such as the amygdala and hippocampus. Neuroendocrine and neural effects of this kind have been well-documented in translational research with rodent and non-human primates exposed to caregiving deprivation and early stress (Liu et al., 1997; Plotsky & Meaney, 1993).

Consistent with stress models, a growing body of work in humans demonstrates structural and functional anomalies in areas of the limbic system among maltreated individuals. With respect to morphological abnormalities, numerous magnetic resonance imaging (MRI) studies indicate reduced hippocampal volumes among *adults* with a history of childhood maltreatment (e.g., Bremner et al., 1997; Teicher, Anderson, & Polcari, 2012). The hippocampus is a posterior limbic structure involved in learning and memory, and with its reciprocal connections to other limbic regions, is thought to also play a role in emotional behavior. Interestingly, evidence has not consistently indicated atypical hippocampal volumes in samples of maltreated children and adolescents, suggesting that hippocampal effects may not emerge until later in development. Some evidence also suggests early adversity may impact hippocampal function in youth. Reduced hippocampal recruitment in youth exposed to early life stress has been reported along with disruptions in aspects of memory and learning (Lambert et al. 2017; Lambert & McLaughlin, 2019).

Findings of structural differences in other stress-susceptible subcortical brain regions, such as the amygdala, in maltreated youth are more mixed. A few studies have reported larger amygdala volumes in maltreated children and early adolescents (e.g., Mehta et al., 2009; Tottenham et al., 2010), possibly suggesting accelerated structural maturation of the amygdalae. Yet, other studies have not observed significant differences in amygdala volume (e.g., Gorka, Hanson, Radtke, & Hariri, 2015; McLaughlin et al., 2014a) or found opposite effects (e.g., Hansen et al., 2015). What has emerged consistently, however, is a pattern of heightened amygdala functional response to threatening stimuli in maltreated youth. This evidence is borne out of neuroimaging studies most commonly employing emotional face paradigms, in which participants view angry, sad, fearful, and neutral faces during functional MRI (fMRI). In one recent meta-analysis investigating differences in brain function between maltreated individuals and healthy controls, higher levels of bilateral amygdala activation in response to negative versus neutral faces were identified in maltreated groups (Hein & Monk, 2017). The amygdala is involved in processing emotionally salient information and is thought to facilitate rapid orientation to and recognition of threatening stimuli (Phelps & LeDoux, 2005). Hypervigilance to threatening stimuli, as mediated in part by amygdala hyperactivity, may functionally prepare an individual to respond quickly to future danger in the environment. While heightened vigilance may be adaptive in the short run, this affective endophenotype may place youth at risk for long-term negative physical and mental health outcomes (McCrory, Gerin, & Viding, 2017).

Moreover, it has been suggested that early adversity may accelerate the development of threat-related neural circuitry, including the amygdala, hippocampus, and medial

prefrontal cortex (Callaghan & Tottenham, 2016). Accelerated neural maturation is postulated to be a functional adaptation to the environment, allowing the organism to reach reproductive maturation more quickly for the sake of species survival and putatively equipping an organism with more mature cognitive capacities to meet the challenges of the environment. Evidence has supported such a hypothesis with respect to accelerated aging, including studies documenting decreased cortical thickness among children exposed to adversity (Busso et al., 2017; Gold et al., 2016) and earlier pubertal timing, and cellular aging (Colich, Rosen, Williams, & McLaughlin, 2020). Despite potential benefits in the short run with respect to enhanced regulatory abilities that may come with accelerated development, Callaghan and Tottenham (2016) posited that “stress acceleration” of neural circuitry may hinder later plasticity, or the ability to adapt to environmental demands and contexts.

A smaller but growing body of evidence also implicates neural pathways involved in reward and motivation, including mesolimbic dopamine pathways in the ventral striatum and medial prefrontal cortex. Several neuroimaging studies in humans find blunted striatal brain activity during the anticipation of monetary rewards in maltreated individuals (Dillon et al., 2009; Goff et al., 2013; Hanson et al., 2015; Mehta et al., 2010). This is consistent with preclinical studies showing evidence of chronic stress effects on dopamine signaling and reward-mediated behavior, including diminished dopamine transmission in the ventral striatum and decreased approach behaviors (e.g., Brenhouse, Lukkes, & Andersen, 2013; Matthews & Robbins, 2003). Altered dopamine signaling may occur via stress-related alterations in HPA axis function (Nusslock & Miller, 2016). From an evolutionary perspective, some have argued that these effects are

advantageous in a threatening or dangerous environment, pushing an individual toward avoidance behavior rather than approach as a means of safety (Teicher & Samson, 2016). Again, however, long-term consequences of this neural endophenotype may include increased risk for poor mental health outcomes, as blunted neural reward processing has been linked with anhedonia and depressive symptoms (Corral-Frias et al., 2015).

Given the heterogeneity of maltreatment experiences, it has recently been proposed that specific dimensions of maltreatment (e.g., threat versus deprivation) may have differential impacts on brain development (McLaughlin, Sheridan, & Lambert, 2014a). Threat or abuse exposures may impact neurobiological stress systems, primarily in the ways discussed above. In contrast however, environmental deprivation, as in the case of severe caregiving neglect or institutional rearing, may uniquely impact cortical development. Specifically, in contexts of severe neglect or deprivation, over-pruning of synaptic connections (especially during sensitive periods of neural development) may occur in order to accommodate low stimulation and low complexity environments (McLaughlin et al., 2014a). This over-pruning is thought to result in lower cortical volume in areas of the association cortex involved in social and cognitive processing. Indeed, children raised in institutional care show overall lower levels of grey matter volume (Mehta et al., 2009) and reduced cortical thickness in regions of the prefrontal, inferior parietal, and superior temporal cortex (McLaughlin et al., 2014b).

Models of Direct Influence of Normative Parenting on Brain Development

A growing body of work, largely within the last decade, has sought to characterize the

effects of what has commonly been referred to as “normative variations” in parenting, or parental behaviors below the threshold of abuse or neglect. Normative variations in parenting, such as individual differences in parental warmth, consistency, and harshness, are consistently related to behavioral development and adjustment in youth and are associated with alterations in offspring neurobiology in preclinical research (Champagne & Cheney, 2005). In human models, recent attention has been paid to the ways in which normative variations in parenting are associated with youth’s structural development and functional neural activation, particularly with respect to the neural substrates of emotion-related reactivity and regulation (see Tan et al., 2020) including networks involved in salience processing (negative and positively valenced systems) and cognitive/regulatory control. Together, this work suggests that a full continuum of the quality of the caregiving environment, from adversity to sensitivity, may be neurally embedded.

Brain Structure. Variations in normative parenting have been linked with differences in subcortical and cortical morphometric indices in youth, with some findings mirroring those from maltreatment studies. Whereas research has documented higher amygdala volumes in maltreated youth, higher levels of positive parenting are associated with lower amygdala volume in adolescents (Whittle et al., 2009, 2014) and infants (age 1, Bernier et al., 2019). Evidence also points to associations between aspects of adaptive caregiving and *larger* hippocampal volumes (e.g., Luby et al., 2012; Rifkin-Graboi et al., 2015) and greater gray matter density (Schneider et al., 2012), complementing research suggesting maltreatment exposure is associated with *smaller* hippocampal volumes in adults. In one longitudinal study, Luby et al (2012) found that young children (ages 3–5 years) exposed to greater

maternal supportive behavior showed larger hippocampal volumes at ages 7–13 years. Interestingly, however, Rao et al. (2010) noted contrasting results, finding that higher maternal support and availability at age 4 predicted *smaller* hippocampal volumes in adolescent youth (ages 13–16). Given that certain subdivisions of hippocampal development follow an inverted u-shape, peaking in early adolescence, authors reasoned that early supportive parenting may set the stage for accelerated hippocampal maturation during adolescence.

Research documenting associations between cortical morphology and parenting have focused mostly on adolescent samples to date. One of the only studies to assess brain structure in children as a function of normative parenting variations found that early positive parental behavior (at age 4) predicted larger total grey matter volumes at age 8 (Kok et al., 2015). Interestingly, age 8 parenting behavior was not associated with structural development, suggesting an earlier sensitive window for parental effects. Findings from this study show that sensitive parenting is associated with *larger* gray matter volumes and complement those from the maltreatment literature, which indicate early adversity is associated with *lower* grey matter volumes in children.

Studies with adolescents, however, evidence a different pattern of effects – specifically that higher levels of sensitive parenting may be associated with accelerated maturation of cortical systems (Schneider et al., 2012; Whittle et al., 2014). For example, findings from the *Orygen Adolescent Development Study* showed that higher frequencies of maternal positive behaviors (e.g., affection, validation) during a parent–adolescent interaction predicted increased cortical thinning from early (approximately 12 years) to later in adolescence (approximately 16 years) (Whittle et al., 2014). Conversely, evidence suggests

maladaptive parenting, such as maternal aggression and inconsistent parenting, is associated with metrics suggestive of immature cortical development, including higher cortical volumes (i.e., ACC, OFC, Whittle et al., 2009), longitudinal increases in the lateral parietal, supramarginal gyrus, and SFG thickness (Whittle et al., 2016), and greater overall cortical thickness and asymmetry (Frye et al., 2010) in adolescent samples. Together, this suggests that while adaptive parenting may impose accelerating effects on normative brain development in adolescence, normative variations in maladaptive parenting may impose delaying maturational effects.

This pattern of findings is interesting given evidence that children reared in extreme cases of deprivation (i.e., institutionalized rearing) also show reduced cortical thickness, suggestive of accelerated synaptic pruning in these youth, as well as evidence and theory described above suggesting that early adversity is associated with accelerated neurobiological development (Callaghan & Tottenham, 2016). This evidence on maltreated youth, however, primarily comes from studies of children; whereas, evidence of delayed cortical development as a function of normative variations in maladaptive caregiving comes from studies of adolescents. This collectively may suggest differential patterns based on stage of development, such that adversity in the caregiving environment is related to accelerated cortical development during childhood but delayed cortical development during adolescence. Along those lines, it has been proposed that adolescence is a period of neurobiological re-adaptation, as neurobiological stress systems resurvey the current environment and reorganize if conditions are significantly different than in earlier periods (Gunnar et al., 2019). Thus, it is plausible that during this sensitive period, neural patterns of adaptation emerge that differ in direction from earlier periods.

Alternatively, this evidence may suggest that maltreatment relative to normative variations in maladaptive caregiving exerts qualitatively different effects on brain development, with moderate levels of adversity linked to delayed maturation and with extreme adversity “tripping” evolutionary mechanisms to accelerate development. More research is clearly needed across childhood and adolescent periods and across a full continuum of caregiving to understand these findings.

Brain Function. Accumulating evidence linking normative variations in parenting with youths’ brain function has primarily emerged from functional neuroimaging studies probing neural recruitment of salience processing systems – negative valence (e.g., threat response), positive valence (e.g., reward anticipation) – and cognitive control (e.g., inhibition). While structural brain studies may be well equipped to characterize relatively stable neural markers of caregiving effects, functional brain imaging studies allow for *in vivo* and implicit neural processes to be captured.

Consistent with maltreatment studies, neural circuits subserving threat or negative valence systems may be sensitive to normative variations in caregiving. This has been demonstrated with respect to individual differences in lower parental warmth (Romund et al., 2016) and maternal negative affect and dysregulation (Pozzi et al., 2019; Turpyn et al., 2018), with these factors related to youths’ higher amygdala and salience network response to negatively valenced stimuli. Moreover, altered patterns of amygdala and salience network resting-state functional connectivity (e.g., amygdala, medial prefrontal cortex) associated with maladaptive parenting have been observed, with coupling between these subcortical–cortical regions suggestive of accelerated development (Kopala-Sibley et al., 2018; Thijssen et al., 2017). This evidence is consistent with proposed models of

emotion-related brain development in the context of parenting (Tan et al., 2020), most consistently with aspects of maladaptive parenting conferring risk for higher levels of negative affect.

Interestingly, while such patterns of heightened amygdala engagement have been linked to internalizing psychopathology and trauma-related disorders in youth and adults (Kuwaja & Burkehouse, 2017), blunted amygdala reactivity has also been linked to externalizing psychopathology (Blair, Veroude, & Buitelaar, 2018). With respect to the latter, findings from one longitudinal study suggest that harsher parenting may confer risk via an externalizing pathway. Specifically, Gard et al. (2017) found that harsher parenting measured at age 2 was related to *lower* amygdala reactivity at age 20, which in turn was associated with higher levels of antisocial behaviors in adulthood in a male sample. Thus, while most evidence supports the fact that harsher parenting is associated with increased salience processing, multiple neural pathways to maladjustment may occur.

Recent research also highlights potential biological sex differences in parental influences on youth brain function in negative valence systems. For example, one recent study observed sex-differentiated neural associations with negative parenting behaviors (Chaplin et al., 2018). For adolescent boys, higher negative parenting predicted blunted neural responses in affective salience regions (i.e., anterior insula, left anterior cingulate cortex); whereas, for girls, negative parenting predicted higher neural responsivity. Other evidence has documented interactions with sex and pubertal timing, finding differential brain responses to threatening stimuli in late developing boys versus girls in the lateral prefrontal cortex for children exposed to higher levels of corporal punishment (Barbosa et al., 2018). These studies highlight important features of biological

context and underscore the need to investigate differential patterns by sex or gender.

Additionally, an accumulating body of evidence links variations in the caregiving environment with the positive valence system or reward processing, including regions such as the ventral and dorsal striatum and medial prefrontal cortex (mPFC). Lower levels of parent–adolescent relationship quality (Qu, Fuligni, Galvan, & Telzer, 2015), lower parental warmth (Casement et al., 2014; Morgan, Shaw, & Forbes, 2014), insecure early attachment history (McCormick, McElwain, & Telzer, 2019) and higher maternal depressive symptoms (Qu et al., 2016) have been linked with youths' increased reward-related striatal or mPFC response (but see Schneider et al., 2012). For example, although Qu et al. (2015) did not observe significant relations between parent–adolescent relationship quality and adolescent reward-related brain function measured concurrently, findings indicated that longitudinal decreases in parent–adolescent relationship quality were associated with longitudinal increases in activity in the ventral striatum and dlPFC in response to monetary rewards. Moreover, increases in ventral striatum response mediated the association between decreased positive parent–adolescent relationship quality and adolescents' greater risk-taking behavior. This suggests that lower levels of warmth or connectedness in the parent–child relationship may sensitize reward circuitry to secondary rewards (i.e., money), and this heightened responsivity may confer risk for negative behavioral outcomes. In contrast to this pattern of results, Tan et al. (2014) documented that higher levels of maternal negative affect, observed during a parent–adolescent interaction, was associated with lower brain response in reward and salience network regions (i.e., nucleus accumbens, amygdala, anterior insula, and sgACC) during a socially rewarding context in which youth

received positive feedback from peers (i.e., peer acceptance). This could suggest that poorer parent–child relationship quality may differentially relate to reward circuitry response as a function of context – orienting youth away from socially relevant rewards and instead toward nonsocial rewards. Further research is needed to explore this potential pattern.

On the whole, however, evidence points to a pattern of increased reward-related striatal sensitivity in youth experiencing elevated levels of maladaptive parenting. This is in contrast to evidence from maltreatment studies, suggesting that maltreatment exposure is related to blunted striatal sensitivity. It may be that maladaptive parenting exerts sensitizing effects on the mesolimbic dopamine system, up until more extreme or chronic circumstances (i.e., maltreatment) are experienced. The latter then results in reward-related blunting effects. This is in line with evidence from rodent models indicating that acute stress enhances reward salience via modulation in the dopamine system (Abercrombie et al., 1989) through increases in glucocorticoids (Rouge-Pont et al., 1998). However, chronic exposure to stressors results in attenuated dopamine release in the striatum (Cabib & Puglisi-Allegra, 1996) and decreased basal dopamine levels (Mangiavacchi et al., 2001).

Finally, some prior work has also shown that aspects of the caregiving environment may be correlated with neural function facilitating cognitive control and decision-making. Higher levels of anterior insula response (Marusak et al., 2018), greater anterior insula connectivity (Guassi Moreira & Telzer, 2018), and higher ventrolateral PFC (McCormick et al., 2015) response during cognitive control and decision-making contexts have been observed in youth with more negative family factors, including parental psychological control, high family conflict, and low family

cohesion. For example, higher levels of family conflict were associated with greater ventral striatum–anterior insula coupling during safe decision-making in the presence of one’s mother in a risky driving task, when adolescents presumably inhibited risky responses in the face of potentially rewarding stimuli. This neural pattern was in turn associated with greater risk-taking behaviors in adolescents longitudinally (Guassi Moreira & Telzer, 2018). The anterior insula is thought to be involved in integrating affective and cognitive processes, and may be a critical relay point for recruiting other networks (Menon & Uddin, 2010). Thus, aberrant anterior insula response and connectivity may disrupt effective cognitive regulation. Parenting may not only influence emotion generating, reward, and motivational neural systems, but may also be tied to how affective and cognitive control systems interface.

Developmental timing of parental influence.

Notably, some research has suggested that parental influences on brain development may vary based on the developmental timing of exposure. The brain undergoes sensitive periods of development, or high levels of plasticity which may render developing neural circuitry vulnerable to environmental inputs (Tottenham & Sheridan, 2010). Identifying these temporal effects is difficult to capture in human research, as maladaptive caregiving exposures rarely occur in isolated windows and cannot be experimentally manipulated. However, initial evidence of sensitive periods of neural development as they relate to parenting in humans largely stems from research on childhood maltreatment and adversity. For example, one study indicated reduced hippocampal volume was most strongly related to maltreatment exposure occurring during early childhood (ages 3–5 years) and early adolescence (ages 11–13 years) relative to other periods in a sample of

adult females (Andersen et al., 2008). Other work has suggested that the preadolescent period may also represent a time of heightened sensitivity, with one study finding peak amygdala effects of maltreatment exposure occurring at ages 10–11 years (Pechtel et al., 2014). Given that different neural regions follow different growth rates across development, it may be that specific circuitry possesses unique windows of sensitivity to the caregiving environment. Due to the preliminary nature of this empirical work in humans however, further research is certainly needed in this arena.

Neural synchrony. The above studies conceptualize the developing brain in the context of direct or long-term influences of parenting, although links between parenting behavior and neural development are inevitably reciprocal, occurring on micro and macro level time-scales. An important example of this comes from evidence characterizing parent–child synchrony (for review, see Feldman, 2012). Synchrony refers to the coordinated response of behavioral and biological processes across close social partners, with this coupled exchange of social signals beginning early in human development and creating a foundation for attachment relationships (Feldman, 2012). Brain-based measures of parent–child synchrony offer one example of these coordinated social responses, with evidence suggesting that parent–child neural synchrony may represent greater social connectedness and have implications for youth adjustment. For example, one study utilizing resting-state functional connectivity found that parents and adolescents with more similar resting-state intrinsic networks showed greater day-to-day emotional synchrony and higher adolescent emotional competence (Lee, Miernicki, & Telzer, 2017). Several studies using fNIRS have also found greater neural synchrony during cooperative parent–child interactions compared to individual activity in prefrontal and temporal-parietal

areas (e.g., Miller et al., 2019; Reindl, Gerloff, Scharke, & Konrad, 2018), and that higher levels of parent–child neural synchrony are associated with adaptive outcomes (e.g., higher emotion regulation, problem solving). These studies show intriguing avenues for understanding brain-based synchrony between parent and child that may complement longitudinal research designs of parental influence on brain development.

Models of Neurobiological Susceptibility to Normative Parenting

Above, we reviewed evidence from models suggesting that parenting directly influences aspects of youths' brain development, although further research is required to clarify contradictory findings and understand longitudinal relations. However, theory has posited that aspects of neural function and structure may not only be affected by environmental inputs but may dictate the extent to which adversity or support influences developmental outcomes. Indeed, the notion that children differ in their susceptibility to the social environment, such as negative psychosocial effects of maladaptive parenting, has long been recognized in the study of child development. One theory, referred to herein as differential susceptibility theory (Belsky & Pluess, 2009; see also biological sensitivity to context, Boyce & Ellis, 2005), posits that individuals differ in their neurobiological susceptibility to the environment in an evolutionarily adaptive manner, such that some children are more susceptible or adapt more highly in response to environmental inputs than others. Central to this theory is that neurobiological susceptibility functions in a “for better or for worse” fashion (Belsky & Pluess, 2009). That is, youth who are most susceptible to their social environments are the same youth most likely to

thrive in supportive social environments or to suffer in harsh social environments.

The majority of research to date on differential susceptibility has investigated behavioral markers of susceptibility, such as temperament or genetic markers of susceptibility (see Belsky & Pluess, 2009). More recently, however, brain function and structure have been discussed as optimal endophenotypic markers of susceptibility, as these measures may offer a summary of genomic expression and may allow one to make inferences regarding specific processes based on neural circuitry involved (e.g., cognitive, affective mechanisms) (Schriber & Guyer, 2016). Thus, while research discussed above illustrates the potential direct effects of parenting, differential susceptibility offers a window into how such differences in neurobiology may interact with one's environment to determine health and behavior.

Neurobiological susceptibility is thought to be a product of both genetic as well as early environmental influences that shape brain development. Thus, ostensibly, parenting may (1) directly impact neurobiological susceptibility (as reviewed in the section above) and (2) interact with individual differences in susceptibility at any point in development to predict adjustment and behavior (see Figure 3.1 for proposed model). Social environmental influences on neurobiological susceptibility are hypothesized to occur most strongly during

early sensitive periods of development, whereas, interactions between the social environment and neurobiology reflective of differential susceptibility may manifest more prominently later in development (e.g., adolescence). Given that adolescence is both a time of significant neurodevelopment and that psychopathology (e.g., depression) tends to emerge during this time, adolescence may represent an especially important development period for individual differences in susceptibility to manifest (Schriber & Guyer, 2016). Not surprisingly then, existing studies investigating neurobiological susceptibility patterns in youth to date have focused primarily on this period.

Brain Structure. A small handful of studies to date have examined youth brain structure as an endophenotypic marker of susceptibility to parenting. In particular, a few studies have examined key limbic structures as differential susceptibility markers in adolescent samples. Findings indicated that under maladaptive family circumstances (e.g., maternal aggression, low family connectedness), larger hippocampal volumes were predictive of depressive symptoms in adolescents. Specifically, in maladaptive family contexts, larger hippocampal volumes were predictive of higher levels of depressive symptoms; whereas, in more adaptive family contexts, larger volumes were associated with lower levels of depressive

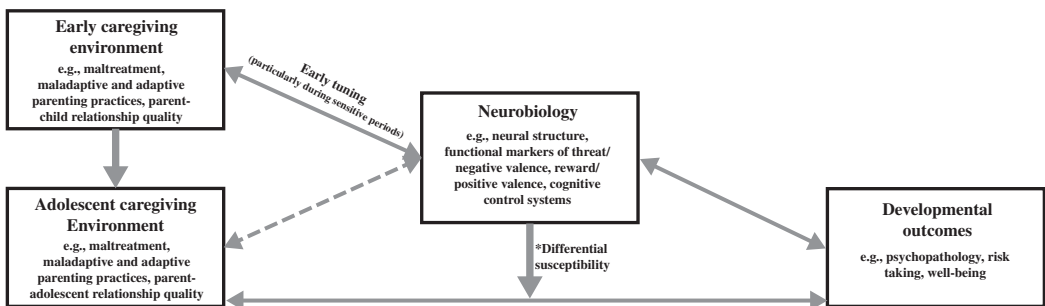


Figure 3.1 Proposed model of caregiving influences on youth brain development

symptoms (Whittle et al., 2011; Schriber et al., 2017). Thus, the same neural profile that conferred risk in one parenting context conferred benefits in another. Further, for adolescents with smaller than average hippocampal volumes, no significant relations emerged between family context and adolescent adjustment, suggesting that these adolescents may be less sensitive to parenting in general.

For amygdala, gender differences in susceptibility profiles were observed (Yap et al., 2008). For girls, smaller amygdala volumes represented a susceptibility factor to maternal aggression, predicting greater adjustment for girls exposed to *high* levels of maternal aggression and poorer adjustment for girls exposed to *low* levels of maternal aggression (Yap et al., 2008). An opposite pattern emerged for boys, with larger amygdala volumes representing susceptibility. Youth were not uniformly affected by maladaptive parenting, however, as maternal aggression was not related to adjustment for adolescents with low (for girls) or average (for boys) amygdala volumes. These initial studies (notably using nonclinical community samples) suggest that neurobiological susceptibility patterns, particularly in key limbic structures known to be susceptible to early adversity and stress more generally, may be evident during the adolescent period of development.

In a recent study of adolescents, differential susceptibility patterns were also identified with respect to cortical thickness (Deane et al., 2019). Under contexts of high maternal aggression, lower levels of cortical thinning were associated with lower well-being. In contrast, under conditions of low maternal aggression, lower cortical thinning was associated with greater well-being. As above, a neural profile (i.e., increased cortical thinning) distinguishing youth less susceptible to maternal behavior emerged. Unlike previous studies, neural susceptibility findings did not extend

to measures of depression. Still, this initial evidence suggests that not only does parenting directly influence cortical development, neural structure may independently interact with parenting, resulting in differential patterns of adjustment and behavior.

Brain Function. Only two studies to our knowledge have tested models of neurobiological susceptibility to the caregiving environment with respect to youths' brain function via MRI. In one study of adolescent girls, higher neural sensitivity to social exclusion in affective salience regions (e.g., anterior insula, dACC, and sgACC) was associated with fewer depressive symptoms in the context of supportive parent–adolescent relationships, but associated with greater depressive symptoms in the context of conflictual parent–adolescent relationships (Rudolph et al., 2018). Girls with lower activation in affective salience regions, however, were less sensitive to poorer parent–adolescent relationship quality, showing low to average levels of depressive symptoms regardless of relationship quality. In another study, youth evidencing higher caudate activation to parental praise reported greater depressive symptoms under contexts of low maternal acceptance and high peer victimization (Sequeira et al., 2019). In contrast, for youth with lower levels of caudate activation to parental praise, peer and maternal factors were not significantly associated with adolescent depression, again perhaps illuminating a marker of reduced susceptibility to social influence more generally. These studies suggest that heightened sensitivity in affective salience regions, in response to either positive or negatively valenced affective stimuli, may represent a neurobiological marker of susceptibility in adolescent youth, once again predicting internalizing symptoms during this period.

Altogether, initial evidence underscores that parenting may not influence youth behavior uniformly, in part as a function of

neurobiological factors. However, it is important to note that this evidence to date is correlational in nature and further research is needed to support the directionality of such associations.

Summary and Integration

Parenting, from its most extreme forms of maladaptation to more normative variations in healthy parental behavior, exerts diverse effects on developing neural structure and function. Extant research suggests that like maltreatment exposure, normative variations in maladaptive parenting impact developing neural circuitry most consistently within stress and negative valence systems. Although no study to our knowledge has examined both maltreatment and normative variations of parenting within one study, this cumulative evidence makes possible the notion that maladaptive parenting may function on a continuum, ostensibly impacting stress and affective systems in the brain in a linear fashion. That is, as negative features of the caregiving environment increase in quality, frequency, and/or intensity, so too do neural impacts.

While these bodies of research show similarities, diverging patterns distinguishing maltreatment from normative variations in maladaptive parenting emerge in some key areas. First, whereas maltreatment exposure has been linked to blunted neural reward sensitivity in mesolimbic systems, normative variations in maladaptive parenting are associated with heightened reward sensitivity as measured at the neural level (e.g., Casement et al., 2014; Qu et al., 2015). Second, aspects of maltreatment have been related to reduced thickness in association cortex and accelerated cortical maturation (e.g., McLaughlin et al., 2014b); however, studies of normative variations in parenting suggest that maladaptive parenting is associated with increased cortical maturation in adolescent

samples (e.g., Whittle et al., 2009, 2014, 2016). On the whole, these diverging findings may suggest that extreme adversities in the caregiving environment, such as maltreatment exposure, relative to normative variations in parenting may produce qualitatively disparate effects on neural function. It is possible that effects on these systems do not function in a linear fashion, with extreme maladaptive parenting associated with blunted (e.g., reward system function) or accelerated (e.g., association cortex structure) development.

Finally, an intriguing and nascent literature in human brain development points out that not only does the quality of parenting impact the developing brain, but aspects of brain structure and function at any given point in development may interact with social inputs such as parenting to predict adjustment. In particular, differential susceptibility theory posits that certain neural endophenotypes may place youth at particular risk for psychopathology when exposed to maladaptive parenting, but may confer benefits when exposed to supportive or sensitive parenting. This framework highlights the complexity of how parenting and the social environment are neurobiologically embedded, as neural systems may be an output, mediator, or moderator when predicting future health and behavior.

Future Directions

Human studies reviewed herein have primarily utilized observational or correlational research methods to investigate the effects of parenting on youths' brain development. This research has undoubtedly enriched our understanding; however, correlational designs in human research may limit causal inferences in understanding the effects of parenting. It is important to note that preclinical studies documenting caregiving effects in animal models have afforded the ability to experimentally

manipulate the caregiving environment and thus complement observational research in humans (Sanchez, Ladd, & Plotsky, 2001). Still, opportunities exist to take steps forward in research design for human models, especially given that animal models do not universally translate to human development. As an example, in a randomized control trial, Milgrom and colleagues (2010) randomly assigned parents of infants to a parent sensitivity training or control intervention, and uniquely measured infant brain volumes and white matter microstructure using DTI following the intervention. Results indicated that infants of parents who received sensitivity training showed greater white matter connectivity and maturation compared to controls. Such controlled designs may be one avenue for future research to empirically test parenting's effect on human brain development.

Further, most research to date has focused on maternal behavior exclusively as it relates to youth brain development. However, youth are likely exposed to other caregivers and diverse family structures. Contextual factors within the family system – including the number and gender identity of caregivers, multigenerational structure, movement of family in and out of the home, and biological relation – are rarely considered. Future research exploring this diversity is needed, and may be relevant to parental influence on neural structure and function as well as youth's susceptibility to such exposures.

Relatedly, most research to date has been done within largely Western, White, higher socioeconomic samples (Qu, Jorgensen, & Telzer, *in press*) without taking into consideration how parenting and its effects on the brain may differ across ethnic, cultural, religious, and economic contexts. Indeed, the goals, values, and parenting strategies of families significantly differ based on ethnic background. For instance, Chinese and other

Asian families traditionally emphasize the importance of family harmony and respect; families from Mexican and other Latin American backgrounds have a strong sense of family connection, interdependence, and support often referred to as familism; and African American families tend to value interdependence and perseverance in the face of adversity (Fuligni, 1998; Hill, 2006; McLoyd, Dodge, & Lansford, 2005). These family values often translate into more authoritarian parenting strategies, including the use of discipline (Hill, 2006). Thus, adolescents from these cultural groups may benefit more from family harmony and suffer less from parenting that utilizes harsh discipline. European American families tend to endorse individualism and individual achievement, family values that often translate into more authoritative parenting strategies that emphasize reasoning and discussion (Hill, 2006). Importantly, within each culture, the parenting practices that are valued are associated with more positive and adaptive outcomes.

While most research on parenting and brain development has been conducted in White samples, emerging theory and research suggests that the brain may be shaped by cultural experiences within the family context (Qu, Jorgensen, & Telzer, 2021). For example, Mexican American adolescents who tend to place a higher value on familism, show greater activation in the ventral striatum when making decisions to contribute to their family that involve self-sacrifice, compared to European American adolescents who show more ventral striatum activation when gaining for themselves and not their family (Telzer et al., 2010). Importantly, heightened ventral striatum activation when contributing to their family predicts longitudinal decreases in risk-taking behavior and depression among Mexican American adolescents (Telzer et al., 2013, 2014), suggesting that culturally shaped neural

processes have implications for youths' adjustment and well-being over time. These findings underscore the importance of broadening samples to include families from diverse backgrounds to augment current theory and support culturally sensitive policies and family-based interventions that are sensitive to variations in cultural values across families. This is particularly important for culturally grounded intervention efforts that may have iatrogenic effects if non-culturally sensitive efforts are implemented (Hecht et al., 2003).

Conclusions

Altogether, this chapter reviewed literature characterizing relations between parenting and human brain development, describing research underscoring direct effects of the full continuum of parenting as well as interactive effects through early childhood to adolescence. Evidence to date makes clear that not only is extreme adversity in the caregiving environment, as in the case of abuse and neglect, related to the function and structure of the developing brain, but also that normative and perhaps more subtle variations in parenting quality are tied to youth's neural development as well. This notion provides optimism that prevention efforts targeting parenting practices and family functioning may have the potential to significantly impact youth development and that even brief preventative interventions may have important cascading effects. Moreover, families experiencing a wider range of hardship, from severe adversity to more normative stressors, can benefit from parenting support, education, and intervention.

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4 Parenting and Children's Social and Emotional Development: Emotion Socialization across Childhood and Adolescence

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It can be argued that the most important influence on children's social and emotional development is the parent-child relationship. Starting before birth, parental history, parenting behavior, and parent-child interactions influence children's development, growth, and adjustment to their ever-changing world. In this chapter, we focus on how parenting influences social and emotional development, with a focus on emotion socialization (ES), the ways in which caregivers teach children about emotions and emotion regulation within the self, others, and relationships. ES lays the groundwork for future social interactions and shapes both normative and atypical development, especially children's emotion regulation abilities (Eisenberg et al., 1998a; Morris et al., 2017a; Spinrad et al., 2020). Indeed, children's successful management of negative emotions such as anger and sadness has been linked to academic achievement, social competence, resilience, mental and physical health, and success later in life. In contrast, children's difficulties regulating emotions have been linked to internalizing and externalizing problems, relationship difficulties, substance abuse, and adjustment difficulties more broadly (Eisenberg & Morris, 2002; Zeman et al., 2006). It should be noted that parents' own emotion regulation is key to caregiving and the ES process, and children's

learning about emotions often occurs by observing parents and modeling their behavior (Morris et al., 2007). Moreover, there is a nature and nurture component of ES due to genetic similarities among parents and children paired with parenting practices and socialization techniques that influence emotional development within both social and biological contexts.

It is also important to note that children are active in the ES process and in their own social and emotional development (Bell, 1979). For example, children's temperamental predispositions toward emotionality and regulation, perceptions of parents and social situations, choice of social interactions, and history with caregivers all influence emotional development and emotion regulation (Morris et al., 2007). The ways in which children interpret ES practices and their perceptions of the parent-child relationship play significant roles in the developmental processes discussed in this chapter.

We utilize a developmental lens to discuss ES in this chapter across different developmental periods: infancy and early childhood (0–6); middle childhood (6–11); and adolescence and young adulthood (11–25). These age ranges are broad and somewhat arbitrary, but provide a general guide for thinking about ES across different developmental stages, and studies discussed in relevant sections generally

include participants in these age ranges. We begin the chapter by discussing the study of ES, defining important constructs and discussing relevant theories and measures. Next, we discuss ES across developmental periods. Then, we have sections on the importance of fathers and culture in the ES process. We end with examples of parent education programs with a focus on ES and emotion regulation and suggest next steps in research, practice, and policy related to social and emotional development.

The Study of Emotion Socialization

When children are young, attachment to a caregiver is one of the primary influences on children's emotional development. Children who are securely attached feel emotionally safe, and caregiver responses are typically predictable and responsive. In contrast, children with insecure attachments may not know what to expect emotionally from a caregiver, resulting in either avoidant behavior that minimizes emotional expression or emotion dysregulation in response to inconsistency which may reflect efforts to get the caregiver to respond to the child's needs (Morris et al., 2017b). As children grow older, parenting is often examined in terms of parenting style, indicative of the emotional climate of the parent-child relationship. Authoritative parents tend to be warm and accepting, typically encouraging independence and acceptance of emotional expressions. In contrast, authoritarian parents are more likely to punish a child for expressions of negative emotions, and may be overly harsh in their response to emotions (Morris et al., 2013). Indeed, parenting style can be thought of as a constellation of parental attitudes toward the child, predicting situation-specific parenting practices (Darling & Steinberg, 1993).

Emotion socialization (ES) is broadly defined as socializing behaviors that influence

a child's learning regarding emotion experience, expression, and regulation, and emotion-related behavior, in a manner consistent with socializers' beliefs, values, and goals about emotion with a goal of functioning well and adapting to society (Eisenberg et al., 1998b). ES practices occur within the context of the parent-child relationship, and parental behaviors may be interpreted differently by children depending on attachment or parenting style. Moreover, there is strong evidence that ES impacts social and emotional adjustment across various domains of development regardless of parenting style (see Spinrad et al., 2020). Gottman et al. (1996) categorized parental ES as either emotion coaching or emotion dismissing (see Table 4.1). Emotion coaching parents tend to be accepting of emotions, help children problem-solve in emotional situations, and label and talk through emotional experiences with their children. Further, emotion coaching parents are more likely to be authoritative in their approach to ES as exhibited by high levels of responsiveness to their children's individual developmental needs and high levels of warmth and support in how they react to their children's negative emotions (Morris et al., 2013). In contrast, emotion dismissing parents typically punish children for expressing emotions or minimize or neglect children's feelings.

Furthermore, emotion regulation develops as the child observes parents' emotion expression and sees parents as social referencing agents in regulating their own emotions (Eisenberg et al., 1998a). Family emotional climate is informed by the parent-child relationship/attachment, parenting styles, family expressivity, and marital/couple relations. Parents also impact children's emotion regulation by directly reacting to children's emotion, supportive coaching, encouraging emotion expression, and teaching emotion regulation strategies. (Morris et al., 2007). Numerous

Table 4.1. *Examples of supportive and unsupportive emotion socialization practices*

Supportive emotion socialization		Unsupportive emotion socialization	
Construct	Example	Construct	Example
Labeling emotions	"I can see that you are angry. Will you try telling me why you feel this way?"	Minimize	"It doesn't matter. Everybody feels sad sometimes, but you're fine."
Expressive encouragement	"It is absolutely okay to feel sad about this. I am always here for you."	Dismiss	"You need to act your age. Crying is for babies."
Problem-solving	"Losing games is tough. Maybe we could go practice kicking the ball together when we get home?"	Punish	"If you don't get over it right now you are going to sit in timeout."
Reward	"I'm sorry you are having a hard day today. Maybe we can go get some ice cream later?"	Magnify	"Don't you dare raise your voice at me young lady!" (parent yelling)
Overjoy	"It is so good to see you smiling! It makes me so happy to see you happy."	Override	"This is nothing to be upset about, everything is great!"
Cognitive reappraisal	"I know you wanted two snacks, but look how happy you made your sister by sharing with her."	Neglect	"Don't talk to me until you are done being angry."

Note. Each construct included can be organized into one of Gottman's four Emotion Socialization Parenting Styles, with the Supportive ES constructs typically fitting with the Emotion Coaching style, and the Unsupportive ES behaviors typically fitting the Dismissing and Disapproving styles (Gottman, 2011).

studies have found that the relation between ES and children's adjustment is mediated by children's emotion regulation, indicating that one important mechanism through which parenting impacts children's development is via its impact on children's emotion regulation abilities (Morris et al., 2017a; Spinrad et al., 2020).

In a recent parent-child emotion regulation dynamics model, Morris et al. (2018) furthered the theoretical understanding of ES by emphasizing the dynamic and dyadic processes involved in the development of emotion regulation, highlighting that moment-to-moment physiological and behavioral dynamics during

real-time parent-child interactions are one core mechanism of how ecological contexts (including culture, neighborhood, school, peer, and family, etc.) are linked to short- and long-term development of emotion regulation and psychosocial adjustment. Indeed, physiological processes (e.g., heart rate, hormonal levels) that occur in a dyadic context are often indicative of synchrony/attunement of physiological and behavioral mechanisms related to emotionality and emotion regulation (see Feldman, 2012).

To assess parental ES practices, researchers have primarily used child reports of parenting practices or parent self-report questionnaires such as the Coping with Children's Negative

Emotions Scale (CCNES; Fabes et al., 2002) and Emotions as a Child Scale (EAC; Magai & O'Neal, 1997). These scales have typically been used among parents of school-age children and adolescents to capture supportive responses to children's negative emotions such as support/reward; emotion- and problem-focused reactions; and unsupportive reactions such as neglect, magnify, and punitive and minimization reactions (see Table 4.1). In addition, other scholars have incorporated Gottman's meta-emotion philosophy and developed open-ended interviews to inquire about parents' feelings, their reactions, and the reasons for their reactions when their children express negative emotions (e.g., Raval & Martini, 2011). Structured observational recordings of parent-child discussions of emotional events have been coded to assess emotion coaching and dismissing behaviors in both the laboratory (e.g., Cassano & Zeman, 2010) and the home (e.g., Havighurst et al., 2010). Researchers have even used naturalistic observations of parent-child interactions (e.g., Cole et al., 2006) and coded recordings of families' daily lives and parent-child interactions to capture parental ES (e.g., Sperling & Repetti, 2018). Recently, researchers have proposed real-time affect dynamics models of parent-child interactions and explored the physiological processes underlying ES (Lougheed et al., 2020; Morris et al., 2018). Generally, supportive and unsupportive ES is linked to different physiological dynamic change patterns (e.g., Zhang et al., 2017).

Emotion Socialization across Developmental Periods

While the ES process occurs across development, various developmental considerations are important to note specifically during infancy, middle childhood, and adolescence and young adulthood. Additionally, each of

these developmental phases has unique themes to consider, including biology, psychopathology, and neurodevelopment.

Infancy and Early Childhood

One-year old Julia is just learning to walk. She toddles on along the sidewalk in front of her house and her foot hits a small rock. She falls and scrapes her knee. Julia immediately looks to mom who quickly picks her up and says, "You fell and scraped your knee, that can be scary. I bet it hurts. Let's go and get you cleaned up and put a band-aid on your boo boo." Before Julia can even start to cry, she is cuddled in mom's arms, feeling safe, and looking forward to her superhero band-aid.

Julia's mother displayed many behaviors involved in ES, specifically emotion coaching: she labeled emotions, problem-solved the situation, and soothed and comforted Julia. When these types of behaviors are used consistently, children feel safe due to the sensitivity and responsiveness of caregivers. In contrast, Julia's mother could have ignored her, said "get up, you are fine," or over-reacted by displaying fear and distress herself. These are examples of unsupportive ES behaviors.

The socialization of emotion regulation begins early in infancy and toddlerhood, with the regulation of emotions and behaviors largely depending on support from parents. As highlighted in the vignette, this process involves early interactions with primary caregivers, which create the foundation for children's ability to eventually develop autonomy and regulate their own emotions (Eisenberg et al., 1998a; Morris et al., 2007). Parents attempt to regulate their infant's emotions by physically soothing, addressing the child's immediate needs, adjusting their own facial expressions, or changing the infant's current

environment (Kopp, 1989). As an infant transitions into early childhood, they begin developing cognitive and emotional skills that allow for increasingly complex emotion regulation strategies. Parents continue to assist in regulating their child's emotions but begin to do so more through parallel practices such as modeling and coaching (Gottman et al., 1996; Morris et al., 2007). Additionally, parents can talk about emotions or emotion-eliciting situations directly, contributing to the child's emotion knowledge (Thompson et al., 2020). Emotion knowledge and newly developed cognitive strategies allow children to internalize, express, and regulate their emotional experiences with greater independence, though at this stage the parent continues to play a primary role in guiding the emotion regulation process (Morris et al., 2007). As such, it is understood that the dynamic process of ES develops and unfolds with increasing experiences and parent-child interactions (Lougheed et al., 2020).

As mentioned previously, parents build and shape their children's schemas about emotion through their own emotional expressions in the home (see Eisenberg et al., 1998a). These schemas allow children to attribute meaning and context to emotion, as well as determine whether an emotion would be considered "appropriate." Parents' discussions of emotion with their children also play a role. When parents talk about emotions both during and following an emotional situation, they can help children understand how emotion fits into general social contexts (Zhang et al., 2020). In addition, parents' reactions to their children's emotions socialize the child's understanding, expression, and regulation of their own emotions. It has been documented that parental reactions can either support and facilitate children's emotional expressiveness through coaching and scaffolding or diminish emotional expressiveness through minimizing and

punishing emotional behavior (Eisenberg et al., 1996; Gottman et al., 1996).

Emotion regulation has also been found to be observable at the physiological level. For example, it has been found that children's parasympathetic regulation of emotion can be influenced by parental ES from the earliest stages of life (Perry et al., 2020; Zhang et al., 2020). Many studies have examined young children's vagal withdrawal as an indicator of heightened arousal (Porges, 1995). For instance, warmth and supportiveness from mothers during early infancy has been associated with enhanced vagal withdrawal in response to challenging or emotion-eliciting tasks at 10 months (Perry et al., 2016). Additionally, mothers who provided high levels of emotional support to their child at age three had children with greater levels of vagal withdrawal during challenging tasks at age four compared to children with mothers who displayed low levels of emotional support (Perry et al., 2013). Similarly, children of mothers who were non-supportive in their reactions to their child's negative emotions at the age of five years were found to have poor physiological and emotion regulation at age 10 (Perry et al., 2020).

ES practices also can influence brain development during early childhood. Indeed, parents' ES practices can be associated with differences in both the structure and function of the neural networks linked to emotion processing and regulation (Kerr et al., 2019; Tan et al., 2020). The development of the infant brain is particularly sensitive to external influences, including interactions with caregivers. High levels of maternal sensitivity in infancy have been associated with smaller volumes in limbic structures, including the amygdala and hippocampus, which likely leads to decreased stress reactivity (Rifkin-Graboi et al., 2015). In addition, maternal sensitivity is positively related to increased functional connectivity

between the hippocampus and regions in the prefrontal cortex (PFC) implicated in emotion regulation (Rifkin-Graboi et al., 2015). As children develop more advanced self-regulation abilities throughout early childhood, explicit ES practices (e.g., emotion coaching) can further impact the function of brain regions involved in emotion regulation. For instance, Grabel et al. (2019) examined the neural impact of an interpersonal scaffolding activity in which four- to six-year-old children were instructed to think about how a character was feeling and how this emotion felt in the character's body. Using functional near-infrared spectroscopy (fNIRS), the authors found that the scaffolding was associated with greater activation of the children's lateral PFC during a frustrating task, suggesting enhanced emotion regulation. On the other hand, harsh parenting practices during early childhood may have adverse effects on emotion-related brain function. For example, the experience of harsh parenting at three years of age has been associated with increased error-related negativity (ERN) at six years of age. ERN is a neural response to mistakes that can be measured by electroencephalography (EEG) and is associated with negative affect and anxiety (Meyer et al., 2015). Thus, the literature suggests that parents' ES practices can meaningfully impact the functional and structural development of brain regions during the foundational period of early childhood, setting the stage for later development.

Middle Childhood

Eight-year-old Julia comes home from school with tears in her eyes. Dad is home, and he gets her a snack and asks what's wrong. Julia tells him about being left out of a game of four-square on the playground. Dad asks Julia about how that made her feel, and then

they talk about what she can do if that happens again. Julia gives the answers while dad listens, and then they enjoy fresh-baked chocolate chip cookies.

Independence and autonomy continue to be important developmental goals during middle childhood, and ES should involve less instruction and more listening and guidance than in early childhood. Julia and her dad clearly have a good relationship, and she is willing to talk with him about her problems. Importantly, Dad is not driving the conversation, but is helping Julia to navigate it on her own.

As children enter middle childhood, they start to explore new environments, such as school and extracurricular activities, and are assigned multiple roles with new responsibilities (Collins & Madsen, 2019). They face increasing academic, social, and emotional challenges, and their relationship with parents is also challenged. During middle childhood, children interact less frequently with their parents, and both children and parents show less overt affection to each other (Collins & Madsen, 2019). Because of this, children in middle childhood learn more emotional self-regulation strategies and are more likely co-regulate with parents (Morris et al., 2007). Generally, authoritative parenting, particularly the emotion-related components such as warmth and support and autonomy granting, predict self-esteem, social skills, and emotional competence. As in early childhood, emotion-related parenting practices such as reactions to children's negative emotions are associated with children's emotion regulation skills and psychosocial adjustment outcomes. For example, Silk et al. (2011)'s found that maternal neglect and punishment were associated with increases in children's internalizing symptoms over one year. Likewise, Valiente et al. (2007) found that parents' supportive reactions to emotions were positively linked to children's effortful control

(a construct closely linked to emotion regulation) whereas unsupportive reactions were negatively related to children's effortful control. Similarly, Suveg et al. (2011) found that maternal supportive reactions were negatively linked to children's externalizing problems, and unsupportive reactions were positively linked to externalizing problems and negatively related to children's emotion regulation.

Similar to infancy and early childhood, research suggests that parenting practices during middle childhood may influence brain regions involved in emotion regulation and processing. A recent functional magnetic resonance imaging (fMRI) study by Pozzi et al. (2020) examined the link between observed maternal parenting practices during a cooperative, problem-solving interaction and school-age children's brain activation during an emotional faces task. The results of this study revealed that greater negative parenting behaviors were associated with increased task-related amygdala activation in children. In another investigation, parental elaboration (i.e., discussing emotional experiences and reactions for a child's benefit) during conversations about a child's negative experiences has been linked to lower levels of neural reactivity to emotions (as indexed by decreased late positive potential [LPP]; Leventon et al., 2019).

Because parents and children continue to engage in co-regulation during middle childhood, studies also have examined the link between parents and children's emotion-related neural activity. Specifically, studies have used hyperscanning (i.e., simultaneous neuroimaging of two individuals) to assess synchrony and cross-brain connectivity in interacting parent-child dyads (Misaki et al., 2020). One such study using fNIRS found that mothers and their school-age children demonstrated increased synchrony within emotion regulation regions (i.e., dorsolateral and frontopolar PFC) during cooperative versus independent

tasks (Miller et al., 2019). Interestingly, the level of synchrony was different for mother-daughter and mother-son dyads (Miller et al., 2019) suggesting that sex-differences may impact the neural mechanisms underlying ES. Together, these neuroimaging studies provide initial evidence that ES influences neural development in middle childhood.

Adolescence and Young Adulthood

Julia is a senior in high school.

Recently, she came home past her curfew by just a few minutes, and her mom thought she smelled alcohol on her breath. The next day, mom and dad talk with Julia about the night before. They ask her what happened, and she admits to having a few beers. Julia and her parents discuss safety and potential consequences of under-age drinking and drinking and driving. They agree that Julia will never drink and drive and will always communicate with them about where she is. They also discuss consequences for drinking and driving and coming home late (i.e., taking away her car for a week). They explain that the reason for taking away her car is not to punish her or because she is a bad person, but to help her learn that her decisions have consequences, and to give her time to think about her behavior. At the end of the conversation, mom and dad check in to be sure she is feeling okay, and let her know that she can always come to them with any problems she has.

In adolescence and young adulthood, ES still involves checking in about emotions and problem-solving, as Julia's parents do in the vignette. But it also involves being aware about the risks and realities of adolescent development. Parents' need to establish safe boundaries to protect teens and help them make

informed decisions about behavior that likely has an emotional component. As adolescents become young adults, parents continue to guide and support decision-making and behaviors through conversations, sharing accurate information, and providing emotional support.

There are a number of neurological and interpersonal milestones relevant to ES and emotion regulation during adolescence (i.e., 11–18 years) and young adulthood (i.e., 18–26 years). For example, the PFC develops greatly during adolescence, which is tied to higher-order cognitive processing, self-control, and reasoning and therefore allows for enhanced emotion regulation abilities (Tottenham & Galván, 2016). During young adulthood, there is further growth and full maturation of the PFC (Eshel et al., 2007). In addition to the developmental changes in the brain, adolescents and young adults experience transformations in their relationships with their parents. Specifically, parent–child relationships become more horizontal and peer-like in nature during adolescence (Branje et al., 2013). Compared to childhood, adolescents also spend less time with their parents (Larson et al., 1996) and develop a more balanced view of their parents as authority figures (Smetana, 2000). During young adulthood, parent–child relationships experience decreases in conflict and increases in closeness, intimacy, and open communication (Morgan et al., 2010). Despite these transformations, parents continue to be important influences in the lives of their children during adolescence and young adulthood.

Perhaps owing to these neurological and interpersonal developmental milestones, research has demonstrated that effective parental ES efforts marked by high levels of supportive ES (e.g., emotion coaching) and low levels of unsupportive ES (e.g., emotion dismissing, psychological control) are linked to increased emotion regulation during adolescence and

young adulthood. For example, Cui et al. (2020) found that high levels of supportive maternal ES were significantly related to decreased negative affect and increased emotion regulation in adolescents. In addition, unsupportive maternal ES was significantly and negatively related to adolescent internalizing problems. Using a college student sample, Buckholdt et al. (2009) reported that unsupportive ES (i.e., punishment, neglect, and magnification; see Table 4.1) was significantly and positively related to difficulties regulating emotions. Manzeske and Stright (2009) found that psychological control was positively and significantly related to young adult emotion regulation difficulties. Other than specific ES practices, parental emotion expression during parent–adolescent interaction also contributes to adolescent emotional competence. For example, Cui et al. (2015) found that observed parent anger during parent–youth conflict discussion was linked to poor vagal regulation (indicated by decreases in youth’s respiratory sinus arrhythmia [RSA] levels) whereas observed parent positive affect was linked to better vagal regulation (increases in youth’s RSA levels) in real time.

Parents’ ES practices continue to play an important role in neural development during adolescence and young adulthood (Tan et al., 2020). In regards to brain structure, negative parenting practices have been associated with decreased volumes in regions involved in emotion regulation. For instance, young adults who have a history of experiencing harsh corporal punishment from parents exhibit significantly lower volumes in the medial PFC (Tomoda et al., 2009). Considering brain function, research on adolescents suggests that supportive parenting practices are associated with decreased activation in the amygdala while viewing emotional faces (Romund et al., 2016). This suggests that, similar to middle childhood, high levels of supportive practices

(and low levels of negative practices) may lead to decreased emotional reactivity in adolescents. Additionally, a variety of fMRI studies have examined the role of mothers' evaluative statements (i.e., praise and criticism) on emotion-related neural activation in adolescence. For instance, Butterfield et al. (2020) found that high levels of warmth were related to decreased activation in emotion processing and regulation regions including the amygdala, insula, anterior cingulate cortex, and ventrolateral PFC (Butterfield et al., 2020). The findings from this study demonstrate that positive ES practices may allow for more adaptive neural responses to emotional stimuli in adolescence.

Simultaneous neuroimaging and hyperscanning have also been used to explore parent-child dynamics in adolescence. Kerr et al. (2020) used simultaneous fMRI scanning to examine brain activation of parent-adolescent dyads during the Testing Emotion Attunement and Mutuality (TEAM) task. The TEAM task is presented as a cooperative game in which both members of a dyad must recreate patterns correctly to avoid losing prize money. However, the task is pre-programmed to occasionally show participants that their partner made a costly error. The results of the study demonstrated that high levels of positive parenting practices were associated with increased activation in the adolescent ventromedial PFC while viewing their parents' costly errors. These findings suggest that youth with warm and supportive parents may engage in more emotion regulation and empathy in response to their parents' errors (Kerr et al., 2020). Further, in another study using fMRI hyperscanning of parent-adolescent dyads, Ratliff et al. (2021) found that parents and adolescents exhibited cross-brain connectivity (i.e., functional connectivity between brain regions of interacting participants) in a variety of emotion processing and regulation regions.

These innovative studies suggest that parents' ES practices may not only influence children's neurobiology longitudinally throughout development but also during real-time social interactions.

The Role of Fathers in Emotion Socialization

Despite the exponential growth in our understanding of ES over the past two decades, the role of fathers in ES processes continues to be understudied. Considering both the similarities and differences found between maternal and paternal ES processes in parenting, additional research is needed on the independent effects of paternal ES, the relationships between ES effects in the co-parenting dyad, and the role of child sex on paternal ES effects (e.g., father-daughter versus father-son). Notwithstanding existing gaps in the literature, recent studies have helped to identify differences in paternal ES processes during various stages of life, including infancy and early childhood, middle childhood, and adolescence and early adulthood.

First, considering infancy and early childhood, findings regarding the role of fathers in ES are encouraging. For example, McElwain et al. (2007) found that fathers who were supportive in their reactions to their child's negative emotions had children with heightened emotional understanding, even after accounting for the mother's ES. Similarly, positive paternal parenting has been linked to children's effortful control after statistically controlling for maternal parenting (Karreman et al., 2008). Attachment security with fathers has been found to be directly associated with children's emotion regulation behavior in frustrating contexts, and children with more secure relationships with their father were found to regulate their emotions better during challenging lab tasks (Boldt et al., 2020). Moreover,

these supportive paternal responses to emotions in early childhood predicted better emotion regulation in early adolescence, while non-supportive father responses tended to predict higher aggression and engagement with delinquent peers (Godleski et al., 2020).

In middle childhood, studies have revealed notable differences between maternal and paternal ES strategies. For example, Cassano et al. (2007) found that fathers tend to respond to their child's sadness by utilizing minimalizing strategies, while mothers tend to use excessive encouragement. Similarly, Eisenberg et al. (1996) found that maternal problem-focused reactions were associated with better coping and social skills and girls' popularity, and minimizing reactions were linked to poor social skills and poor coping skills. Surprisingly, paternal minimizing and problem-focused reactions were associated with poor social skills and low popularity, which may indicate that daughters' problems in social competence trigger paternal use of problem-focused reactions. Regarding child outcomes, studies have shown that fathers who are highly balanced (i.e., moderate warmth/responsiveness and non-supportive reactions, below average expressive encouragement) in their responses to their child's emotions tend to have children who engage in high levels of cognitive appraisal of emotions in middle childhood (Wang et al., 2019). Considering the unique contributions of paternal ES, it has been argued that while mothers and fathers tend to contribute to more similarities than differences in emotion regulation during middle childhood, each parent offers unique stances on emotion that, when adaptive, ultimately afford beneficial experiences in the emotional development of the child (McElwain et al., 2007).

During adolescence and early adulthood, although mothers tend to be closer to their children and more involved in their lives compared to fathers (Branje et al., 2013), fathers

still play a critical role in emotional development. For example, McEwen and Flouri (2009) reported that high levels of father warmth and low levels of father overprotection and psychological control were related to low levels of difficulties in emotion regulation among adolescents (aged 11–18 years). Using a college student sample, Lugo-Candelas et al. (2016) found that high levels of father supportive ES and low levels of father unsupportive ES were related to poor student mental health (e.g., depression, anxiety, substance use). Few published studies have examined whether father socialization is incrementally related to adjustment after controlling for mothers' parenting. However, one study using a longitudinal design in which mother and father parenting were analysed simultaneously, Van Lissa et al. (2019) found that the intercept for father support was more strongly related to adolescent emotion regulation compared to maternal support. In addition, the intercept for paternal psychological control was significantly and negatively related to emotion regulation after controlling for the intercept for maternal psychological control. This evidence suggests that fathers' emotion-related parenting may afford unique socialization benefits above and beyond that of mothers during adolescence, though this may vary depending on the dimension of parenting.

Cross-Cultural Considerations

Understanding the cultural meaning of parental ES practices is critical, as ES practices occur within context and culture. Studies adopting ES theories and measures developed in U.S. populations to other cultures have revealed both similarities and differences regarding the importance of supportive and unsupportive ES. For example, minimizing and invalidating reactions were not significantly linked to emotion regulation or

internalizing or externalizing problems in Chinese children and adolescents in several studies (e.g., Tao et al., 2010). It is possible that certain ES practices may not be perceived as negative or intrusive in the Chinese culture when combined with other parenting behaviors. Indeed, Wang et al. (2019) found that paternal minimization was positively linked to Chinese adolescents' emotion regulation skills when it was used in combination with high problem-focused and emotion-focused responses. Further, parental ES efforts marked by expressive encouragement and minimizing reactions/responses may have different cultural implications in China compared to other cultures (Wang et al., 2019), as inhibition or suppression of emotion expression is valued in Eastern culture for maintaining social harmony (Matsumoto et al., 2008). Using a college student sample, Lugo-Candelas et al. (2016) reported that maternal supportive ES was negatively related to student mental health symptoms among Latino Americans but not European Americans or African Americans. Moreover, maternal unsupportive ES was positively related to student mental health symptoms among European Americans and African Americans, but not among Latino Americans. These findings suggest that certain parental ES efforts may be more or less salient in different cultural or ethnic groups.

Researchers also have expanded ES theories and adapted ES measures to identify culturally salient parental ES practices. For example, researchers adapted the CCNES to Chinese and Indian parents by adding more culturally relevant emotion-eliciting scenarios and identified culturally relevant responses, including supportive responses such as explanation-oriented and reflection/moral-enhancement responses and unsupportive responses such as scolding and not talking to the child for a brief period (e.g., Chan et al., 2009; Raval et al., 2018). Among urban, educated, middle-class

families in China and India, parents may establish goals such as relatedness/connectedness and autonomy/independence, which are used to train their children to understand the impact of their behavior and explain how their emotion regulation influences relationships with others (Raval et al., 2018). Such socialization efforts are referred to as *reflection-enhancement* (Chan, 2012), which derives from Chao (1994)'s concept of *Guan* (training) and involves socializing emotions with moral rules. Asian parents are likely to use secondary responses such as psychologically controlling strategies, scolding, or not talking to the child when their primary training practices do not work (Raval & Martini, 2011; Raval et al., 2014). Indeed, Chan (2012) found that among Hong Kong Chinese mothers, high levels of endorsement of the value *Guan* were linked to greater use of both supportive and unsupportive ES practices. Such findings demonstrate that the Hong Kong mothers' endorsing style of *Guan* may adopt coaching strategies when their children show negative emotions. However, these mothers may utilize punitive and minimizing strategies when supportive strategies fail.

Cultural differences regarding the use of ES may be due to that fact that parental values and goals influence emotion-related parenting styles and behaviors. For example, Chan et al. (2009) found that mothers emphasizing the relational emotional competence of promoting interpersonal harmony were more likely to use both the coaching and emotion-dismissing approaches and less likely to use the emotion encouraging approach. Mothers emphasizing individualistic emotional competence of promoting self-esteem use both coaching and emotion encouraging approaches and were less likely to use the emotion dismissing approach. Further, Chan (2012) found that mothers who believed that children should be exposed to different emotional experiences (more of a

Western value) were more likely to use supportive strategies and less likely to use unsupportive responses. In contrast, mothers who did not believe that parents should play an active role in emotion teaching used more unsupportive responses than supportive responses. Indian mothers were more likely to emphasize interconnectedness than autonomy in the socialization goals and on the CCNES, and endorsed the explanation-based responses more than the problem-focused responses (Raval et al., 2014). Regarding the effects of ES, overall findings among Eastern populations were consistent with those found among Western samples (Raval et al., 2018). However, the implications of some specific responses on adjustment outcomes may differ across cultures. For example, expression encouragement in Asian samples was positively related to unsupportive responses, and it was not related to child outcomes (Raval et al., 2018). Problem-focused responses were not significantly related to youth outcomes among Indian families (Raval et al., 2014). Such findings suggest that broader social, political, and economic factors may shape parents' socialization goals and meta-emotion beliefs (Raval & Walker, 2019), which in turn may influence socialization practices (Raval, et al., 2018). Further, the links between ES practices and child outcomes may also vary by cultural contexts.

To better understand ES across cultures, researchers have made efforts to adopt a grounded theory approach involving parent-child conversations about emotions. For example, home visitors of the same cultural backgrounds were sent to Tamang and Brahman village households in Nepal to observe three-, four-, and five-year-old children (Cole et al., 2006). Social interactions between children and their primary caregivers were coded based on the observation of narratives. Nepali caregivers' responses to children's

anger and shame included ignoring, teaching, scolding, teasing, cajoling/coaxing, and nurturing. A similar narrative approach was taken to study emotion talk among native and immigrant Chinese families as well as American families. In a series of studies by Wang, the scholar asked mothers and their three-year-old children to talk about shared memories of positive and negative emotional events or asked mothers to tell a story with their children using a picture book (Wang, 2013). Coded mother and child emotion talk demonstrated a variety of major ES strategies, i.e., emotion attributions (language ascribing an emotional state or reaction to the child or the main character of the story), emotion explanations (language explaining the causes and consequences of emotions), and emotion criticism (language focusing on installing proper behavior by using moral lessons). Such bottom-up approaches capture nuances of parental ES practices in different cultural contexts.

Taken together, there are both similarities and differences in the ES goals and practices across cultures. Researchers adopting and applying existing ES theoretical models and measures to a different culture should bear in mind that both parents' understandings of the same ES practices and the implications of some ES practices on child developmental outcomes may differ. Thus, parental values and goals in the specific cultural context should be considered. Further, the endeavor to search for culturally salient ES goals and practices is warranted, particularly using grounded theory to understand ES across contexts and cultures.

Programs and Interventions Targeting Emotion Socialization and Regulation

Considering the impact of parental ES behaviors on child emotion regulation and

adjustment, a number of parent and child interventions have been developed featuring content and practices designed specifically to target parents' ES behavior and practices (Morris et al., 2020). Common themes found within these interventions include teaching parents how to aid their child in identifying and labeling emotions, improving children's emotion coping strategies, and instruction on how parents can respond to their children's emotional expressions in ways that are supportive, rather than dismissive or punitive (Porzig-Drummond et al., 2014).

One notable aspect found to contribute to the success of these interventions is the inclusion of ES and emotion regulation as a foundational intervention priority, rather than as an add-on to a previously established program (Maliken & Katz, 2013). With this finding in mind, it may be advisable for those developing interventions and programs related to emotion regulation to build outward, with goals related to ES and emotion regulation at the core (Hajal & Paley, 2020). The number of ES and emotion regulation-based interventions continues to grow, as does the body of evidence indicating that interventions targeting parents' emotion understanding and emotion regulation are capable of predicting improvements in parental emotion-related socialization behaviors and child emotion regulation capacities (Eisenberg, 2020; Hajal & Paley, 2020). Indeed, parents who possess greater competence in regulating their own emotions tend to be better prepared for assisting their children's emotional abilities (Zachary et al., 2019). For example, parents with emotion regulation difficulties experienced challenges in employing newly acquired skills with their children (Maliken & Katz, 2013). Thus, asking emotionally vulnerable parents to implement emotion-based training with their children can be ineffective or even counterproductive (Hajal & Paley, 2020).

In Table 4.2, we provide examples of interventions designed primarily for or featuring components related to enhancing parents' ES behaviors and subsequent child emotion regulation. These interventions were shaped by various theoretical perspectives (e.g., family systems, attachment) and were designed to target various individuals/systems in the family, including the child, the parent, the parent-child relationship, or the family system. This list is not exhaustive. The interventions were identified in reviews of parenting interventions designed to cultivate secure parent-child relationships and enhance child adjustment through emotion regulation-based practices (Hajal & Paley, 2020). It should be noted that this overview does not include therapeutic methods such as parent-child interaction therapy, cognitive behavioral treatments, or trauma-focused cognitive behavioral therapy. For additional information on these treatment approaches, see a review by Hajal and Paley (2020).

Attachment Biobehavioral Catch-Up (ABC; Dozier et al., 2009) was developed to assist caregivers of infants and toddlers who have experienced early emotional adversity, while improving the child's attachment and emotional expression. ABC emphasizes teaching parents how to interpret and respond to their young child's inconsistent emotional cues. This intervention helps parents become aware of their own emotional responses to their child's potentially ambiguous emotions and to reframe the responses in an adaptive way by considering the child's emotional history. One key feature of this intervention is an emphasis on in-the-moment feedback from program facilitators to allow teaching and practice opportunities in real time. Primary outcomes associated with the ABC intervention include improved parent-child attachment, increased child emotion expression, and enhanced emotionally supportive parenting (Berlin et al., 2014).

Table 4.2. *Overview of empirically supported programs that include an emotion socialization focus*

Intervention	Delivery method	Languages available	Targeted caregivers	Selected child and family outcomes	Example of evaluation studies
Attachment biobehavioral catch-up abcintervention.org	Home Visit; 10 weekly sessions	English German Mandarin Norwegian Russian Spanish	Parents of young at-risk children ages 0–2	• Improved child attachment • Increased child emotion expression • Enhanced emotionally supportive parenting	Berlin et al., 2014; Lind et al., 2014
Circle of security circleofsecurityinternational.com	Group; 20 weekly sessions	Danish English Italian Japanese Spanish Swedish	Parents of children ages 0–5	• Decreased unsupportive emotional responses by mother • Enhanced emotional functioning • Improved child attachment	Cassidy et al., 2017
Minding the baby Medicine.yale.edu/childstudy/communitypartnerships/mtb	Home visit; weekly to biweekly for 24 months	Danish English Spanish	High-risk first-time parents of children 0–2	• Increased maternal reflective functioning • Improved infant attachment	Slade et al., 2020
Positive parenting program – enhanced www.triplep.net	Group; 10 weekly sessions	Chinese English French German Japanese Spanish	Parents of children ages 0–12	• Decreased child disruptive behaviors • Improved parental emotion regulation	Sanders et al., 2007

Reflective parenting program <i>reflectivecommunities.org</i>	Group; 12 weekly sessions	English Spanish	Parents of children ages 0–12	• Decreased parent depressive symptoms • Decreased parenting stress	Grienenberger et al., 2015
Tuning in to kids <i>tuningintokids.org.au</i>	Group; 6 weekly sessions	Arabic Cantonese English Somali Vietnamese	Parents of school-aged children	• Improved parental emotion coaching • Increased child emotion knowledge • Improved parent/child emotion regulation	Havighurst et al., 2010; Havighurst et al., 2013
Tuning in to teens <i>tuningintokids.org.au</i>	Group; 6 weekly sessions	Arabic English German Norwegian Vietnamese	Parents of preadolescents and adolescents	• Increased child emotion communication skills • Improved parental emotion socialization • Decreased internalizing and externalizing behavior in children	Kehoe et al., 2020

Note: This is not necessarily an exhaustive review or a deep comparison of programs, rather a description of the evidence for some of the well-studied programs that are widely available.

Circle of Security (COS; Powell et al., 2013) is an attachment-based intervention that is delivered in small groups with a primary focus of helping children (ages 0–5 years) build healthy and secure attachment relationships with their caregivers. This program focuses on helping parents separate their own personal attachment experiences from the current attachment needs of their child. Parents are also taught skills for self-calming to respond to their child in a more supportive and sensitive manner. COS has been found to contribute to decreased unsupportive emotional parental responses and enhanced parent–child attachment and child emotional functioning (Cassidy et al., 2017).

Minding the Baby (MTB; Slade et al., 2005) is another attachment-based program developed for mothers of infants and toddlers who have had traumatic experiences. MTB is delivered one-on-one in the home by either a nurse or social worker. One primary goal of this intervention is to increase mothers' awareness of both their own and their child's mental and emotional states. With special attention to the potentially vulnerable states of the mother, MTB attempts to improve the mother's ability to read her child's basic emotional cues and respond sensitively without potentially becoming overwhelmed by previous trauma. Evaluation of MTB has revealed program-based increases in maternal reflective functioning and improved parent–child attachment (Slade et al., 2020).

The *Positive Parenting Program* (Triple-P; Sanders et al., 2000) was primarily developed to prevent or reduce child behavior problems in children under the age of 12. Triple-P offers a number of intervention variations depending on the child's age and specific target symptoms. For emotion-related purposes, *Triple-P Enhanced* was designed to address both parent and child behavior, with a focus on helping parents manage their own and their child's

emotions and distress. Outcomes associated with Triple-P Enhanced include fewer reported disruptive child behaviors, and improved emotion regulation in parents (Sanders et al., 2007).

The *Reflective Parenting Program* (RPP; Grienberger et al., 2015) was designed for parents of children ranging from infancy to preadolescence, and focuses on training parents to separate their child's emotions from their own. RPP teaches parents how to reflect the child's emotions back through coaching and modeling as a means for developing the child's emotion regulation skills. In addition, parents are taught mindfulness practices with the intent of helping them monitor their emotional responses to their child and develop greater sensitivity. RPP has been found to contribute to decreases in parental depressive symptoms and stress (Grienberger et al., 2015).

Tuning in to Kids (TIK; Havighurst et al., 2013) and *Tuning in to Teens* (Kehoe et al., 2014) are group-based interventions designed with a core focus on enhancing parents' ES capacities. These interventions are delivered over six sessions and teach parents how to regulate their own emotions so that they can be more supportive of and responsive to their child's emotional dysregulation. In addition, parents are instructed on how to be emotion coaches for their children by increasing their awareness of their own emotions, communicating with their child or teen about their emotions, and assisting them in both regulating and recovering from emotional distress. Finally, each of these programs instructs parents in developing emotional self-care skills including a focus on proactive emotion regulation to aid parents in self-regulating during moments of high emotional intensity (Kehoe et al. 2014). Studies have shown these programs to be associated with outcomes including improved use of emotion coaching, increased knowledge of emotions in children, and improved emotion regulation in both

parents and children (Havighurst et al., 2013; Kehoe et al., 2020).

Conclusions and Future Directions

In conclusion, research on ES has indicated that mothers and fathers play critical roles in shaping their children's emotional, behavioral, and neurological outcomes during infancy and early childhood, middle childhood, and adolescence and young adulthood. Across these different developmental periods, evidence has demonstrated that high levels of supportive ES efforts and low levels of negative and emotion dismissing parenting behaviors are related to more adaptive emotion regulation and fewer adjustment difficulties. More recent investigations have demonstrated parental socialization efforts, expression of emotions, and the family emotional climate to be associated with the development of brain regions tied to emotion regulation and processing. In addition, although mothers and fathers may use different approaches to emotion-related parenting practices, evidence from the literature has demonstrated that each parent affords unique ES experiences for their children, albeit more research on the specific role of paternal ES is needed. Perhaps, the different relational attributes that characterize children's relationships with mothers and fathers may create unique socialization experiences critical for healthy emotional outcomes. Further, parental use of ES strategies and implications of ES may differ across cultures contingent on specific parental ES values and goals. Grounded theory studies have revealed some interesting culturally salient ES goals and practices. Moreover, interventions focusing on ES and regulation have produced important insights into effective programs that can be utilized with families of children at varying ages, including the importance of parental control of their own emotions during the socialization process. Collectively,

the literature has demonstrated that despite the emotional, cognitive, behavioral, and neurological transformations that occur during the first two decades of life and beyond, parents consistently play critical roles in shaping the emotional well-being of their offspring. It is important to acknowledge that as children age, parents likely need to adapt their specific ES techniques to meet the unique developmental needs of their children. That is, how a parent helps a 5-year-old with negative emotions is going to be qualitatively different than how they help a 15-year-old. Thus, authoritative parenting, which is defined by high levels of responsive parenting (Morris et al., 2013), would provide a helpful approach to adapting ES practices during children's development.

Future research would benefit from an exploration of the unique role that context plays in the ES process. Specifically, researchers should consider the parent-child relationship, family dynamics, culture, parent/child psychopathology and gender, and the history of moment-to-moment interactions in understanding ES more broadly. The perspective of the child/adolescent is also important, as children's own beliefs about emotions and relationships can shape the ways in which they interpret ES efforts. Future research also would benefit from cross-disciplinary methods in the study of ES, bringing together neuroscientists, sociologists, psychologists, clinicians, educators and others and utilizing multiple methods of analysis and data points across development. Importantly, given the importance of ES and emotion regulation in children and adolescents' social and emotional development, intervention and prevention programming designed to address these behaviors and abilities should be widely available. We argue for a public health approach to providing universal education regarding successful ES strategies and emotion regulation skills (see Morris et al., 2017c), supporting caregivers

across socio-economic divides and meeting parents where they are: through home visiting programs; group classes at childcare centers, schools, community centers and religious facilities; and providing information in doctors' offices, through social media campaigns, and in places where children and families regularly gather. Such efforts, in combination with increasingly nuanced research to understand the most important factors in supporting ES and emotion regulation in context, will likely yield promising results for children and families, impacting positive development and reducing psychopathology in future generations.

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5 Parents Matter: The Cornerstone for Children's Cognitive and Language Development

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Two-year-old Sasha heard the word “elephant” for the first time at her birthday party. The next day at the zoo, the new word flowed out of her mouth when she saw the giant elephant near the pond. Her learning was evident, and even generalizable to another elephant that looked nothing like the stuffed animal she got from her grandfather.

Language learning leaps forward when children are toddlers and preschoolers, with some studies suggesting that they learn up to nine new words a day (Carey, 1978; but see Goldfield & Reznick, 1990). Imagine being able to learn a foreign language at that rate. And other cognitive skills are not far behind. At two, Sasha struggles to fit the triangle into the shape sorter. At three, she will do this effortlessly and by five, she will even put together a simple puzzle upside down. What we might not realize is that these everyday moments harbor rich learning experiences if we know what to look for. Did you know, for example, that gobbling up words is a powerful predictor of later reading, writing, mathematics, and social skills (Pace et al., 2019)? Would you guess that simple spatial learning games couched in shape sorters and puzzles are the backdrop for later STEM (science, technology, engineering, mathematics) learning – setting trajectories that prepare children to enter into formal schooling and beyond (Verdine et al., 2014; 2017)?

In this chapter, we dive into the elusive construct of learning. What does learning *mean*, *how* do we offer children opportunities to flex their learning muscles, and *what* do children need to learn to become effective members of

our twenty-first-century society? Dubbed the *science of learning*, we recognize that children are not empty vessels to be filled with knowledge (Golinkoff & Hirsh-Pasek, 2016; Pellegrino & Hilton, 2013; Sawyer, 2006). Rather, as Piaget argued years ago (1962), children are active explorers and discoverers (Hirsh-Pasek, 2003; Sawyer, 2006), who actively construct their knowledge (Mayer, 1992). In teaching, some suggest that it is time for the “sage on the stage” to give way to the “guide on the side” (King, 1993). Think coach, not director.

To do so, we must recognize that all learning begins as social, as we are a fundamentally social species (Tomasello, 2014) with a socially prepared brain (Kuhl, 2007). Reading offers a prime illustration. Reading is not a suite of isolated skills to be learned, as a “sage” would deliver it, through drills of letters and words. First the child must develop social skills such as turn-taking, which infants demonstrate very early (e.g., Bateson, 1975; Dominguez et al., 2016). Without this parent–child collaboration, the learning of language would be impossible (Hirsh-Pasek, Adamson, et al., 2015). But the social aspect does not stop there, children who

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learn spelling-to-sound rules as a kindergartener, but who do not have the rich language base, learned through collaborative turn-taking, to support their translation from print to word, often fail to become strong readers by third grade (Chall et al., 1990; Neuman & Kaefer, 2018). Thus reading itself, a skill much touted as central to learning, is impossible if it is not built upon the backs of other skills such as collaboration and communication that are all interconnected in a whole-child perspective of development (Hirsh-Pasek, Zosh et al., 2015).

Through engaging children in socially interactive environments, we enhance the probability that they will optimize learning of all sorts. One key way to organize and support learning is through play (Sahlberg & Doyle, 2019; Sobel & Jipson, 2016; Weisberg et al., 2013; Zosh et al., 2018). Research suggests that play is central to the foundation of learning. However, there is no single path to any single cognitive outcome. These revelations afford both practical implications and allow us to recommend policies that support social, cognitive, and language development for all children. A prime lesson throughout our discussion is that the old-fashioned divisions between so-called “soft skills” such as communication, listening, and the ability to collaborate and “hard skills,” such as math and vocabulary, are no longer relevant. They are inextricably intertwined.

Learning, Cognition, and Language: What Do These Words Really Mean?

Defining Learning

Sasha did not learn the word “elephant” because she was drilled on it. It was a simple experience. She opened a package and as she found the fluffy toy amidst the wrapping paper her grandfather uttered, “It’s an elephant. Do you like it?” Her mother said, “What a lovely elephant!” Sasha’s learning happened the moment she opened the package, yet she did not say the

word then. Her use of the word the next day at the zoo assured us she was paying attention.

Learning can be dually defined as “knowledge or skill acquired by instruction or study” and “modification of a behavioral tendency by experience” (“Learning”, n.d.). These definitions distinguish two dominant approaches to learning. On the one hand, we hear famous psychologists like Pavlov (1927) and Skinner (1950) in the background. If we reward children, it will change their habits and behaviors and the result will be learning. On the other hand, we hear Piaget (1962), Vygotsky (1967), Bruner (1983), Montessori (1964), Siegler, (2000), and Oakes and Rakison (2019). If we set up the environment so that children can discover and explore, they will teach themselves. The difference is not that stark, but in caricature, we can think of these as opposed positions that appear throughout our philosophies of parenting and education. Grading systems are compatible with the Skinnerian reward system, while so-called progressive education is more tilted towards experiential learning.

Defining Cognition

Cognition is best defined as that which is mental processing and storage – or “what you know, what you remember, and what you think” (Reisberg, 2016, p.3). Under this banner, we generally find content like mathematics, language, and literacy, but also the *learning to learn* skills, including memory, attention, persistence, and problem solving. Indeed, these processes can be bundled together under the term *executive function skills* (Carlson, 2003; McClelland et al., 2007; Zelazo & Müller, 2011). Executive function skills can actually be nurtured. Dancing and clapping (Tominey & McClelland, 2011), Tae Kwon Do (Lakes & Hoyt, 2004), and “Simon Says” (Zelazo et al., 2018) have all been used in interventions that increase executive function skills. These skills seem to reside in the

frontal lobe and harnessing them has proved important for later learning in formal school (Blair, 2017; Diamond et al., 2007; Verdine et al., 2014).

Defining Language

The term *language* captures the words and combinations that we use to talk about and represent the world we live in. Young children are not explicitly taught language. They derive language from the experience of hearing it in context from caregivers who are responsive to their attempts at communication (Hudson et al., 2015; Tamis-LeMonda et al., 1996), share attentional focus with them (Adamson et al., 2019; Tomasello & Farrar, 1986), and repeat what they are saying or simplify language for them (Golinkoff et al., 2015; Weisleder & Fernald, 2013). This is no easy task. Even learning one word can be challenging (Bloom, 2002). Who is to say, as philosopher Quine (1960) once pointed out, when you label a glass by its English “glass” you are referring to the transparent container that you drink out of rather than to the transparency of the substance or the clunk the glass makes when you return it to the table? Recognizing that a blue wine goblet might also be referred to as a glass makes us all marvel at how children begin to gather up words by the time they are 12–18 months of age (Fenson et al., 1994).

Of course, having strings of words would do the child little good if they could not distinguish the difference between “mom biting dog” and “dog biting mom” – which generally occurs in the child’s expressive repertoire sometime between 16 and 24 months (Brown, 1973; Hirsh-Pasek & Golinkoff, 1996). Children gain exponentially more power when they can communicate the contents of their mind and talk about a future and a past or even an imaginary world beyond the here and now. Now that we know the key features of learning, we can first examine contexts for

learning and then begin to unpack *how* children learn and *what* children need to know to be successful in this complicated world.

Cultural Context

There are many cross-cultural similarities in parent–child interactions, such as the game “Peekaboo” which is played across a wide variety of languages and cultures (Fernald & O’Neill, 1993). However, culture-specific beliefs around infants’ role in interaction, their ability to communicate, and their autonomy, as well as beliefs about play influence these parent–infant interactions as well (Fogle & Mendez, 2006; Kuchirko & Tamis-LeMonda, 2019; Roopnarine, 2011; Weber et al., 2017). For example, during a book-sharing interaction between 4-year-olds and their mothers from European American, African American, Dominican, Mexican, and Chinese backgrounds in the United States, mothers showed culture-specific book-sharing styles, differing in content focus (e.g., emphasis of individual goals) and the extent to which they asked questions to engage children in the activity (Luo et al., 2014). Interestingly, differing beliefs and learning goals can lead to the same outcome. Western parents’ use of mental state talk relates to children’s theory of mind development (Ruffman et al., 2002), whereas parents’ use of behavior clarifications relates to Chinese children’s theory of mind development, suggesting culture-specific pathways (Liu et al., 2016).

Under-Resourced Environments

There is much research to suggest that less stimulating environments do not support strong learning (e.g. Hart & Risley, 1995; Starkey et al., 2004). By way of example, children from under-resourced environments are said to hear less language addressed to them, which negatively impacts vocabulary growth (Hart & Risley, 1995; Golinkoff et al., 2019; but see

Sperry & Sperry, 2019). This is the case within socio-economic contexts (Masek et al., 2020). Yet, this perceived deficit in language exposure may result from a variety of causes (e.g. Pace et al., 2017). Further, whether from under or more resourced environments, the same characteristics predict later language growth, such as interactions where the parent responds to their child's actions, such as if a child points into the air and the parent says "that's right! That's an airplane!" (Masek et al., 2020).

Fathers

What about fathers? Many studies lump both fathers and mothers in a "parent" group (e.g., Adamson et al., 2019) or study only mothers (e.g., Hirsh-Pasek et al., 2015). However, the existent literature on language development suggests that fathers impact their children's language development in unique and complex ways that complement mothers. Fathers speak differently to their children than mothers do. They use more *wh*-questions like who, what, and where (Rowe et al., 2004), which challenge children to practice their reasoning skills and to provide complex verbal responses (Rowe et al., 2017). Just consider the possible answers to "what is an elephant" compared to "is this an elephant?" This may be why Rowe et al. (2004) found that children spoke more to their fathers than their mothers when playing one-on-one with similar toys. More words were needed to answer the father's complex questions. Furthermore, the quality of father-child engagement during play contributes uniquely to their child's cognitive and language outcomes, independent of mother-child engagement (Tamis-LeMonda et al., 2004). There is also evidence that one parent's higher sensitivity can buffer the effects of the other parent's lower sensitivity on language outcomes (Malmberg et al., 2016).

While more research is needed in diverse samples, we suggest that our models of *how*

and *what* children learn are applicable across cultures, socioeconomic status, and who is interacting with the child, be they mothers, fathers, grandparents, or teachers.

How Children Learn

Learning is optimal when a learning experience is active, engaged, meaningful, socially interactive, iterative, and joyful (Hirsh-Pasek, Zosh et al., 2015; Zosh et al., 2018). These pillars of learning describe the kinds of experiences that make a difference for children's learning – whether it is language, executive function, math and reading, or other skills. Importantly, as we progress through these various pillars, we understand that they are interrelated and overlapping. But they nonetheless provide some structure that highlights how to be "guides on the side."

Active Learning

Children must actively approach problems, be they learning a new word or constructing a tower. A two-year-old who learns how to get a ball out of a mostly closed box is problem solving. A four-year-old who lines up two sets of objects – coins, shells, or bottle tops – to compare which is longer is solving a math problem. These children are actively constructing solutions in the process of learning.

Active learning experiences lead to better learning outcomes. Zosh et al. (2013) found that preschoolers learned new words better when they were presented with a known and unknown object and were asked to find the "glark." This presentation indirectly asked them to actively deduce that "glark" referred to the unknown object, and worked better for learning than when children were just told that the new object was a glark. Kersey and James (2013) found that children who actively wrote letters as compared to children who watched someone else write the letters produced greater

activity in the area of the brain associated with letter perception when later shown those letters. In *dialogic reading* (Zevenbergen & Whitehurst, 2003), children are actively engaged with the content of a book. Parents ask questions, offer commentary about the story, and prompt children to connect the story to their own lives. This strategy builds children's vocabulary (Whitehurst et al., 1994; Zevenbergen & Whitehurst, 2003).

The benefits of active learning can be found in a variety of settings. Children who chose to play with puzzles during naturalistic observations in the home between two and four years of age did better at a spatial task at age 4.5 than children who did not (Levine et al., 2012), suggesting the active use of spatial skills through puzzle play may support the development of spatial learning. When children five to ten years old visited a museum with their families, children in the families who asked more questions and made more observations learned more than children whose families did not (Borun et al., 1996). Even apps can be active when they are minds-on (Hirsh-Pasek et al., 2015). The "Bedtime math" study offers one prime example. First graders who answered interactive math questions on an iPad with their parents at bedtime had increases in math achievement compared to a control group (Berkowitz et al., 2015).

Engaged Learning

According to Fredricks et al. (2004), being engaged consists of acting appropriately for the learning environment (such as sitting during circle time), feeling invested in the subject or learning environment, and being motivated to learn for learning's sake. A related aspect of engagement is attention. Children who are engaged will have more focused and sustained attention on their task. Both engagement and attention are associated with later

academic success in preschool (Razza et al., 2012), through adolescence (Barriga et al., 2002).

Conversely, when children encounter disruption, distraction, or extraneous information, this results in shorter engagement times and a reduction in learning. Background music in a learning video reduces infant learning (Barr et al., 2010) and even books with manipulative features (such as lift-the-flap and textured pieces) tend to be more disruptive to learning than books without these features (Chiong & DeLoache, 2012). This is true also of distractions in the environment. Background TV has negative effects on length of play and focused attention (Ribner et al., 2021; Schmidt et al., 2008). In one study by Fisher et al. (2014) classroom walls were decorated with posters, artwork, and other items commonly found in elementary school classrooms. Kindergarteners who attended a class in the decorated classroom spent more time off-task and learned less than children in an undecorated classroom. There is a great amount of overlap between the pillars of learning, and we see that engagement is entwined with not only attention, as mentioned above, but also with how meaningful the subject is to the child.

Meaningful Learning

When learning is meaningful, children link what they are learning to their current knowledge and build upon it (Novak, 2002; Shuell, 1990). For example, when children realize that cars, airplanes, and even boats are vehicles they build connections between words and gain a deeper understanding of vocabulary (Neuman et al., 2007). They build not just a dictionary, but a base which becomes the well-spring of knowledge as they move from *learning to read* to *reading to learn* (Adams, 1980).

This applies not only to language, but to other areas as well. For quite some time,

scientists have pointed out that over-reliance on abstract materials can lead to use of those materials without understanding them (Nathan, 2012). Parents and teachers can instead use the implicit knowledge children already have as a springboard for instruction. Fyfe et al. (2014) suggest that mathematics should first be taught in a concrete context (such as using a pizza for fractions), and then instruction should “fade” into the traditional mathematical forms (such as $\frac{1}{4}$). Indeed, a preschool mathematics curriculum that helped children see mathematics in their everyday activities was more effective than the previous curriculum (Clements & Sarama, 2007).

This can be accomplished across subjects by making target information comprehensive and integrated into familiar contexts (Grabinger & Dunlap, 1995). One study by Hudson and Nelson (1983) found that children remembered more about a story when it was about a familiar event. Indeed, full curricular programs are based on the idea that teaching new vocabulary in a way that builds on children’s background knowledge allows them to learn the new vocabulary more effectively (e.g., Neuman & Kaefer, 2018).

Socially Interactive Learning

COVID-19 showed us how learning depends on social interaction and how much it cannot be replaced by remote instruction. That is not to say children learned nothing from their digital experiences. Teachers were herculean in their attempts. Yet, a pre-COVID study just reported from Indiana showed that children who transferred into an online charter school in grades 3–8 experienced large, negative impacts on their math and English achievement that lasted at least three years compared to matched peers who transferred to a brick-and-mortar charter school (Fitzpatrick et al., 2020). However, it must be noted that the

children in this study self-selected to move to online schooling, unlike during the COVID pandemic. We will only see the true outcomes of children moving online as the research is published in upcoming years.

We are beginning to understand why children learn more when their learning experiences occur with a social partner who guides them through learning, rather than leaving children to discover purely on their own (Mayer, 2004). Infants imitate adults within an hour of birth (Meltzoff & Moore, 1977) and later learn better when given social cues (Wu et al., 2011). This is particularly true for language learning (Adamson et al., 2004, 2019; Hudson et al., 2015; Kuhl, 2007). Caregiver responses to a child’s vocalizations and actions in an appropriate and timely manner sets supports learning language (see Tamis-LeMonda et al., 2014, for a review). In four- to six-year-olds, engaging in more conversations relates to increased connectivity in the language center of the brain (Romeo et al., 2018).

Socially interactive learning may be disrupted by what has been called technofence, where our smart phones, computers, tablets, and TVs break the delicate learning interactions (McDaniel & Radesky, 2018). In one study, electronic books that read a story aloud, played music, and had supplemental activities reduced the amount parents talked with their three-year-old (Parish-Morris et al., 2013). Another study showed that parents were less responsive to their infant’s vocalizations and gestures when using electronic toys compared to more traditional non-electronic toys (Miller et al., 2017). Zosh et al. (2015) found higher-quality parental language and greater use of spatial language when playing with a traditional compared to an electronic shape sorter. All of these studies suggest that technofence is not just about the attention our cell phones demand (Reed et al., 2017), it applies to the toys that children play with as well.

Iterative Learning

Piaget proposed that children learned through generation, testing, and revision of hypothesis, building each hypothesis on the outcome of the last, and thus building knowledge (Piaget, 1962). This has borne out in the recent research (Baillargeon & DeJong, 2017). Not only is there the opportunity to learn something new with repeated exposure, iterative learning also improves retention, and is effective across the lifespan (e.g., Baillargeon & DeJong, 2017; Cepeda et al., 2009; Dunlosky et al., 2013). Baillargeon and DeJong (2017) showed 11- and 12-month-old infants a series of events that were aimed to build upon each other to teach the infant why an object would fall off another object. When these events were not iterative in that they did not show different examples, or they changed the color of the object, the infants did not learn the rule (i.e., that a triangle would fall off of a ledge if the large end hung over the edge). However, in the iterative condition in which they showed the infant different examples of different orientations of the same object, and in what cases it would fall off the ledge, the infants did learn the rule. This is well-studied with college students, where research supports that revisiting information periodically throughout a longer period while revising previous mistakes is much more effective for long-term learning than cramming for tests (Cepeda et al., 2009; Dunlosky et al., 2013).

Joyful Learning

Learning is fun. Infants from a variety of languages and cultures gleefully play “Peekaboo,” and preschoolers delight in counting yet another taxi. Neuroscientists have found that positive affect and surprise are related to learning (Betz et al., 2017), perhaps due to increased dopamine levels in the brain (e.g., Cools, 2011). Fun also

increases learning; Bisson and Luckner (1996) identify four benefits of fun in learning: motivation, suspension of social reality, stress reduction, and relaxed alertness. This allows the child to indulge in their curiosity and tendency towards exploration. When an activity is fun, children are more motivated, and this may be rooted in the joy found in curiosity and exploration (Zosh et al., 2018). In a review of executive function skills, Diamond (2012) observed that “children devote more time and effort to activities they love” (p. 335) and gains in executive function are dependent “on the amount of time children spend doggedly working on those skills” (p. 337).

As we can see there are certain elements that help children master math, reading, language, and executive function which need to be included in any pedagogical approach to increasing learning. In fact, there are a number of initiatives that are already in place. Among them is the Montessori preschool. Dr. Montessori consciously constructed *active, engaging, meaningful, socially interactive, iterative, and fun* elements as her classroom pedagogy. Research by Lillard and Else-Quest (2006) demonstrates in a number of cases that when Montessori education is compared to a control group of children who have education in a “business as usual” environment, these children outscore their peers in reading, writing, and mathematics in both preschool and elementary. Montessori is not the only example of successes that have used these principles; others include the Tools of the Mind curriculum (Bodrova & Leong, 2007).

What Do Children Learn? Introducing the 6Cs

There are two models of evaluating success in learning outcomes (Golinkoff and Hirsh-Pasek, 2016). In the “traditional model” success is judged by how a person performs on a

test. Another perspective suggests that success is “becoming collaborative, creative, competent, and responsible citizens of tomorrow” (Golinkoff & Hirsh-Pasek, 2016, p. 30). When given a choice, most parents opt for the latter perspective, which includes the learning of content but does not preclude the learning of a broader suite of skills (Golinkoff & Hirsh-Pasek, 2016; Winthrop, 2018). Current models of education and parenting remain largely traditional, rooted in the notion that the key to success is mastering particular content (such as reading, writing, and math).

If learning is about breadth and mastery of skills, then one must decide which skills are paramount. Some learning models emphasize one competency over others, such as social skills (Tomasello, 2014) or grit (Duckworth et al., 2007). We selected six skills that are evidence-based, malleable, and measurable: collaboration, communication, content, critical thinking, creativity, and confidence (the 6Cs; Golinkoff and Hirsh-Pasek, 2016). Unlike the traditional model, in which *content* is independent from its context, the 6Cs model proposes a suite of competencies that allow us to build content and use it effectively. Here we elaborate on each of the 6Cs, and provide a table on how parents can support the development of each competency.

Collaboration

Working and learning in teams is an important part of life. From the classroom to the board room, being able to collaborate is part of being human. In a recent Education Week article, executives were asked to design a high school course, and collaboration and teamwork were a common thread that ran through the majority of the designs. In the words of Michele Freed, general manager of education experiences at Microsoft, “To be successful students must ... most importantly, collaborate with each other to succeed” (Davis, 2020).

Collaboration begins to develop in infancy, with mother–infant dyads experiencing biological synchrony of heart rhythms and brain activity during one-on-one interactions that are full of mutual eye contact, maternal utterances, and infants’ sustained attention to objects of joint play (Feldman et al., 2011; Piazza et al., 2020). Through this dyadic synchrony, parents support their children’s development of empathy (Levy et al., 2019) and self-regulation, a skill that contributes to later collaborative efforts and school readiness (Feldman et al., 1999; Walker & MacPhee, 2011). During infancy, children engage increasingly in social learning, building on their understanding of others as intentional agents (Leong et al., 2019; Meltzoff, 1995). By age two, children begin to work together collaboratively to solve problems (Warneken et al., 2006).

Children’s understanding of and engagement in collaborative activities continues to develop into the school years (Young et al., 2019). It is influenced by prior social experiences (Schuhmacher & Kartner, 2015) and parental expectations shaped by culture (Alcalá et al., 2018). In early childhood children’s prosocial behaviors include cooperating with others, accepting others’ ideas for activities, helping others, showing sensitivity to others’ feelings, and sharing with others. Cooper et al. (2014) found that kindergarteners who had good social skills in kindergarten, but low or average reading skills, performed on later academic assessments similarly to their peers who had good reading skills in kindergarten. Social skills were just as important as reading skills for academic outcomes in fifth grade! Similarly, Caprara et al. (2000) found third graders who rated highly in prosocial behaviors did better in eighth grade, and that third-grade academics were no longer predictive of performance in eighth grade when prosocial behaviors were added to the analysis.

Parents can support collaboration by building their children's social skills, and by encouraging children to help with chores for the sake of the family and including them in housework from an early age (Alcalá et al., 2018). Recent brain work shows that social skills form the foundation for a wide range of developing skills (e.g., language, reading, math, executive function); therefore, as a fundamentally social species, we grow everything we do from the fertile soil of our social relationships (Meltzoff et al., 2009).

Communication

In the *Education Week* article by Davis (2020), Bruce Gardner, North America program director for IBM stated, "There is such a differentiation today between the small percentage of students who can give a firm handshake, make eye contact, and converse with confidence and those who cannot." The way we interact and communicate with others is the way that we operate in the world. Imagine trying to do something as simple as ordering a coffee without speech or gesture, could you? While this is only an example of simple communication, more advanced forms are needed to collaborate, learn effectively, and even just play well with others.

Communication is a basic need, and even before language skills are developed infants communicate with parents and other adults in meaningful ways through gesture, gaze, vocalizations, and facial expressions (Brooks & Meltzoff, 2005; Bruner, 1983). Later in early childhood parent-child interactions that involve balanced, back-and-forth turn-taking build a communication foundation for the child, contributing to language outcomes across a range of cultural and societal groups, including low-SES children (Hirsh-Pasek, Adamson, et al., 2015) and in both English and Spanish (Ramírez-Esparza et al., 2017)

and building structural connectivity across language regions in the brain (Romeo et al., 2018). In fact, recent evidence suggests communication is so integral to learning that young children learn little from pre-recorded television (Madigan et al., 2020). One lovely study exposed nine-month-old English-speaking children to Mandarin Chinese either in-person or through audio and audio/visual recordings. They found that the infants learned Mandarin speech-sounds only in the in-person, interactive condition, the recordings had no effect (Kuhl, Tsao, & Liu, 2003).

Children's communicative interactions evolve and become more complex as conversational skills develop. Communications with parents become increasingly symbol-infused and decontextualized (Adamson et al., 2004; Rowe, 2012), and children help each other build elaborate stories as they become more advanced communicators (Nicolopoulou et al., 2015). Stories are narratives, and these narratives are rich webs of language that help to define knowledge and support children's later learning. Children's communication skills in turn support a cascade of other positive outcomes, including math (Purpura et al., 2011), reading (Duff et al., 2015), academic trajectories in elementary school (Pace et al., 2019), and high school performance (Snowling et al., 2001).

Parents can support communication skills in their children by prioritizing back-and-forth communication in which their child can contribute, even if only through gestures or smiles, over one-way communication such as television. Parents can also include their children in conversation, encouraging them to contribute what is developmentally appropriate to stories told together, conversations about what they observe in the world around them, and an infinite number of topics.

Communication, the soul of language, reading, and writing, is really the way in which our species allows us to represent worlds that do

not yet exist, go backwards in time, and think about the future. It is important to realize that language is not locked into a particular culture or frame. In fact, most of the world speaks more than one language, and there are many studies that suggest that those that do end up having better skills and executive function (see Bialystok, 2015 for a review; but see Valian, 2015; Paap & Greenberg, 2013).

Content

Content areas – particularly the three Rs: reading, writing and arithmetic – are key ingredients necessary for success. Yet, other content areas are critical as well, including science, technology, engineering, the arts, and learning-to-learn skills. It is striking how much of the early content knowledge across domains is based upon early communication skills. To be a good reader, one has to have a sufficient vocabulary. Likewise, to learn mathematics skills, one has to have the language of mathematics as that propels children forward in understanding how to best learn about units and sets (Ribner et al., 2020; Purpura et al., 2019). Spatial skills are malleable and are built up through experiences including block and puzzle play and by hearing spatial language (Bower et al., 2020; Uttal et al., 2013) and, in turn, are foundational for building early math knowledge (Verdine et al., 2014); they are reliable predictors of adolescents' later achievement in science, technology, engineering, and math fields (Wai et al., 2010).

Critically, Ralph et al. (2020) found a difference in how much boys and girls are exposed to spatial language *before* kindergarten, even though girls hear more spatial language *in* kindergarten (Ralph et al., 2020), which suggests there is a crucial period in which spatial language is most helpful to building a strong basis of math and science learning. This difference, as well as other factors such as adult encouragement, may contribute to sex differences in math

performance and all the way to which jobs are pursued by boys and girls.

Executive function skills have also surfaced as a content area. Building these skills is foundational for building early literacy and math skills (Clark et al., 2010; McClelland et al., 2007). These skills are malleable through training, such as the Tools of the Mind curriculum (Blair, 2017; Bodrova & Leong, 2007; Diamond et al., 2007), dramatic pretend play (Goldstein & Lerner, 2018); and playing games in which the rules change, such as versions of the game “Simon Says” in which children have to do what one puppet says but ignore direction from another puppet (Zelazo et al., 2018), or “red light, green light” in which preschoolers must run fast when the light is purple and slow when the light is orange, and then be able to adjust when the rules switch (Tominey & McClelland, 2011).

Almost all of these studies can inspire applications in the home. During block and puzzle play, use spatial language such as “this cow goes next to the pig” and “put this small green block on top of the big red block.” Perhaps evolve this, as your child grows, into discussions about the angle of the sun or the steepness of a hill. Play executive-function supporting games such as “Simon Says,” “Red Light, Green Light,” and “Head-Shoulders-Knees-and-Toes” and change up the rules: make the colors of the lights purple and orange and make knees the first body part even as you sing “head.”

Critical Thinking

Three-year-old Josh asked: “Do all vehicles have wheels?” Mom answered: “Some vehicles have wheels, but not all of them. Boats don’t have wheels.” He returned with a Babar book showing the monkey Zephir receiving a boat wheeled on a trailer. He was pulling upon his content knowledge to build a theory, and was able to find evidence to support it.

Critical thinking is a tool of inquiry which is dependent on content knowledge and allows humans to move beyond “seeing is believing” to weighing evidence for and against one’s beliefs (Facione, 1990). Comparing and contrasting, analyzing, and synthesizing information are skills that comprise critical thinking (Ennis, 1987). Doing so is key to learning, as children are provided with a wealth of information and must sort through what is true and what is not. Theory of mind, or the ability to infer and represent the mental states of others, is crucial for developing this skill, as it enables children to mentally represent and compare multiple points of view (Kuhn, 1999). Once children achieve this, they can learn to analyze and synthesize different points of view and draw conclusions on the basis of evidence rather than their own personal realities (Kuhn, 1999), just as Josh analyzed visual proof and his mother’s statement, and made his own conclusion.

Development of critical thinking skills is often evaluated by assessing which sources children choose to believe regarding new information. As early as age three, children distinguish different information sources by their credibility (Koenig & Harris, 2005), taking into account whether sources have a history of accuracy or knowledge of the relevant information (Heyman, 2008). However, even though they know the credibility of these sources, children can still be swayed by other factors. For example, Danovitch and Mills (2014) found that children believe both objective and subjective claims made by familiar cartoon characters (such as Dora the Explorer) above unfamiliar characters, even when the familiar character was unreliable. Also, MacDonald et al. (2013) found that children are more trusting of an unreliable ingroup member than a reliable outgroup member. Through experience, children learn to account for these biases when evaluating new information, though even adults struggle at times to

reason about new information appropriately (Mills, 2013). Critical thinking skills are predictive of academic performance (O’Hare & McGuinness, 2015) and parents may support the development of these skills by discussing these biases with children and helping them identify particular situations, such as when children lie to further their own self-interest, in which it is beneficial to be skeptical about sources of information (Heyman, 2008).

Creativity

Creativity is critical in today’s technologically-advanced world; it enables us to move from basic knowledge or a mastered foundation to the development of new knowledge. How does a child become creative? There is some evidence that this can be accomplished through play, although some of these findings are mixed and some studies suffer from methodological flaws (Evans et al., 2021). Of those connecting play to creativity, Garaigordobil and Berruoco (2011) found that a play intervention increased measures of verbal creativity (e.g., ability to contribute new ideas) as well as graphic creativity (e.g., originality of drawings). Creativity may also be built up through an interaction between play and vocabulary; Holmes et al. (2015) found that children with higher vocabulary participated more frequently in complex social play, and the complexity of social play was positively correlated with creativity. It is important to note, though, that in this study, one site had more children who spoke different languages and, in that site, there was less complex social play. Therefore, the connection between vocabulary and complex social play may, in fact, be due to language differences becoming a barrier to play.

Some recent and exciting work more narrowly confines the broad definition of creativity to better understand its components and outcomes (Carr et al., 2016; Evans et al.,

2021). In a recent work from our lab, Evans et al. (2021) proposes that creativity may go through a series of levels, first engendering curiosity, which leads to exploration and then to creativity. Children who are better able to handle ambiguity, who have a need to know, and who ask a lot of questions are more likely to try many ways to solve a particular problem such as getting a ball out of a tube. Together, curiosity and exploration are what we might call the divergent aspects of creativity because they allow children to generate a number of solutions and put them on the idea board for consideration. Creativity, or convergent thinking, then emerges as children choose the best solution from among those assembled. This approach to creativity has opened a new frontier in research and is an area currently overflowing with new studies (e.g., Jirout, 2020; Shah, 2018). However, this study highlights the point that barriers to play need to be overcome so that children can reap the benefits of social play with peers.

This line of reasoning suggests that parents can foster creativity by encouraging exploration and curiosity. For example, take your child on a walk and follow their interests. Perhaps they notice something that they find interesting, such as flowers, cars, or clouds. Explore the variation between these; can your child name some similarities and differences between three Subarus, or count the number of petals on a collection of flowers?

Confidence

The confidence to try new things, to take informed risks, and to persist and learn in the face of failure is essential in this era of the knowledge worker. Young children are typically over-confident regarding their abilities (Lipko et al., 2009). However, over-confidence in later childhood and adulthood, while common, can have negative consequences, interfering with learning, memory (Dunlosky

& Rawson, 2012) and job performance (Tracey et al., 1997). This is related to having a fixed or growth mindset (Ehrlinger et al., 2016). People who have fixed mindsets believe that intelligence is inflexible and attach their confidence to a static identity. They believe in statements such as “I do well because I am good at math” and tend to be over-confident and avoid taking risks. Those with growth mindsets believe intelligence can be developed and therefore leave opportunity for personal growth. They believe in statements such as “I do well because I work hard at math” and are willing to try activities or challenges in which they are not certain they will succeed.

Children’s development of a fixed or growth mindset, as reflected in their effort and perseverance on tasks, is shaped in part by their experiences receiving praise from their parents. A child who hears “I can see you worked hard to figure that out” will have more of a growth mindset than a child that hears “wow! You’re so good at math!” (e.g., Gunderson et al., 2013), as will a child that sees a parent struggle to complete a task, but persevere (Leonard et al., 2017). Having a growth mindset is a strong predictor of academic achievement, likely because this mindset encourages children to pursue rather than avoid situations in which they may need to struggle and fail in order to learn. In fact, Claro et al. (2016) found that having a growth mindset buffered children from some of the negative associations between socioeconomic status and academic performance.

Parents can support confidence in their children throughout the lifespan by modelling struggle. With small children, this may look like letting them see you try every shape on the shape sorter before finding the right one. For older children, this might look like them observing you while you learn a new skill or accomplish something you find difficult, such as baking a cake or learning to play an instrument. Using growth mindset language is also

important for confidence, and can be modelled on yourself as well as used with your children and others around you. Saying things like “I found this difficult, but I did it” or “Wow! I know you put a lot of time in to learn that” can help your children have the confidence to work through difficulties. Confidence and growth mindsets are necessary components of grit, the determination and perseverance to take calculated risks in the face of failure in order to reach long-term goals, which in turn predicts academic success and goal achievement (Duckworth et al., 2007). Furthermore, confidence is required for children to have creativity – you have to believe that taking risk is acceptable in order to be creative (see Table 5.1).

Play: Linking the How and What of Learning

Together, the 6Cs form an integrated, systematic model, because these skills work in tandem to help create learning opportunities and competencies for young children. With this model, one can construct the best environments for learning, be it in a school, museum, or home environment. So how do we grow the *how* and the *what* of learning? What is the best medium in which to learn these skills? The answer is playful learning. Supporting playful learning at home is just as important as doing so in schools. Children spend 80 percent of their time outside of school (Meltzoff et al., 2009), so parents have an out-sized opportunity to implement strategies supporting the skills we have presented.

Play can be the perfect medium for parents to routinely make use of the pillars of how children learn and, in doing so, support their development of the 6 Cs. While play can be difficult to define, it can be conceptualized as a spectrum that ranges from child-initiated and child-led free play (e.g., when children spontaneously play an imaginary game together) to

playful instruction (e.g., teaching preschoolers animal sounds by encouraging them to “walk like the animal and make its sound”) – all of which can be active, engaging, meaningful, socially interactive, iterative, and joyful for the child (Zosh et al., 2018). While play across the spectrum can assist learning, guided play is the sweet spot in the middle where the child reaps the learning benefits of play while targeting a specific learning goal (e.g., an adult and child practicing colors through block play) (Fisher et al., 2013; Toub et al., 2018; Weisberg et al., 2013). Therefore, parents and children can use the full spectrum of play to support the development of all of the 6Cs.

Children who play freely, most often at home, reap a host of benefits. Playing with blocks increases mathematical and spatial awareness skills (Schmitt et al., 2018; Verdine et al., 2017), pretend play may improve children's sensitivity to social cues (Lillard, 2017), and creative play may even boost a child's ability to learn from direct instruction in school, as Töugu et al. (2017) found when studying four- to nine-year-olds solving an engineering problem.

Guided play with a parent or teacher is often the situation in which children learn best when there is a targeted learning goal (Toub et al., 2018; Weisberg et al., 2013). Fisher et al. (2013) found that four- to five-year-old children learned shapes better through guided play than both free play and direct instruction, and Cavanaugh et al. (2017) found that kindergarteners learned literacy skills and executive function better through guided play than an assigned activity. In a block-play intervention, Schmitt et al. (2018) found that children of parents with low educational attainment who engaged in a semi-structured block play intervention experienced greater gains in numeracy, cognitive flexibility, and executive function than did their peers with higher-educated parents, revealing the

Table 5.1. *Parent-child activities that support the 6 C's*

Activity	Ages	Collaboration	Communication	Content	Critical Thinking	Creativity	Confidence
Peek-a-boo	Infants		Back-and-forth anticipation and timing				
Storytelling	Infants, children	Building a narrative together	Back-and-forth conversation, vocabulary			Creating a story	
Book reading	Infants, children		Question asking/ answering, vocabulary				
Block and puzzle play	Infants, children	Completing a task together	Back and forth conversation with a goal, vocabulary	Spatial language, math	Applying properties of one object to other objects	Generating new constructions and solutions	
Baking and cooking	All ages	Following instructions together	Understanding and giving instructions	Math, science	How to adjust/ adapt instructions to changes	Exploration of the properties of food and how they are combined, developing own recipes	Producing a tangible product, self-reliance
Family Meals	All ages		Back-and-forth discussion, communication of thoughts and feelings	Social norms. Vocabulary of daily tasks and family matters.			
Music	All ages	Working with others to produce complex outcomes no one could produce individually.			Understanding complex rules and the variation allowed within them.	Composition of new music	Ability to voice reality through development of lyrics or composition.

Games such as Simon says, board games, sports	All ages		Following directions	Executive function		Coming up with own instructions	
Outside exploration	All ages		Question-asking, vocabulary	Math, science	Analyzing similarities and differences of discovered objects	Foundational to creativity	Discovery
Chores	All ages	Working with family members towards a shared goal	Vocabulary, conveying/ understanding intentions				
Communication about relationships	Children, adolescents	Discussion and communication with a shared goal in mind.	Clearly conveying meaning	Vocabulary of emotion	Understanding others, feelings, motivations (Theory of Mind)		Ability to communicate one's thoughts, feelings, and actions.
Learning to drive	Adolescents	Negotiating independence within parental guidelines and rules	Following directions		Assessing risks		Autonomy and independence

potential power of play to mitigate SES-related disparities in executive function and school readiness.

Research has found that several aspects of parents' playful interactions with children relate to the development of executive function, including autonomy, support scaffolding, (Hammond et al., 2012) and parental responsiveness (Merz et al., 2017). For example, when Polish mothers "scaffolded" a game of "Peekaboo" with their four-month-olds by providing prolonged periods of "preparation" where they stopped halfway through covering their face for a moment before progressing to full coverage, their infants were more engaged in the game two months later, revealing ways in which parents can support social coordination through play (Nomikou et al., 2017).

Bottom line is that there are many ways to use the 6Cs. You can adapt them to different cultures and languages and the content can vary widely. What this chapter is not saying is that one should abandon core curricula in favor of playful learning through guided play; it is asking us to rethink how playful pedagogies and rich curricula can work in tandem. There is no age at which guided play ceases to be useful. Lifesize board games that boost STEM learning (Bustamante et al., 2020) can be modified for classroom use, learning through theater can be applied to any age, and even adults benefit from active learning (e.g., Schwan & Riempp, 2004). There has to be breadth not only of content, going beyond math and reading, but also of context. Learning does not only occur at school, it happens at home, in our community, and in all the spaces we inhabit. In a revolutionary way of thinking about this, we apply these same principles to public spaces. Imagine a public space that invites critical thinking and imagination, that has the 6Cs built into its very core.

Practical Implications and Policy

Playful learning can be supported by communities in a wide variety of relevant, low-cost ways within community spaces (Bustamante et al., 2019) and should be implemented as a partnership with the community and its citizens (Bustamante et al., 2019; Schlesinger et al., 2020) in order to meet the unique needs of the particular community. Interventions can take place in spaces frequented by families within the community, such as supermarkets (Ridge et al., 2015), libraries (Hassinger-Das et al., 2020), playgrounds, bus stops, and blocked-off streets (Bustamante et al., 2019; Schlesinger et al., 2020) and to fit into the daily routines of parents and children, rather than necessitating a dedicated time and destination. Interventions in these spaces boost language, parent-child engagement, and physical activity (Bustamante et al., 2020; Hassinger-Das et al., 2020; Ridge et al., 2015; Schlesinger et al., 2020) all of which are linked to positive academic outcomes.

Conclusion

Social skills and interactions are not ancillary to building cognitive and language skills but are in fact foundational to how children learn (Kuhl, 2007; Meltzoff et al., 2009). Current research and theories of cognitive and language development now argue for the need to cultivate a broader set of skills that are foundational for how children learn (Golinkoff & Hirsh-Pasek, 2016; Winthrop, 2018). By capitalizing on informal playful learning opportunities, parents and caregivers can play a central role in setting up that foundation for young children across communities and cultural contexts and still keep social relationships at the core. Those social relationships formerly known as soft skills, are anything but soft! They are the very heart of what it means to learn.

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6 Discipline and Punishment in Child Development

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Overview of Current Research on Discipline and Punishment

Main Types of Discipline and Punishment

Classic frameworks for characterizing discipline strategies describe three categories of discipline (Hoffman & Saltzstein, 1967). Inductive reasoning involves explaining to children how other people are affected by their actions. Love withdrawal involves parents' manipulation of children's negative emotions through expressions of anger or disapproval. Power assertion involves parents' exertion of power and authority over the child. Each of these types of discipline may be invoked either reactively in response to misbehavior that has occurred already or proactively to try to prevent future misbehavior. However, inductive reasoning tends to be more proactive, whereas power assertion more often is reactive in punishing a child for misbehavior that has occurred already.

Within these three overarching categories, parents use a number of specific strategies, such as removing privileges, offering rewards for good behavior, implementing time-outs, and spanking. In a longitudinal study in the United States, the most frequently used strategies were letting the child know the parent was monitoring the situation and was aware of any misbehavior, talking with children, using distraction, and modeling desired behaviors; corporal punishment was among the least frequently used strategies with preschoolers (Socolar et al., 2007), which is the age group

that receives the most corporal punishment (Straus & Stewart, 1999). Similarly, in a study of nationally representative samples in 24 low- and middle-income countries, parents of 2- to 4-year-olds reported that the most commonly used discipline strategy was explaining why something was wrong (Lansford & Deater-Deckard, 2012). Corporal punishment was used less frequently, but across the 24 countries, 63 percent of parents reported that the child had been corporally punished in the previous month.

When parents use inductive reasoning, they discuss with children how other people are affected by children's behaviors. For example, parents may help children understand why certain behaviors are wrong and encourage children to take others' perspective (e.g., "How would you feel if Sarah wouldn't share her toys with you?"). A key advantage of inductive reasoning is that it proactively attempts to socialize children to improve their behavior in the future rather than reactively punishing children for misbehavior that has occurred already.

A large body of research has demonstrated that inductive reasoning, compared to love withdrawal or power assertion, is related to better child outcomes in a variety of domains. For example, a meta-analysis of 41 studies found that parents' teaching, encouraging, and guiding the child, all features of inductive reasoning, are related to better self-regulation among preschoolers than are power assertive methods (Karreman et al., 2006). Inductive

reasoning is especially important in the development of empathy and moral and prosocial behavior (e.g., Krevans & Gibbs, 1996). Mothers' use of inductive reasoning predicts adolescents' moral identity, which involves ascribing moral characteristics such as kindness and fairness more than non-moral characteristics such as intelligence and athleticism as being key to their identity (Patrick & Gibbs, 2012). By contrast, mothers' love withdrawal and power assertion are not related to adolescents' moral identity. Adolescents regard inductive reasoning as being a more appropriate form of discipline than love withdrawal or power assertion and respond better to inductive discipline than to other forms of discipline (Patrick & Gibbs, 2012).

In contrast to inductive reasoning, which attempts to manage children's behavior by appealing to children's sense of right and wrong and focusing on how children's behavior affects other people, love withdrawal involves children's fear of being abandoned or unloved if they misbehave, and power assertion relies on establishing parents' authority over children and often induces children to behave out of fear of being punished if they do not obey. The most frequently studied form of power assertion is corporal punishment, which typically is defined as the use of physical force to cause pain, but not injury, to modify children's behavior. Meta-analyses have demonstrated that corporal punishment is related to a range of negative child outcomes, including externalizing and internalizing problems, worse social relationships, poorer academic achievement, and lower moral internalization (Gershoff, 2002; Gershoff & Grogan-Kaylor, 2016). These negative effects have been found regardless of a variety of study design features (Gershoff & Grogan-Kaylor, 2016) and regardless of the severity of corporal punishment and whether parents perceive its use as justified (Alampay et al., 2017).

Predictors of Discipline and Punishment

Characteristics of children, parents, and communities all predict which types of discipline parents use and the circumstances in which parents use such discipline. For example, a study in which mothers in Hong Kong and Taiwan provided in-depth, open-ended responses to hypothetical vignettes that manipulated a number of circumstances surrounding children's misbehavior found that how parents said they would respond depended both on what was entailed in children's misbehavior and on the context in which children misbehaved (Fung et al., 2017). For example, mothers indicated that they would discipline differently in public versus at home, if only family members versus friends or acquaintances were present, and if children violated rules related to safety versus morals or social conventions. Mothers also often described not just one disciplinary response but also contingency plans of what they would try first and what they would do next if the initial response did not work (Fung et al., 2017). Thus, child, parent, and community factors may predict parents' discipline, but so do micro features of specific situations.

Child factors. Characteristics of children that make their behavior more difficult to manage are likely to elicit harsher and more frequent discipline of a variety of forms (Larzelere, 2000). For example, parents are more likely to use harsh discipline with children who are noncompliant (Patterson, 2002) and who have problems with attention (Alizadeh et al., 2007), than with children who are compliant and easier to control. Similarly, parents are more likely to use inconsistent and hostile discipline with children who have irritable temperaments and high levels of negative emotionality, in part because these children are less likely to comply with parents' socialization attempts (Bates et al., 2014). Children's difficult behaviors and parents'

harsh discipline are transactional over time, so although children's behavior problems elicit harsher parental discipline, harsh parental discipline also increases children's behavior problems over time (Lansford et al., 2011; Patterson, 1982). Different types of child misbehavior also elicit different forms of discipline. For example, when children fail to act prosocially, parents tend to respond with inductive reasoning, but when children behave aggressively or in other antisocial ways, parents are more likely to respond with punishment (Grusec & Kuczynski, 1980).

In contrast to parents' increased likelihood of using corporal punishment with children with externalizing problems, parents are less likely to use corporal punishment with children with internalizing problems (Grogan-Kaylor & Otis, 2007). For children who are depressed, fearful, or anxious, parents may be aware that using corporal punishment can be especially disruptive to parents' socialization attempts (Kochanska et al., 2007). When parents do use corporal punishment, their children's risk of future internalizing problems is heightened (as is the risk of future externalizing problems) (Gershoff, 2002; Lansford et al., 2014).

Parent factors. A primary reason that children's externalizing behavior problems increase the likelihood that parents will use harsh forms of discipline is that managing children's difficult behaviors increases parents' stress. When parents are experiencing stress, physiological, emotional, and cognitive mechanisms make it more likely that they will use harsh and inconsistent discipline (Deater-Deckard, 2004). Stress has physiological effects on brain structure and function that impair psychological functioning (Lupien et al., 2009). Stress also increases parents' anxiety, anger, and other negative emotions and decreases parents' ability to regulate these emotions (Deater-Deckard, 2004). Cognitively, stress increases parents' propensity to make

hostile attributions for others' behavior and contributes to other biases in social information processing (Pinderhughes et al., 2000). Together, physiological arousal, negative emotionality and dysregulation, and cognitive biases all contribute to parents' use of inconsistent and harsh discipline.

A number of other experiences besides children's difficult behaviors can contribute to parents' stress and, therefore, their use of harsh and inconsistent forms of discipline. For example, parents with less income and education are more likely to use corporal punishment and less likely to use inductive reasoning than parents with more income and education (Ryan et al., 2016), perhaps because of higher levels of stress that often accompany lower socioeconomic status, but also because of beliefs about childrearing that change during the course of formal education (e.g., Davis-Kean et al., 2019). Parents also often model their own parenting behaviors on their experiences as children, so parents who were corporally punished when they were children, for example, are more likely to corporally punish their own children than parents who were not corporally punished as children (Wang et al., 2014). Parents who believe that they are responsible for parent-child interactions (i.e., an internal locus of control) are less likely to use harsh discipline than are parents who believe that their child is responsible for parent-child interactions (i.e., an external locus of control) (Rodriguez, 2010). In addition, parents who tend to make hostile attributions about others' intentions in ambiguous social situations and who evaluate aggressive responses positively are more likely than parents without these social information processing patterns to use corporal punishment (Lansford et al., 2014), as are parents who make negative attributions regarding their own children's behaviors (Berlin et al., 2013).

Studies that compare mothers' and fathers' discipline generally find either no differences (e.g., Feldman & Klein, 2003) or that mothers discipline more than fathers (e.g., Hallers-Haalboom et al., 2016). For example, in a study of families in nine countries, mothers were found to use corporal punishment more frequently than fathers in seven countries (Lansford, Alampay et al., 2010). There were no differences between mothers and fathers in the other two countries (Sweden, which outlawed corporal punishment in 1979, and Thailand, where corporal punishment was rarely used by either mothers or fathers). Mothers have also been found to use more non-coercive verbal discipline than fathers in the same families (Volling et al., 2006). Because mothers still, on average, spend more time with children than do fathers, mothers may witness more child misbehavior than fathers and be present to respond (Huerta et al., 2013).

Community-level factors. Much of the variation in parents' discipline can be accounted for by community-level factors rather than characteristics of individual parents or children. For example, countries differ in laws and policies related to discipline, including legal bans on corporal punishment (Global Initiative to End Corporal Punishment, 2022). Even in the absence of laws and policies, communities differ in norms and expectations regarding discipline. For example, corporal punishment is more common and accepted in "cultures of violence" in which intimate partner violence, homicides, and other forms of violence also are more common (Ember & Ember, 1994). In a longitudinal study in nine countries, parents' use of corporal punishment was predicted by community-level factors, such as whether corporal punishment was accepted in the community, as well as by individual-level factors, such as the child's externalizing behaviors (Lansford et al., 2015).

Moderators of Links between Discipline and Child Development

Child gender, child age, temperament, overall climate of the parent-child relationship, and culture have been examined as predictors of parental discipline as well as moderators of associations between different forms of discipline and child outcomes. Each is discussed in the sections that follow.

Child gender. Few differences have been found in parents' use of different forms of discipline with sons and daughters. When gender differences are found, they generally show that parents are more likely to use corporal punishment and harsh verbal discipline with sons than daughters, but the differences tend to be small and inconsistent, with many studies finding no gender differences at all (Jansen et al., 2012). In 32 low- and middle-income countries, 2- to 4-year-old boys experienced more corporal punishment than girls, but the effect sizes were small and not practically meaningful (Deater-Deckard & Lansford, 2016). Overall, parents discipline sons and daughters more similarly than differently.

Despite mean level similarities in discipline, gender may still moderate associations between discipline and child outcomes if boys and girls respond to discipline in different ways. For example, corporal punishment may increase boys' risk for externalizing behavior and girls' risk for internalizing problems, a possibility suggested in a meta-analysis that showed stronger links between corporal punishment and children's externalizing problems if the sample included a larger proportion of boys than girls (Gershoff, 2002). It is unclear whether gender moderates associations between other forms of discipline and children's development.

Child age. As with child gender, child age may have main effects on parents' use of different forms of discipline, and child age may

moderate associations between discipline and child outcomes. In terms of main effects, parents generally recognize the need to tailor discipline to children's age and developmental status. For example, although simple distraction and redirecting infants and toddlers from an undesired behavior or object to an alternative one may be all the discipline needed, as children develop cognitively, inductive reasoning becomes more important as children are able to understand more complex explanations and others' perspectives (Collins et al., 2002). Parents also decrease their use of corporal punishment from childhood to adolescence. For example, in a national sample from the United States, 49 percent of parents reported spanking their 0- to 9-year-old children in the previous year, whereas 23 percent of 10- to 17-year-olds reported having been spanked in the previous year (Finkelhor et al., 2019).

In a meta-analysis, the relation between corporal punishment and children's externalizing behavior was stronger when the sample included 10- to 12-year-olds than when the sample was younger or older (Gershoff, 2002). Gershoff (2002) suggested that child effects may explain the curvilinear relation; 10- to 12-year-olds with externalizing problems may elicit more corporal punishment from their parents than younger children because parents expect older children to be able to control their own behavior and react more negatively when older children fail to do so. She further hypothesized that the relation between corporal punishment and externalizing behavior may be weaker for adolescents because parents rarely use corporal punishment with adolescents, and adolescents' externalizing behaviors may be more strongly influenced by peers (Gershoff, 2002).

Another reason that links between parental discipline and child outcomes may change over the course of development is that children

increasingly regard parents' knowledge and skills rather than ability to administer rewards and punishments as the basis of parents' authority, which increases the importance of inductive reasoning (Braine et al., 1991). For example, 4-year-olds are more likely than 6-year-olds to say they would behave in socially conventional ways to avoid punishment because authority figures prohibit some behaviors and less likely to refer to reasons for the acceptance of the social conventions (Yau & Smetana, 2003). The nature of the transgression is also important to understanding children's perceptions of what parents' disciplinary responses should be, though, because even young children regard moral transgressions as more serious than violations of social conventions (Padilla-Walker, 2008). As children develop, it is important for them to internalize parents' socialization messages so that they will behave according to an internalized set of standards rather than just in the presence of an authority figure to avoid punishments or obtain rewards.

Parental discipline may also change in its relation to other parenting domains as children age. For example, as children move into adolescence and spend more time in peer contexts outside of parents' direct control, distal forms of parental monitoring take on increasing importance (Lionetti et al., 2019). Parents can set rules and limits and solicit knowledge from their adolescents, yet parents also rely on adolescents to disclose information about their activities, companions, and whereabouts (Kapetanovic et al., 2020). Part of discipline during adolescence involves parents' communication with adolescents to ensure that adolescents understand their parents' expectations, know what consequences of misbehavior will be, and know that parents will follow through with consequences if adolescents break rules. Limit-setting and consequences for misbehavior both often involve access to

screen time. For example, parents often attempt to set limits on their children's and adolescents' screen time and access to particular games, apps, or forms of social media. Furthermore, a consequence for misbehavior is often a loss of screen time or electronics for a specified period of time (or a reward of screen time for good behavior) (Hawi & Rupert, 2015).

Child temperament. Bidirectional main effects have been found between child temperament and parental discipline. Children with more difficult temperaments elicit harsher parental discipline than children with easier temperaments, and harsh discipline increases children's negative emotionality, fearfulness, and irritability (Lengua & Kovacs, 2005). Temperament also moderates the association between discipline and child outcomes. For example, socialization messages are better internalized when parents use gentle forms of discipline and de-emphasize power assertion with toddlers who are temperamentally fearful and anxious, whereas for toddlers who are temperamentally fearless, being securely attached and using this attachment relationship as the basis of cooperative discipline matters more than minimizing children's anxiety (Kochanska, 1995). Children who are high, but not low, in resistance to control engage in fewer externalizing behaviors when their parents use more restrictive control, perhaps because these children are at risk for externalizing problems and benefit from having fewer opportunities to engage in them (Bates et al., 1998).

Overall climate of the parent–child relationship. Many parenting theories have distinguished between parenting styles and parenting practices, with styles characterizing the broad emotional climate of the relationship and practices characterizing specific behaviors (Grusec & Goodnow, 1994). If the overall climate of the parent–child relationship

is warm and loving, children will be more open to their parents' socialization attempts through discipline than if the parent–child relationship is hostile or neglecting, and the climate of the parent–child relationship may moderate the association between discipline and child outcomes (Fletcher et al., 2008). For example, for children who were insecurely attached to their parents at the age of 15 months, parents' power assertive discipline predicted the development of more resentful opposition and anti-social behavior between the ages of 25 and 67 months, but these associations were not found for children who were securely attached at 15 months (Kochanska et al., 2009). The overall climate of the parent–child relationship continues to be important during adolescence. For example, in a sample of Mexican-origin adolescents and their parents, the overall climate of the parent–adolescent relationship (e.g., warmth, hostility) and specific socialization practices (e.g., in relation to respect and independence) were jointly predictive of adolescents' subsequent adjustment (Kim et al., 2019).

Whether a parent–child relationship that is generally warm and loving can moderate the detrimental effects of corporal punishment has yielded somewhat mixed findings. Although some studies report no association between corporal punishment and negative child outcomes in the context of a warm and loving parent–child relationship (McLoyd & Smith, 2002), other studies have shown that corporal punishment predicts negative child outcomes even in the context of warm and loving parent–child relationships (Lee et al., 2013). A study of mother–child relationships in eight countries found that corporal punishment was more strongly related to children's anxiety in the context of generally warm and loving parent–child relationships, perhaps because children have a hard time reconciling parents' warmth with their use of corporal punishment

and feel uncertain from moment to moment about how their parents will behave (Lansford et al., 2014).

Culture. Different forms of discipline are more common in some countries and cultural groups than others. In a comparison of different forms of discipline in nationally representative samples of parents of 2- to 4-year-olds in 24 low- and middle-income countries, 11–18 percent of the variance in non-violent forms of discipline such as offering explanations or giving the child something else to do, 14–19 percent of the variance in psychological aggression (yelling at the child or insulting the child), and 27–38 percent of the variance in parents' reports of whether they believe it is necessary to use corporal punishment to rear a child properly was accounted for by country of residence (Lansford & Deater-Deckard, 2012). The range of differences was stark. For example, 4 percent of parents in Albania believed it was necessary to use corporal punishment to rear a child properly compared to 93 percent of parents in Syria. One percent of parents in Belarus, Kazakhstan, and Ukraine reported that their child experienced severe forms of corporal punishment (hitting on the head or beating with an object) in the previous month compared to 40 percent of parents in Mongolia and Yemen. Despite these differences across countries, there were also commonalities. For example, the most frequently used form of discipline across countries was explaining to the child why something they did was wrong (Lansford & Deater-Deckard, 2012).

Qualitative research has taken a different approach to understanding forms of discipline that may be specific to particular cultural groups. For example, "shaming" has been described as a form of discipline used by Chinese parents to instill shame after children misbehave with a goal of teaching children right from wrong and socializing them to

"confess and repent" after misbehaving (Fung, 1999, p. 201). In a comparison of shaming in samples of 7- to 14-year-olds in Canada and China, children in China perceived shaming as being more common than children in Canada, but children in both countries believed that inductive reasoning is more effective than shaming or love withdrawal and believed that shaming harms children's self-worth and psychological well-being, especially as they get older (Helwig et al., 2014). Shaming is related to psychological control and can be a form of love withdrawal if children perceive that their parents' love and affection are conditional on children's behaving and thinking in ways prescribed by parents.

Cultural differences in discipline strategies may arise in part because of differences in broader supports for desired behaviors. For example, in many cultural contexts it is common for children to contribute to the family's well-being by caring for younger siblings or doing chores, such as fetching water, that benefit the entire family (de Guzman et al., 2005). When children are taking care of other family members, they may learn prosocial behavior implicitly. In cultural contexts in which children have fewer opportunities to behave prosocially in directly contributing to others' welfare, parents may rely more on inductive reasoning as a way to socialize children's prosocial behavior (Hastings et al., 2007).

Culture also has been found to moderate associations between parental discipline and child outcomes. For example, although corporal punishment was related to more child aggression and anxiety in China, India, Italy, Kenya, the Philippines, and Thailand, the association was weaker in countries where corporal punishment was more normative (Lansford et al., 2005). In the same countries, mothers' expressions of disappointment, timeouts, and shaming were related to more child

anxiety, and expressions of disappointment and yelling were related to more child aggression; perceived normativeness of these forms of discipline moderated some of the links between discipline and child aggression and anxiety (Gershoff et al., 2010). Effect sizes did not differ by country in a meta-analysis of associations between corporal punishment and 17 child outcomes (Gershoff & Grogan-Kaylor, 2016).

Even if a form of discipline is culturally accepted and normative, that does not mean that it is not detrimental to children. For example, at an individual level, if parents use corporal punishment in a cultural context in which it is normative, children may not perceive that parents are personally rejecting them in a way that might be the case if parents use corporal punishment in a cultural context in which it is not normative. At a societal level, however, cultural contexts in which corporal punishment is accepted also have higher acceptance of other forms of aggression, and societal rates of violence are higher (Lansford & Dodge, 2008).

Mediators and Mechanisms through which Discipline is Related to Children's Development

Discipline does not only have direct associations with child outcomes. Instead, discipline is often related to child outcomes indirectly via social, cognitive, or emotional mediators that serve as intervening mechanisms. Key mediators of links between discipline and child outcomes are children's perceptions of parental love and affection, social information processing, development of empathy and conscience, and neurocognition.

Children's perceptions of parental love and acceptance. One mediator of links between parental discipline and child outcomes is children's perceptions of being loved and accepted

by their parents. For example, the association between corporal punishment and psychological maladjustment was mediated by children's perceptions of parental rejection in a low-income sample from the United States (Rohner et al., 1996). Likewise, children who experienced corporal punishment and harsh verbal discipline were more likely to perceive their mothers as being hostile, which in turn predicted more child aggression and anxiety in a sample from China, India, the Philippines, and Thailand (Lansford, Malone, et al., 2010). These findings suggest that parental discipline may not only have direct effects on children's behavioral and psychological adjustment but also indirect effects through information discipline conveys to children about their parents' love and acceptance versus hostility and rejection. Thus, the overall climate of the parent-child relationship may both mediate and moderate the association between discipline and child adjustment.

Children's social information processing. Social information processing refers to the ways that children encode information in social situations, make attributions about others' intentions, and generate and evaluate possible behavioral responses (Crick & Dodge, 1994). One mechanism through which corporal punishment has been found to predict subsequent child aggression is through the development of aggressogenic patterns of social information processing, such as a propensity to make hostile attributions about others' intentions (i.e., to believe that others acted in a particular way to be mean or because they do not like someone, even in situations that are ambiguous so that a person may have acted in a particular way by accident or for another benign reason) (Yaros et al., 2016). For example, children who experience corporal punishment are more likely to make hostile attributions about others' intent in ambiguous social situations, more likely to

generate aggressive response options, and more likely to evaluate aggression positively than are children who do not experience corporal punishment (Weiss et al., 1992). Hostile attributions and positive evaluations of aggressive responses are, in turn, predictive of children's aggressive behavior (Weiss et al., 1992).

Development of empathy and conscience.

A main reason that inductive reasoning is an effective form of discipline is that children's empathy develops when parents explain the rationale for why children should or should not behave in particular ways and encourage children to think about how their behavior affects other people. The relation between inductive reasoning and children's prosocial behavior is mediated by children's empathy (Krevans & Gibbs, 1996). By contrast, power assertive discipline during childhood is related to less empathy in both childhood and adulthood (Lopez et al., 2001). The development of children's conscience also is associated with the extent to which parents discuss emotions, rather than rules or consequences, during discipline episodes (Laible & Thompson, 2002). Conscience, including whether children feel guilt and anxiety about misbehaving and whether they can inhibit prohibited behavior, mediates the link between discipline and child outcomes, particularly moral behavior (Kochanska, 1993), suggesting the importance of parents' use of inductive reasoning.

Neurocognition. Another mediator of relations between discipline, particularly corporal punishment, and child outcomes is brain structure and function (Gershoff, 2016). For example, less gray matter in the prefrontal cortex, which is an area of the brain important for processing emotions and engaging in high-level executive functions such as decision-making, has been found in young adults who experienced chronic corporal punishment (at least once a month in any form, and at least once a year with an object) during childhood

than young adults who did not (Tomoda et al., 2009). The same areas of the prefrontal cortex that are affected by corporal punishment are also involved in post-traumatic stress disorder (Bremner et al., 1997), depression (Fitzgerald et al., 2008), and other mental health problems (Gershoff, 2016). Cortisol production is related to corporal punishment, which affects the brain through the hypothalamic–pituitary–adrenal (HPA) axis (Kohrt et al., 2014), indicating that one way that corporal punishment harms development is through children's stress response systems.

Implications of Research on Discipline and Punishment for Practice and Policy

Research on discipline and punishment has informed the international agenda related to protecting children from all forms of violence. The 1989 Convention on the Rights of the Child (CRC) asserts children's rights to protection from abuse and exploitation and to participation in decisions affecting their lives (United Nations, 1989); it has been ratified by all UN countries except the United States. The Sustainable Development Goals were adopted by the United Nations General Assembly in 2015 to guide the international agenda through to 2030, through 17 broad goals that are each operationalized in specific, measurable targets (United Nations, 2017). The explicit goal of outlawing corporal punishment is set forth in the Sustainable Development Goals Target 16.2: "End abuse, exploitation, trafficking and all forms of violence against and torture of children" (Global Initiative to End Corporal Punishment, 2022). Progress toward Target 16.2 is measured as the percentage of children aged 1 to 17 who experienced any corporal punishment or psychological aggression (e.g., calling the child lazy, stupid, or other insults; yelling at the

child) by caregivers in the previous month (Danish Institute for Human Rights, 2018). In part as a response to this international agenda, there has been exponential growth in the number of countries that have outlawed corporal punishment. As of June 2022, 63 countries have outlawed corporal punishment in all settings, including at home (Global Initiative to End Corporal Punishment, 2022; see Table 6.1).

Several parenting interventions aim to reduce parents' use of corporal punishment and increase parents' use of non-violent forms of discipline. For decades, researchers and practitioners alike have recognized the importance of changing parents' discipline as a way of ultimately improving children's behavior (e.g., Patterson, 1982). Because parent-child interactions are reciprocal, as children's behavior improves, parents also become less likely to use harsh forms of discipline (and need to discipline less often overall because children are misbehaving less). Many comprehensive parenting programs include modules that are designed to reduce parents' use of corporal punishment and verbal aggression (e.g., yelling at or disparaging the child) and increase parents' use of inductive reasoning (see www.blueprintsprograms.com for a review of evidence for different programs). In general, interventions are more effective at changing parental discipline if they directly target parents' behaviors, rather than just knowledge and attitudes, and give parents opportunities to role play and practice using different discipline strategies with their own child in the presence of trained facilitators who can provide feedback and suggestions for improvements (UNICEF, 2017). These kinds of interventions can be expensive and time-intensive, however, and may not be feasible. The examples that follow illustrate a range of interventions that vary in cost and time-intensiveness.

Research-based versus biblical reinterpretation intervention. Corporal punishment is used more frequently by parents who identify with conservative Protestant religious groups than by parents who are not religious or who identify with other religious groups (Gershoff et al., 1999). The "spare the rod, spoil the child" idea is often invoked by conservative Protestant groups to justify the use of corporal punishment. An intervention in which students at a conservative Christian university were randomly assigned to a research-based intervention (that presented students with scientific evidence about the detrimental effects of corporal punishment on child development), a biblical reinterpretation intervention (that presented students not only with scientific evidence but also with a reinterpretation of "spare the rod, spoil the child" biblical passages), or a no intervention control group found that students' endorsement of corporal punishment decreased the most in the group exposed to both the biblical reinterpretation and research (Perrin et al., 2017). These findings suggest the importance of addressing motivations and beliefs regarding the necessity and acceptability of using corporal punishment in attempts to reduce its use. However, although changing beliefs may be necessary, it may not be sufficient to change parents' behaviors because the proportion of parents who report using corporal punishment far exceeds the proportion of parents who believe using corporal punishment is necessary to rear children properly (Lansford & Deater-Deckard, 2012).

Video Feedback Intervention to Promote Positive Parenting and Sensitive Discipline (VIPP-SD). The VIPP-SD is an intervention for families of children with externalizing problems that is designed to help parents set limits and discipline in a sensitive way (see Juffer & Bakermans-Kranenburg, 2018, for a detailed description of the program).

Table 6.1. *Countries that have outlawed all forms of corporal punishment, including in the home*

Country	Year of legal ban	Brief description of the law passed
1. Sweden	1979	"Children are entitled to care, security and a good upbringing. Children shall be treated with respect for their person and individuality and may not be subjected to corporal punishment or any other humiliating treatment."
2. Finland	1983	"A child shall be brought up in the spirit of understanding, security and love. He shall not be subdued, corporally punished or otherwise humiliated. His growth towards independence, responsibility and adulthood shall be encouraged, supported and assisted."
3. Norway	1987	"The child must not be subjected to violence or in any other way be treated so as to harm or endanger his or her mental or physical health."
4. Austria	1989	"The minor child must follow the parents' orders. In their orders and in the implementation thereof, parents must consider the age, development and personality of the child; the use of force and infliction of physical or psychological suffering are not permitted."
5. Cyprus	1994	Law prohibits "any unlawful act or controlling behaviour which results in direct actual physical, sexual or psychological injury to any member of the family."
6. Denmark	1997	The child "may not be subjected to corporal punishment or any other degrading treatment."
7. Latvia	1998	"A child cannot be treated cruelly, cannot be tormented and physically punished, and his/her dignity and honour cannot be offended."
8. Croatia	1999	"Parents and other family members must not subject the child to degrading treatment, mental or physical punishment and abuse."
9. Germany	2000	"Children have the right to a non-violent upbringing. Corporal punishment, psychological injuries and other humiliating measures are prohibited."
10. Israel	2000	"Corporal punishment of children, or humiliation and derogation from their dignity as a method of education by their parents, is entirely impermissible, and is a remnant of a societal-educational outlook that has lost its validity. The child is not the parent's property and cannot be used as a punching bag the parents can beat at their leisure, even when the parents honestly believe that they are fulfilling their duty and right to educate their child. The child depends upon the parents, is entitled to parental love, protection and the parents' gentle touch. The use of punishment which causes hurt and humiliation does not contribute to the child's personality or education, but instead damages his or her human rights. Such punishment injures his or her body, feelings, dignity and proper development. Such punishment distances us from our goal of a society free of violence."

Table 6.1. (*cont.*)

Country	Year of legal ban	Brief description of the law passed
11. Bulgaria	2000	"Every child has a right to protection against all methods of upbringing that undermine his or her dignity; against physical, psychical or other types of violence; against all forms of influence, which go against his or her interests."
12. Turkmenistan	2002	"Humiliation of the child's dignity, corporal punishment, other physical abuse harmful for the child's mental or physical health are inadmissible."
13. Iceland	2003	"It is the parents' obligation to protect their child against any physical or mental violence and other degrading or humiliating behaviour."
14. Romania	2004	"(1) The child has the right to be shown respect for his or her personality and individuality and may not be made subject to physical punishment or to other humiliating or degrading treatments. (2) Disciplinary measures concerning the child can only be taken in accordance with the child's dignity, and under no circumstances are physical punishments allowed, or punishments which relate to the child's physical and mental development or which may affect the child's emotional status."
15. Ukraine	2004	"Physical punishment of the child by the parents, as well as other inhuman or degrading treatment or punishment are prohibited."
16. Hungary	2005	"The child shall not be subjected to torture, corporal punishment and any cruel, inhuman or degrading punishment or treatment."
17. Greece	2006	"Physical violence against children as a disciplinary measure in the context of their upbringing brings the consequences of Article 1532 of the Civil Code."
18. Togo	2007	Revision of penal code to make it a criminal offence to assault a child even lightly.
19. Spain	2007	"Parents/guardians must exercise their authority with respect for the child's physical and psychological integrity."
20. Venezuela	2007	"All forms of physical and humiliating punishment are prohibited."
21. Uruguay	2007	"Parents and guardians, as well as all persons responsible for the care, treatment, education or supervision of children and adolescents, are prohibited from using corporal punishment or any type of humiliating treatment as a means of reprimanding or disciplining them."
22. Portugal	2007	"Whoever repeatedly, or not, inflicts physical or psychological ill-treatment, including corporal punishment, deprivation of liberty and sexual offences, is punished with 1–5 years of imprisonment."
23. New Zealand	2007	"Nothing in subsection (1) or in any rule of common law justifies the use of force for the purpose of correction."
24. Netherlands	2007	"Caring for and raising one's child includes the care and the responsibility for the emotional and physical wellbeing of the

Table 6.1. (*cont.*)

Country	Year of legal ban	Brief description of the law passed
		child and for his or her safety as well as for the promotion of the development of his or her personality. In the care and upbringing of the child the parents will not use emotional or physical violence or any other humiliating treatment.”
25. Liechtenstein	2008	“Children and young people have the rights outlined in the Convention on the Rights of the Child and to the following measures: (a) protection notably against discrimination, neglect, violence, abuse and sexual abuse; (b) education/upbringing without violence: corporal punishment, psychological harm and other degrading treatment are not accepted.”
26. Luxembourg	2008	Law “prohibits physical violence and inhuman and degrading treatment within families and educative communities.”
27. Republic of Moldova	2008	Children have the right “to be protected against abuse, including corporal punishment by his parents or persons who replace them.”
28. Costa Rica	2008	“Parental authority confers the rights and imposes the duties to orient, educate, care, supervise and discipline the children, which in no case authorises the use of corporal punishment or any other form of degrading treatment against the minors.”
29. Albania	2010	“Corporal punishment or punishment of any other form entailing consequences on the physical and mental development of the child shall be prohibited.”
30. Congo	2010	“It is forbidden to use corporal punishment to discipline or correct the child.”
31. Kenya	2010	Every child “has the right . . . to be protected from abuse, neglect, harmful cultural practices, all forms of violence, inhuman treatment and punishment, and hazardous or exploitative labour.”
32. Tunisia	2010	Revision of Penal Code “making it a criminal offence to assault a child even lightly.”
33. Poland	2010	“Persons who exercise parental authority and who provide care of or custody over a minor may not apply corporal punishment.”
34. South Sudan	2011	“Every child has the right: . . . to be free from corporal punishment and cruel and inhuman treatment by any person including parents, school administrations and other institutions.”
35. Cabo Verde	2013	“In exercising the right to correction parents must always keep in mind the rights of children and adolescents to an upbringing free from violence, corporal punishment, psychological harm and any other measures affecting their dignity, which are all inadmissible.”
36. Honduras	2013	“It is prohibited to parents and every person charged with the care, upbringing, education, treatment and monitoring [of children and adolescents], whether on a temporary or

Table 6.1. (cont.)

Country	Year of legal ban	Brief description of the law passed
		permanent basis, to use physical punishment or any type of humiliating, degrading, cruel or inhuman treatment as a form of correction or discipline of children or adolescents.”
37. North Macedonia	2013	“All forms of sexual exploitation and sexual child abuse (harassment, child pornography, child prostitution), forced procuring, selling or trafficking children, psychological or physical violence and harassment, punishment or other inhuman treatment, all kinds of exploitation, commercial exploitation and abuse of children that violates basic human freedoms and rights and rights of the child, are prohibited.”
38. Andorra	2014	“Whoever corporally mistreats mildly or harms physically, a person, shall be punished by imprisonment or a fine up to 6.000 euros. If the mistreatment consists of a corporal punishment, a sentence of imprisonment shall be imposed.”
39. Estonia	2014	“It is prohibited to neglect a child, to mentally, emotionally, physically or sexually abuse a child, including to humiliate, frighten or physically punish a child, and also to punish a child in any other way that endangers the mental, emotional or physical health of a child.”
40. Nicaragua	2014	“The father, mother, or other family members, guardians or other persons legally responsible for the son or daughter have the responsibility, the right and duty to provide, consistent with the child’s evolving capacities, appropriate direction and guidance to the child, without putting at risk his or her health, physical integrity, psychological and personal dignity and under no circumstances using physical punishment or any type of humiliating treatment as a form of correction or discipline.”
41. San Marino	2014	“Children have the right to protection and security, and shall not be subjected to corporal punishment or other treatment harmful to their physical and psychological integrity.”
42. Argentina	2014	“All forms of corporal punishment, ill-treatment and any act that physically or mentally injures or impairs children and adolescents are prohibited.”
43. Bolivia	2014	“(1) The child and adolescent has the right to good treatment, comprising a non-violent upbringing and education, based on mutual respect and solidarity. (2) The exercise of the authority of the mother, father, guardian, family members and educators should use non-violent methods in parenting, education and correction. Any physical, violent and humiliating punishment is prohibited.”
44. Brazil	2014	“Children and adolescents are entitled to be educated and cared for without the use of physical punishment or cruel or degrading treatment as forms of correction, discipline,

Table 6.1. (*cont.*)

Country	Year of legal ban	Brief description of the law passed
		education or any other pretext, by their parents, by the members of their extended family, by persons responsible for them, by public officials implementing social and educational measures or by any other person entrusted with taking care of them or treating, educating or protecting them.”
45. Malta	2014	Revision of Criminal Code to clarify “that the infliction of any kind of corporal punishment constitutes a crime against the person.”
46. Benin	2015	“The State shall ensure that discipline within the family, at school and in other public or private institutions does not involve corporal punishment or any other form of cruel or degrading treatment.”
47. Ireland	2015	Children granted equal rights to protection from assault in repeal of “reasonable chastisement” law that previously allowed parents to use corporal punishment.
48. Peru	2015	“To prohibit the use of physical and humiliating punishment against children and adolescents. This prohibition applies in all areas where children and adolescents are, including the home, school, community, workplaces and other related places.”
49. Mongolia	2016	“Children have the right to be protected from crime, offences or any forms of violence, physical punishment, psychological abuse, neglect and exploitation in all social settings.”
50. Montenegro	2016	Children “shall not be subjected to corporal punishment or any other cruel, inhuman or degrading treatment.”
51. Paraguay	2016	“All children and adolescents have the right to good treatment and for their physical, psychological and emotional integrity to be respected. This right includes the protection of their image, their identity, their autonomy, their thoughts, their feelings, their dignity and their values. Corporal punishment and humiliating treatment of children and adolescents is prohibited as a form of correction or discipline, especially when it is imparted by parents, tutors, guardians or anyone responsible for their education, care, guidance, or treatment of any kind. Children and adolescents are especially entitled to receive guidance, education, care and discipline by implementing guidelines for positive parenting.”
52. Slovenia	2016	“Corporal punishment of children is prohibited.”
53. Lithuania	2017	“Parents and other legal representatives of the child may appropriately, according to their judgment, discipline the child, for avoiding to carry out his duties and for disciplinary infractions, with the exception of corporal punishment and any other form of violence.”

Table 6.1. (cont.)

Country	Year of legal ban	Brief description of the law passed
54. Nepal	2018	“Each child has a right to be protected against all types of physical or mental violence and punishment, neglect, inhumane behaviour, gender based or discriminatory abuse, sexual abuse and exploitation committed by his/her father, mother, other family members or guardian, teacher or any other person.”
55. Georgia	2019	“Corporal punishment, torture or any other cruel, degrading or inhuman treatment or punishment of children shall be prohibited in the family, preschool or general education institutions, alternative care services, medical and/or psychiatric institutions, penitentiary facilities and any other places.”
56. South Africa	2019	Repealed law recognizing parents’ power “to inflict moderate and reasonable chastisement on a child for misconduct.”
57. France	2019	“Parental authority is exercised without any physical or psychological violence.”
58. Republic of Kosovo	2019	“Corporal punishment and disciplinary measures that would harm and reduce the dignity of the child, including forms of physical and mental violence and behaviours that degrade, disgrace and put child in inappropriate situations are prohibited in each environment at home and family, educational institutions, in child care institutions, bodies of law and the justice system, in the premises of work and the community.”
59. Japan	2020	“A person who exercises parental authority over a child shall not discipline the child by inflicting corporal punishment upon him/her.”
60. Seychelles	2020	“No child shall be subjected to corporal punishment.”
61. Guinea	2020	“All forms of corporal, physical or verbal punishment, cruel, inhuman, degrading or humiliating treatment of children are explicitly prohibited.”
62. Colombia	2021	“Parents or individuals who exercise parental authority over children and adolescents have the right to educate, raise and correct their children according to their beliefs and values. The only limit is the prohibition to use physical punishment, cruel, humiliating, or degrading treatment and any type of violence against children and adolescents.”
63. Republic of Korea	2021	The protector of children (i.e., parents and other adults with parental authority) shall not “inflict physical pain or psychological pain, including violent language, on the children.”

Note: All information is from country reports produced by the Global Initiative to End All Corporal Punishment of Children, available at <https://endcorporalpunishment.org>

Families engage in six sessions, generally at home, with an interventionist; each session lasts about an hour and a half. At the beginning of each session, the interventionist video records the parent and child interacting with one another. Later in the session, the interventionist watches previously recorded video segments with the parent, periodically stopping the recording to comment on the child's or parent's behavior by emphasizing enjoyable parts of the interaction and reinforcing positive parenting. In later sessions, interventionists also help parents analyze ways in which they could become more sensitive when interacting with their child. In a randomized controlled trial with mothers of 1- to 3-year-olds, the program was found to improve mothers' use of distraction, inductive reasoning, and attempts to understand children's perspectives and to reduce mothers' use of commands, expressions of disapproval, physical intervention, and giving in (Van Zeijl et al., 2006), findings that have been consistent across other randomized controlled trials with different populations (Juffer & Bakermans-Kranenburg, 2018).

Positive Discipline in Everyday Parenting Program. As more countries have outlawed corporal punishment, many have sought educational materials and interventions that are relatively portable across cultures to teach parents about forms of discipline to use instead of corporal punishment. The Positive Discipline in Everyday Parenting Program focuses on promoting sensitive, responsive caregiving and parents' use of inductive reasoning and other positive forms of discipline rather than corporal punishment and other types of punitive parenting (Durrant, 2020). Generally, the program is delivered during eight sessions, each lasting two hours, plus a follow-up session held two or three weeks after the conclusion of the program. Trained facilitators work with 5–20 participants at a time in a highly interactive format that involves a series of activities and

guided discussions. Preliminary evidence is promising, as parents who completed the program reported less positive views of corporal punishment, more self-efficacy in their role as parents, and more confidence in their ability to control their anger in situations involving parent–child conflict than they did prior to participating in the program (Durrant et al., 2014). The program has been adapted for use in many diverse countries (e.g., Australia, Canada, Gambia, Georgia, Guatemala, Japan, Kosovo, Mongolia, Palestine, Paraguay, Philippines, the Solomon Islands, Venezuela) (Durrant et al., 2017).

No Hit Zones. In the United States, which has not outlawed corporal punishment, some locales have established No Hit Zones as a way of raising awareness of the importance of not hitting children and providing an entry point for talking with parents about alternative forms of discipline (Frazier et al., 2014). Hospitals have been the most common locales for No Hit Zones, although they also have been established in district attorney's offices, public parks, and even some entire cities (Mastrangelo & Lansford, 2020). No Hit Zones have the potential to decrease staff members' acceptance of corporal punishment and increase perceived efficacy to be able to intervene when they witness hitting (Gershoff et al., 2018). Thus, even when national bans are not implemented, individual organizations may take intermediate steps by establishing No Hit Zones that can ban corporal punishment in specific settings (Gershoff, 2020).

Limitations of Current Research, Directions for Future Research, and Conclusions

Limitations of current research on parental discipline with respect to understanding child effects, neuroscience, and gene x environment interactions suggest three directions for future

research. First, previous research has established that children with externalizing behavior problems and other characteristics that make them more difficult to parent elicit harsher and less consistent discipline than children who are easier to parent. However, many studies, especially those with cross-sectional designs, do not take into account child effects, which can lead to overestimates of the extent to which parents' use of discipline affects subsequent child behavior (versus the extent to which children's behaviors affect subsequent discipline; see Larzelere et al., 2004). Future research will benefit from attempts to understand bidirectional relations between child behavior and parents' discipline to draw stronger conclusions about the discipline strategies that are most effective for particular children in particular situations. Second, neuroscience is a rapidly developing area of research, particularly as new technologies and methods make it possible to understand brain structure and function in ways that were not possible in the past. Future neuroscience research can contribute to understanding of mechanisms through which parents' discipline is related to children's behaviors and emotions. For example, neuroscience research has begun to document ways that parents' use of punishment and rewards (including non-tangible rewards such as praise) are related to children's brain structure and function associated with response inhibition, emotion regulation, and the like (e.g., Tomoda et al., 2009). Third, meta-analyses of studies with genetically-informative designs have concluded that parents' genetic and environment factors (Klahr & Burt, 2014) as well as children's genetically-influenced behaviors (Avinun & Knafo, 2014) are related to parenting. For example, children's genotype contributes to their responsiveness to positive discipline, which in turn increases children's compliance (Kok et al., 2013). Future research with

genetically-informative designs will advance the understanding of genetic moderation of links between parental discipline and child outcomes as well as clarify the extent to which relations between parental discipline and child outcomes are attributable to genetically transmitted characteristics (e.g., aggression as manifested in parents' use of corporal punishment and in children's aggression toward peers).

Taken together, the large body of research on parental discipline and punishment suggests that inductive reasoning that focuses on helping children understand how their behavior affects other people is related to better child development outcomes than are power assertive forms of discipline, particularly corporal punishment. Inductive reasoning is related to the development of prosocial behavior, in part through the development of children's empathy and conscience, whereas corporal punishment is related to the development of internalizing and externalizing problems, in part through the development of children's social information processing biases and influences on brain structure and function. Parental discipline is predicted by child factors, such as difficult child behaviors that elicit harsher discipline; parent factors, such as stress, which makes it harder for parents to control their own behavior to discipline consistently and effectively; and community factors, such as norms about the acceptability of different forms of discipline. The international agenda has increasingly emphasized the importance of protecting children from all forms of violence, including corporal punishment, leading a growing number of countries to outlaw corporal punishment. A number of parenting interventions have been demonstrated to be effective in reducing parents' use of harsh discipline and improving positive parenting. Bidirectional relations between child behavior and parental discipline suggest the benefits for

both children and parents of increasing inductive reasoning and eliminating corporal punishment.

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7 Parenting From a Cultural and Global Perspective: A Review of Theoretical Models and Parenting Research in Diverse Cultural Contexts

Qing Zhou and Sara Chung

Culture is broadly defined as language, customs, beliefs, and practices that are shared by or associated with a social group (Becher & Trowler, 2001). The intersection of culture and parenting has been a heated topic for several reasons. First, while many commonalities exist in parenting styles and practices of caregivers across cultures and societies, cross-cultural studies have revealed some group differences in parenting practices by nation, race, ethnicity, or culture groups (e.g., Cheung et al., 2016; Cole, Tamang, & Shrestha, 2006; Jackson-Newsom, Buchanan, & McDonald, 2008). These differences are often theorized to reflect cultural differences in values, beliefs, norms, and socialization goals. Parenting (or family socialization more broadly) is thus a critical social context through which the ingredients of culture (e.g., language, values, customs, daily routines) are transmitted from generation to generation. Researchers have also found cohort differences in parenting values and practices as nations/societies underwent rapid economic, social, and political reforms (Chen et al., 2005). The study of parenting can thus provide a snapshot of culture within unique historical periods. Second, with the increasing worldwide need for developing, adapting, and disseminating evidence-based family interventions to reduce children's maladjustment and promote resilience, there is a high demand for culturally

relevant measures of parenting and empirical studies evaluating the socialization functions of parenting practices in the focal culture. Because the core theoretical frameworks for the science of parenting, including its key constructs, dimensions, and measurement instruments, were predominantly developed based on Western, educated, industrial, rich, and democratic samples (Nielsen et al., 2017), testing the generalizability of these frameworks and methods to other cultures is a crucial step to advance the science of parenting and its translations into clinical practice.

Theoretical Frameworks for Understanding Cultural Influences on Parenting

Various developmental and cultural psychology theories have been proposed to characterize the mechanisms by which cultural contexts shape parenting practices. The bioecological model of human development (Bronfenbrenner & Morris, 2006) views culture as a macrosystem that shapes a child's development through the exosystems (e.g., neighborhoods, parent workplaces, mass media), mesosystems (e.g., interactions among family, school, and peer network), and microsystems (e.g., family, school, peer network). This theory frames parenting as a mediating mechanism explaining culture-related variations in child

developmental outcomes. Other theories have focused on unpacking “culture” and characterizing the specific processes in which various elements of culture shape parenting goals, parental attributions, and practices. For example, Trommsdorf (2012) theorized that cultural differences in worldview (i.e., views of the self and its relationship to the world) serve as the foundation for cultural differences in the goals of self-regulation (or agentic regulation) and values towards specific forms of self-regulation, which in turn shape how parents foster these behaviors or characteristics in children. In independent cultures (such as most European American communities), the self is viewed as a fixed entity and the key goal of self-regulation is to achieve individual-oriented goals (e.g., expressing one’s unique self). Accordingly, traits such as emotion self-expression, assertiveness, self-esteem, and self-efficacy are more highly valued and emphasized in socialization. In contrast, in interdependent cultures (such as East Asian, South Asian, or Southeast Asian communities), the self is viewed as malleable (to personal effort and social influence), and the goal of self-regulation is to change the self to achieve harmony with the group. Characteristics such as self-restraint, self-awareness, self-discipline, and emotion regulation are more highly valued and encouraged in socialization. This theory has received some empirical support from studies examining cultural variations in parents’ and children’s emotion expression and emotion discussion (e.g., Cameras, Kolmodin, & Chen, 2008; Wang & Fivush, 2005).

Chen and French (1998) proposed a socio-cultural theory to characterize a critical process through which cultural values and beliefs shape parenting practices: parental attitudes, evaluations, or attributions of and reactions to children’s behavioral traits. By showing approval and encouragement of culturally valued traits and disapproval and discouragement of

culturally undesired traits, parents are creating the structure for children to modify their behaviors, which can shape the stability of these traits, as well as their adaptive functions in the given cultural context. Empirical supports for this theory were found in a study comparing cohort differences in relations between the trait of shyness/sociability-cooperation and peer and social competence in China (Chen et al., 2005). While shyness was associated with higher social competence in the 1990 cohort (before the implementation of the open-door policy in China and the subsequent social and economic reform in 1980s), it was unrelated to social competence in the 1998 cohort, and shyness was associated with more peer and mental health problems in the 2002 cohort (Chen et al., 2005). Furthermore, Ding et al. (2020) found that the relations between shyness and school and social adjustment problems were more evident in non-migrant children than migrant children in urban China, likely because the rural-to-urban migrant families tended to maintain more traditional socialization values than non-migrant families.

Harkness and Super’s (1994) developmental niche theory defined children’s cultural environment as consisting of three interrelated systems: the physical and social settings of children’s everyday life, customs (e.g., child-care and education), and the caretakers’ psychology (e.g., caretakers’ values/beliefs or ethnotheories). The three systems work in harmony and provide cultural environments for children. Parents’ ethnotheories or culturally influenced beliefs are shaped by socio-cultural context and guide parenting practices in specific settings. For example, qualitative interviews of parental beliefs revealed that European American parents viewed play as an important vehicle for early development, whereas Asian parents placed the importance of academic training above play. Consistent

with the cultural differences in parental beliefs, European American parents provided more toys for fun and pleasure and European American preschool-aged children spent more time in pretend and physical play at home, whereas Asian parents provided more educational toys and Asian children spent more time in pre-academic learning activities at home (Parmar, Harkness, & Super, 2004). The developmental niche theory is useful in guiding the research on daily routines and parenting practices across cultures.

To summarize the various theoretical frameworks discussed above, we propose a conceptual model of mechanisms of cultural influences on parenting and its implications for child development in Figure 7.1. It is important to highlight two points. First, there are both direct and indirect pathways through which socio-cultural factors (e.g., cultural values and norms) shape parenting practices and behaviors. For example, cultural values on emotion expression (e.g., emotion display rules) can have direct influences on parental expressions of emotions or indirect influences on parental emotion expression via parental beliefs on emotion socialization. Second, there are bidirectional relations between many cultural elements and parenting, which reflect dynamic processes that undergo constant transformations. For example, as immigrant parents adopt or adapt the values, norms, and customs of the host culture while navigating the environmental demand of the new host society, their parenting practices change, which in turn may shape and create new cultural norms and customs that differ from their heritage cultures (e.g., the adaptive culture perspective, Lau, 2010).

In two meta-analysis studies, Pinquart and colleagues examined the associations between parenting styles (authoritative, authoritarian, permissive, and neglectful parenting) and children's and adolescents' behavior problems,

academic achievement and self-esteem across cultures (Pinquart & Gerke, 2019; Pinquart & Kauser, 2018). A total of 428 studies were included in the meta-analysis on parenting and child behavioral problems and academic achievement and 116 studies were included in the meta-analysis on parenting and child self-esteem. Results suggested that in both Western and non-Western cultures, authoritative parenting was associated with better child outcomes (lower internalizing and externalizing problems, higher academic achievement and higher self-esteem), whereas authoritarian and neglectful parenting was associated with poorer child outcomes. There was some evidence of cultural differences in the *strengths* of associations between parenting and child outcomes. For example, the associations between authoritarian parenting and maladaptive outcomes were weaker in countries higher on individualism, and the associations between authoritative parenting and academic achievement were also weaker in more individualistic countries. Some differences in strengths of relations between parenting and child outcomes were also observed across ethnic groups within Western countries. These findings suggest that despite cultural differences in norms and values towards authoritative and authoritarian parenting styles, their overall socialization functions are similar across cultures.

There have been several thoughtful reviews of research on parenting in diverse cultural contexts. For example, Chen, Fu, and Zhao (2015) reviewed theories and cross-cultural research on socialization processes in parent-child relationships and peer relationships. Wang and Chang (2010) provided a qualitative review of research on parenting and socialization in contemporary China. Labella (2018) conducted a systematic review of research on emotion socialization (i.e., socialization practices related to regulation and expression of emotions) in African American families. In

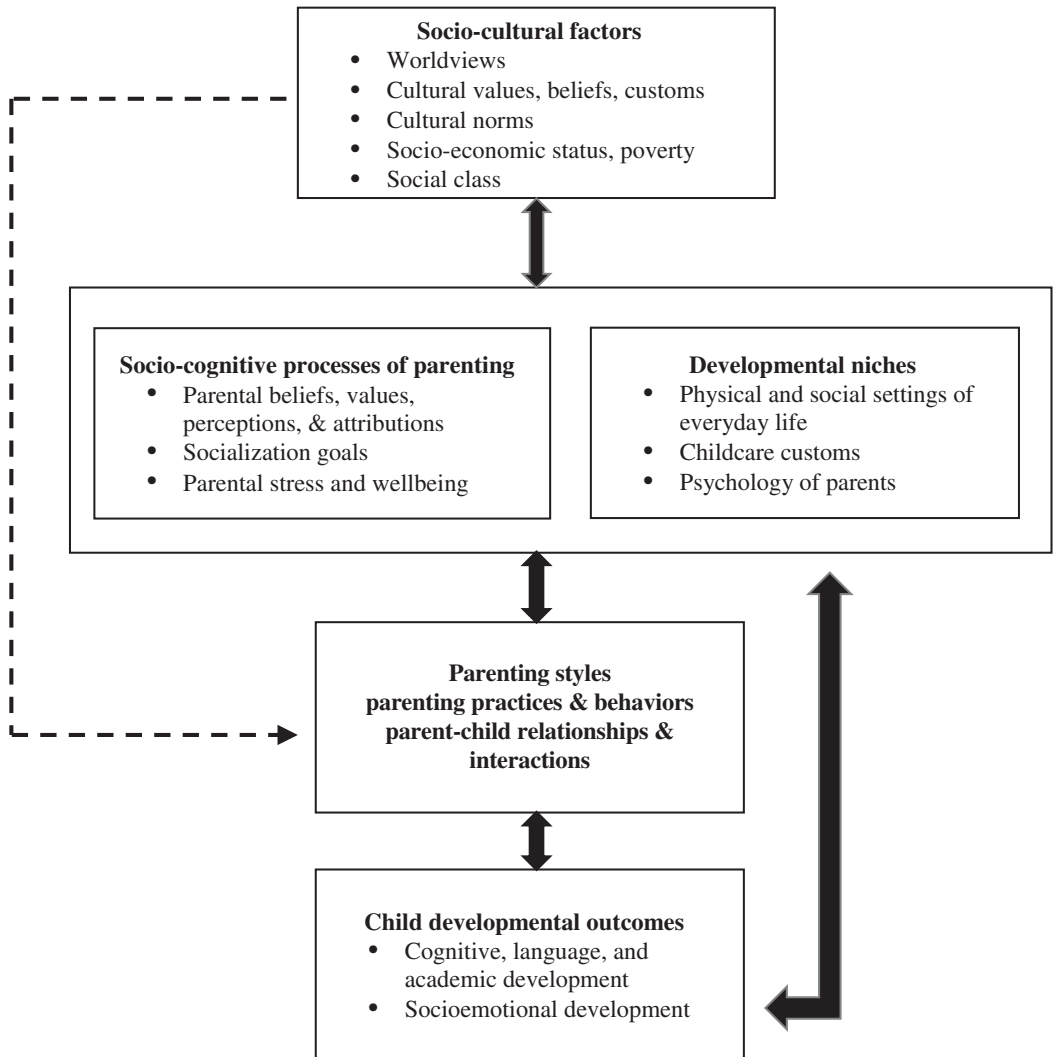


Figure 7.1 A conceptual model on mechanisms of cultural influences on parenting, Chen & French (1998), Harkness & Super (1994), and Trommsdorf (2012). *Note:* Solid arrows depict direct relations and dotted arrows depict indirect relations.

the sections below, we provide a selective qualitative review of empirical studies on parenting in four cultural contexts: (1) low- to middle-income countries of sub-Saharan Africa, (2) Southeast Asian countries, (3) refugee families from sub-Saharan Africa and Southeast Asia, and (4) refugee and immigrant families in Western destination countries. These four cultural contexts were selected

because: (1) parenting practices in these cultural contexts have been understudied, despite the strong need for research-based interventions to prevent or reduce risks and promote resilience in children living in those communities; and (2) the unique socio-cultural (e.g., socioeconomic, political, or religious) features of these cultural contexts create several new directions for parenting research.

Parenting in Low- to Middle-Income Countries of Sub-Saharan Africa

Effective parenting is a challenge to achieve under ideal circumstances and is thus made even more difficult under inadequate familial, societal, and national resources (Edin & Lein, 1997). Successful caregiving is undermined by the multitude of day-to-day stressors of living in dangerous or deteriorated circumstances, which directly impact the ability that parents of low socioeconomic status (SES) have to meet the physical, socioemotional, and cognitive needs of their children (Bornstein & Putnik, 2016). Compared to children in high-income countries, those in low- to middle-income (LMIC) countries are at considerably higher risk for adverse physical and psychological outcomes, such as maltreatment, malnutrition, and mortality (Houweling & Kunst, 2010; Knerr, Gardner, & Cluver, 2013). According to a meta-analysis conducted by McCoy et al. (2016), an estimated 81.0 million children aged 3 and 4 years in LMICs experienced low socioemotional and/or cognitive development in 2010, with the highest proportion of affected children in sub-Saharan Africa (29.5 million; 44.0 percent of children ages 3 and 4 years), followed by South Asia (27.8 million; 37.8 percent). Low development scores were associated with poverty, rural residence, and lack of cognitive stimulation (McCoy et al., 2016).

Despite the well-recognized need to bolster children's developmental outcomes in LMICs, most of the empirical work and theories in developmental science emanates from North America and Western Europe and little data has been gathered in resource-constrained settings (Abubakar & van de Vijver, 2017). Much of the research conducted in LMIC countries has been focused on the dissemination of parenting interventions, which were developed based on research conducted in

high-income Western countries (e.g., Knerr et al., 2013). Thus, empirical research evaluating whether authoritative parenting – the “optimal” parenting style in middle-class European/European American families – is associated with adaptive child outcomes in sub-Saharan African communities is critical to inform the dissemination and/or cultural adaptation of parenting interventions in these communities.

The United Nations classifies sub-Saharan Africa as a group of African countries of four regions, Central, East, Southern, and West Africa (United Nations, 2022). This classification excludes Algeria, Egypt, Libya, Morocco, Tunisia, Western Sahara and Sudan, which form the subregion of Northern Africa. In comparison to other regions in the world, the contextual influences on parenting and child development unique to sub-Saharan Africa include its very low per capita income, burden of diseases, and numerous wars and conflicts resulting in displacements of its inhabitants. Moreover, children and adolescents in sub-Saharan African countries had the lowest levels of school attendance and completion in the world: 81 percent of children in sub-Saharan Africa (versus 91 percent world-wide) were enrolled in primary school and only 63 percent (versus 85 percent world-wide) of children completed primary school; 63 percent of adolescents (versus 84 percent world-wide) in sub-Saharan Africa were enrolled in secondary school and only 38 percent of adolescents (versus 71 percent world-wide) complete lower secondary education (UNESCO, 2019). The low rate of school attendance is not only a risk factor for exposing children to other vulnerabilities in the environment, but also heightens the critical roles of caregivers (e.g., parenting) in child development. Although little data have been gathered in this region, several local researchers have contributed work to the existing parenting literature.

Several studies conducted in sub-Saharan Africa have illuminated the prevalence of parenting styles in individual countries and examined the associated socio-demographic characteristics, although this body of work is very limited in its geographic representation. In South Africa, a retrospective study of college students found that authoritative parenting was significantly more common than authoritarian and permissive parenting among mothers and fathers (Roman, Makwakwa, & Lacante, 2016). This relationship was moderated by ethnicity and gender, such that Black fathers scored significantly lower on both authoritative and authoritarian parenting than White fathers, and mothers were more likely to practice authoritative parenting with their female offspring than male offspring. Similarly, another study of adolescents recruited from South African public schools found that authoritative parenting was more common than authoritarian parenting among both mothers and fathers, and mothers scored significantly higher than fathers on all types of parenting (Roman et al., 2015). These authors noted that these findings may reflect a closer relationship that South African mothers have with their children than fathers. Because Black African fathers had to work away from their families historically during apartheid, they were less involved in their parenting role (Roman et al., 2015). In a cross-sectional study of community- and school-based samples of parents of primary school-based children in Uganda, Huang and colleagues (2017) found that 28 percent of parents were depressed. Moreover, parental depression was associated with parents' higher endorsement of corporal punishment and practice of more harsh/inconsistent discipline, which in turn was associated with poorer physical health and behavioral functioning in children. Importantly, Huang et al. (2017) identified several socio-demographic

risk factors for parental depression, including low parental education, food insecurity, low social support, and a high number of children in the family.

Parents in Nigeria and Kenya were also found to utilize authoritative parenting at higher rates than authoritarian, permissive, or uninvolved parenting (e.g., Efobi & Nwokolo, 2014; Kugbey, Mawulikem, & Atefoe, 2015; Njagi, Mwanja, & Manyasi, 2018). Efobi and Nwokolo's (2014) work found that a sample of secondary students in Nigeria endorsed experiencing slightly more authoritative parenting than authoritarian parenting, and uninvolved parenting the least. These findings were mirrored in Embu County, Kenya (Njagi et al., 2018). In Mombasa County, Kenya, 85.2 percent of a sample of secondary students also endorsed authoritative discipline while fewer students (60.3 percent) endorsed authoritarian discipline (Kilonzo, 2017). These studies, however, failed to measure for a significant difference between authoritative and authoritarian parenting. A study conducted among adolescents in Ghana indeed found a non-significant difference between the two parenting styles, which these and other authors attributed to the possibility of parents shifting their approach based on situational demands as well as the difficulty in differentiating authoritative and authoritarian parenting styles in Ghanaian culture because punishment is seen as such a typical way of life (Kugbey et al., 2015; Nyarko, 2014).

Another recent piece of important work regarding parenting in sub-Saharan Africa is a meta-analysis of 44 studies conducted in 13 countries that examined the associations of parenting styles and dimensions with child and adolescent mental health outcomes (Devlin, Wight, & Fenton, 2018). Their study found that, overall, similar associations were found between parenting practices and child outcomes compared to those documented in high-income

countries. Authoritative parenting style and related qualities, including parental warmth, support, and monitoring were associated with positive outcomes, including fewer risky sexual behaviors, lower substance use, and better psychosocial and educational outcomes. Authoritarian parenting style and negative behavior control, including physical punishment and psychological maltreatment, were associated with a higher risk of antisocial behaviors and worse educational and mental health outcomes. The authors argued that the similarities across sub-Saharan Africa and high-income countries in the associations between parenting and child outcomes supports the adaptation of existing parenting interventions to LMICs.

Devlin and colleagues (2018) noted, however, that most of the studies included in their analyses were conducted in only three countries: Kenya, Ghana, and South Africa, which limits the cross-country generalizability of their conclusions, and the conceptual framing of parenting and its dimensions were diverse and markedly inconsistent across studies, which impedes cross-country comparisons. These reflections, in addition to Kugbey and colleagues' (2015) observation that parents appear to shift their parenting styles based on the demands of the current situation, indicate that perhaps an examination of daily behaviors and practices may better illuminate parental influences on children's development rather than a reliance on measuring overall parenting dimensions and styles.

In order to monitor developmental trajectories of children in sub-Saharan Africa, Bornstein and colleagues (2017) examined multiple domains of childhood experiences from population-based survey data gathered from 14 sub-Saharan African countries representing 72,000 0- to 4-year-old children. In terms of caregiving practices aimed at promoting children's socioemotional and cognitive

development practiced in the previous three days, they found that fewer than 10 percent of caregivers read to their children, fewer than 20 percent told stories, fewer than 35 percent of caregivers named, counted, and drew with their children, fewer than 50 percent sang songs and took their children outside, and about 50 percent of mothers and fewer than 20 percent of fathers played with their children. The countries with the highest percentage of caregivers who engaged in most of these practices were Nigeria, Somalia, Sierra Leone, Ghana, and Cameroon. Burkina Faso, Guinea-Bissau, Djibouti, and Mozambique generally had the lowest percentage of caregivers engaging in most of these practices.

Bornstein and colleagues (2017) also found that more than 70 percent of children of the sample had received discipline characterized as nonviolent (e.g., explaining why something was wrong, removing privileges) and psychological aggression (e.g., yelling, name-calling), more than 65 percent had received physical punishment (e.g., spanking, shaking), and over 15 percent had received severe physical punishment (e.g., being hit on the head or beaten with an implement). Over 35 percent of caregivers endorsed the belief that physical punishment is necessary to raise a child properly. Gambia and Guinea-Bissau had relatively high proportions of children experiencing nonviolent discipline and low proportions experiencing psychological aggression or any form of physical punishment. In contrast, Cameroon and Togo had relatively low proportions of children experiencing nonviolent discipline and high proportions receiving psychological aggression and both types of physical punishment.

The authors observed that individual caregiving activities and discipline practices vary drastically in their prevalence across the region, ranging from 0 percent to 87 percent of caregivers (Bornstein et al., 2017).

Nonetheless, in line with other literature on parenting styles common in the region (e.g., Kugbey et al., 2015; Njagi, Mwanja, & Manyasi, 2018), a sizable proportion of caregivers endorsed belief in the use of physical punishment as necessary discipline. Despite this belief, however, a large majority of caregivers engaged in reasoning and explaining to their children as one of their means of discipline, indicating that the value of socializing children with explanations and rationales for misbehaviors cuts across countries (Bornstein et al., 2017).

Researchers have long documented that regional practices for socioemotional and cognitive development have been largely derived from indigenous practices such as folklore, idioms, riddles, proverbs, and participatory learning rather than only books or toys (e.g., Nsamenang, & Lo-oh, 2010). Some authors argue, however, that core practices like storytelling, during which older extended family members told folktales that depicted attributes such as generosity and greed to the children to teach morals and values, are no longer an aspect of daily life in African communities (e.g., Amos, 2013). Urbanization has led to the prevalence of nuclear family systems rather than extended systems and has shifted socialization away from interpersonal interactions to modernized learning via technology. Other authors have highlighted the persistence of some traditional daily practices in other areas of sub-Saharan Africa. Among the Yoruba in Nigeria, for example, families reside in compounds – circular structures that create an open environment with an entire community in which elders provide childcare and socialization to children of other families through the use of lullabies and proverbs as well as delivering immediate discipline or correction when a child misbehaves rather than waiting for their parents to discipline them at home (Babatunde & Setiloane, 2014). Similar

reliance on extended family and other community members for the socialization and discipline of children is documented in other studies, including a qualitative study of migrant parents from Ethiopia, South Sudan, Zimbabwe, Nigeria, Sierra Leone, Eritrea, Ghana, and the Democratic Republic of Congo (Mugadza et al., 2019). Children in these communities are taught specific physical and verbal mannerisms for interacting with adults that reflect the values of obedience and respect for elders, such as greeting their parents on their knees at the beginning of the day in the Yoruba community (Babatunde & Setiloane, 2014; Mugadza et al., 2019).

In summary, the research reviewed above examined the overall characteristics and individual differences of parenting in sub-Saharan African regions. Researchers found some support for the generalizability of Western parenting constructs (e.g., authoritative and authoritarian parenting) and their socialization functions in sub-Saharan African countries. Researchers also highlighted the need to consider indigenous parenting practices and culturally salient socialization contexts (e.g., extended family, community) when studying parenting/caregiving practices in these regions. One implication of these findings for intervention is that while parent education programs promoting supportive parenting practices (e.g., reasoning, consistent discipline, warmth) can benefit these communities, adaptations are needed to deliver the programs to diverse caregivers (e.g., parents, grandparents, adult family members, community caregivers) and in diverse settings.

Parenting in Southeast Asian Countries

Southeast Asia comprises of eleven countries, Brunei, Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, Philippines, Singapore,

Thailand, Timor-Leste, and Vietnam. The region is characterized by rich ethnic and language diversity, with countries that vary widely in their demographic profiles, wartime and migration experiences, political systems, and economic development (Rao & Sun, 2010). While Brunei, Malaysia, Singapore, and Thailand are classified as upper-middle- and high-income countries, for example, the others are LMICs.

Much like their neighbors in East Asia, Southeast Asian culture has been greatly influenced by Buddhism, Confucianism, and Taoism, which have shaped cultural traditions as well as the family structure and roles (McHale, Dinh, & Rao, 2013). Interpersonal harmony, conforming to norms, respecting elders, and filial piety are values that permeate parenting practices in the region (e.g., McHale et al., 2013). The saying, "Spare the rod, spoil the child," encapsulates the perception of harsh emotional and physical discipline as necessary to raise well-behaved children (Beazley et al., 2006). Authoritarian parenting, as such, and the practice of harsh discipline with the emphasis on academic achievements and family obligations may be more in line with cultural values and developmental expectations in some Southeast Asian countries such as Vietnam (Nguyen, Chang, & Loh, 2014). Very little research on parenting styles has been conducted in Cambodia and Laos, and the small body of existing work largely focuses on parenting among immigrants and refugees to other countries. Similar to other Southeast Asian regions, the proverbs of, "Bend steel while it is hot" and "If you love your children, do not show it," underlie parenting in Cambodia, and parents traditionally adopt an authoritarian style, valuing obedience and favoring punitive disciplinary strategies (Czymoniewicz-Klippel, 2019; Lim & Pot, 2015). In Laos, one study found that only a quarter of their adolescent sample

endorsed feeling understood by their parents, while almost half experienced high parental monitoring, indicating parenting characteristics that are also in line with an authoritarian style (Lee et al., 2019).

Research in other areas of Southeast Asia, however, has revealed different trends in parenting styles. In the countryside of the Philippines, the large majority (over 87 percent) of parents endorsed using permissive parenting, with authoritarian parenting as a distant second at a rate of 11.4 percent (Lanzarrote, et al., 2013). This finding was similar to those found in the Philippine's metropolitan area of Cebu, in which the majority of adolescents in the sample reported experiencing permissive parenting (Hock et al., 2018). The study also found a gender difference in which girls were more likely to report authoritative and authoritarian styles than boys, which the authors note is in line with higher limitations of social freedoms (e.g., romantic relationships, sex) parents often place on girls (Hock et al., 2018).

One of the most diverse countries in Southeast Asia, Indonesia, consists of 250 native ethnic groups and over 700 different languages and dialects and is home to the world's largest Muslim population. Indonesian cultural identity has been historically influenced by Indian, Malay, Arabic, Chinese, and European sources and shares many Asian lifestyles and values. In their review, Riany, Meredith, and Cuskelly (2017) discussed that the shifting parenting styles across children's development among Indonesian parents is similar to that in other Asian countries, which begins with permissive style in early childhood, followed by more authoritarian style after their sixth birthday. The implementation of a more permissive style is owed to the belief that children before age 6 are not able to understand behavioral expectations and rules, and thus strict discipline and

an expression of authority via physical distance is not adopted until children turn 7 (Riany et al., 2017).

The limited work examining daily parental practices has revealed some similar trends among Southeast Asian regions for discipline, maintaining family hierarchy, and reliance on extended family members. Quantitative studies in Cambodia have found that the majority of parents and children reported non-violent and positive parenting behaviors, including praising children and verbal corrections of children's misbehavior as well as the reliance on psychological aggression rather than physical discipline (Lim & Pot, 2015). Qualitative work by Czymoniewicz-Klippel (2019), however, found that among a smaller sample of parents, hitting was a commonly used form of discipline in Cambodia, and when a combination of "polite" words, cursing, and hitting were not effective, parents resorted to increasing authoritarianism and surveillance in order to maintain power over their children. One parent stated in an interview, for example, that she chained her son to a bed behind her market stand so she could monitor him while she worked to prevent him from playing video games. Parents and adolescents in Myanmar and Vietnam similarly endorsed physical punishment as a widely used disciplinary tool (Falb et al., 2020; Lee et al., 2018; Mestechkina, Duc Son, & Shin, 2014). Data from a recent study found that physical abuse was quite prevalent in Northern Vietnam among children ages 12 to 17, with boys being more likely to be victims (with a lifetime prevalence of 42.8 percent) than girls (34.6 percent) (Tran et al., 2018). In qualitative interviews, parents and adolescents in Myanmar discussed parental use of physical force as a means of gaining compliance because parents are blamed for children's disobedience, which could result in social isolation from the community (Falb et al., 2020; Lee et al., 2018).

Over 50 percent of a sample of 4,482 young children in Laos were found to have received hitting, slapping, or spanking as punishments (Tran, Luchters, & Fisher, 2017). Indonesian parents, however, have been found to rarely physically punish their children, and instead may rely on cajoling, shaming or embarrassing their child, or scaring their child by telling stories of supernatural punishments (Riany et al., 2017). Despite the prevalence of physical aggression in the region, some researchers have found that community members play a role in curtailing the extent and prevalence of the discipline strategy. In Myanmar, for example, community members would tell parents to "not overdo" discipline in cases of extreme violence that resulted in injury and/or screaming or crying that could be overheard by neighbors (Falb et al., 2020). Similarly, in Cambodia, teachers have promoted the use of non-violent discipline (Czymoniewicz-Klippel, 2019).

Southeast Asian parents across the region appear to similarly adopt the teaching of family hierarchy and parental authority to their children. Lanzarrote and colleagues (2013) found that parents in the Philippines endorsed socialization goals similar to other areas of Asia, including teaching how to show respect for elders at a young age. Indonesian parents impart the responsibility of *bekti* to their children, which means never being rude, disobedient, or argumentative with parents, and always withholding emotional expressiveness and praying for their parents to receive good things (Riany et al., 2017). Parents in Vietnam similarly teach their children to be quiet, to listen, and to avoid asking questions that may be perceived as challenging when interacting with an authority figure (Mestechkina et al., 2014). In an interview, parents in Cambodia discussed limiting their children's behaviors in order to maintain parental power, especially when Western

teachings have become increasingly prevalent in the context of globalization, which is perceived to threaten the hierarchical structure of families (Czymoniewicz-Klippel, 2019). One parent discussed saying no to their child joining in a dance, and another reported threatening their child with violence to get them to abstain from watching hip-hop on TV and wearing Western-style clothing.

Families' reliance on extended family members is prevalent in Southeast Asia, although there are regional differences in the types of relatives that reside with each other. Grandparents often live under the same roof or close by in Vietnam (Jayakody & Phuong, 2013). Indonesian families are likely to live with extended family that includes second- and third-degree relatives, who are considered as an important aspect of one's identity (Riany et al., 2017). In Cambodia, however, distrust among communities still persist after becoming widespread in the 1970s when the violent Khmer Rouge regime separated children from their parents in attempt to dismantle the family unit (Field, Om, Kim, & Vorn, 2011). Because Cambodian parents often still find it challenging to maintain trusting relationships with neighbors and even kin, poor accessibility to childcare and education generates a reliance on biological grandparents for household support and, oftentimes, adolescent siblings for childcare (Czymoniewicz-Klippel, 2019).

In summary, research on parenting in Southeast Asia has highlighted the critical impact of local cultural factors (e.g., religion, cultural values and norms, family structure, and indigenous parenting beliefs) in shaping parenting practices. Due to the vast heterogeneity in ethnic, language, socioeconomic, and political contexts of the Southeast Asian region, diverse patterns of parenting practices have emerged across different communities. An important implication for developing or disseminating parenting interventions in this

region is the need to evaluate and adjust the cultural fit between the target intervention and the local community. Partnership between researchers and community stakeholders can be a promising strategy to build the "cultural bridge" (e.g., connection or partnership that aligns the goals and content of the intervention with the goals and values of the local community).

Parenting in Refugee Families from Sub-Saharan Africa and Southeast Asia

The number of refugees forcibly displaced due to war, civil conflict and violence, and political instability is estimated to be 26.4 million by the United Nations High Commission for Refugees (UNHCR, 2021). A large majority of refugees and displaced individuals have historically migrated from Africa and Southeast Asia, with South Sudan and Myanmar as two of the top five countries of origin for refugees. South Sudan's pursuit of independence resulted in 1.9 million civilian deaths over the course of 15 years, and the country continues to experience frequent violent events between armed groups (Johnson, 2014). Other sub-Saharan African countries such as the Democratic Republic of Congo have also experienced armed conflict for decades. The violent conflict between the Myanmar military and ethnic groups has been ongoing since its independence in 1948. In Cambodia, survivors of the genocide perpetrated by the Khmer Rouge regime have continued to suffer from threats to their safety, health, and property over the last 30 years (Field et al., 2011).

Exposure to traumatic experiences, such as mass killings, deaths of loved ones, and physical assaults, prior to and during migration among refugees is strongly linked to psychological distress, which can persist for decades after resettlement (e.g., Marshall et al., 2005).

Parents' exposure to trauma and their subsequent psychopathological symptoms have reverberating effects on the mental health of subsequent generations via dysfunctional parenting practices and disruptive family relationships (e.g., Cobham & Newman, 2018; Sangalang & Vang, 2017). Indeed, among adolescent children of Khmer Rouge regime survivors, parents' trauma exposure was positively predictive of role reversing (e.g., parental elicitation of the child's care and attention) and overprotective parenting behaviors, which in turn were related to increased depressive and anxiety symptoms (Field et al., 2011). Other recent studies have similarly found that among refugees, associations between parents' trauma exposure and youth's delinquent behaviors were mediated by lack of parental engagement (Sangalang & Vang, 2017) and parents' post-traumatic stress disorder (PTSD) symptoms were directly associated with their children's insecure or disorganized attachment style (van Ee et al., 2016). This body of research indicates that parents' experience of trauma has both direct and indirect influences on children of refugees. An implication for policy and intervention is the need to provide trauma-based mental health evaluation and counseling services to refugee families.

Parenting among Refugees and Immigrant Families in Western Destination Countries

A plethora of risk factors such as poverty, lack of social support, limited familiarity with and access to services, and heightened cultural conflicts with Western host cultures are experienced by refugees and immigrants (Dennison et al., 2019; Salami et al., 2017). These threats to livelihood directly exacerbate parenting stress and competence, which can in turn lead to higher rates of behavior problems, depression, and anxiety in their children via increased

hostility toward their child and marital conflict (Field et al., 2011; Go & Le, 2005). Among culturally oriented stressors, qualitative interviews of African refugees in Canada identified multiple forms, including cultural conflicts with parenting, gender role conflicts, lack of culturally appropriate services, inability to perform cultural traditions, and language barriers (Stewart et al., 2015). Salami and colleagues' (2017) review described cultural conflicts with parenting among African immigrants include the inability to rely on physical punishment as a disciplinary tool, which threatened parents' sense of authority, and the difficulty in transmitting traditional African values and practices to their children as well as supporting their education. Similar cultural conflicts with parenting have been reported among Southeast Asian parents who immigrate to Western host cultures. Southeast Asian immigrants have also reported difficulty navigating the parenting goals of teaching their children their native language and engaging with their schooling (Nguyen et al., 2014; Uy, 2015). Despite having higher expectations for their children's academic attainment, Southeast Asian immigrants were found to be the least involved with their children's schools and academic assignments and to feel the least prepared to help their children academically compared to other immigrant and racial groups (Uy, 2015).

Many immigrant parents face discrimination and inequality in their destination countries. In interviews, African refugees reported that they perceived racial prejudice to be a barrier to employment opportunities and that their children have been the target of discriminatory behaviors in day care centers (Stewart et al., 2015). As Dennison and colleagues (2019) eloquently stated, in the United States the omnipresence of discrimination and racism toward immigrants and people of color make it difficult for families to know who to trust

and from whom to seek help. Many immigrant families refrain from utilizing social services despite the threats of food scarcity and lack of housing and transportation out of fear of being ineligible for the services and even deportation. Being victims of racist behavior has been associated with a reduced sense of belonging, depression, and diminished self-efficacy, resulting in poorer health outcomes and increased family dysfunction (Dennison et al., 2019). Children of immigrant families who experience intersecting inequalities are at increased risk for poorer academic, economic, and mental health outcomes (Suárez-Orozco, Yoshikawa, & Tseng, 2015).

A unique psychological process that is especially relevant for studying parenting in immigrant families is acculturation, or changes that occur as a result of contact with culturally dissimilar people, groups, and social influences (Gibson, 2001). Because an immigrant can simultaneously adopt elements of the host culture while retaining elements of the heritage culture, immigrant families can adopt and blend socialization beliefs and practices from both cultures and form a unique developmental niche. In a longitudinal study of Chinese American immigrant families living in the San Francisco Bay Area, Zhou and colleagues found that parents' American orientation (acculturation to the host culture) was associated with a higher use of authoritative parenting and lower use of intrusive parenting, higher overall emotion expressions and greater expressions of positive affect in parent-child interaction, and greater involvement at children's schools (e.g., volunteering in classrooms, participating in school events) (Anicama, Zhou, & Ly, 2018; Chen et al., 2014; Chen, Main, Zhou, & Lee, 2015; Kho et al., 2019). Parents' Chinese orientation (maintenance of the heritage culture) was also associated with higher authoritative parenting, but it was associated with parents' lower use of

emotion talk and lower emotion expressions (Chen et al., 2014, 2015; Tao et al., 2013). Moreover, as children make contact with peers and adults from different cultural groups through their extended family, school, peer network, and community, they develop independent cultural orientations. Researchers found that children's cultural orientations can directly impact parenting or interact with parents' cultural orientations in shaping parenting. For example, children's American orientation was associated with parents' higher use of authoritative parenting and intrusive parenting, and children's Chinese orientation moderated the link between parents' Chinese orientation and authoritative parenting (Chen et al., 2015; Kho et al., 2019). These findings suggest that acculturation is a dynamic and interactive process in immigrant families, and parenting and family relationships are shaped by family members' acculturation experiences and their complex intersections.

In summary, the research on parenting among refugees and immigrant families in Western destinations has highlighted the significant impact of migration-related risk and resilience processes (e.g., acculturation and acculturation stress, discrimination, socioeconomic inequality) on parenting and child development in these communities. A policy implication is that agencies and programs serving refugee and immigrant families need to monitor families' experience of migration-related stressors and provide support or resources to help them navigate these transitions (e.g., language-competent parent education programs and family counseling services for new immigrant families).

New Themes in the Research on Culture and Parenting

From our review, we have observed several new themes in the research on culture and

parenting. First, researchers have moved beyond asking the question of what is different and similar in parenting across cultural groups, nations, or societies to examining the cultural processes or mechanisms that might account for observed cultural group differences or similarities in parenting. To examine cultural processes, researchers have moved beyond relying on group status (e.g., nationality, race, ethnicity) as a proximal indicator of culture and are increasingly using direct measures of cultural processes (e.g., cultural values, acculturation, ethnic identity ...). This allows researchers to test complex relations between cultural processes and parenting beliefs and practices. Second, researchers examining the socialization functions of parenting on children's developmental outcomes in non-Western cultures have moved beyond testing the generalizability of Western socialization models and are increasingly incorporating local sociocultural factors, indigenous parenting beliefs and practices, and culturally salient child developmental outcomes into the theoretical model. Third, cross-cultural research is increasingly using multi-method/multi-informant measures or mixed methods approaches to capture parenting. Because different methods of parenting reflect different cultural influences (e.g., parents' and children's reports of parenting are differentially shaped by parents' and children's cultural values), assessing parenting from multiple methods allows researchers to investigate multiple pathways through which culture shapes parenting. Fourth, researchers are increasingly utilizing longitudinal designs or intensive longitudinal designs (e.g., ecological momentary assessment, or moment-to-moment dynamic interactions). These approaches allow researchers to study the dynamic interactions between culture, parenting, and child development across time. Fifth, the growing research on cultural adaptations of parenting interventions to

diverse cultural groups (e.g., Lau, Fung, Ho, Liu, & Gudino, 2011) provide valuable data on whether and how cultural barriers and cultural assets to successful parenting and child development can be modified by intervention.

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Part II

Parenting across Development: Social, Emotional, and Cognitive Influences

8 Bolstering the Bond: Policies and Programs That Support Prenatal Bonding and the Transition to Parenting*

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Developmental and Neurobiological Research on Prenatal Bonding and Attachment

Infants are born predisposed to develop strong relationships to those most likely to protect them; this emotional connection from the child to the protective adult is described as *attachment* (Ainsworth, 1979; Bowlby, 1983; Crittenden, 2006; Spierling et al., 2019). In turn, parents' behavioral and physiological responses prime them to respond to attachment behaviors, such as crying, with protective behaviors (Ainsworth, 1979; Bowlby, 1983; Cong et al., 2015). This emotional connection from the attachment figure to the child is described as *bonding* (Scatliffe et al., 2019). Parental bonding is more often studied in biological mothers, but similar processes of bonding can occur in fathers and other caregivers who act in the role of parents (Bowlby, 1983; Cong et al., 2015; Dayton, Malone, & Brown, 2020). Relationships are a dyadic experience, influenced by both the parent and the child, dynamically changing over time, and shaped by the family context (Ainsworth, 1979; Crittenden, 2006; Wilson et al., 2000). *Bonding* and *attachment* are distinct concepts, even though the labels are sometimes used interchangeably (Habib & Lancaster, 2006; McNamara, Townsend, & Herbert, 2019).

Parental bonding is shaped by a person's own experiences of being parented, beginning

in early childhood (Bowlby, 1973, 1983; Crittenden, 2006; Fonagy, Steele, & Steele, 1991; Slade et al., 2020). These attachment relationships influence brain systems involved in future parenting (Abraham et al., 2016; Hays-Grudo & Morris, 2020). In childhood, both traumatic and supportive experiences can alter the trajectory of relationships (Ainsworth, 1979; Bowlby, 1973; Fonagy et al., 1991). In adulthood, positive support from a partner can protect and buffer against other risk factors that may impact parental bonding (Alhusen et al., 2012; Cuijlits et al., 2019; Thomas et al., 2017). The following sections describe how this dynamic bonding relationship is influenced over time by neurobiological, social, and environmental factors unique to this period of parental and infant development.

Neurobiology of Prenatal and Postnatal Parent–Infant bonding

Although parent–infant bonding mainly takes place in the postnatal period, neurobiological changes during pregnancy contribute to parent–infant bonding and are influenced by

* The findings and conclusions in this article are those of the authors and do not necessarily represent the official position of the Centers for Disease Control and Prevention.

attachment relationships. Such changes continue during the postpartum period and are sensitive to experiences both prenatally and in the early postpartum period; thus, interventions early in the transition to parenting have potential to positively influence early parent–infant bonding through neurobiological pathways (Kim, Strathearn, & Swain, 2016; Swain et al., 2014; Vittner et al., 2018).

Neurological Changes Associated with Parental Bonding

Changes in brain structure and activity during pregnancy and the postpartum period are associated with parent–infant bonding. Increases in gray matter volumes in mothers from the prenatal to postnatal period were found in brain areas associated with motivation and reward, and emotion regulation (Kim et al., 2010a, 2016). These increases are associated with the mother's positive emotions about her infant (Kim et al., 2010b). Cortical responses related to attention allocation show plasticity from the pre- to postpartum period. In mothers, cortical changes associated with increased attention to infant faces, including distress, during pregnancy are associated with increased postpartum bonding and greater postpartum maternal sensitivity (Dudek et al., 2020; Pearson, Lightman, & Evans, 2011). Prenatal to postpartum changes in neural activity in response to infant crying have also been seen among new fathers, though research is still emerging in this area (Alyousefi-van Dijk et al., 2020).

Maternal brain changes continue in the postpartum period and are influenced by sensory cues from the infant (Kim et al., 2016; Swain et al., 2014). Amygdala activity related to an infant's cries is linked with negative responses in non-parents, but with positive responses from their own mothers (Kim et al., 2016). Maternal brain activity associated

with anxiety increases in response to infant cries and visual cues, and may influence the neural reward pathway associated with maternal behavior (Kim et al., 2016; Swain et al., 2014).

Neuroendocrine Factors Associated with Parental Bonding

Neuroendocrine factors, including oxytocin and cortisol, play important roles in prenatal and early postnatal bonding. Oxytocin production increases during pregnancy and lactation (Keverne, 2006; Scatliffe et al., 2019) and is positively associated with maternal–fetal bonding and postnatal maternal behavior (AAP, 2011; Feldman et al., 2007; Levine et al., 2007). Higher levels of oxytocin in parents are associated with greater responsiveness and behavioral synchrony in parent–infant interactions (Scatliffe et al., 2019; Vittner et al., 2018), which contribute to early bonding and future social behavior (Abraham et al., 2016; Feldman, 2017; U.S. Department of Health and Human Services, 2011). Oxytocin increases in mothers, fathers, and infants with skin-to-skin contact and infant suckling (Scatliffe et al., 2019; Vittner et al., 2018), while cortisol decreases (Scatliffe et al., 2019). Thus, oxytocin may reduce parental stress and anxiety (Scatliffe et al., 2019) and promote parent–infant bonding (Coyle, Roggman, & Newland, 2002; Thul et al., 2020). Maternal and infant cortisol levels are positively related, and infants of emotionally available mothers have lower levels of cortisol, suggesting that more maternal care toward the infant and reduced maternal stress can decrease infants' stress responses (Philbrook et al., 2014; Thomas et al., 2017).

The hormones dopamine and prolactin may also influence early parent–infant bonding. In response to infant stimuli, oxytocin and dopamine are co-activated, which may make

synchronous parent–infant interactions more rewarding (Feldman, 2017). The hormone prolactin is stimulated and maintained by infant suckling, like oxytocin (Buhimschi, 2004), and can play a role in inducing maternal caregiving behaviors (Russell, Douglas, & Ingram, 2001).

Early maternal relationship experiences and neurobiological pathways. Positive parenting environments and early adversity (e.g., child maltreatment) have contrasting impacts on parental and infant brain development, epigenetics (i.e., changes in gene expression), hormonal levels, and responses to stress (Hays-Grudo & Morris, 2020). Adults' self-reports of their early relationships with parents show positive associations between early positive maternal care and adult oxytocin levels (Strathearn et al., 2009; Swain et al., 2014), increased gray matter brain volumes in areas related to emotion regulation and social and sensory information processing, and higher activation of brain activity in frontal areas in response to infant cries (Kim et al., 2010b). In contrast, self-reports of early child maltreatment are associated with lower levels of oxytocin in women (Heim et al., 2009). Although these early experiences influence brain development that can shape cross-generational parenting patterns, they also suggest the potential for interventions early in the transition to parenting to improve outcomes for the next generation.

Development of the Bonding Relationship and the Effects of Risk and Protective Factors

Parent factors that affect the bonding relationship. The emotional experience of pregnancy and childbirth and the transition to parenting represents a major change that is shaped by multiple parental characteristics beyond the parent's own childhood experiences (Alhusen et al., 2012; Slade et al., 2020; Spierling et al., 2019). Parent stress, anxiety, and depression

are associated with poor parent–infant bonding (Alhusen et al., 2012; Coyl et al., 2002; Habib & Lancaster, 2006; Pawluski, Lonstein, & Fleming, 2017; Rollè et al., 2020; Nath et al., 2019). Current stress, such as from lack of social support, mental and physical health concerns, and financial stress can compound prior trauma and add to the strain that may accompany pregnancy and parenting (Alhusen et al., 2012; Cuijilts et al., 2019; Feldman, Sussman, & Zigler, 2004; McNamara et al., 2019; Rollè et al., 2020; Spierling et al., 2019). The psychosocial and structural stressors associated with racial/ethnic status are additional sources of trauma and stress for parents, which have been linked to poor maternal and infant outcomes (National Academies of Sciences, Engineering, and Medicine [NASEM], 2019). Further, lack of resources and supportive policies can limit parents' ability to take leave from work to engage in pregnancy-related self-care, recover from childbirth, develop the breastfeeding relationship, and spend time with the baby, which negatively affects the bonding process (Feldman et al., 2004; Spierling et al., 2019; Van Niel et al., 2020). In contrast, supportive relationships and evidence-based interventions can mediate the effect of risk (Cuijilts et al., 2019; Mascheroni & Ionio, 2019; McNamara et al., 2019; Shah, Muzik, & Rosenblum, 2011; Slade et al., 2020). Parenting interventions that also address parent psychopathology, if present, can improve parent–infant bonding and have long-term benefits for both the parent and the infant (NASEM, 2016; Nath et al., 2019).

Prenatal bonding. Because relationships are based on mental and emotional representations, the parents' bond with the child can begin before birth and is shaped by expectations and emotional experiences (Ainsworth, 1979; Condon, 1993; McNamara et al., 2019; Rollè et al., 2020; Spierling et al., 2019). The

physiological experience of pregnancy, including fetal movement, may intensify the process of prenatal bonding (Cuijilts et al., 2019; Spierling et al., 2019). The strength of the prenatal bond affects health and preparation behaviors during pregnancy that support fetal development and predispose parents toward nurturing behavior that shapes the developing infant–parent attachment relationship (Ainsworth, 1979; Habib & Lancaster, 2006; McNamara et al., 2019; Rollè et al., 2020; Spierling et al., 2019).

Childbirth experience. Childbirth represents a powerful moment of separation, transition, and new connection, and also confers considerable risk to both the mother and child (Feldman et al., 1999; Lawn et al., 2010; Spierling et al., 2019). The emotional experience can vary greatly and can depend on factors such as mothers' feelings of support and control, level of risk, and the experience of medical interventions. These factors can cause physical stress and trauma and therefore limit the emotional resources that parents need to take on the role of primary protector (Feldman et al., 1999; Jordan et al., 2014; Spierling et al., 2019). Premature birth may pose a risk for bonding because it unexpectedly interrupts the pregnancy and increases risk for short- and long-term complications (Shah et al., 2011). The premature infant may not participate in the bonding experience like a full-term infant; for example, the features and cries of premature infants are perceived as more distressing than those of full-term newborns (Endendijk et al., 2020; Shah et al., 2011). Healthcare inequities impact risk for premature birth and subsequent prenatal bonding (Alhusen, 2008). Moreover, separating premature infants from parents to address medical needs can disrupt the bonding process (Feldman et al., 1999; Jordan et al., 2014). However, negative effects on bonding can be mitigated (Cong et al., 2015; Korja, Latva, &

Lehtonen, 2012). Supportive care and hospital policies that encourage breastfeeding and physical contact with the baby can decrease stress and enhance the bonding experience (e.g., (Campbell-Yeo et al., 2015; Cong et al., 2015).

Postnatal bonding. In addition to parental neurological and neuroendocrine changes (Cong et al., 2015), the first year of life brings rapid changes in infants' needs and abilities to communicate and interact, which shape parental responses (Ainsworth, 1979; Bowlby, 1983). Clear preference for attachment figures develops in the second half of the first year, when infants become mobile and actively seek proximity when danger is perceived (Ainsworth, 1979; Bowlby, 1983). However, parents' ability to adapt to the ever-shifting needs of their child may be limited by competing employment-related tasks if they return to work (Van Niel et al., 2020).

The extreme vulnerability and dependence of newborns combined with their limited ability to communicate their needs is associated with vigilance on the part of the parent. This enhances responsiveness but can also lead to chronic heightened arousal (Feldman et al., 1999). Such parental arousal can lead to anxiety, depression, and hopelessness, particularly if the parent does not feel they can adequately care for both their child and themselves (Feldman et al., 1999; Shah et al., 2011). In addition, the physical stress of caring for an infant around the clock is often exacerbated by sleep deprivation, which can decrease sensitivity to the child's signals. This process can become a negative feedback cycle when parents do not have adequate resources or receive sufficient support (Bai, Whitesell, & Teti, 2020). However, higher levels of social support from partners during pregnancy are associated with lower maternal depression in the postpartum period, which in turn, is associated with higher quality maternal–infant interaction (Thomas et al., 2017).

Environmental influences: supports and interventions for bonding. Parents' ability to nurture and manage stress can be influenced by social and environmental factors such as cultural expectations, support from other caregivers, and financial resources that allow parents to spend time with their children (Cuijlits et al., 2019; Feldman et al., 2004; Spicer, 2011; Spierling et al., 2019; Van Niel et al., 2020). Supportive policies, programs, and interventions can assist parents in the task of nurturing and protecting their child (Campbell-Yeo et al., 2015; Feldman et al., 2004; Mascheroni & Ionio, 2019; NASEM, 2016, 2019; Shah et al., 2011; Slade et al., 2020). Parenting interventions may not only improve parenting behaviors and infant attachment, but may also change the neurobiology of the parent's brain (Iyengar et al., 2014; Kim et al., 2016). The following sections will highlight some policies, programs, and interventions that can support parental bonding and improve long-term outcomes for children.

Policies That Support Prenatal Bonding and the Transition to Parenting

Family Leave

Family leave (i.e., paid and/or unpaid job-protected maternity, paternity, or medical leave) policies are intended to allow parents time to bond with their new infant, settle into new parenting routines, and focus on their infant without the additional stress of competing work demands (NASEM, 2019). Research indicates associations between family leave and maternal and child outcomes related to bonding, including maternal stress, depression, self-esteem, postpartum rehospitalizations, marital satisfaction, breastfeeding, infant physical health, and child externalizing behaviors (Berger, Hill, & Waldfogel, 2005;

Feldman et al., 2004; Hamad, Modrek, & White, 2019; Jou et al., 2018; Mandal, 2018; Pihl & Basso, 2019; Van Niel et al., 2020).

Duration of family leave has been associated with parenting outcomes as well; however, evidence on the optimal leave duration for maximum maternal and infant benefits is unclear (Ahammer, Halla, & Schneeweis, 2020; Baker & Milligan, 2010; Feldman et al., 2004). Research indicates that women taking less than 12 weeks of leave afforded by the Family and Medical Leave Act (discussed below) have more negative affect and lower responsiveness (Clark et al., 1997), whereas women taking all 12 weeks and fathers taking up to 14 days have positive impacts on marital satisfaction, family centrality, and infant preoccupation (Feldman et al., 2004). Data from multiple countries suggest increased involvement from fathers who took more than two weeks of family leave (Huerta et al., 2013). The relationship between family leave and postpartum depression may be U-shaped (from 6 weeks to 12 months) with benefits maximized at 6 months (Dagher, McGovern, & Dowd, 2014).

The interpretation of family leave research is limited by the potential effects of self-selection bias on parent-child outcomes; parents from higher socioeconomic status (SES) may have more opportunity to take longer family leave in the U.S. In addition, parents who value parent-infant bonding time more may also elect to take longer family leave which may be associated with parent-child relational outcomes. Overall, the research on family leave indicates potential positive impacts on multiple family relationships (maternal-infant, paternal-infant, and marital) along with maternal and infant health and wellbeing.

Federal Family Leave Policies

Among the Organization for Economic Cooperation and Development (OECD)

41 member countries in 2018, the U.S. was the only country that did not require paid leave for new parents (Livingston & Thomas, 2019). All other OECD countries provided a combination of paid maternity leave, paternity leave, and/or parental leave. The U.S. federal Family and Medical Leave Act (FMLA) of 1993, 29 U.S.C. § 2601, entitles some employees up to 12 weeks of unpaid leave (within a 12-month period) for the birth and care of a newborn, adopted child, or foster child. Employees are eligible for FMLA-protected leave if they have worked for a covered employer (that employs at least 50 employees within 75 miles of its location) for at least 12 months, or at least 1,250 hours. FMLA protects the employee's employment and provides for continued health benefits coverage during the 12-week period. In addition, the Federal Employee Paid Leave Act, effective October 1, 2020, entitles federal employees covered by FMLA up to 12 weeks of paid parental leave for the birth, adoption, or foster placement of a child, without being required to exhaust FMLA leave beforehand.

State Family and Medical Leave Laws Applicable for New Parents

As of 2019, 12 jurisdictions (California (CA), Connecticut (CT), Hawaii (HI), Maine (ME), Minnesota (MN), New Jersey (NJ), Oregon (OR), Rhode Island (RI), Vermont (VT), Washington (WA), and Wisconsin (WI) and the District of Columbia (DC)) have enacted laws to complement the federal FMLA allowing leave to care for a newborn child, adopted child, sick family member, or to address their own medical needs. Some jurisdictions' laws differ slightly from federal FMLA protections. For example, the RI Parental and Family Medical Leave (RIPFML) Act provides job protection for employees for up to 13 weeks across any two calendar years (RI ST § 28-48-2). Additionally,

nine of these jurisdictions (CA, NJ, RI, NY, WA, DC, MA, CT, OR) offer paid leave. California, the first U.S. jurisdiction to create a paid family and medical leave benefit, requires covered employers to provide 12 weeks of leave in a 12-month period (CAL. GOV'T CODE § 12945.1, 12945.2, 12945.6). The CA Paid Family Leave Law provides partial wage replacement benefits – up to 60–70 percent of wages earned – under the state's disability insurance program (CAL. UNEMP. INS. CODE § 3300 et. seq.).

Federal or state family leave policies do not cover all employers or all employees. In 2019, only 18 percent of private industry workers had access to paid family leave, with only 8 percent of the lowest-wage earners having access to paid leave (U.S. Bureau of Labor Statistics, 2019). Even when an employer is subject to FMLA or a state law, families might not be able to afford to take the full 12 weeks of unpaid leave. Such gaps may limit how many and which families will benefit from these types of policies to promote parental bonding.

Breastfeeding Support

Breastfeeding may enhance the mother–infant relationship by facilitating sustained, consistent interaction and physical contact between a mother and her infant (Stern, 1997), supporting maternal postpartum and long-term health (U.S. Department of Health and Human Services, 2020), triggering physiological and neurological changes associated with maternal caregiving behaviors (Kim et al., 2016; Levine et al., 2007), increased sensitivity and secure attachment (Tharner et al., 2012; Weaver, Schofield, & Papp, 2018), and decreased maternal postpartum anxiety, depression (Ritchie-Ewing, Mitchell, & Christian, 2019), and stress (Groer, Davis, & Hemphill, 2002). Given the many child and maternal health benefits

associated with breastfeeding, the American Academy of Pediatrics (AAP), and the World Health Organization (WHO) recommend that infants be exclusively breastfed for the first six months followed by continuation of breastfeeding with the introduction of appropriate complementary foods for one year or longer (AAP Section on Breastfeeding, 2012; WHO & United Nations Children's Fund [UNICEF], 2003). However, many mothers face significant barriers to breastfeeding, especially those with limited access to resources and social support. Disproportionate social and practical barriers to breastfeeding contribute to disparities in breastfeeding initiation, exclusivity, and duration (Chiang et al., 2021; Jones et al., 2015).

Pressure to breastfeed, or challenges related to sustaining breastfeeding among those who initiate it, has been associated with anxiety, stress and frustration, and may increase the risk of symptoms of depression (Demirci et al., 2018; Watkins et al., 2011). However, professional support, breastfeeding education, and support from fathers or other parenting partners can play an important role in mothers' breastfeeding decisions, behaviors, and self-efficacy and may contribute to overall improved breastfeeding outcomes (Mahesh et al., 2018; Munn et al., 2016; U.S. Department of Health and Human Services, 2020). The benefits of breastfeeding on bonding are enhanced when new mothers receive adequate support for achieving their breastfeeding goals (Diez-Sampedro et al., 2019; Kitamura, 2019; Watkins et al., 2011).

Federal Breastfeeding Support

The Fairness for Breastfeeding Mothers Act of 2019 requires public federal buildings to provide lactation rooms (that are hygienic, shielded from view, and not a bathroom) (40 U.S.C. § 3318), not only for employees but also the public, if the building has public

bathrooms. Additionally, the Fair Labor Standards Act, 29 U.S.C. 207(r) provides that a covered employer must provide reasonable breaks for covered employees to express breast milk for one year after the child's birth and a breast milk pumping place, other than a bathroom, that is shielded from view and free from intrusion. This law applies to public and private employers across the country and allows states to provide even greater protections to employees.

Some federal agencies actively work to support breastfeeding strategies for families. For example, the Center for Disease Control and Prevention (CDC) promotes breastfeeding in healthcare and community settings by collecting national and state data on breastfeeding and breastfeeding support, through state partnerships to support breastfeeding mothers and their employers, and by developing toolkits and educational materials for healthcare providers and communities (Centers for Disease Control and Prevention, 2019). CDC supports WHO's Baby-Friendly Hospital Initiative (WHO & UNICEF, 2018) as a standard to support breastfeeding in hospital settings (Centers for Disease Control and Prevention, 2019).

State Laws That Support Breastfeeding

Several types of state laws support breastfeeding and breast milk pumping practices. Since 2018, all U.S. states have expressly authorized breastfeeding in public and private places (ChangeLab Solutions, 2018). Some states have accomplished this by affirmatively protecting a person's right to breastfeed. For example, Alabama law provides, "A mother may breastfeed her child in any location, public or private, where the mother is otherwise authorized to be present" (ALA. CODE § 22-1-13). New York state passed the "Breastfeeding Mothers' Bill of Rights," which includes:

“You have the right to breastfeed your baby in any location, public or private, where you are otherwise authorized to be. Complaints can be directed to the New York State Division of Human Rights,” and “You have a right to breastfeed your baby at your place of employment or child day care center in an environment that does not discourage breastfeeding or the provision of breast milk” (N.Y. PUB. HEALTH LAW § 2505-a).

Other states have allowed breastfeeding in public by exempting it from public indecency laws. For example, in Wyoming, “[t]he act of breastfeeding an infant child, including breastfeeding in any place where the woman may legally be, does not constitute public indecency” (WYO. STAT. ANN. § 6-4-201). Laws such as these can facilitate breastfeeding in various locations, may support breastfeeding initiation and duration, and may be particularly beneficial for mothers from ethnic minority groups who may be disproportionately impacted by stigma and cultural norms (Hawkins, Stern, & Gillman, 2013).

Baby-Friendly Hospitals

To support breastfeeding, WHO and the United Nations Children’s Fund (UNICEF) developed the Baby-Friendly Hospital Initiative (BFHI) in 1992 (WHO & UNICEF, 2018). BFHI focuses on ten evidence-based guidelines that include pre- and postnatal breastfeeding support and education, and facilitation of breastfeeding initiation with skin-to-skin contact between mothers and infants within the first half-hour after delivery (WHO & UNICEF, 2018). In a Japanese study, hospital care that included skin-to-skin contact followed by early breastfeeding initiation consistent with BFHI steps was associated with improved mother-to-infant bonding (Kitamura, 2019). A meta-analysis of breastfeeding support strategies

identified the BFHI as the most effective intervention at promoting exclusive breastfeeding for at least six months postpartum (Kim et al., 2018), with notable positive outcomes among women with fewer years of formal education, those living in poverty, and among racial/ethnic minority women (Hawkins et al., 2015; Philipp et al., 2001; Vasquez & Berg, 2012). As such, a 2020 Surgeon General’s Call to Action to Improve Maternal Health highlights BFHI as a health system approach to support breastfeeding (U.S. Department of Health and Human Services, 2020).

Several U.S. states have enacted statutes or regulations that require or encourage hospitals to adopt baby-friendly policies such as the BFHI, but criteria vary. For example, an Illinois statute indicates that hospitals, “consider guidance provided by the Baby-Friendly Hospital Initiative” when developing an infant feeding policy (210 ILL. COMP. STAT. ANN. 81/5, 81/10). Florida encourages the adoption of baby-friendly policies by allowing a facility to advertise as “baby-friendly” if it has complied with 80 percent of the Department of Health’s requirements, which are based on BFHI (FLA. STAT. § 383.016). Eighteen states have laws that address hospital practices that support breastfeeding among maternity patients (ChangeLab Solutions, 2018). Future research on state-level BFHI policies could investigate population-level impacts on maternal–infant bonding.

In sum, family leave, breastfeeding, and baby-friendly hospital policies may support the parent–infant relationship in the transition to parenting through neurobiological and social mechanisms (Feldman et al., 2004; Kim et al., 2016; Kitamura, 2019; Weaver et al., 2018). Although policies may be nationally or even globally implemented, the accessibility of the policies may vary by economic and demographic factors.

Programs and Interventions That Support Prenatal Bonding and the Transition to Parenthood

In this section we highlight select programs that support prenatal bonding and the transition to parenthood through the first year of life; Table 8.1 includes links for these programs. We focus on programs with national or state reach, relevant research since 2005, prenatal components, bonding or early attachment as an evaluation outcome, availability of RCT data, meta-analysis data or a comprehensive systematic review, and evidence for risk factors associated with bonding (i.e., parental psychopathology, trauma, prematurity) or implementation with nonmaternal caregivers through clinic or hospital-based and community-based programs.

Clinic or Hospital-Based Programs That Support Prenatal Bonding and the Transition to Parenthood

Delivery hospitals and prenatal care clinics are locations for many programs designed to support prenatal bonding and early parenting outcomes. Three hospital- or clinic-based, parenting programs with broad implementation across both place and time include Kangaroo Care, Birth Attendants, and the Family Nurture Intervention.

Kangaroo Care

Kangaroo Care was developed to provide incubating warmth for low birthweight infants by creating prolonged and intentional skin-to-skin time with caregiver and infant. The Kangaroo Care model has been implemented widely in low- and high-resourced countries, and with adoptive and biological mothers, fathers, and other caregivers (Ruiz-Peláez, Charpak, & Cuervo, 2004). Many studies have

examined Kangaroo Care's influence on biological and parent–infant bonding outcomes. A study with preterm infants in the Neonatal Intensive Care Unit (NICU) showed increased oxytocin and decreased cortisol levels for both mother and father during skin-to-skin contact with their infant compared to the time immediately before contact (Cong et al., 2015). A randomized trial of Kangaroo Care with neonates measured significantly higher levels of maternal attachment at three months in the Kangaroo Care group compared to controls (Gathwala, Singh, & Balhara, 2008). A meta-analysis noted decreased infant mortality when Kangaroo Care was initiated in the first week of life (Lawn et al., 2010). Kangaroo Care is associated with improved neonatal behavioral outcomes (Ohgi et al., 2002), and randomized trials comparing skin-to-skin contact to incubator care at birth showed improved infant cardio-respiratory stability (Bergman, Linley, & Fawcus, 2004). A longitudinal study using a quasi-experimental design suggested more positive mother-child engagement at nine years of age among children who received the skin-to-skin intervention than among controls (Bigelow et al., 2018).

Perinatal Support from Birth Attendants

Health organizations recognize continuous maternal support during childbirth as an addition to standard medical obstetric care and an evidence-based strategy for improving the overall safety of childbirth (Association of Women's Health & Obstetric and Neonatal Nurses, 2018). In addition to improving birth outcomes and reducing medical costs, studies across cultural, ethnic, and social contexts suggest that trained labor support may also improve mother–infant interactions and opportunities for bonding during the early postpartum period (Campbell et al., 2007;

Table 8.1. *Where to find more information on programs to support prenatal and postpartum bonding*

Program name	Program description	Type	Where to find more
Kangaroo Care	Kangaroo Care is an approach to provide incubating warmth for low-birthweight infants through intentional skin-to-skin time between the caregiver and infant.	Clinic- or hospital-based	World Health Organization, program overview: https://www.who.int/publications/i/item/9241590351
Birth Attendants	Birth attendants or doulas are professionals who provide psychosocial and physical support during birth and/or postpartum. Supports can include assistance during labor, preparing for the baby's arrival, and breastfeeding assistance.	Clinic- or hospital-based	Selected examples: https://icea.org/about/#top www.dona.org/ https://cappa.net/ www.blackdoulas.org/
Family Nurture Intervention (FNI)	FNI aims to promote the emotional connection between parent and infant in the NICU through facilitated calming routines.	Clinic- or hospital-based	https://nurturescienceprogram.org/family-nurture-intervention/
Healthy Start	Healthy Start grantees support families prenatally through 2 years of child age with the goals of reducing infant mortality, reducing health disparities, and connecting families to the services available in their communities.	Federal- HHS/HRSA and implemented in the community	https://mchb.hrsa.gov/maternal-child-health-initiatives/healthy-start
Circle of Security (COS) International™ Perinatal Protocol	COS is an attachment-based parenting model that targets parents of young children who are at risk for an insecure attachment relationship by supporting responsive parenting practices; the COS Perinatal Protocol is implemented in the third trimester of pregnancy.	Community-based	www.circleofsecurityinternational.com/

Infant Massage	Infant massage allows for focused, parent–child connections through skin-to-skin massage techniques to calm the neonate through gentle physical contact with a caregiver.	Community-based	Selected examples with broad reach: www.infantmassageusa.org/ www.iaim.net/
Baby Triple P- Positive Parenting Program (Baby TP)	Triple P is tiered, based on need (from primary prevention to targeted intervention), promotes behaviorally based positive parenting strategies and prevents behavioral problems. Baby TP is a component of the Triple P approach for expectant and new parents.	Community-based	www.triplep.net/glo-en/home/
Legacy for Children™ (<i>Legacy</i>)	<i>Legacy</i> is a public health approach to prevent developmental delays associated with poverty for young families (prenatally and postnatally) through positive parenting.	Community-based	www.cdc.gov/ncbddd/childdevelopment/legacy.html www.uclahealth.org/mattel/legacy-for-children/

Note. HHS = U.S. Department of Health and Human Services; HRSA = Health Resources and Services Administration; NICU = Neonatal Intensive Care Unit

Hans, Edwards, & Zhang, 2018; Hans et al., 2013; Sosa et al., 1980). Support from a trained birth attendant, or doula, is associated with aspects of bonding and parenting, including positive maternal perceptions of the infant, feelings of happiness and ease with becoming a mother, and self-reported knowledge and skill related to infant attunement and bonding (Campbell et al., 2007; McLeish & Redshaw, 2019). Community-based programs have offered doula services to enhance social support and promote maternal–infant bonding and positive parenting among communities of women vulnerable to social risk factors, such as adolescent, racial/ethnic minority, and low-income mothers (Glink & Attfeld, 1999; McLeish & Redshaw, 2019). In a randomized control trial (RCT) of the Chicago Doula Project, mothers who received a doula intervention demonstrated increased positive engagement with infants, responsiveness to infant distress, child-centered parenting values, and less infant distress (Hans et al., 2018; Hans et al., 2013).

Family Nurture Intervention (FNI) for Parents of Infants in the NICU

The goal of FNI is to establish and strengthen the bond between a mother and her preterm infant as soon as possible after birth. FNI aims to facilitate infant autonomic regulation through “calming routines” that promote mothers’ abilities to calm their infants and counteracts potential negative effects of the NICU environment (Beebe et al., 2018). FNI is implemented several hours a week in the NICU throughout the family’s hospitalization (approximately six weeks) by supporting mothers and their infants in interactions that involve scent cloth exchange, eye contact and communication during infant massage, and calming parent–infant skin-to-skin contact facilitated through a trained nurturing

specialist (Welch et al., 2013). Research indicates that the intervention is associated with both positive maternal parenting outcomes and infant neurophysiological outcomes (Beebe et al., 2018; Hane et al., 2015; Myers et al., 2015; Porges et al., 2019). Specifically, compared to standard care, mothers participating in FNI demonstrated more maternal sensitivity and responsive behaviors during a holding and feeding session in the hospital (Hane et al., 2015) and more positive, contingent, and calming touch interactions with their infants at four months, and their infants displayed more positive vocalizations (Beebe et al., 2018).

All three interventions described in this section demonstrate evidence for supporting positive parenting outcomes and the parent–child bond during the potentially stressful and vulnerable birth and neonatal period. Kangaroo Care and FNI leverage the neurobiological effects of touch on parenting, and birth attendants focus on psychosocial support for parents in the perinatal period.

Community-Based Interventions for Specific Populations and Risk Factors

Health Resources and Services Administration’s (HRSA) Healthy Start Program to Reduce Health Disparities

Healthy Start is a community-based, federally funded program that serves mothers and young children, prenatally through two years of age, with the goals of reducing infant mortality, reducing health disparities, and connecting families to the services available in their communities (Escarne et al., 2017). Healthy Start programs promote child development in areas where infant mortality rates are above the national average. Select Healthy Start programs explicitly focus on parenting and

bonding, for example, the Healthy Start Fatherhood initiative. Experienced Healthy Start Fatherhood program coordinators report that Healthy Start Fatherhood programs contribute a previously unavailable “safe space” for fathers who participate, while program sustainability was often threatened by retention of participants with transient addresses (Obure et al., 2020). Documentation of parenting outcomes related to Healthy Start programming is sparse. One study notes modest evidence of decreased parental neglect among Healthy Start participants (Duggan et al., 2004). Most evaluations focus on infant outcomes, with some showing evidence for improvement in outcomes such as birth weight and preterm delivery (Salihu, et al. 2009).

Circle of Security International™, Perinatal Protocol for Attachment Risk

The Circle of Security (COS) is an attachment-based parenting model that targets parents of children (0–5 years) who are at risk for an insecure attachment relationship. COS teaches parents to respond sensitively to their child's emotional needs by reading their child's cues and enhancing their own reflective functioning (Powell et al., 2014). COS has been implemented in Denmark with pregnant women at risk for mental disorders as part of an early intervention program to promote maternal sensitivity (Aarestrup et al., 2020). The COS Perinatal Protocol (COS-PP) (Cooper, Hoffman, & Powell, 2003) has demonstrated positive impacts on infant attachment security and maternal sensitivity and reductions in maternal depression among mothers with a history of substance abuse enrolled in a jail-diversion program (Cassidy et al., 2010). Intervention reviews suggest that COS-PP may be a promising intervention for pregnant women who have experienced trauma such as

child maltreatment (Erickson, Julian, & Muzik, 2019; Stephenson et al., 2018).

Infant Massage for Fathers

Studies on infant massage, mostly with mothers, suggest the intervention can promote parental bonding, attachment, and improve infant outcomes (i.e., day–night circadian rhythms, bilirubin production, and early growth and development) along with decreasing postpartum depression symptoms (Cleveland et al., 2017). In addition, infant massage has demonstrated positive impacts on infant weight gain, length of stay, and the experience of pain among infants in the NICU along with positive impacts on mother–infant attachment (Pados & McGlothen-Bell, 2019; Shoghi, Sohrabi, & Rasouli, 2018). However, a Cochrane meta-analysis and systematic review of RCTs on infant massage concluded that more rigorous research is needed to determine the impact on low-risk families (Bennett, Underdown, & Barlow, 2013).

Fathers have reported that early physical contact is an effective strategy to promote father–infant bonding (Scism & Cobb, 2017) and early skin-to-skin contact has been associated with increased oxytocin in fathers (Vittner et al., 2018). Although the evidence for father–infant massage research is promising, research related to infant massage with fathers is scant (Magill-Evans et al., 2006). Some of the previously discussed benefits of breastfeeding and Kangaroo Care as they relate to skin-to-skin contact may be achieved through infant massage with a variety of caregivers (Cleveland et al., 2017). Two smaller studies with fathers showed that infant massage increased father–infant connections, reduced parental stress (Cheng, Volk, & Marini, 2011), and increased positive, expressive, and responsive father–infant play interactions (Cullen et al., 2000).

Baby Triple P Positive Parenting Program (Baby TP) Research among Parents with Mental Disorders and Parents of Preterm Infants

Baby TP is an intervention for expectant and new parents developed as part of the multilevel Triple P Positive Parenting Program (Sanders, 1999) series. The needs-based levels of Triple P (i.e., Levels 1–3 primary and secondary prevention; Levels 4–5 are treatment for behavior problems) have shown promise as a prevention and intervention program to reduce risk of trauma and childhood behavioral issues through promoting positive parenting practices and parental competence (Morris et al., 2017). Mothers with severe mental illness reported their involvement in Baby TP postnatally increased their sense of achievement (Butler et al., 2014). However, an evaluation of Baby TP implemented prenatally and postnatally, found no significant intervention effects other than fathers' reported program satisfaction (Mihelic, Morawska, & Filus, 2018). An RCT of Baby TP for very preterm infants also revealed no significant intervention effects, with insufficient program dosage proposed as a potential contributing factor (Evans et al., 2017). Thus, it is unclear whether Baby TP can replicate the positive impact on parenting outcomes shown for Triple P.

Future research with larger sample sizes, including fathers and other caregivers, and replication studies could foster our understanding of interventions that support parental bonding for fathers, parents who have experienced trauma, and parents with mental disorders.

Legacy for Children™

Public health prevention programs are designed to address risk factors before they lead to disorders (Morris et al., 2017). The

CDC's Legacy for Children™ (*Legacy*) program was designed to prevent the negative impacts of poverty on child health and development by supporting maternal bonding and positive parenting through three mechanisms: promoting sensitive, responsive mother–child interactions, promoting a maternal sense of belonging to a community, and enhancing maternal self-efficacy (Perou et al., 2012). The *Legacy* program has two tested curriculum versions; one is prenatal through age three (University of California at Los Angeles, UCLA), and the other is birth through age five (University of Miami, UM). The UCLA prenatal sessions focus on preparing the mothers for birth, discussing feeding choices such as breastfeeding, practicing relaxation techniques, and creating connections and a system of support among the mothers. Postnatal sessions across both curricula facilitate reading and responding to infant cues, understanding and fostering a healthy attachment relationship, and reducing barriers to maternal infant bonding (e.g., stress, maternal self-care, negative thought attributions, mothers' own early childhood experiences, and partner support).

Results from the set of RCTs of *Legacy* indicate the program has a positive impact on child emotional and behavioral outcomes across sites at intervention end and two years post-intervention (Kaminski et al., 2013) and child IQ, achievement, and externalizing behaviors at six years post-intervention in Los Angeles (Barry et al., 2021; Perou et al., 2019). These outcomes are presumably achieved through intervention impacts on maternal bonding and positive parenting behaviors, as suggested by the program model and qualitative data (Hartwig et al., 2017; Kotzky et al., 2020). However, parental mediation analyses from the RCTs have yet to examine this relationship.

To address a gap in evidence-based programs for Spanish-speaking parents, the

Legacy UCLA curriculum was translated and adapted for Spanish-speaking parents and found to be culturally acceptable (Beasley et al., 2020; So et al., 2020). Spanish-speaking mothers also reported improvements in the parent–child relationship, parenting stress, and parental confidence (Kotzky et al., 2020). The *Legacy* model has been implemented by Head Start, Healthy Start, pediatric healthcare settings, and other service delivery locations within local communities with the aim of reducing barriers to access for families from low-income households such as child care, transportation, or schedule limitations (Robinson et al., 2019).

Next Steps for Bolstering the Parent–Infant Bond

Research over the past ten years has expanded our understanding of some of the neurobiological mechanisms of prenatal and postnatal bonding relationships (Kim et al., 2016; Swain et al., 2014; Vittner et al., 2018) that were shaped by foundational attachment research years before (Ainsworth, 1979; Bowlby, 1973). Although our understanding of these important early relationships has increased and intervention programs are available, gaps still exist in programs and interventions to support parents through the transition to parenting. In this final section, we will first discuss gaps in the parent–infant bonding literature and then conclude with a discussion of some integrated systems approaches and policies that address social determinants of health and may hold promise for filling gaps in support of families.

Research Gaps

The family household structure has changed significantly over the last 50 years, with fewer children living in traditional two-parent, married households and more children living

in single mother, single father, grandparent-headed or co-habiting households (NASEM, 2019). More than 30 percent of families include adopted, step, or foster children and 16 percent of same-sex couples were raising children, according to 2014–2016 U.S. Census data (Goldberg & Conron, 2018). Recent nationally representative data suggest that caregivers in grandparent-headed households were less likely to report emotional support for parenting when compared to parent-headed households (Rapoport et al., 2020) and poor mental health in fathers/male primary caregivers is associated with poorer child mental health and child physical health (Wolicki et al., 2021). However, these changes in family structure and needs are not yet reflected in evidence-based interventions that meet those needs; both parenting research and interventions for fathers and nonbiological parents are limited (Dayton et al., 2020; NASEM, 2016).

In addition, few parental bonding interventions are tailored for specific cultures or for diverse families. Listening to the needs of diverse families (Spicer, 2011) and recognizing the unique cultural contexts within and across cultural groups can support parenting program effectiveness (Teti et al., 2017). Although some aspects of parenting may be universal, as described by a *Legacy* Spanish-speaking mother, “It is in all cultures, how we are worrying about our babies”; programs may need tailoring, and factors that may facilitate program utilization may differ (Beasley et al., 2020). For example, evaluation of *Legacy* revealed that the program’s focus on every mother’s ability to positively impact their child’s development (rather than on mental health supports) appeared to align with the value system of the Spanish-speaking mothers and may have reduced stigma for trying a new program (Beasley et al., under development). Moreover, programs may also need to consider stressors such as racism that impact the health,

wellbeing, and parental bond in distinct ways across cultures (Alhusen et al., 2012; Malawa, Gaarde, & Spellén, 2021; Rollé et al., 2020). Future intervention research with families from diverse cultures could help answer questions on the needs and resources related to parental bonding.

Integrated Systems and Policy Supports

Population-level impacts on early relationships may be enhanced by broader systems-level solutions for supporting families. Although the bonding relationship can begin before birth and continues throughout development (Condon, 1993; Siddiqui, Hagglof, & Eisemann, 2000; Spierling et al., 2019), prenatal and postpartum medical care may be siloed, with different specialties attending each period. Access to quality prenatal care along with the availability of paid family leave may support the parent–infant bond and eliminate early disadvantages experienced by ethnic and racial minority groups (Alhusen, 2008; NASEM, 2019). Innovations in prenatal, pediatric, and family healthcare approaches that embed parenting supports may contribute to improving prenatal bonding, parenting, and child outcomes (Ickovics et al., 2019; NASEM, 2016).

A “two-generational approach,” supports and builds both parent and child resources and is being incorporated into health systems (Brundage & Shearer, 2019; Jones et al., 2015; NASEM, 2019). For example, an integrated family approach proposes addressing the needs of all family members, behavioral and physical, through coordinated and integrated services in health systems. A pilot study bundled insurance coverage of prenatal and pediatric care and embedded evidence-based maternal and infant health interventions through the first year to promote the parent–child relationship and overall family health

(Brundage & Shearer, 2019). The prenatal and postnatal programs CenteringPregnancy and CenteringParenting, implemented in obstetric care and pediatric care respectively, are integrated healthcare approaches that support prenatal care, preterm birth, breastfeeding, and parenting knowledge and efficacy outcomes while supporting the psychological, social, and behavioral health of mothers; these approaches may even have stronger outcomes for African American women (Ickovics et al., 2019; Jones et al., 2018; Robinson, Garnier-Villarreal, & Hanson, 2018). They hold promise for reaching large numbers of diverse families with prevention efforts through trusted relationships (Brundage & Shearer, 2019).

Policies and programs that support the economic security of parents may also have impacts on parenting and child development outcomes by reducing stress, promoting responsive family environments, and increasing available family resources for stimulation of child development (NASEM, 2019). Research could investigate if policies and programs addressing social determinants of health such as the Earned Income Tax Credit that has evidence for improved maternal mental health (Evans & Garthwaite, 2014), outcomes for low birthweight infants (Hoynes, Miller, & Simon, 2015) and prenatal care utilization (Hamad & Rehkopf, 2015) may also improve parental bonding and parenting outcomes. Examining income security programs that support parents, and in particular families most at risk for poor health and development such as those eligible for Supplemental Security Income, may help identify policies and strategies that could address gaps in parenting supports and child development disparities.

Conclusions

The parent–infant bond, often beginning prenatally, can protect the vulnerable infant from

harm and support the infant's healthy development (Ainsworth, 1979; Habib & Lancaster, 2006; McNamara et al., 2019; Spierling et al., 2019). Evidence suggests that some federal and state laws, healthcare interventions, and community-based parent support programs may promote parental bonding. Systems-level supports and approaches to addressing social determinants of health are theoretically indicated avenues that may benefit from additional research. A combination of parenting policies and programs may be more likely to achieve population-level impacts than single approaches (Morris et al., 2017; Teti et al., 2017).

This chapter was written in 2020 in the midst of an unprecedented public health emergency from a global pandemic and with social circumstances that could impact prenatal and postnatal bonding on a population level, such as social distancing impacting family connections during pregnancy, labor, delivery, and the postpartum period, along with increasing risk factors for bonding such as anxiety and depression (Hermann, Fitelson, & Bergink, 2021; Yoshikawa et al., 2020; Zanardo et al., 2020). Calls have been issued for a comprehensive public health pandemic response inclusive of health equity strategies, targeted interventions, mental health screening, surveillance and research (Veldhuis, Stuart, & Fallin, 2021). Future research questions may include impacts on bonding for racial and ethnic groups disproportionately affected by the pandemic. Research collaborations, such as the consensus-based and standardized efforts from the National Institute of Health's funded PhenX Toolkit¹, have the potential to investigate and buffer the potential adverse impacts of these circumstances on parent–infant bonding and reduce racial and ethnic health disparities, to better support future generations of parents and babies.

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¹ www.phenxtoolkit.org/index.php

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9 Parenting during Infancy and Early Childhood

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Introduction

Parenting in infancy and early childhood (from birth to age five) is a key factor that can enhance or undermine children's concurrent and long-term well-being and achievement depending on the quality and context in which it occurs. The role of parenting during this phase of life is considered of paramount importance because: (a) infants and young children are completely reliant on adults to meet their basic needs; (b) young children typically spend more time with their parents than with other adults or socializers (e.g., teachers, peers); (c) even when young children spend time in childcare, the quality of parenting or other home/parental characteristics, explain more variance in cognitive and social-emotional outcomes than do features of childcare (NICHD ECCRN, 2003), and (d) the skills acquired and relationships formed in the context of parent-child interaction during early childhood have long-term implications for outcomes throughout the lifespan (e.g., Raby et al., 2015). As illustrated in Figure 9.1, our primary goals are to review the impact of specific dimensions of parenting on child outcomes and how parental characteristics and experiences contribute to individual differences in parenting quality and child outcomes. In doing so we consider some child characteristics that moderate these relations. Although most relevant literature focuses on mothers, we include literature on fathers and note when differences exist. Likewise, although much of the literature is based on

European American families, we attend to possible racial and cultural differences in parenting, predictors of parenting, and child outcomes. We conclude with implications of this work for policy, prevention, and intervention. Importantly, the developmental tasks, phases and transitions children undergo during early childhood set the context for parenting. Thus, we begin there.

Key Developmental Milestones, Stages, and Transitions during the First Five Years

The first two years of life, or infancy, is a period of accelerated physical growth also marked by the rapid acquisition of motor, cognitive, language, social, and emotional skills (Lamb, Bornstein & Teti, 2002). Infants are uniquely vulnerable and dependent on parents for their survival given they are unable to meet even their most basic needs. Neonates demand round the clock care that is often unpredictable (e.g., frequent, short feedings). Furthermore, they are highly dependent on parents to help them regulate their emotions, in particular to soothe them when they are distressed. Across infancy, eating and sleeping patterns become more consistent and aligned with adult routines and self-regulation skills emerge. During the second half of the first year, many infants begin to experience separation anxiety and stranger anxiety which typically increase in intensity into the second year of life, stabilize, and then diminish (Brooker

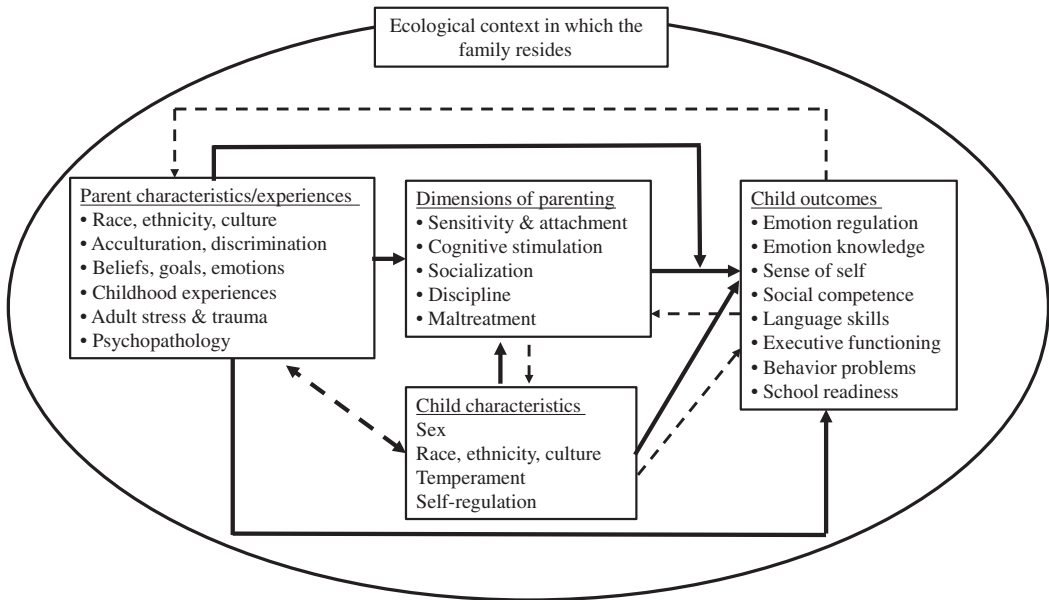


Figure 9.1 Conceptual model of the role of parenting and parent characteristics in infant and early childhood development

et al., 2013). Importantly, there are vast individual differences in the presence, intensity, and timing of these emotional stages that may present challenges to parents.

The second year of life is characterized by the onset of walking, rapid gains in language production and comprehension, and children's growing need for independence. This phase is often described as the "terrible twos," because there is a normative increase in both tantrums and aggressive behavior (Potegal & Davidson, 2003). During the preschool period, approximately three to five years, children's logical thinking increases, vocabulary and grammar skills encounter explosive growth, sense of self (i.e., gender, race) becomes more stable, and children rapidly acquire new knowledge essential for school entry (Barbarin & Wasik, 2009). Preschoolers also develop theory of mind or the awareness that others have thoughts and feelings of their own, begin to display empathy more consistently, and behave in an increasingly socially competent or prosocial manner.

Finally, as children enter kindergarten, they face increased expectations for self-control and behaving according to social norms, but often require guidance to engage in these behaviors in challenging situations and for sustained periods of time.

We turn next to specific dimensions of parenting that are critical during the first five years given these developmental tasks, including: parental sensitivity and parent-child attachment, cognitive stimulation and scaffolding, key aspects of parental socialization, and parental discipline. We also describe the impact of child maltreatment.

Dimensions of Parenting and Relevant Child Outcomes

Parent Child Attachment and Sensitive Parenting

The role of early parental sensitivity and parent-child attachment on children's

functioning is one of the most frequently studied aspects of parenting in early childhood. According to Bowlby (1969/1980), infants whose parents consistently and appropriately respond to their needs develop a secure working model in which they feel worthy of love and perceive the world as a safe place in which their needs will be addressed. In contrast, infants who experience non-responsive, harsh, or inconsistent parenting develop an insecure working model in which they feel unworthy of love and perceive the world as a relatively unsafe place and doubt their needs will be addressed. Bowlby asserted these models inform children's expectations for and evaluations of their own and others' behaviors over time.

Ainsworth and colleagues provided the first empirical evidence to support these propositions by carefully observing mother–infant interaction in the home over the first year of life and identifying three infant attachment classifications based on infant behavior during a standardized stress-inducing laboratory task (Ainsworth et al., 1978). Secure infants had mothers who responded to them quickly, consistently, and appropriately, a set of behaviors labeled as maternal sensitivity. Secure infants sought closeness and contact with their mothers and appeared happy and comforted by their mothers following separation. Avoidant infants had mothers who ignored and rejected them, and subsequently did not involve mothers in their play, tended not to become distressed during separations from their mothers, and avoided or ignored their mothers upon return from separation. Resistant infants had mothers who engaged in inconsistent caregiving behavior. Resistant infants engaged in little exploration and maintained close proximity to their mothers, became highly distressed during separations, engaged in angry and resistant behavior towards their mothers following separations,

and were not soothed by their mothers' behaviors. Years later, Main and Solomon (1990) described an additional attachment classification: disorganized. Disorganized infants displayed unusual and repeated behaviors that did not have a recognizable or clearly organized goal to avoid, resist, or gain proximity to their mothers. Disorganized infants are highly likely to have experienced maltreatment, trauma, atypical parenting and/or to have parents diagnosed with mental illness or their own history of trauma (Cyr et al., 2010).

Research has consistently replicated the finding that sensitivity is a significant and consistent predictor of children's attachment to both mothers and fathers (Verhage et al., 2016). In addition, how mothers respond to infant cues related to safety and comfort, particularly crying, fear, and bids to be picked up, is a stronger/unique predictor of attachment security over and above sensitivity to other cues (Woodhouse et al., 2020). In contrast, father behavior during play and the extent to which and conditions under which fathers are involved in different aspects of caregiving appear to be particularly important predictors of father–child attachment (Brown et al., 2018).

A good deal of research supports the view that internal working models of “self” and “other” formed in the context of early parent–child relationships has important implications for children's subsequent social-emotional well-being. Securely attached children exhibit fewer externalizing behaviors (e.g., aggression), less anxiety and depression, better understanding of emotions and emotion regulation skills, more compliance to parental requests, and more social competence with siblings and peers during the first five years of life compared to insecure children; and disorganized children are at greatest risk for subsequent mental health problems compared to all other attachment classifications (see Leerkes &

Bailes, 2019). Secure attachments to parents also predict important biological outcomes such as more adaptive stress reactivity and regulation, epigenetic modifications, and immune function (Ehrlich et al., 2016). And, although studied less frequently, there is evidence that attachment security predicts better cognitive and academic outcomes in early childhood (e.g., Bernier et al., 2020), perhaps because securely attached children are more confident to explore their environment and better able to attend to learning opportunities given trust in self and other. Recent evidence supports the notion that the impact of parent-child attachment in early childhood has long-reaching implications on social-emotional (Boldt, Goffin & Kochanska, 2020) and academic adjustment (Dindo et al., 2017) into adolescence via cascade effects.

Although most of the research to date has focused on the positive outcomes associated with a secure attachment to mother, a secure attachment to father has yielded similar benefits for child adaptation (e.g., Boldt et al., 2020; Ramchandani et al., 2013). Recently, Dagan and Sagi-Schwartz (2018) underscored the need for a family network approach to the study of attachment in which child outcomes could be compared across four possible configurations: two secure attachments, secure to mother only, secure to father only, and two insecure attachments. Such studies are limited at this point and results are inconclusive. For instance, in one study preschoolers' attachment to fathers only was particularly important in relation to subsequent conduct problems (Bureau et al., 2017), but in another, focused on stress reactivity as measured by cortisol, a secure attachment to mother only but not father only was linked with more typical/adaptive cortisol reactivity across tasks (Kuo et al., 2019). Differences in the primacy of a secure attachment to mother versus father may be a function of child age, the nature of the outcome, some

combination, or other factors. Additional research on this issue is warranted and attention to parental role/relative time investment rather than gender may be useful and has the added benefit of being more inclusive of less traditional family structures (e.g., same-sex couples; multi-generational households).

Although sensitivity was first examined as a predictor of attachment, consistent evidence indicates that sensitivity predicts a wealth of other important child outcomes. Moreover, in some studies, sensitivity predicts child outcomes over and above attachment security, indicating that sensitivity influences children's development via pathways other than a child's internal working model (e.g., Wright et al., 2018). As elaborated in Leerkes and Bailes (2019), sensitivity to infant distress cues may teach children important lessons about the meaning, effective communication of, and regulation of negative emotions and how to respond to others' needs appropriately – all of which may prevent children from engaging in maladaptive behaviors. Indeed, maternal sensitivity to distress or in distressing contexts predicts lower stress reactivity, better behavioral and physiological emotion regulation, fewer behavior problems, and enhanced social competence. Sensitivity to non-distress cues, which may help promote physiological homeostasis, normative patterns of stress reactivity, and psychological well-being has also been a consistent predictor of children's adaptive social-emotional outcomes (see Leerkes & Bailes, 2019). Additionally, sensitive parenting, or similar constructs, early in life predicts later attentional control and executive functioning as well as their neural underpinnings (Swingler et al., 2018; Swingler et al., 2017) and language and pre-academic skills, school readiness and early school success over and above the contribution of parental cognitive stimulation (Leerkes et al., 2011; Vallotton et al., 2017). It is probable that children whose emotional and

social needs are met are better able to attend to and learn from their parents and other socialization agents and to engage in self-directed exploration and learning, all of which may enhance the acquisition of these skills. Importantly, early parental sensitivity is a particularly salient predictor of (a) adaptive social-emotional outcomes among children at greater risk by virtue of a negative temperamental disposition (e.g., Leerkes, Blankson & O'Brien, 2009) and (b) adaptive academic outcomes among children at greater risk by virtue of low pre-academic skills (e.g., Leerkes et al., 2011). Finally, early sensitive caregiving has lasting effects on later outcomes (e.g., Raby et al., 2015).

Many have questioned the universality of attachment theory and validity of common measurement approaches among families who are not of European heritage. Mean differences in diverse samples generally suggest less favorable outcomes among minority groups, further raising questions regarding construct validity. However, careful synthesis of studies suggests that core predictions of attachment theory (i.e., security is linked with adaptive outcomes; sensitivity predicts security) are supported across a wide variety of cultures (see Mesman et al., 2018 for a review). Additionally, apparent racial/ethnic differences in sensitivity typically become non-significant once socio-economic differences are statistically controlled, and there are typically no differences in the pattern of association between sensitivity and key outcomes such as attachment (e.g., Malda & Mesman, 2017). Nevertheless, there is a strong need to remember that what constitutes sensitive caregiving may not be uniform across groups, given different belief systems, norms, and contextual demands (including discrimination); thus indicators of sensitivity must be culturally appropriate (Mesman et al., 2018). For example, during parent-child interactions, European

American mothers are more likely to use suggestions to structure children's behaviors, and these are typically viewed as sensitive behaviors, whereas Latina mothers tend to use directives and physical manipulation, which some have viewed as intrusive/insensitive behavior. However, consistent with Latino cultural values and socializations goals, the function behind such behaviors may be to provide supportive guidance, rather than to control and undermine children's development. A culturally informed observational measure of Latino parenting during early childhood indicated that intrusiveness and guidance could be distinguished, and while parental intrusiveness was an indicator of negative parenting, parental guidance was an indicator of positive/sensitive parenting and had positive implications for children's adjustment (Benito-Gomez, 2020). Next, we turn to the specific messages parents convey to their young children which become increasingly important as infants transition to toddlers, given maturing motor, language, and cognitive skills.

Parental Socialization

Socialization refers to multiple processes through which children learn social, emotional, and cognitive skills to participate in their social communities (Grusec & Davidov, 2010). For example, parents shape their children's sense of gender and engage in gender socialization behaviors by the materials and messages they provide to their children and others through their selection of toys, books, and the division of household chores (Leaper & Farkas, 2015). Additionally, parents engage in emotion socialization behaviors such as expression, modeling, and discussion of emotions which have implications for children's emotional development and well-being during the first few years of life. For example, teaching toddlers and preschoolers about emotions

increases their knowledge and ability to discuss emotions, which has been associated with better emotion regulation and adaptive social behavior and reduced behavior problems over time (Brophy-Herb et al., 2015; Denham, Bassett, & Brown, 2013). It is important to note that parental socialization goals and the impact of specific socialization practices sometimes vary as a function of race and culture (Cole & Tan, 2007). For example, compared to European American parents, African American parents are more likely to punish and minimize the display of negative emotions; however, these behaviors are not associated with negative outcomes for this group (Nelson et al., 2012). These racial differences in emotion beliefs and socialization practices may be an adaptive response for parents who experience discrimination in an effort to protect their children, suggesting that emotion socialization may be a key component of racial socialization practices (Dunbar et al., 2017). Racial socialization practices, such as positive messages about one's own race, are beneficial for young children's cognitive and social development and may serve as a buffer against discrimination (Caughy & Owen, 2015).

Cognitive Stimulation and Scaffolding

Young children's language and cognitive development, as well as school readiness, benefit from parental stimulation and scaffolding. For example, parents' provision of a language-rich environment (e.g., speaking, singing, oral narratives), stimulating home materials (e.g., books, cause-and-effect toys/objects, puzzles, art supplies) and activities (e.g., shared reading, trips to libraries/museums, pretend play), as well as information and autonomy, support while interacting is associated with children's better vocabularies, reading skills, performance on achievement tests, and executive function skills (Bradley et al. 2001; Melzi, Schick,

& Bostwick, 2013; Meuwissen & Carlson, 2018; Price & Kalil, 2019; Suor et al., 2019). The way parents engage with their children and provide information plays an important role. Parental warmth, praise, encouragement, and acknowledgment of frustration as opposed to criticism, negativity, and highly controlling behavior promote more adaptive child outcomes in these domains (Valcan, Davis & Pino Pasternak, 2018). A recent meta-analysis of studies of children from birth to age eight demonstrated that parental cognitive support has a stronger effect on younger versus older children's executive functioning, suggesting that this type of parenting is especially salient in early childhood (Valcan et al., 2017). Both maternal and paternal, including non-residential father, engagement in these behaviors benefits young children (e.g., Choi, Palmer, & Pyun, 2014; Towe-Goodman et al., 2014), and fathers' engagement in these behaviors is especially important when maternal cognitive support is limited (Martin, Ryan & Brooks-Gunn, 2010).

Parental Discipline and Guidance

Another important aspect of parenting in early childhood is parental discipline and guidance, which allow children to internalize rules and engage in compliant behaviors; however, the extent to which children engage in these behaviors depends on the quality of parental behaviors (Grusec & Davidov, 2010). Whereas parental discipline can include harsh and intrusive behaviors characterized by negative control, parental guidance often includes behaviors such as encouragement, modeling, and making the task fun. Children whose parents engage in harsh, intrusive, and overly controlling discipline are more likely to show maladaptive emotional and behavioral-related outcomes, including low effortful control, defensiveness, non-compliant behaviors, and

avoidant attachments (Carlson & Harwood, 2003; Guzell & Vernon-Feagans, 2004; Ispa et al., 2004; Taylor et al., 2013). This type of parenting undermines children's opportunities to develop adaptive regulatory processes and engage in learning and socialization experiences (Graziano, Keane, & Calkins, 2010). Accordingly, infants' physiological reactivity to stress has been consistently examined as a moderator between harsh and intrusive discipline and children's adjustment problems, such that young children with less adaptive physiological reactivity and regulation are more likely to develop internalizing and externalizing behaviors in the context of intrusive parenting (Rudd, Alkon, & Yates, 2017). This type of parenting also has implications for cognitive development. Indeed, children with mothers' and fathers' who were more intrusive at 24 months demonstrated poorer language and cognitive development at 36 months (Tamis-LeMonda et al., 2004). Coercive parenting practices during early childhood have been associated with long-term negative outcomes, such as higher deviance during adolescence (Patterson, 2016). In contrast, children whose mothers engage in gentle guidance and less negative control engage in more compliant and self-assertive behaviors (Donovan, Leavitt, & Walsh, 2000).

Even though a source of considerable debate, corporal punishment or spanking is a common discipline technique that parents use during early childhood around the world (Gershoff, 2013). Even though cultures differ in their perceptions on whether this technique is considered normative, adaptive, or abusive, many professionals and research studies discourage using corporal punishment or spanking as a discipline technique (Korbin, 2003). However, approximately half of parents of children between 0 and 9 years old in the United States engage in this behavior, with a peak during the preschool years (Finkelhor

et al., 2019), and similar rates were reported in a 33-country survey by UNICEF (UNICEF, 2010). In a recent meta-analysis that included over 160,000 parents and their children, spanking was linked to children's aggression, non-compliance, internalizing symptoms, and later substance abuse (mean effect size = 0.33) (Gershoff & Grogan-Taylor, 2016). Fifteen of the cited studies included children under the age of two, and these findings were not moderated by age or by the country where the study was conducted, suggesting that significant effects are present in early childhood and across cultures.

Given differences in the sociocultural context in which families are embedded, some have argued that for African American and Mexican American children, controlling behaviors and spanking may be less detrimental for children's adjustment, particularly when they are combined with high levels of warmth (Halgunseth, Ispa, & Rudi, 2006). However, a few studies have found that parental race and warmth do not moderate the links between spanking and child outcomes in early childhood (e.g., Barajas-Gonzalez et al., 2018; Berlin et al., 2009), given that very young children may not yet have the cognitive skills to interpret harsh and overly controlling behaviors as caring rather than punitive. In a recent study of Latina mothers, toddlers whose mothers were bicultural and engaged in parental guidance showed the lowest levels of defiance (Wood & Grau, 2018). Given mixed findings in the literature, and during early childhood in particular, it is important to consider parental discipline within the racial/cultural context in which it is embedded to best understand implications for young children's adjustment.

Child Maltreatment

Child maltreatment, including abuse and neglect, is highly prevalent in early childhood.

Nearly 700,000 children experience substantiated maltreatment each year and more than a quarter of the victims are under the age of three (U.S. Department of Health & Human Services [DHHS], 2020). The vast majority of children (92 percent who experience maltreatment are maltreated by one or both of their parents, and mothers are more likely to be the perpetrator of maltreatment than fathers (U.S. DHHS, 2020). Infants in the first year of life are particularly likely to be maltreated and nearly half of fatalities resulting from maltreatment occur during this time (U.S. DHHS, 2020). Children in poverty disproportionately experience maltreatment, and racial and ethnic differences in rates of maltreatment are driven by differences in socio-economic status (Kim & Drake, 2018).

Decades of research underscore the damaging effect of maltreatment on nearly all domains of child well-being, including physical, mental, and behavioral health, and these effects are presumed to occur through several emotional, relational, cognitive processing, and biological mechanisms (Jaffee, 2017; Oh et al., 2018). Maltreatment in infancy contributes to problems with physical health, including conditions such as failure to thrive and head trauma, cognitive development and developmental delay, the development of secure attachment bonds, and social-emotional functioning (Cyr et al., 2010; Harden, Buhler, & Parra, 2016). These effects extend to toddlerhood and the preschool period. In a population-based sample of over 19,000 children, maltreatment was a risk factor for poor school readiness across several domains of cognitive, social, emotional, and physical development (Bell et al., 2018). In addition to direct effects on physical, cognitive, and social-emotional well-being, child maltreatment is presumed to impact young children's development through biological mechanisms. Maltreatment in infancy and

early childhood is associated with alterations in brain development, structure, and function (Harden et al., 2016), hypothalamic-pituitary-adrenal axis functioning (Koss & Gunnar, 2018), and epigenetic processes (Parade et al., 2016), and emerging evidence suggests that these biological processes mediate associations of early maltreatment and behavioral outcomes in children (Bernard, Zwerling, & Dozier, 2015; Parade et al., 2016). However, relatively few studies have rigorously tested these mechanisms, and more research is needed to understand these processes in early childhood, particularly in infancy.

Despite evidence of the consequences of maltreatment on several domains of young children's development, not all children who are maltreated experience negative outcomes. In a sample of 943 children pooled from research projects across five sites in the United States, 42 percent of children with a child protective services report of maltreatment prior to age four were found to be resilient across behavioral, social, and developmental domains of competence when assessed at four and six years of age (Dubowitz et al., 2016). Although the factors that promote resilience among young children exposed to maltreatment are not well understood (Cicchetti, 2013), recent research suggests that cognitive stimulation and emotional support in the home environment are associated with resilience at the time of school entry in children with child protective service involvement prior to one year of age (Sattler & Font, 2018). Interestingly, child self-regulation and temperament, assessed using parent reports, were not associated with resilience. More research utilizing independent observations of child behavior is needed. In sum, parental behavior and parent-child relationships in early childhood have important implications for children's subsequent development and functioning. Next, we elaborate

on how parental characteristics and experiences contribute to individual differences in parenting and child outcomes.

Individual Differences in Parental Characteristics and Experiences

Parents vary tremendously in parenting-specific beliefs and emotions, cultural values and experiences, developmental experiences prior to parenthood, and current well-being and circumstances, each of which have the potential to influence parenting and child outcomes (Tarraban & Shaw, 2018). As illustrated in Figure 9.1, the effects of parent characteristics on child outcomes may be (a) direct via mechanisms including genetic transmission, contagion, and modeling; (b) indirect via their effects on parenting; and/or (c) the result of joint/interactive effects between parent characteristics and parenting behavior (see Leerkes & Augustine, 2019).

Normative Parental Cognitions and Emotions

Individual differences in normative parental cognitions and emotions predict parenting in a variety of domains. For example, parents who believe spanking is an appropriate discipline technique are more likely to engage in spanking (Pinderhughes et al., 2010), and parents with more democratic/child-centered as opposed to parent-centered/controlling beliefs are observed to be more sensitive both concurrently and longitudinally (Schofield & Weaver, 2016). Additionally, the attributions parents make about their children's behavior predict parental emotion and behavior as well as child outcomes. For example, parents who believe child misbehavior is deliberate and caused by negative child traits are more likely to feel angry in response, respond harshly, and have children with more behavior problems in

the long run (see Bugental & Corpuz, 2019). Likewise, parental negative emotion and physiological arousal and dysregulation while parenting are associated with negative parental cognitions about the child, harsh, insensitive and asynchronous parental behavior, and negative child outcomes including attachment disorganization, behavior problems, and regulatory difficulties (see Leerkes & Augustine, 2019). In contrast, parental positive emotion, empathy, and emotion regulation while parenting are associated with sensitive parenting, better perceived parent-child relationships, and adaptive child outcomes (see Leerkes & Augustine, 2019). Importantly, parental cognitions and emotions are influenced in part by child characteristics; for instance, children higher in temperamental negative emotionality and behavior problems often elicit more negative parental cognitions and emotions as noted in the above-cited reviews.

Additionally, there are important between-culture differences in parenting emotions and cognitions. Parental belief systems about socialization goals and child rearing are culture specific and have implications for parenting and child development (Harkness & Super, 1992). For example, parents from individualistic cultures are more likely to emphasize independence and self-confidence, whereas within collectivistic cultures the role of obedience, proper demeanor, and harmony are highly valued (Harwood et al., 1996). European American mothers believe in individual autonomy and are more likely to use suggestions to structure their toddlers' behaviors, whereas Puerto Rican mothers believe in connected interdependence and are more likely to use commands and physical restraints (Harwood et al., 1999). Thus, to understand parental cognitions and parental behaviors, as well as their impact on child outcomes, researchers need to situate families in the

cultural context in which they are embedded (Bornstein, 2015).

Within-Group Cultural Variation

Within-group cultural factors also predict differences in parenting behavior and related child outcomes. One such factor is acculturation, which refers to the extent to which individuals endorse cultural practices and behaviors of the dominant culture (Grau et al., 2015). Immigrant parents have different beliefs about child rearing and become more acculturated the longer they stay in their cultures of destination (Bornstein, 2015). Acculturation processes influence parenting behaviors and in some cases moderate associations between parenting and child outcomes. For example, within the United States, Latina immigrant mothers who are less acculturated are more likely to engage in controlling behaviors and use spanking as a disciplinary strategy (Barajas-Gonzalez et al., 2018); however, they also show greater levels of maternal warmth (Ispa et al., 2004). Among less acculturated parents, parenting behaviors such as parental control are associated with fewer negative or with positive repercussions for child development, compared to more acculturated parents (Barajas-Gonzalez et al., 2018; Ispa et al., 2004; Wood & Grau, 2018). A key limitation of studies that examine within-group cultural variation is that generational status, country of origin, and language are often used as proxy measurements of acculturation. Given that acculturation processes are dynamic and change over time, future studies should conceptualize acculturation as a multidirectional and multidimensional process. Parents' experiences of structural and institutional inequalities (e.g., discrimination, anti-immigrant legislation, sociopolitical climate around immigration) places them under high stress, which may undermine parenting quality

(Rodriguez-Jenkins, 2014). For example, parental exposure to discrimination is associated with young children's lower well-being directly and via coercive parenting behaviors (Anderson et al., 2015). Due to their dependence on caregivers, barriers to accessing resources, experiences of discrimination, and social disparities are more likely to affect young children as these factors often result in parents' underutilization of community and other public services (Rodriguez-Jenkins, 2014). This work suggests that the cultural and racial context in which parent-child interactions occur, and individual experiences within these contexts, gives different meanings and functions to parenting behaviors.

Childhood Experiences

Scholars and clinicians have long been interested in the intergenerational transmission of parenting, or the idea that how one was parented may be replicated in the next generation via genetic transmission, modeling, or other mechanisms. Indeed, evidence suggests that negative parenting (e.g., harshness, spanking, maltreatment) and positive parenting (e.g., warmth, support) are correlated across generations using both retrospective and prospective designs (Kerr & Capaldi, 2019). These associations are often mediated by the impact of early parenting on factors such as emotion regulation, social competence, and academic achievement, which then enhance or undermine subsequent parenting. For example, mothers' recollections of how their own mothers responded to their emotions in childhood predicted their parenting beliefs, attributions, and emotional reactions to crying, which in turn predicted their emotion socialization behaviors with their own toddlers (Leerkes, Bailes & Augustine, 2020). Finally, adults who have autonomous (i.e., secure) states of mind about childhood attachment

experiences are more sensitive to their infants, and their infants are more likely to have a secure attachment than infants of mothers with an insecure adult attachment (Verhage et al., 2016). Although significant, each of these intergenerational effects are small to moderate in magnitude, and there is great interest in identifying the characteristics and experiences that can break this chain for intervention purposes. Importantly, negative intergenerational effects are diminished among adults for whom supportive romantic relationships and competent co-parents are present and among adults with positive traits such as self-control and coping skills (Kerr & Capaldi, 2019). When experiences of abuse and trauma are passed across generations, this is often described as being the result of “ghosts in the nursery,” a metaphor illustrating that the mechanisms of transmission may be beyond conscious awareness, and as such may be difficult to alter (Lieberman, & Harris, 2010). Parent-child psychotherapy has been demonstrated as one effective approach to alter this legacy by helping parents engage as “angels in the nursery.”

Adult Experiences of Stress and Trauma

In addition to childhood experiences, experiences of stress and trauma in adulthood have the potential to shape parental behavior when caring for young children. Chronic stressors that are often experienced by parents living in poverty, such as food insecurity, housing instability, and living in an unsafe neighborhood, have the potential to consume parental psychosocial resources and undermine sensitive parenting (Blair & Raver, 2016). Traumatic events, such as experiencing intimate partner violence (IPV), may also contribute to parenting behavior. Indeed, a recent meta-analysis demonstrated that IPV victimization was positively associated with parental neglect

and physical aggression, and negatively associated with positive parenting (Chiesa et al., 2018). However, parental stress and trauma do not always undermine parenting, and other individual and contextual risk and protective factors may exacerbate or mitigate these associations. Among families of preschoolers with a history of early adversity, IPV was associated with more punitive discipline, but only when parents reported high levels of household chaos, and punitive discipline was in turn associated with preschoolers' externalizing behavior problems (Coe et al., 2019). Parental stress and trauma also directly contribute to child outcomes (McIntosh et al., 2021) yet there is evidence that the timing of stress exposure is important, with some research demonstrating that prenatal exposure to IPV is more predictive of infant behavior problems and physiological stress reactivity than postpartum exposure to IPV (Levendosky et al., 2016). This work is consistent with prenatal programming models that suggest that the prenatal environment has the potential to shape child stress reactivity, behavior, and health through epigenetic and metabolic mechanisms (Conradt et al., 2018). Recent research demonstrates that responsive parenting buffers preschoolers from the effects of maternal trauma symptoms (Greene et al., 2020), underscoring the complexities of these processes. The extent to which parents cope with stress and trauma, and the impact of these factors on parental psychopathology, is also important.

Parental Psychopathology

Most of the research focused on parental mental health/psychopathology and parenting behavior in early childhood has focused on parental depression, particularly maternal depression. This body of literature suggests that maternal depression is a risk to parenting, yet not all mothers who experience depression

exhibit negative or less sensitive parental behavior. Meta-analytic reports suggest that there is a small yet significant negative association between maternal depression and maternal sensitivity, with the strongest effects observed in studies that compare groups of mothers with and without clinical levels of depression (Bernard et al., 2018). In addition to methodological characteristics, there is evidence that individual child characteristics moderate this effect. For example, maternal depression is negatively associated with maternal sensitivity, but only when infants are highly negative in their interactions with mothers (Newland et al., 2016), and maternal depression is positively associated with maternal unresponsiveness, but only when infant negative emotionality is high (Norcross, Leerkes, & Zhou, 2017). Paternal depression is also highly prevalent and is associated with fathers' parenting behavior and subsequent child outcomes (Sweeny & MacBeth, 2016). More recently, emerging work suggests that maternal anxiety is a risk factor for challenges with bonding, mother–infant interactions, and infant social-emotional development (Field, 2018). Likewise, research on maternal Post Traumatic Stress Disorder suggests it is associated with more negative maternal behavior and infant attachment insecurity; however, these effects are not consistently replicated and much of this research has been conducted using maternal self-reports or observational assessments with small sample sizes (Christie et al., 2019).

Implications for Policy and Practice

The science of parenting in early childhood has contributed to a range of prevention and intervention practices to enhance child social, emotional, and cognitive development. Basic mechanisms research, as well as applied research that has tested the efficacy of

prevention and intervention programs, has also shaped policy. For example, in the United States the federal government financially supports a range of strategies to strengthen parenting and child well-being. Below, we present examples of three approaches to support parents and young children, including screening to identify children and families who may benefit from additional support, home visiting as a strategy to promote parenting and child well-being, and evidence-based relational interventions for children with mental health and behavioral difficulties.

Settings that serve young children have gained an understanding of the role that parental history, mental health, stress, and relationships have in shaping parental behavior, and several of these settings utilize standardized assessments to screen children and families as part of their routine practice. For example, the American Academy of Pediatrics Bright Futures Guidelines recommends routine screening for maternal depression and social determinants of health during well-child visits (Hagan, Shaw, & Duncan, 2017). Screening tools are becoming available to meet these guidelines, such as the Survey of Well-being of Young Children that includes items to assess factors that contribute to parenting including parental depression, interparental relationship functioning, and socio-economic status (Perrin et al., 2016). However, integration of screening into pediatric primary care and other settings is challenging and requires significant resources and buy-in from providers to achieve sustainability.

Families who are identified through screening efforts that may benefit from support are often referred to evidence-based home visiting programs. Maternal, Infant, and Early Childhood Home Visiting (MIECHV) is a federally funded initiative designed to support expectant parents and families with young

children, and includes the promotion of positive parenting, reductions in child maltreatment, and improvements in maternal and child health and well-being (Duffee et al., 2017). The home visiting models supported by MIECHV often include curricula that directly teach parenting skills and strategies to strengthen the parent–child relationship. Parents receive education and guidance on a range of topics ranging from basic infant care to the promotion of learning in the home. Screening assessments are routinely administered to identify parents who may benefit from additional support related to mental health, substance abuse, domestic violence, and other threats to parenting. Furthermore, there is a strong focus on addressing family risks due to poverty. Most families served through MIECHV receive an evidence-based model that has demonstrated effectiveness across several parent, child, parenting, and family domains.

Relationship-based interventions to support young children with mental or behavioral health difficulties are also grounded in theory and research focused on parenting and parent–child relationships. Several intervention approaches have drawn on tenets of attachment theory to utilize the parent–child relationship as a vehicle for improvements in child well-being. For example, Child–Parent Psychotherapy (CPP) is an intervention strategy for children from birth to five years that have experienced trauma or have mental health or attachment difficulties (Lieberman, Ippen, & Dimmler, 2018). CPP has demonstrated effectiveness in improving the parent–child relationship, child attachment security, and child mental health symptoms across racially and ethnically diverse samples (Lieberman et al., 2018). Parent–Child Interaction Therapy (PCIT) is another relationship focused intervention for children aged two to seven years with behavior problems

(McNeil, & Hembree-Kigin, 2010). PCIT includes a trained clinician who observes parent–child interactions and uses a bug in the ear to coach the parent. A recent meta-analysis demonstrated that PCIT is an effective strategy to reduce child externalizing behaviors and parenting stress, and also to increase child compliance (Thomas et al., 2017).

Taken together, there are now several strategies to support parenting, the parent–child relationship, and child well-being that arose from the research described above. Several strategies that support both mothers and fathers have demonstrated positive outcomes for young children and families, yet effect sizes are often small in magnitude (Supplee & Duggan, 2019). Understanding why some families benefit from these approaches more than others continues to be important. Further, emerging evidence suggests that parent, family, provider, and program characteristics contribute to variability in parent engagement in these supports (Bower et al., 2020). However, relatively few studies have examined factors that contribute to parent engagement, results are inconsistently replicated across studies, and therefore these processes are not well understood. Better understanding of challenges, with family engagement, is critical for ensuring that evidence-based policies and practices reach their full potential.

Summary and Conclusion

The extent to which parents shape the outcomes of their children has been of interest to lay persons, philosophers, scientists and clinicians for over a century. As such, the empirical literature in this area is vast. Research in recent decades has contributed to especially important insights related to the role of both mothers and fathers, the processes that explain

associations with child outcomes, how findings vary as a function of child characteristics, and the factors that support or undermine adaptive parenting and child outcomes. Parenting research has become more inclusive in terms of culture, race, and ethnicity; and most of this research suggests that the parenting dimensions considered in this chapter operate more similarly than differently across groups, with discipline being a possible exception, although findings are mixed in this domain. There has also been increased attention to within-group variability that has shed important light on the roles of discrimination and acculturation processes in relation to parenting and effects on children. And, although not the focus of this chapter, longitudinal studies have more effectively teased out child effects on parents, bi-directional effects between parents and children within time, and transactional effects whereby parents and children each contribute to change in the other over time (noted by dashed lines in Figure 9.1).

Despite these advances, important limitations remain. First, observed associations between parenting or parent characteristics and child outcomes can be a function of untested processes such as shared genetics or prenatal programming. Although some studies (e.g., adoption designs) demonstrate clear effects of parenting over and above these factors, this is often not considered. Second, although fathers and other parental figures (e.g., grandparents, fictive kin) have been studied more, most parenting studies in early childhood, our own included, tend to focus on the mother, rather than the network of parents in a child's life. Benefits of a parental network approach would include understanding the relative importance of different parental figures, as well as the totality of a child's socialization experiences across caregivers in relation to key outcomes. Likewise, scholars often focus on a single dimension of parenting,

neglecting crucial questions about the relative importance of different dimensions or the extent to which dimensions operate together to influence child outcomes. Recent advances in person-centered approaches make this an exciting time to consider the impact of patterns of parenting, by a single parent or within a network of parents, on child outcomes (e.g., Carpenter & Mendez, 2013).

Although the field is ripe for further inquiry, sufficient evidence exists to draw three clear conclusions. First, children acquire important relationships, skills and knowledge over the first five years of life that have longstanding implications for their well-being and success, and parents play a critical role in these processes. Second, parents' experiences and characteristics, along with child characteristics, influence the quality and nature of parenting, all of which in turn predict child outcomes. Third, the nature of many of these associations varies as a function of child characteristics, such that more vulnerable children may be especially susceptible to both positive and negative effects of parenting. As such, continued efforts to enhance the quality of parenting during infancy and early childhood are warranted. Several strategies, including home visiting and parent-child relationship-based interventions, have demonstrated positive outcomes for young children, their parents, and the parent-child relationship. Important financial investments have been made at the federal level to increase the reach of these strategies, yet significant unmet need remains, and only a small percentage of families who could benefit from support are able to access services (Meisch & Isaacs, 2019). Strengthening the bridge between research and policy, and further underscoring the need for additional investment in these efficacious approaches, may improve the lives of children and families.

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10 Parenting That Promotes Positive Social, Emotional and Behavioral Development in Middle Childhood

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Marco was excited to start first grade. He loved to read and learn, was enthusiastic about sharing interesting facts with his friends, and was excited when he knew the answers to the teacher's questions. He was hurt and embarrassed when she gave him a yellow card every time he forgot to raise his hand to answer questions or when he got out of his seat to share something new with a friend. While Marco's parents had worked with him at home to play gently with his toys and be well-regulated in the house and with friends, grade school presented a new context where he would need to learn to manage his behaviors. They were determined to support his love of learning and a positive sense of self, although they were unsure how.

Middle childhood corresponds to the elementary school years (ages 6–11 years) and is recognized as a key period in the development of cognitive, social, emotional, and behavioral capacities that establish the foundations of mental and physical health and academic success into adolescence and adulthood (e.g., Landry & Smith, 2010; Marceau et al., 2019). There are increasing demands and expectations on children coinciding with their growing capacities for self-regulation, coping, problem solving, social responsibility, and academic competencies, and it is perhaps the establishment of these capacities that contributes to their self-esteem, confidence, and happiness, or instead, social, emotional and behavioral problems. During this time, children increasingly interact with individuals

and systems outside the family (i.e., school, peers, media), and parents are faced with the complex task of providing adequate guidance and support while also promoting and respecting increased individuation.

During this time, parents increasingly engage with their children's expanding developmental contexts. Their children enter grade school, where they are expected to regulate their behaviors and emotions in the classroom and work independently on their school work. Parents become involved in school settings through their interactions with teachers, parent organizations, and through supporting academic activities at home (e.g., Garcia Coll et al., 2002). Children are also navigating peer relationships both in and outside of school, and both in person and on social media. These relationships become more complicated as children's opportunities to connect grow increasingly independent (e.g., Gifford-Smith & Brownell, 2003). Further, access to television and online media has dramatically expanded, increasing children's exposure to and socialization by images and ideas other than those offered by parents (Odgers & Jensen, 2020). These extra-familial contexts increase the need for parental warmth and acceptance to maintain a trusting and engaged relationship with children, and increased requirements for involvement, limit-setting and monitoring to protect children and promote prosocial engagements. Further, these relationships and interactions occur in the context of broader economic, socio-political and

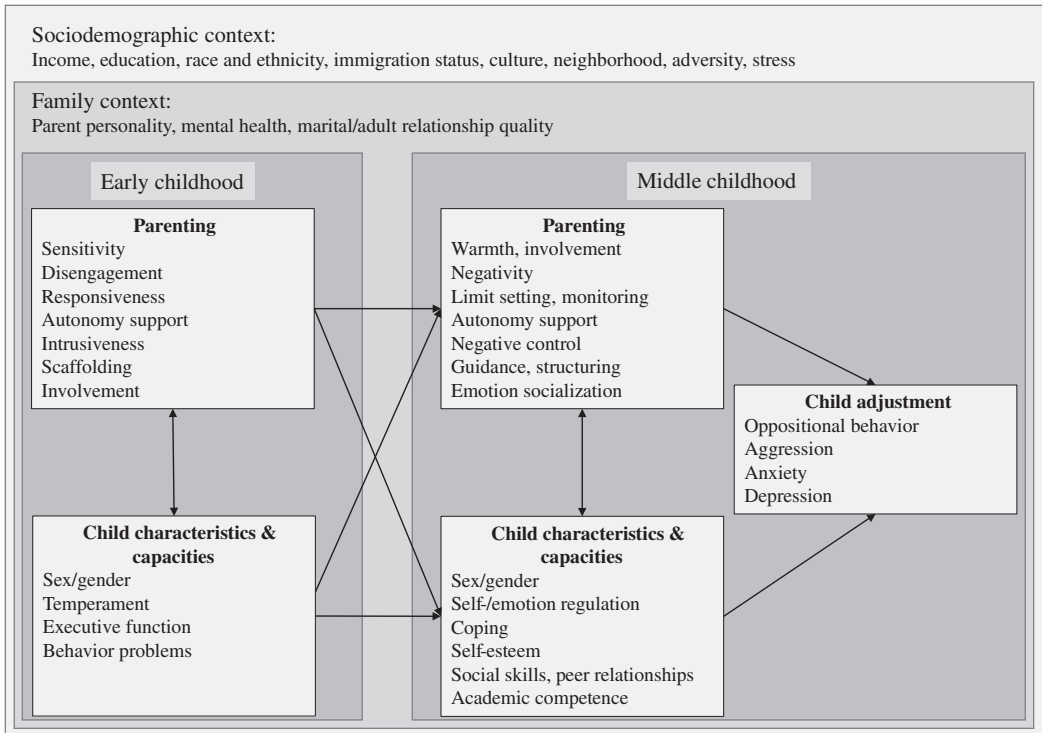


Figure 10.1 Middle-childhood parenting and child adjustment in bioecological and developmental context

cultural contexts that can have profound effects on the risk and protective factors in children’s developmental context.

This chapter outlines how parents promote positive middle-childhood development in connection with wider and intersecting contexts, consistent with a bioecological model of children’s development (see Figure 10.1). Given that middle childhood is theorized to be a period during which children’s personalities and behaviors coalesce into forms that endure into adolescence and adulthood (Collins, 1984), parenting during middle childhood is essential in promoting positive adjustment. We review the emerging or evolving parenting behaviors in middle childhood and their role in effectively supporting middle-childhood development.

Middle-childhood Parenting in Context

Evolving and Emerging Parenting in Middle childhood

Max is an eight-year-old who loves building with Lego. His mother joins him for 15 or 20 minutes a day to build with him. Usually, they just talk about what they are building and engage in imaginary play. Sometimes Max tells his mother about things happening at school, and occasionally he shares a struggle he is having. She listens, validates his feelings and how difficult the situation is, and helps him consider ways to deal with the problem. Max’s mother is building a warm, accepting relationship with Max and supporting emotion regulation and coping skills development,

reflecting common parenting behaviors during middle childhood.

Parenting behaviors most commonly studied during middle childhood include warmth, involvement, control, monitoring, and emotion socialization, most likely reflecting the contextual and developmental demands on parenting. Although middle childhood has unique demands on parenting, early developmental periods form the foundations of parenting and children's behaviors during middle childhood, which are reciprocally related across early and middle childhood (e.g., Bradley & Corwyn, 2007), making it challenging to separate one period from another. Both child behavior and parenting during early childhood have implications for middle-childhood parenting. For example, evidence suggests that early childhood conduct problems increase the likelihood of negative parenting behaviors in middle childhood (e.g., Bradley & Corwyn, 2007). However, this occurs in the context of early parenting behaviors such as lower positive parenting (Akcinar & Shaw, 2018) and corporal punishment (Zulauf et al., 2018) contributing to those conduct problems. Conversely, early childhood self-regulation elicits more warmth, scaffolding and consistent limit-setting (e.g., Klein et al., 2018), and these early caregiving practices predict children's capacity for self-regulation in middle childhood (e.g., Colman et al., 2006). Hence, middle-childhood parenting might reflect both continuity and change from early childhood parenting and child behaviors.

Parental warmth indicates the quality of parents' relationship with their children. It is often defined as including acceptance, positive affect, affection, and responsiveness to children's needs and emotions. It is contrasted with parental rejection, negativity, or hostility. This facet of parenting is commonly examined from infancy through adolescence, and it

remains an important foundation of parent-child relationship across development. In middle childhood, warmth appears to be a continuation of early childhood indicators of parents' sensitivity and responsiveness (e.g., Chang et al., 2017), support for autonomy (Bradley et al., 2021), and parenting self-efficacy and attitudes (e.g., Rominov et al., 2016). Some evidence suggests relative stability in responsiveness from early to middle childhood (Wilson & Durbin, 2013), while there is also evidence that fluctuations in warmth over time may be associated with external factors (Zvara et al., 2018).

Parent involvement, that is social or dyadic time with children, is also examined as an indicator of the quality of the parent-child relationship. It peaks in early and middle childhood (Lam et al., 2012) declining in adolescence. Parental involvement in middle childhood has its roots in parental satisfaction, warmth and involvement in early childhood (Livingston et al., 2018; Racz et al., 2019; Rominov et al., 2016) and also expands to include involvement in academic support (Kohl et al., 2000; Weiss et al., 2005) and monitoring behaviors (Racz et al., 2019) that become relevant as children engage in extra-familial settings more independently.

Parental control refers to parental efforts to shape child behaviors, encouraging prosocial and adaptive behaviors and reducing problem behaviors. These include effective control strategies such as consistent rule- or limit-setting, scaffolding, monitoring and contingency management, as well as ineffective strategies that include hostile, intrusive and psychological control, and physical punishment. As options for behavior increase in middle childhood, parents' need to set appropriate boundaries also increases. Externalizing behaviors in early childhood longitudinally predicted less effective limit-setting (Wang et al., 2011; Yates et al., 2010).

Parental scaffolding behaviors, balancing guidance, and structuring with support for autonomy, are often studied in early childhood as supporting the development of self-regulation. Scaffolding can be broadly considered as strategies to transition regulation from parent to child (Leith et al., 2018). This is generally comprised of accommodation to meet children's capabilities, encouraging autonomy, child-led decision making, and facilitating successful interaction while limiting intrusiveness. In middle childhood, scaffolding continues to be necessary to aid in the development of problem solving and emotion management. Middle-childhood respect for autonomy has its roots in early childhood autonomy support and shows an increase, particularly from fathers, in middle childhood (Bradley et al., 2020).

Parental monitoring reflects parenting behaviors regarding attention to, knowledge, and tracking of child's location, peer contacts, and activities (Dishion & McMahon, 1998). In early childhood, parental monitoring is limited to the home, early learning environments, and parent-facilitated peer interactions. In middle childhood, however, children's behavior and potential environments are less directly controlled by parents and increased monitoring often occurs. For instance, children may begin or increase participation in extracurricular activities, spend more time at school, and broaden their social networks during this period.

Parent warmth, involvement, and feelings of parental satisfaction and efficacy both concurrently and prospectively predict parental monitoring (Lac et al., 2009; Racz et al., 2019). Low self-regulation (Petit et al., 2007) and conduct problems in both early childhood and middle childhood prospectively predict increased monitoring in middle childhood (Patrick et al., 2005; Racz et al., 2019), suggesting that children elicit different levels of

monitoring, and that higher levels of externalizing may make monitoring more difficult (e.g., via deceit, lack of disclosure) (Kerr & Stattin, 2000).

Monitoring and knowledge of media exposure is an area that becomes particularly relevant at this age as children grow independent in their use of the internet and social media (Coyne et al., 2017). Given the increased accessibility and use of technology and media by children in middle childhood (Bleakley et al., 2014; Kim & Davis, 2017), the call for parental monitoring of youth interaction with technology and media has increased (American Academy of Pediatrics, 2018). Parental media monitoring can include restrictive monitoring, such as limiting content and amount of media, and active monitoring, such as physical observation, online monitoring, or discourse with children, peers, or other parents (Holmgren et al., 2019; Page Jeffery, 2020). Co-use, jointly engaging in media seeking to connect with children, is another monitoring avenue and is associated with increased engagement with educational content and family connection (Lee & Chae, 2007; Padilla-Walker et al., 2018).

Emotion socialization encompasses parent behaviors related to children's experience, expression, and modulation of emotion. These behaviors typically include parental responses to children's emotions, discussions with children about emotions, and to some extent parents' personal emotion regulation (Eisenberg et al., 1998). While it is also key in early childhood, emotion socialization in middle childhood continues to be a necessary factor in helping children understand and appropriately respond to emotion in their increasingly complex social and learning environments. Emotion socialization may be conceptualized as supportive behaviors, such as reflecting or encouraging expression, and nonsupportive behaviors marked by, for

example, minimization or punitive approaches (Eisenberg et al., 1999). Parents' beliefs about emotions (Lozada et al., 2016), emotional competence goals (Chan et al., 2009), and emotional awareness (Gottman et al., 1996) have been associated with emotion socialization.

In summary, evidence suggests that early childhood parenting carries forward into middle childhood through transitioning parenting behaviors, predicting new parenting behaviors that emerge in middle childhood, and through children's early childhood self-regulation and adjustment problems, which, in turn, elicit or shape parenting behaviors. In addition, parenting occurs with the broader social, economic, and cultural contexts of the family.

The Bioecological Context of Parenting

Parenting operates in transaction with child and parent characteristics, as well as with broader social, economic, and cultural contexts. A Bioecological Model (Bronfenbrenner & Morris, 2006) suggests that parenting behaviors occur within a web of systems, where they influence and are influenced by intrapersonal, interpersonal, and contextual factors.

Parenting mediates and interacts with parent self-regulation, personality, and mental health. Parent mental health, particularly maternal depression, is associated with less effective parenting behaviors (e.g., Lovejoy et al., 2000). Various aspects of parenting, such as warmth, consistency, and appropriate limit-setting have been demonstrated as mediators of the association between maternal depression and child emotional and behavioral problems in childhood (e.g., Riley et al., 2009, Sellers et al., 2014). Parent hostility in middle childhood has also been shown to mediate the relation between parents' postnatal emotional distress and children's emotional and

behavioral difficulties in middle childhood (Giallo et al., 2014). Looking beyond mood disorders, parent ADHD symptoms related to higher levels of home chaos, which mediated the effects of ADHD on inconsistent discipline and nonsupportive responses to children's negative emotions (Mokrova et al., 2010). Additionally, parent substance use disorders were related to less parental support, which in turn predicted more internalizing and externalizing symptoms (Bountress & Chassin, 2015).

Additional evidence points to parent personality relating to parenting (Prinz et al., 2009). For example, extraversion, agreeableness, conscientiousness, and openness and lower neuroticism are related to more warmth and behavioral control, whereas higher agreeableness and lower neuroticism are related to more autonomy support. Also, parent self-regulation is related to effective parenting (e.g., Sanders et al., 2019). Parental executive function, that is, cognitive processes that include attention regulation, inhibitory control, and flexibility, is particularly relevant for engaged and responsive parenting (Crandall et al., 2015), and may underlie consistent contingency and environment management, as well as the ability to provide non-harsh discipline (Deater-Deckard et al., 2012; Sanders & Turner, 2018) and to engage in scaffolding (Mazursky-Horowitz et al., 2018).

Parenting also is shaped and conditioned by child factors. Child internalizing and externalizing behaviors prospectively predict less effective parenting and are reciprocally related to them (Zvara et al., 2018), while child prosocial behavior predicts increased warmth (Putnick et al., 2018), highlighting that bidirectional associations between parenting and child behaviors must be considered. In middle childhood, inconsistent discipline has been related to increases in negative emotionality

in children, while child irritability has also predicted increases in inconsistent discipline by parents (Lengua & Kovacs, 2005). Lower inhibitory control predicted increased maternal hostility and was also predicted by maternal internalizing symptoms (Leve et al., 2019). There is even evidence of evocative effects of early childhood characteristics on middle-childhood parenting. Infants classified as low reactive had mothers who were more likely to employ reasoning as a response to verbal conflict when children were in middle childhood (Hardway et al., 2013). Conversely, preschool “hard to manage” temperament predicted more power-assertive parenting in middle childhood, particularly when families were low in resources (younger, lower education, low income; Kim & Kochanska, 2020).

Further, parenting mediates and interacts with stress and adversity in its impact on children. Economic hardship (Conger et al., 2002), neighborhood disadvantage (Pachter et al., 2006), marital stress (e.g., Elam et al., 2017), and other familial risk factors have been associated with less effective parenting. These family risk factors often co-occur and may have cumulative and cascading effects on children’s development (Evans et al., 2013). For example, Jackson and colleagues (2019) found that maternal depression and stress in early childhood mediated the relation between economic hardship and harsh parenting in middle childhood in a sample of African American boys. On the other hand, positive parenting in the face of such risk factors buffers the effects of risk. For example, greater positive engagement by fathers in early childhood mitigated the positive association between family poverty and child internalizing and externalizing problems in middle childhood and disrupted continuity in children’s internalizing problems from early to middle childhood in families living below the poverty level (Lee & Schoppe-Sullivan, 2017).

Racial, Ethnic, and Cultural Aspects of Parenting in Middle Childhood

Parenting occurs embedded in the context of cultural and ethnic traditions, beliefs and practices which vary both across and within ethnic groups. While there has been an increasing call to move beyond descriptive studies of parenting to studies of mediators and moderators of parent behaviors and child outcomes, the literature examining more sophisticated mechanisms of parenting in families of diverse ethnic and cultural backgrounds is lacking. Here, we raise some concerns with this area of research before discussing evidence for race, ethnic, or cultural difference in parenting in middle childhood.

Despite within-group diversity in SES, families from racially or ethnically diverse backgrounds disproportionately experience poverty, which may lead to difficulty disentangling race or ethnicity differences from diminished opportunities for family investment (Conger & Donnellan, 2007) or increased family stress (Conger et al., 2002) without explicit study design considerations. Consideration of income and the stress of poverty among the many contexts of parenting is critical.

Next, broadly evaluating majority–minority differences or characterizing a parenting “group” obscures within-group heterogeneity. For example, the majority of U.S. immigrants are people of color. Consequently, a portion of parents of color are simultaneously navigating acculturation and biculturalism. First-generation immigrant parenting often reflects values of family obligation, self-reliance, productivity, and ethnic identity. Thus, generational status and acculturation are important considerations to understand parenting behaviors in families of color. In addition, the task of ethnic-racial socialization, or families’ communication of specific messages and practices that are relevant to and provide information

concerning the nature and status of race and ethnicity, may be acutely central to children of color's identity formation in middle childhood. In middle childhood, children both increase their social awareness and begin to consolidate their personal identities. It may also be the first time children recognize their experiences of exclusion, invisibility, and discrimination (García-Coll & Szalacha, 2004).

In addition, it is critical to recognize that minority groups often include people of diverse countries, ethnicities, cultures, languages, and paths to residing within the United States; and the topics studied (e.g., the absence of African American fathers, parenting and academic performance in Asian families, the patriarchal nature of Latinx families) are themselves questions rooted in a Western perspective. Indeed, residential and non-residential African American fathers may be highly involved in parenting (Yeung et al., 2001), and a better understanding about parenting and the adjustment of Asian American children might emerge if outcomes such as social-emotional adjustment were studied along with academic ones (Parke 2004).

Three recent studies contribute to a more nuanced understanding of middle-childhood families of color. In a large sample of African American families ($N = 867$), Murry (2019) tested pathways from family context, parental psychological functioning, parenting, and the mother child-relationship to self-pride (self-esteem and racial identity) in middle childhood. Murry found support for a family stress model in which higher stress predicted parental psychological functioning (anxiety and depression). In turn, parental psychological functioning predicted parenting (involvement, adaptive racial socialization, monitoring, and consistent discipline) and the quality of the mother-child relationship. Relationship quality predicted children's self-pride as well as their positive worldview. The study not only affirms

pathways of healthy functioning within African American families, but gives specific focus to adaptive, strength-based processes (e.g. racial socialization, positive parenting) that confer resilient outcomes among African American youth in middle childhood.

Jabaghourian et al. (2014) examined parenting styles and practices of Latinx ($N = 73$: 75 percent Mexican immigrant, 18 percent Hispanic origin born in the United States) mothers and fathers in relation to their middle-childhood outcomes. Parental authoritativeness (both maternal and paternal) was positively associated with teacher reports of children's grades, academic engagement, social competence, self-regulation, perspective taking and negatively associated with children's aggression. Of note, parenting styles were reported by the middle school-age children, who generally endorsed authoritative and not authoritarian practices for both their mothers and their fathers. In fact, the authors note the authoritative pattern to be consistent with the Latinx cultural values of harmonious interpersonal relationships (*respeto*), training for responsibility (*educación*), and dedication to academic success.

Yang et al. (2020) examined emotion socialization among Chinese-immigrant and European American mothers in relation to coping and adjustment in middle childhood. Mothers across backgrounds reported the same level of supportive responses to children's negative emotions. Supportive responses to negative emotions were linked with better psychological adjustment and adaptive coping. Rates of non-supportive responses to negative emotions varied across racial groups, with Chinese-immigrant mothers endorsing more non-supportive responses. However, while these non-supportive responses were linked with behavior problems and maladaptive coping in European American children, there was no evidence of a deleterious effect among the

Chinese-immigrant children. (In fact, there was a trend for positive effects on social-emotional adjustment.) The authors speculated that the practice of punishing or minimizing negative emotions might be instructive in social display rules that allow the children to better fit into their culture. They conclude by highlighting that the construct of “non-supportive reactions” to children’s emotions may itself be culturally bound.

Taken together, these three studies highlight some important considerations for future research among diverse families. In particular, there is tremendous potential for studies of the pathways of parenting to elucidate the practices that parents can engage in to promote the social-emotional milestones of middle childhood (e.g., a positive sense of self, including one’s racial identity). Further, negative stereotypes may be false or they may be falsely interpreted through a Euro-centric lens, construct, and research questions.

The Role of Parenting in the Developmental Tasks of Middle Childhood

Maria got her first smartphone and started engaging with peers on social media. She sent some messages to friends before going to bed but didn’t get a response. When she awoke in the morning, she checked her phone and noticed posted photos of a group of her friends who had a sleepover the night before. Maria’s father noticed that she was sad and quiet in the morning, and she shared what happened. Although her parents were supportive and validated how hurtful it was to be excluded, they struggled to know what advice to offer, how to manage Maria’s exposure to social media, and to support positive friendships in a social media context.

Parents not only support the development of middle-childhood skills and capacities, but

also represent relationships of critical value and importance to children in middle childhood – at this point, still more so that peers or teachers. Parental acceptance and consistency are key to children developing a positive sense of self, of their being loveable, of their environments as being predictable and controllable, and of a positive worldview and optimistic approach to life.

Self-Regulation and Emotion Regulation

Children in this age enter and advance through early grade school with growing expectations of self-regulation of their behavior and emotions in the classroom and at home, increasing expectations of engaging in focused, independent work, and utilizing effective instrumental and social problem solving. These capacities are undergirded by core cognitive, behavioral, and emotion regulation skills.

Effortful control is one key aspect of self-regulation and is conceptualized as the executive-based core of self-regulation that includes executive attention, inhibitory control, and delay ability (Rothbart & Bates, 2006), facilitating the regulation of attention, emotions, and behavior to match the demands of a situation (Rothbart et al., 2000). There is considerable overlap with executive function (Zhou et al., 2011), which includes higher-order functions such as planning, decision making, and problem solving. While the most marked developmental increase in effortful control or executive function occurs during the preschool years, it continues to grow at a moderate rate during middle childhood (e.g., Lengua, 2006; Poon, 2018). Effortful control predicts both positive and problematic adjustment into adolescence, including academic and social competence, and internalizing and externalizing problems (e.g., Lengua et al., 2015; Olson et al., 2017; Poon, 2018).

Parental warmth or support and consistent limit-setting predict higher levels and increases in effortful control (e.g., Ato et al., 2014; Hughes & Ensor, 2009; Karreman et al., 2008; Lengua, 2006), whereas negative parenting, and perhaps inconsistent discipline in particular, predict lower effortful control (Karreman, et al., 2008; Valcan et al., 2018). These associations are not thought to be dependent of child gender or ethnicity (Colman et al., 2006). Research on the role of fathers' parenting on preschoolers' effortful control suggests a contribution above the effects of maternal parenting (Karreman et al., 2008).

Parenting in middle childhood also predicts the development of other aspects of self-regulation, including children's increasingly independent emotion regulation. Evidence suggests that parent sensitivity and emotion socialization in early childhood predicts emotion regulation in middle childhood (e.g., Godleski et al., 2020). Parental negativity and inconsistent discipline predicted greater emotional reactivity (Lengua & Kovacs, 2005; Vertsberger et al., 2019), whereas positive parenting buffered youth negative reactivity during negative peer experiences (Oppenheimer et al., 2016). Parent behaviors aimed at the socialization of emotional understanding and regulation in particular have been shown to relate to children's emotional reactivity and regulation. Supportive reactions to negative emotions are related to children's emotion recognition skills (Castro et al., 2015) and better emotion regulation, and, in turn, lower adjustment problems (Godleski et al., 2020)

Coping and Problem Solving

Children in middle childhood are increasingly able to understand multiple perspectives of a situation, which can facilitate problem solving and coping. Children's coping efforts are conscious, volitional efforts to regulate emotion,

cognition, behavior, physiology, and the environment in response to stress (Compas et al., 2001). Coping can be differentiated between active or approach and avoidant strategies. Active coping approaches are problem-focused and involve thinking about solving the problem or thinking more positively about the problem. Avoidant coping strategies involve trying to stay away from the problem or repressing thoughts about it. Active coping strategies are related to lower levels of psychological symptoms and higher self-esteem (Eisenberg, et al., 2000; Sandler et al., 1994), and avoidant strategies tend to be related to higher levels of internalizing and externalizing problems (Causey & Dubow, 1992; Eisenberg et al., 2000). However, there is some research suggesting these relations may be dependent on stressors or contexts (Valentiner et al., 1994; Gonzales et al., 2002).

In middle childhood, the primary sources of socialization of coping responses are thought to be parents (Compas et al., 1992). In the theoretical model of parent and family influences on children's coping, Kliever et al. (1996) articulated and found empirical support that parents directly coach coping strategies, model coping strategies, and, as a function of family structure or environment, are more or less physically and emotionally available to receive and respond to children's bids for support-seeking in ways that predict children's coping in middle childhood.

Self-Esteem and Self-Concept

Self-esteem reflects a positive estimation of oneself or a positive self-concept. This can be both a general self-worth or a positive self-concept in specific areas of functioning such as school and physical self-concept (Harter, 1987). Self-esteem is thought to result from social interactions with personally important people in which the individual feels supported and reinforced for his or her characteristics,

interests, and behaviors (Gecas & Schwalbe, 1986; Harter, 1987), highlighting the potential importance of parents in children's developing self-esteem (Lam et al., 2012). Self-esteem is related to peer relationships, life satisfaction, and lower anxiety and depression (Guhn et al., 2013). However, despite the potentially important role that self-worth plays in children's mental health and academic achievement, there is relatively little research on the role of parenting in its development, at least during middle childhood.

Studies show that children with higher levels of psychological well-being or self-esteem have closer and more involved relationships with their parents (e.g., Ackard et al., 2006; Kerr et al., 2009). Both warmth and effective control aspects of parenting, sometimes combined into a measure of authoritative or constructive parenting, have been shown to relate to children's self-esteem (e.g., Murray, 2012) and self-pride (McBride, 2019). Conversely, negative conditional regard, a measure of parental rejection, was related to higher shame (Smiley et al., 2019). In addition, children who spent more time with fathers had higher general self-worth (Lam et al., 2012). Self-esteem appears to be differentially affected by mothers' and fathers' parenting. For example, child self-esteem was positively related to maternal warmth but negatively related to father over-protectiveness (Jia et al., 2016). This might also be specific to the gender-based power structures in different cultures, as one study showed that for Indian children living in West London, paternal negativity was related to lower self-esteem, whereas for English children, maternal negativity was related to lower self-esteem (Atzaba-Poria & Pike, 2015).

Social-Emotional Competence

Social-emotional competence in middle childhood is characterized by prosocial behaviors,

social initiation, and self- and social-awareness or self-monitoring. Deficits in social-emotional competence are marked by difficulty with perspective taking, challenging negotiating conflict constructively, social rejection, and loneliness (Rydell et al., 1997; Thomson 2018). This highlights the close ties of social-emotional competence to children's self-regulation, where difficulty in monitoring, modulating, and inhibiting may impede competence. Indeed, self-regulation underlies social-emotional competence in middle childhood (e.g., Penela et al., 2015).

There is robust evidence of the effects of parenting on children's social competence. Parental warmth, support, responsiveness, and sensitivity predict higher social competence (e.g., Criss et al., 2009; Newton et al., 2014; Ruiz-Ortiz et al., 2017), as does autonomy support (Joussemet et al., 2005). Involvement or parent-child time is associated with better competence (McDowell et al., 2003), particularly when fathers have increasing dyadic time with their children (Lam et al., 2012). These associations are also captured in measures of parenting style or overall positive parenting. Authoritative parenting is associated with greater social competence (e.g., Jabaghourian et al., 2014). In a study of families in eight countries, maternal positive parenting was associated with prosocial behaviors in all countries (Pastorelli et al., 2016). The converse is also evident. Inconsistent, coercive, or negative parenting practices predict lower social competence (e.g., Acar-Bayraktar et al., 2019; Slagt et al., 2016).

Parental emotion coaching or coping support might have particular relevance to children's social competence. For example, parents' supportive responses to children's emotions fully mediated the relation between positive parenting and children's prosocial behaviors (Acar-Bayraktar et al., 2019). Also, parents own emotion knowledge or awareness

was associated positively with children's social competence (MacCormack et al., 2019). Parental coping suggestions that support engagement coping were associated with fewer social problems (Abaied & Stanger, 2017). It is possible that a combination of the quality of the parent-child relationship and coping support promotes social competence. In one study, observed parental advice-giving style (warm versus controlling) during a challenging conversation predicted social competence one year later above and beyond content of the parents' advice. Fathers' advice giving predicted social competence above mothers' advice giving (McDowell et al., 2003).

Social Relationships and Peer Acceptance/Rejection

Peer relationships, being accepted, and having supportive, trusting relationships are increasingly important to children's mental health in middle childhood. Parenting plays a key role in children's peer acceptance or rejection, and the effects of parenting are, in part, accounted for by its impact on children's social competence and self-regulation (e.g., Ato et al., 2014; Oppenheimer et al., 2016). For example, parental hostility (Leve et al., 2019) and poor discipline practices (Dishion, 1990) were related to higher peer problems or rejection. In addition, parents' coping suggestions that included more disengagement coping strategies predicted lower friendship quality (Abaied & Stanger, 2017). One study showed that facilitative parenting, which included warmth, low over-directedness, low parent-child conflict, encouragement of socializing with peers, coaching social skills, and parent-school communication, discriminated between children who were bullied or not, where bullied children had poorer peer relationships and engaged in more reactive aggression (Healy et al., 2015).

Academic Competence

Middle childhood marks the start of grade school education and the acquisition of numerous academic achievements, including reading and arithmetic. Positive parenting, including acceptance, warmth, and supportive parenting, relates to higher scores on measures of receptive vocabulary, reading, and math, (Murray, 2012; Wang & Fletcher, 2016). These associations have been replicated in large, national longitudinal surveys of youth that control for child and maternal demographics (Han et al., 2004). Parental support for autonomy is shown to relate to better academic achievement, and, interestingly, supports both social and academic competence, whereas one study showed that an emphasis on school performance promoted academic achievement but was related to lower social adjustment (Joussemet et al., 2005). Paternal parenting practices in early childhood also predict academic outcomes (reading and math), even when accounting for maternal parenting practices, and child and family demographic variables. These findings hold true for resident as well as non-resident fathers and across African American as well as Hispanic families (Coley et al., 2011). The effects of positive parenting on academic achievement may operate in part through parents' promotion of concrete academic skills or children's school engagement (Dishion, 1990; Jabaghourian et al., 2014). They may also operate more indirectly through promoting children's self-concept (McBride, 2019; Murray, 2012) and self-regulation (Jabaghourian et al., 2014; Kim & Kim, 2019), which are both predicted by parenting and known to promote academic achievement. Conversely, negative parenting practices that engender academic shame (Smiley et al., 2019) or damage self-concept may serve as a pathway to academic difficulties in middle childhood. In these ways,

parents are important contributors to children's competence, even during this time of increased academic aptitude.

Parental involvement and monitoring are often examined in relation to academic achievement. Involvement in academic activities or school includes several dimensions such as contact and the quality of the relationships between parents and teachers, parent involvement in school activities, and parent involvement in educational activities at home (Kohl et al., 2000). One study found that authoritative parenting had an indirect effect on academic achievement through more parental monitoring of school involvement (Murray, 2012).

Externalizing and Internalizing Problems

Perhaps the most studied outcomes in middle childhood, particularly in relation to parenting, are children's externalizing and internalizing problems. Externalizing behaviors in middle childhood primarily include oppositional behaviors and aggression. There is extensive evidence showing that children's development of externalizing problems is related to lower warmth, acceptance, and sensitivity (Godleski et al., 2020; Han et al., 2004; Livingston et al., 2018; Loeber et al., 2009; Marceau et al., 2019; Mullineaux et al., 2009; Olson et al., 2017; Slagt et al., 2016), lower involvement or monitoring (Livingston et al., 2018; Racz et al., 2019), higher hostility or negativity (Han et al., 2004; Jackson et al., 2019; Mullineaux et al., 2009; Ruiz-Ortiz et al., 2017; Slagt et al., 2016), inconsistent discipline (Cheung et al., 2018; Dishion, 1990; Lengua & Kovacs, 2005; Ruiz-Ortiz et al., 2017), and power-assertive, harsh or physical discipline (Kim & Kochanska, 2020; Loeber et al., 2009; Olson et al., 2017). The fact that most parenting behaviors examined

are related to externalizing problems suggests that children's externalizing behaviors represent a very generalized response to experiences during middle childhood, perhaps because responding with anger, oppositional behaviors, and aggression reflects children's limited, but growing, self-regulation, emotion awareness, and communication skills.

While the preponderance of research examines maternal behaviors in predicting changes in children's aggressive behaviors across middle childhood, research also shows that fathers' parenting predicts children's adjustment in middle childhood (e.g. Cheung et al., 2018; Rominov et al., 2016). There is evidence of both comparable effects of maternal and paternal parenting behaviors on children's externalizing symptoms, as well as evidence of differential effects of maternal and paternal parenting behaviors on these behaviors in middle childhood (Elam, et al., 2017; Lee & Schoppe-Sullivan, 2017; Ruiz-Braza et al., 2017). The effects of paternal involvement are evident even when children were not living with their fathers (Gold et al., 2020). Further, with increasing salience of peer influences, including reciprocal deviancy training, that is, when peers provide opportunities for and reinforce antisocial behavior, (Forgatch et al., 2016), parents' interaction with their children's peers (e.g. supervision of their children's activities and companions) may decrease the likelihood of children's involvement with antisocial peers (Pettit et al., 1999). Research supports that parents who are more involved with their children's peer contacts have children who exhibit less externalizing in middle childhood (Pettit et al., 2007).

Internalizing problems include anxiety, depressive, and somatic symptoms. It is particularly important to understand the development and presence of anxiety symptoms in middle childhood, given that depressive disorders tend to developmentally follow or

co-occur with anxious symptoms. Cole and colleagues (1997) propose a competency-based model of child depression in middle childhood: Whereas healthy development in middle childhood involves the construction of a positive sense of self, negative appraisals of competence by parents, peers, and teachers may be internalized by children and serve as a mechanism for promoting depression. The authors found that the most significant source of self-perceived competence in third grade girls was from parents (followed by teachers and peers; Cole et al., 1997), whereas these contributors were reversed among sixth grade girls. These findings highlight not only parental contributions to depressive symptoms, but also the multiple domains of child development and the shifting relevant influence of these domains as children navigate middle childhood (Wang et al., 2018).

Although less commonly studied, consistent evidence links internalizing, including anxiety and depressive symptoms, with parenting behaviors, including lower acceptance or warmth (Han et al., 2004; Kuhlman et al., 2014; Marceau et al., 2019), negative conditional regard (Smiley et al., 2020), lower positive parental control or higher negative control (Pougnnet et al., 2011; Smiley et al., 2020), lower involvement, particularly by fathers (Gold et al., 2020; Lee & Schoppe-Sullivan, 2017), and parenting styles that combine affective and control dimensions, such as lower authoritative or constructive parenting (Kerr et al., 2009). Conversely, autonomy support is shown to relate to a sense of well-being indicated by positive affect (van der Kaap-Deeder et al., 2017). These parenting behaviors might impact internalizing through their effects on children's HPA-axis reactivity (Kuhlman et al., 2014), emotion regulation (Schweizer et al., 2018); social competence (Marceau et al., 2019); and shame (Smiley et al., 2020).

Summary and Conclusions

Given that middle-childhood self-concept, social and academic competencies, peer relations, and emotional-behavioral competencies are critical foundations for later well-being, the role that parenting plays in the development of these areas during middle childhood has potential lifelong implications. Parental warmth and acceptance, along with effective control or behavior management strategies, are relevant predictors of almost every aspect of children's adjustment in middle childhood. In addition, parenting predicts children's competencies and adjustment even when economic, family, and contextual risk factors are taken into account.

It is notable that there is little specificity in the relations of parenting behaviors with different indicators of children's competencies and adjustment, but rather that there is robust evidence of parenting impacting most of the areas of child adjustment. This might reflect the broad impact of parenting at this age – that is, parental influences in middle childhood are extensive. It might also reflect the interdependence of aspects of children's competencies and adjustment. Children's self-regulation and self-concept might set the stage for behavioral control, social-emotional and academic competencies, and positive peer relationships, which in turn support lower internalizing and externalizing problems. This would suggest a model in which earlier behavioral, social, and emotional competencies form the building blocks of later adjustment outcomes, and are shaped starting in early childhood. Evidence of early childhood parenting predicting child self-regulation and behavior problems which in turn predict middle childhood parenting and adjustment suggest such a building block model that could be tested more explicitly across early and middle childhood. The evidence reviewed here offers a picture of the

developmental progression of both parenting and children's adjustment from early childhood through middle childhood.

There is some evidence that parenting behaviors may differ depending on the sex of the child. For example, monitoring and emotional support are greater for girls than boys (Patrick et al., 2005; Root & Rasmussen, 2017). Overall, though, there is little or inconsistent evidence of child gender differences in parenting or child behavior and emotional problems during middle childhood reported across studies. Although this might reflect minimal attention to gender differences, it might also indicate that parent behaviors are more responsive to children's emotional and behavioral evocative effects than they are to gender in particular, as might be suggested by fairly substantial evidence that children's temperament emotionality and self-regulation elicit different parental behaviors. This is an important direction for future research to clarify developmental models of parenting across boys and girls, to better understand the potentially different effects of mother and father parenting, and to more adequately inform clinical practice supporting effective parenting.

Children's worlds expand during middle childhood, necessitating that parenting evolves to meet the demands of these expanded contexts and support their children's increasing capacity to navigate their world independently. While there is evidence of continuity in dimensions of parenting that continue to be relevant, such as warmth, there is also evidence of parents' need to provide developmentally appropriate supports for their particular child at that particular stage. As broader ecological spheres become more influential or have more direct influences on children's development, positive and supportive parenting may buffer children from the potential effects of stress and adversity that arise from risk factors in children's contexts. However, those same risk

factors can adversely impact parenting, compounding their effects and increasing children's likelihood of developing adjustment problems. This highlights the critical importance of supporting parents' mental health and effective parenting when families experience stress and adversity. Parenting is malleable, and when positive parenting is maintained in the face of risk, it has protective effects on children's adjustment.

Several parenting programs have ample evidence of their effectiveness in promoting positive parenting, reducing child behavior problems, and improving parent well-being, including in economically and racially diverse samples. Examples include Triple P - Positive Parenting Program (Sanders, 2008), Helping the Noncompliant Child (McMahon & Forehand, 2005), and the Strengthening Families Program (Kumpfer et al., 1996). A valuable next step in examining the effects of parenting programs would be to understand their impact on child capacities that might mediate the effects on reduced behavior problems, including self-regulation, social-emotional competence, coping skills, and self-esteem.

There are several important directions for research on parenting in middle childhood. First, to adequately examine how parenting changes across different developmental periods more specific conceptualizations of the dimensions of parenting, along with consistent terms and operationalization, are needed. Studies often use different terms for similar parenting behaviors, or conversely the same terms operationalized differently, making comparisons across studies and developmental periods difficult. There is also a need for better understanding of parenting at the intersection of ethnicity/race and contextual experiences of income, stress, and adversity, with studies designed to examine these intersections more directly. Further, it is critical to interrogate our research

questions and constructs to de-center white, European American perspectives and values and reduce the likelihood of false and negative interpretations.

Middle childhood, a period of growth in core social, emotional, and behavioral capacities, is an organizing period of development with a potent opportunity for supporting positive growth or reversing early developmental challenges or behavioral difficulties. By adolescence, aspects of personality and psychopathology become more solidified and are rooted in the foundations of social-emotional competence and well-being established in middle childhood. In light of the overwhelming evidence of the effects of parenting on well-being in middle childhood, it is clear that supporting parents and families by reducing stress and adversity, promoting parent mental health and well-being, and enhancing effective parenting is critical for laying the foundation for children's mental health and well-being into adulthood.

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11 Parenting Adolescents

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Parenting Adolescents

If parents searched for parenting books during the teen years, they would be faced mostly with hyperbolic warnings about delinquent or depressed adolescents, mean girls, and sexually uncontrollable boys – preparing parents for what will seemingly be the scariest decade in their child's life. The stereotype of teenage angst and strained parent–child relationships is alive and well, despite the fact that it is largely not supported by research. Indeed, though relationship quality fluctuates somewhat across adolescence, the majority of adolescents aged 11–18 enjoy quality relationships with their parents (Ebbert et al., 2018). In the current chapter we will discuss the unique aspects of adolescent development that lead to transition and restructuring of the parent–child relationship in response to developmental needs. We will then provide an overview of parenting research during adolescence, including global aspects of parenting (e.g., parenting styles, warmth, control) and nuances of parenting that are particularly salient during the teen years (e.g., parental monitoring of media and peers, parental socialization of race, parent–child sex communication). We will then highlight the ways in which parenting adolescents varies as a function of demographics. Finally, we will explore policy application of the research on parenting adolescents and suggest a number of avenues for future research.

Transitions and Changes during Adolescence

The adolescent years are a time of transition for both children and parents, which contributes to the need for adjustments in the parent–child relationship that are sometimes met with reluctance or conflict. Adolescents experience physical, emotional, and social changes that influence how they interact with others both directly and indirectly, and also shape their expectations for autonomy within the parent–child relationship. The onset of puberty includes a variety of physical and hormonal changes that can lead to emotional challenges and adjustments in social relationships (Kerig, 2019), including the parent–child relationship. Parent–child interactions may also be impacted by continued brain development during the adolescent years, as decision making, executive functioning, and impulse control are not yet fully developed (pre-frontal cortex) (Steinberg, 2010). These changes, along with social shifts toward one's peer group (Larson & Richards, 1998), may result in increases in exploration and risk taking behaviors (Chen & Jacobson, 2012) that lead to the need for adjustment in the parent–adolescent relationship.

According to Self-Determination Theory (SDT) (Deci & Ryan, 2000), adolescents also experience an increased desire for autonomy, which is more than just a movement toward independence, but is also adolescents' need to

function volitionally while simultaneously feeling connected to parents, peers, and others. During adolescence, volitional functioning is characterized by teens feeling that their values and behaviors are self-motivated and authentic, and they are not merely doing what parents or other authority figures desire of them. So as the need for both independence and autonomy increase during adolescence, and because optimal development and internalization of values depends upon volitional functioning and motivation, parents ideally become less directly influential in determining the behavior of their teens and instead influence children indirectly through the adolescent's own self-chosen values, beliefs, and behaviors (Padilla-Walker, 2014; Soenens et al., 2019). Though these changes can lead to some minor growing pains, for the most part adolescents continue to rely on their parents despite a desire for increasing autonomy, and most relationships remain strong and stable across adolescence (Jagers et al., 2017).

Parenting during Adolescence

Exploring different aspects of the parent–child relationship provides further insight into specific ways in which parents support or hinder their child's development during this time. It should be noted that adolescents are active participants in the socialization process, despite the majority of research considering unidirectional relations from parenting to child outcomes. Thus, whereas we will be primarily discussing the influence of parents on children, it should be noted that this process is clearly dynamic and bidirectional.

Parenting Styles, Parental Warmth, and Parental Control

Historically, researchers have explored parenting styles and dimensions as they relate

to adolescent outcomes. Parenting styles represent the overarching emotional climate of the parent–child relationship, and are made up of parenting dimensions including warmth/support, and control/demandingness (Darling & Steinberg, 1993). It has been generally established that authoritative parenting, or parenting that is high on both warmth/support and control/demandingness is the most adaptive at promoting a variety of adolescent outcomes. More specifically, authoritative parenting has been associated positively with prosocial behavior, academic efficacy (Carlo et al., 2018), and intrinsic academic motivation (Tang et al., 2018); and negatively with externalizing problems (Pinquart, 2017). Research has also consistently found that neglectful parenting styles (low on both warmth and control) are associated with the most negative adolescent outcomes, including behavioral problems (Di Maggio & Zappulla, 2014), low academic competence (Johnsen et al., 2018), and low prosocial behavior (Carlo et al., 2018). That being said, there is less consistency in the effectiveness of authoritarian parenting, or parenting high on control and low on warmth, probably because the ways in which adolescents interpret parental warmth may vary significantly by ethnicity and culture (Jackson-Newsom et al., 2008).

In terms of parenting dimensions, research has found that maternal warmth is relatively stable across adolescence, while paternal warmth declines slightly (Jagers et al., 2017). Parental warmth has been associated with a variety of adaptive adolescent outcomes, including lower levels of depression (del Barrio et al., 2016), loneliness (Chen et al., 2000), and aggression; and higher levels of self-worth (Chen et al., 2000), social and academic competence, and filial behavior toward parents (Cheah et al., 2012). Clearly parental warmth is an important protective factor for adolescents across a variety of cultures, and

importantly can also act as a buffer against harsh parenting (Wang, 2019). While this is not surprising or innovative, this body of research supports the notion that one of the most important things parents can do is to maintain a positive relationship with their adolescent child.

The maintenance of this positive relationship is in part achieved by parent–child affection and warmth, but it is also clearly undermined by parental control during a developmental time period that is characterized by a desire for increased autonomy. Parental behavioral control is characterized by parents placing developmentally reasonable limits on a child's behavior (e.g., curfew, household rules), and is seen as an adaptive part of authoritative parenting in younger children and teens, that is associated with positive outcomes. That being said, even behavioral control can begin to feel overly controlling during adolescence and the transition to adulthood as demands for autonomy increase (Padilla-Walker & Nelson, 2019), especially if parents' lack of effective behavioral control as adolescents are less often in the parental home leads to psychological control. Levels of parental psychological control (subtly manipulating the child's emotions and feelings to match those of the parent) (Barber, 1996) are generally low, but slightly increase over time for the majority of adolescents (Rogers et al., 2020). Maternal psychological control is consistently associated with higher levels of adolescent depression, anxiety, disordered eating (Eun et al., 2018), and aggression (He et al., 2019) and lower levels of self-esteem (Boudreault-Bouchard et al., 2013) and prosocial behavior toward friends and strangers (Fu & Zhang, 2019). Perhaps the reason that parental psychological control is so consistently associated with negative outcomes is because it impacts adolescents' self-regulation (Rogers et al., 2019) and ability to cope emotionally (Inguglia, 2020), though

more research on potential mechanisms is clearly needed (Laird & Frazer, 2019). Taken together, there is little doubt that parental control, especially psychological control, during adolescence is maladaptive in most cases. That being said, parental monitoring, which may also be seen as a form of parental control, if done in a way that maximizes the adolescent's autonomy and promotes child disclosure, can be relatively more effective in the context of a high-quality parent–child relationship.

Parental Monitoring and Child Information Management during Adolescence

Parental monitoring is defined as “a set of correlated parenting behaviors involving attention to and tracking of the child's whereabouts, activities, and adaptations” (e.g., parental solicitation, behavioral control) (Dishion & McMahon, 1998, p. 61; see Table 11.1 for a breakdown of autonomy supportive and restrictive monitoring strategies across contexts). Adolescents play an active role in parental monitoring by employing a wide range of information management strategies, or strategies children use to manage the amount of information provided to parents. Various forms of adolescent information management strategies fall under two broad categories of disclosure (e.g., voluntarily sharing information with parents) and concealment strategies (e.g., secret keeping), which are distinct yet negatively related constructs (Keijsers & Laird, 2010). Developmental trajectories of parental monitoring and information management generally show slight normative declines in parental solicitation and behavioral control (Lionetti et al., 2019), parental knowledge (Keijsers et al., 2016), and adolescent disclosure (Padilla-Walker, Son, et al., 2018), and an increase in secrecy (Lionetti et al., 2019) across adolescence, though these trends differ

Table 11.1. *Autonomy-supportive and restrictive parental monitoring*

Autonomy-supportive			Autonomy-restrictive		
General monitoring	Media monitoring	Peer management	General monitoring	Media monitoring	Peer management
Parenting that promotes voluntary child disclosure, including parental warmth, trust, and open communication	Active monitoring: actively discussing media use and the content of media with children in an effort to foster critical thinking and conscious consumption of media	Consulting: proactive effort at helping children problem-solve regarding a friendship/peer problem	Parental solicitation: May vary from parents asking one another, asking the child, or asking friends or acquaintances for information	Restrictive monitoring: restricting or regulating children's access to and use of media	Guiding: regulation of children's peer interactions
Behavioral control: jointly agreed upon restriction of children's behavior in order to keep track of whereabouts and activities	Co-use/co-play: engaging in media use with children in an attempt to strengthen the relationship		Psychological control: using love withdrawal or subtly manipulating children's emotions and feelings to match those of the parent		Prohibiting: prohibition children from associating with a certain peer or peers

somewhat for boys and for girls (Keijsers & Poulin, 2013).

Correlates of parental monitoring and information management. Adolescent information management has consistently been found to be a stronger predictor of increased parental knowledge (Keijsers et al., 2016) and lower levels of adolescent delinquency (e.g., Kapetanovic et al., 2019) and substance use (e.g., McCann et al., 2016) compared to parental monitoring strategies. The associations between parental monitoring strategies and parental knowledge and youth outcomes have produced more inconsistent findings, with parental solicitation being moderately related to greater parental knowledge (Keijsers et al., 2016) but not to lower delinquency (Kapetanovic et al., 2019), and parental behavioral control showing nonsignificant (Willoughby & Hamza, 2011), negative (Kerr et al. 2010), and positive (Liu et al., 2020) relations with parental knowledge and delinquency (Keijsers et al., 2010). Parental monitoring behaviors seem to have more indirect effects on parental knowledge and adolescent outcomes through promoting greater disclosure (Willoughby & Hamza, 2011).

Given the transactional nature of parent–adolescent communication, parental monitoring and adolescent information management should be understood within the contexts of the parent–adolescent relationship and family environment. Quality parent–adolescent relationships characterized by warmth (Gondoli et al., 2008) and support (Keijsers et al., 2016) facilitate open communication between parents and children. Parent–adolescent relationship quality also moderates the associations between communication efforts and adolescent outcomes (Laird & Marrero, 2010) such that parental control, parental solicitation, and adolescent disclosure are associated with less externalizing problems only when adolescents perceived high levels of parental

warmth and support (Micalizzi et al., 2019). These findings thus point to the need to adopt a positive parent–adolescent relationship and family climate in order for parental monitoring strategies to be effective in fostering adolescents’ volitional disclosure, preventing adolescent delinquency, and promoting positive adaptations in adolescents. These principles also apply to parental monitoring in specific contexts, including media and peers.

Parental Media Monitoring

Parental media monitoring (i.e., parental mediation) refers to specific parenting and parent–child interactions about children’s media use. The most commonly studied parental strategies of media monitoring are active monitoring and restrictive monitoring (Padilla-Walker et al., 2018). Parental active monitoring involves actively discussing media use and the content of media with children in an effort to foster critical thinking and conscious consumption of media (Fikkers et al., 2017). Active monitoring has been found to be protective against problematic media use (Hefner et al., 2019), aggression, substance use (Radanielina Hita et al., 2018), and sexual outcomes (Collier et al., 2016) concurrently, and against aggression and externalizing behaviors longitudinally (Padilla-Walker et al., 2016). This autonomy-supportive approach to monitoring media is especially important during adolescence when parental control is more likely to lead to secrecy regarding media use instead of open disclosure (Padilla-Walker, Stockdale, et al., 2020).

Indeed, restrictive monitoring involves restricting or regulating children’s access to and use of media. On one hand, parental restrictive monitoring has been negatively related to aggression (Khurana et al., 2019), depression (Fardouly et al., 2018), delinquency (Martins et al., 2017), online risky behaviors

(Cabello-Hutt et al., 2018), and sexual behaviors (Collier et al., 2016). On the other hand, restrictive monitoring has also been associated with greater problematic online behaviors and media use across diverse samples (e.g., Benrazavi et al., 2015; Cabello-Hutt et al., 2018), and greater levels of aggression and externalizing behavior (Padilla-Walker et al., 2016). One explanation may be that restrictive monitoring is effective in the prevention of negative outcomes through less media access (Khurana et al., 2019) and use (Fardouly et al., 2018), thus providing direct intervention and control, but is also not effective in promoting positive outcomes or consistently predicting adjustment because it does not foster critical thinking (Harvey & Manusov, 2020), self-regulation, and sympathy (Padilla-Walker et al., 2016).

These inconsistencies in the findings on parental monitoring strategies and adolescent outcomes may be due to the lack of taking the parent–child relationship into account. More recent studies have differentiated autonomy-supportive, controlling, and inconsistent parenting styles in conjunction with specific monitoring strategies, and have found autonomy-supportive active and restrictive monitoring strategies to be associated with greater prosocial behavior (Holmgren et al., 2019), less aggression (Meeus et al., 2018), and less anxiety and depression (Padilla-Walker et al., 2019), while controlling and inconsistent styles of monitoring were associated with greater physical and relational aggression (Holmgren et al., 2019) and greater online victimization and aggression (Katz et al., 2019) through greater parent–adolescent conflict (Martins et al., 2019) (see Table 11.1). While autonomy-restrictive forms of monitoring may be necessary in extreme circumstances (e.g., if the child is involved in dangerous online relationships), this body of research again points to the importance of the

overarching climate of the parent–child relationship, and future research should continue to explore the manner in which parents monitor adolescent media use (autonomy-supportive or restrictive), as well as how parents utilize a combination of various strategies to effectively monitor their children’s use of media (Padilla-Walker, Coyne, et al., 2018).

Parental Peer Management

Another specific way that parents monitor adolescent behavior is through management of peer relationships. Although peer relationships *do* influence adolescents, scholars argue that peers do not replace the influence of parents (Brown & Bakken, 2011), especially when parents participate in peer management behaviors that impact adolescent relationships (Mounts, 2007). Parental peer management refers to direct, specific parenting behaviors that focus on particular goals, such as improving peer relationships (Mounts, 2011), encouraging prosocial behavior in the adolescent (Gerardy et al., 2015), and interrupting delinquent peer influence (Tilton-Weaver et al., 2013). Specific peer management techniques include: consulting, or a parent’s proactive effort at helping their adolescent problem-solve regarding a friendship (Mounts, 2004); guiding, or parental regulation of peer interactions (Mounts, 2011); and prohibiting, wherein parents prohibit their adolescent from associating with a certain peer or peers (e.g., peers who are engaged in delinquent behaviors; Mounts, 2001). Parental consulting is the strategy that most clearly maximizes adolescent autonomy, while prohibiting is the most controlling (see Table 11.1).

Different outcomes are associated with the various parental peer management behaviors. Adolescents who report higher levels of parental consulting also report higher levels of prosocial behavior (Gerady et al., 2015),

and lower levels of delinquency and drug activity (Mounts, 2004). Implementing the guiding technique, however, has produced mixed results, as regulating peer interactions is associated with higher friendship quality (Mounts, 2004), but also with higher relational aggression (Gerardy et al., 2015). This could be explained by evidence suggesting that the outcomes of peer management techniques can vary across different parenting styles (Mounts, 2002), as well as by the autonomy support offered by the parent (Mounts, 2011). For example, adolescent drug use decreased when authoritative or authoritarian parents used the guiding technique, but drug use increased when uninvolved parents used the same practice (Mounts, 2002).

Similarly, prohibiting an adolescent from interacting with certain peers has shown mixed results. Whereas moderate levels of parental prohibiting were linked with decreased levels of drug use and higher academic achievement in one study (Mounts, 2001), another study showed that higher levels of prohibiting were linked to increased levels of drug use (Mounts, 2002). Again, this may be explained by parenting style, as autonomy-supportive prohibition has been shown to be less maladaptive than controlling prohibition (Soenens et al., 2009). Although navigating an adolescent's time spent with friends versus family may always be a "balancing act" for parents, practicing peer management techniques can positively impact the developing adolescent and their peer relationships, especially in the context of a high-quality parent-child relationship.

Parental Socialization of Race

Another area in which parents proactively socialize their adolescent children is in regards to race and culture. For racial minority groups experiencing racism, parental influence and racial socialization efforts can be especially

protective for children's socioemotional (Becares et al., 2015), psychological (Bynyn et al., 2007), and academic (Varner et al., 2018) outcomes. Racial socialization entails parenting methods and messages aimed at teaching children to not only value their race (Berkel et al., 2009), but also to navigate the structural disadvantage that children of color face (Hagerman, 2017). Research indicates that effective racial socialization practices include cultural socialization, where parents focus on communicating the importance of their cultural values and traditions (Liu & Lau, 2013), as well as sharing positive messages about their race and culture (Neblett et al., 2008). Other techniques involve including strategies for environment and image management (Dow, 2016), and utilizing frameworks such as Critical Race Parenting, which supports critically minded parents in early, frequent, and open communication about race and racism (Montoya & Sarcedo, 2018).

Regarding the racial majority, parents can also have a significant influence on the way their children think about race and racism. Racially progressive White families may participate in a type of racial socialization in order to support antiracist values (Matlock & DiAngelo, 2015), but it is a fundamentally different process from the socialization in which minority families participate, as visibly White families teach their children how to navigate structural *advantage*, rather than *disadvantage* (Hagerman, 2017). One prominent but unhelpful ideology in many White families (Matlock & DiAngelo, 2015) is a "color-blind" approach to race and racism (Bonilla-Silva, 2015). Dominating the narrative since the Civil Rights movement, this approach purports that racism is in the past (Hagerman, 2017), yet the color-blind approach has only resulted in the subtle reinforcement of racist societal systems (Adjei et al., 2018) and has

failed to support socialization processes in families. Antiracist White parents need to remain cognizant that their own privilege may unintentionally reinforce the racial status quo if they shield their own children from explicit racism and the realities that children of color face (Hagerman, 2017), rather than using their privilege to enact change through political action (Matlock & DiAngelo, 2015).

Parent–Child Sex Communication

An important milestone in development is the onset of puberty, involving physical and sexual maturation, as well as the beginnings of adolescent sexual desire (Moore et al., 2015). The sexual socialization of adolescents can impact the likelihood of adolescent sexual risk behaviors (Leland & Barth, 1993), influencing the child's sexual beliefs and decisions (Jaccard et al., 2002), attitudes (Cox et al., 2015), behaviors (Overbeek et al., 2018), and values (Flores & Barroso, 2017). A key component of sexual socialization is parent–child sex communication (Jerman & Constantine, 2010), characterized as a reciprocal, enduring process between parents and children (Flores & Barroso, 2017). Parents who communicate with their children about sexuality can provide a protective influence against sexual risk behaviors (Moore et al., 2015), perhaps because accurately informed youth can make better decisions about their sexual behavior (Liu et al., 2017).

Despite evidence linking parent–child sex communication to positive adolescent sexual health outcomes, such as delayed sexual initiation (Aspy et al., 2007), and dependable STI/HIV prevention (Teitelman et al., 2008), other evidence suggests the opposite, with increased parent–child sexual communication linked to more adolescent sexual risk behaviors (Clawson & Reese-Weber, 2003). These mixed results may be explained by the quality of the

conversation (Deptula et al., 2010). For example, adolescents were more likely to have had sexual intercourse if parents lectured them against sexuality (Rogers et al., 2015), whereas open and warm communication was linked with less risky adolescent sexual behaviors (Malacane & Beckmeyer, 2016). This effective parent–child sex communication is facilitated by the parent–child relationship (Wilson et al., 2010a), as a supportive parent–child relationship is correlated with more conservative sexual attitudes (Cox et al., 2015), as well as higher amounts of sex communication (Flores & Barroso, 2017).

Although most parents assume heterosexuality when engaging in parent–child sex communication (Flores & Barroso, 2017), all children can benefit from non-heterosexist language when discussing sexual development, as it can create a safe space for an undisclosed sexual-minority child to continue to participate in open communication (Majied, 2013). Lesbian, gay, bisexual, transgender or questioning (LGBTQ) children are at higher risk for early abuse (before age 18; Friedman et al., 2008), and negative sexual health outcomes (Flores & Barroso, 2017), with parental rejection after “coming out” increasing LGBTQ rates of homeless and mental health disorders (Rhoades et al., 2018). Parents who utilize non-heterosexist language (Majied, 2013) and do the work required to fully accept their potential LGBTQ youth can provide the supportive environment necessary for the emotion well-being of their children (Tyler & Abetz, 2020).

Inconsistencies in Parenting as a Function of Child and Parent Demographics

The majority of research on parenting has been conducted using Western, white, middle-class mothers, raising questions regarding

whether parenting is consistent for boys and girls, mothers and fathers, and across different racial/cultural groups and families from different socioeconomic statuses. In terms of differences as a function of the child's biological sex, our careful review of the literature suggested there were not consistent mean differences in positive parenting (e.g., warmth) for boys and girls (e.g., Jagers et al., 2017), and very few associations of any kind between parenting and child outcomes were consistently moderated by child gender (e.g., He et al., 2019; Liu et al., 2019). That being said, there were consistent mean differences in parental control and communication as a function of the child's sex. For example, harsh and psychologically controlling parenting (Chen et al., 2000; Di Giunta et al., 2020; Rogers et al., 2020) and restrictive media monitoring (Top, 2016; Martins et al., 2019) were generally higher for boys compared to girls. In turn, behavioral control and parental monitoring/solicitation (Hawk et al., 2016; Willoughby & Hamza, 2011) were higher for girls compared to boys. These differences may be due to (or a result of) steeper decreases in parental knowledge (Laird et al., 2003) and disclosure to parents (Masche, 2010), and increased levels of secrecy (Keijsers et al., 2010) in boys compared to girls. Parents are also consistently more likely to communicate with their daughters compared to their sons, especially about sexuality (Boyas et al., 2012), and parents are more likely to talk about health consequences and sexual norms with girls (Deutsch & Crockett, 2016; Kapungu et al., 2010), while permissive messages are more often communicated to boys (Murphy-Erby et al., 2011). In short, there appear to be few consistent differences in parenting as a function of the child's sex, but girls are more likely to be the recipients of appropriate parental control and frequent parental communication than are boys, which highlights the need for continued intervention

encouraging parents to engage more often and in less controlling ways with their sons.

In terms of differences as a function of parents' biological sex, few studies directly compare mothering and fathering – most present separate models for mother and father (Carlo et al., 2018, Chen et al., 2000; Di Giunta et al., 2020; Tang et al., 2018), use mother reports only (Keijsers et al., 2016; Padilla-Walker et al., 2018), or combine mother and father reports (Johnsen et al., 2018; Martins et al., 2019). For those studies that did directly compare the effect of mothers and fathers, there were few consistent identifiable differences, especially when considering parent sex as a moderator of parenting on child behavioral outcomes. There were, however, a few consistent mean differences in the frequency or quality of parenting as a function of parent sex. Namely, parents differ in how they gain parental knowledge and monitor their adolescent child, and in communication about sexuality. More specifically, when monitoring, mothers use more direct approaches (e.g., asking directly), whereas fathers are more likely to use indirect approaches such as observing or asking their spouse for information (Rote & Smetana, 2018). Additionally, mothers are consistently more likely to discuss sexuality with their children than are fathers (Sneed et al., 2013), again suggesting that while there are few consistent differences in parenting between mothers and fathers, those that do exist suggest that fathers are relying on mothers for information and sexual education of children.

As with child and parent gender, research exploring differences in parenting as a function of race, ethnicity, culture, or SES mostly focus on one group (Chen et al., 2000; Fu & Zhang, 2019; Jagers et al., 2017), with few comparing across multiple groups. That being said, research that systematically compared findings as a function of race/ethnicity/SES did find

that parental control was more frequent in ethnic minority families (e.g., Black, Hispanic) compared to White families, and in low-income compared to high-income families (e.g., Eun et al., 2018; Mounts, 2004). However, it is notable that although overall control is higher for minority and low-income families, this is not the case when it comes to monitoring of the media, where Hispanic (Top, 2016), Black (Khurana et al., 2019), and low-income (Cabello-Hutt et al., 2018) families had lower levels of restrictive monitoring than Asian American, White, and high-income families. Research also consistently finds that all parents struggle to talk with their children about sexuality, but that Chinese (Liu et al., 2017) and Hispanic (Wilson et al., 2010b) parents report feeling the most uncomfortable with these conversations because of cultural taboos on sexuality before marriage, while Black parents are more likely to make parent–child conversations about sexuality a priority (Regnerus, 2005)

Research has also found a few meaningful and consistent moderations as a function of race, culture, and SES in the relations between parenting and child outcomes. Whereas one study found that harsh parenting was associated with internalizing and externalizing problems similarly across nine cultures (Di Giunta et al., 2020), a meta-analysis of parental monitoring found that the link between behavioral control and parental knowledge was stronger among Eastern cultures compared to Western cultures (Liu et al., 2020), perhaps because parental control is perceived differently by adolescents from different cultures. Research has also found that Black adolescents conceptualize parental warmth differently than do White adolescents (Jackson-Newsom et al., 2008), suggesting that what might be seen as controlling to a White teen is seen as loving to a Black teen. For example, whereas European American adolescents consistently display

negative outcomes in response to authoritarian parenting, that is not always the case for adolescents from ethnic minority groups (Morris et al., 2013) and in countries outside the United States (e.g., Italy) (di Maggio & Zappulla, 2014). Taken together, it is clear that more research is needed on how parenting might differ as a function of race, ethnicity, culture, and SES. Relying on parenting research that is primarily conducted using Western, white, middle-class, maternal samples misses considerable and essential nuance. Remedying this should be a continued and intentional direction for future research.

Policy Implications of Parenting Adolescents

Research on parenting has a variety of important implications for intervention and prevention programs for adolescents that include parental involvement. Indeed, the most effective interventions often involve parents and family (Karam et al., 2017; Martínez-Muñoz et al., 2018), so the applications of the above research are numerous. While acknowledging this broad application, we opted to focus on three specific policy implications of basic parenting research. Namely, the role of parents in reducing risky driving, the role of parents in the juvenile justice system, and parent–child sex education.

Parents and Adolescents' Risky Driving

Automobile accidents are one of the leading causes of death among adolescents (National Center for Injury Prevention and Control, 2019) and adolescents have four times higher crash rates than those over the age of 20 (Insurance Institute for Highway Safety, 2020). A variety of U.S. states have instituted graduated driver's licenses, where teenagers are required to practice driving under supervision

before they obtain their license, and this has been associated with a reduction in risk (Williams & Shults, 2010), but generally requires significant parental involvement. Research suggests that parents who are normally involved and engage in appropriate levels of parental monitoring are also involved and monitor their child's driving, suggesting that established patterns of parenting influence parental reactions to this new developmental task (Laird, 2014). Parental involvement is significantly associated with lower levels of risky driving among teens (Simons-Morton & Quimet, 2006), and parents who have strict rules about driving with peers and night driving have adolescents who report lower levels of risky driving (Laird, 2011), especially if parental restrictions occur in the context of a supportive parent-child relationship (Zeringue & Laird, 2018). Despite the positive impact parents can have on adolescents' safe driving, many parents lack the knowledge required to promote safe driving (Mirman & Kay, 2012), are not adequately involved in monitoring and control of their child's driving behavior (Hartos et al., 2002), or do not openly establish and define driving rules (Hartos et al., 2004).

Clearly, continued research and policy on how to best promote parental involvement in driver's education are important and intervention studies provide some direction on the most appropriate next steps. For example, interventions focused on family feedback and parent training (Taubman-Ben-Ari et al., 2017) and that provide concrete follow-up tools (e.g., reports, logging tools) and foster direct and sustained parental engagement (Curry et al., 2015) are the most successful at reducing risky driving over time. Research also has found that parents are more likely to grant driving autonomy to older adolescents, even if they are just beginning the driving process. Given that in the last decade many U.S. states have increased the age of full

licensure to age 18, this may mean that parents will offset safety gains by allowing adolescent autonomy prematurely (Laird, 2014), which has implications suggesting that intervention efforts might be most effective for younger adolescents.

Parents and Adolescents in the Juvenile Justice System

Adolescents, and disproportionately adolescents of color, are arrested for an alarmingly high number of crimes proportional to the percentage of teens that make up the U.S. population (Pinderhughes et al., 2011). Furthermore, how adolescents navigate the juvenile justice system is significantly impacted by a number of aspects of parenting (Tolou-Shams et al., 2018), which are strained by parents' mental health and reduced family functioning. It is clear that not all teens have the benefit of optimal parenting and those who do not are more likely to commit crimes (Sattler & Thomas, 2016) and are less likely to navigate the juvenile justice system favorably (Williams & Steinberg, 2011).

Parental participation in the probation process is critical for cessation from crime, but mothers who know the least about the legal system (Cavanaugh & Cauffman, 2017) and who have the most negative attitudes about legitimacy of the justice system (Cavanaugh & Cauffman, 2015) are the least involved in their child's legal process, which is associated with a higher likelihood of reoffending. Because the juvenile justice system is primarily focused on adjudication rather than providing supportive mental health services, youth who commit low-level offenses likely receive limited effective help and without the needed parental and community support, often end up reoffending. Research highlights the need for increased assistance and education for parents, especially those who live in resource-deprived

communities and who lack adequate social support (Richardson et al., 2014).

Structural inequalities in societies contribute to economic, social, and cultural disadvantages that significantly impact parenting and subsequently adolescent violence and delinquency (Sattler & Thomas, 2016). Policy efforts should include prevention aimed at increasing positive parenting among high-risk samples prior to youth offense, but also intervention for those youth who are already in the system, as research has found that positive parenting (e.g., warmth, monitoring, low hostility) benefits juvenile offenders (Williams & Steinberg, 2011). Comprehensive interventions should target parenting behaviors, as well as neighborhood social organization and peer delinquency (Chung & Steinberg, 2006), and should seek to foster and increase high-quality communication between parents and children to increase parental knowledge and supervision (Flanagan et al., 2019), which are associated with lower recidivism rates.

Parent–Child Sex Education

Notwithstanding both parent (Lagus et al., 2011) and adolescent (Somers & Surmann, 2004) preference that parents should be a child's primary source of sexual information, many parents are uncomfortable about these conversations and reluctant to communicate (Malacane & Beckmeyer, 2016). Parents often cite embarrassment (Wilson et al., 2010a), lack of knowledge (Crichton et al., 2012), discomfort (Elliott, 2010), and developmental concerns (Jerman & Constantine, 2010) as barriers to communication. The resulting lack of sexual health knowledge leads to sexually unprepared youth who are therefore less able to make informed choices about sexual behaviors (Liu et al., 2017), or consider the consequences of these behaviors (de Paul Kanwetuu et al., 2018).

It may be effective for policy makers to focus on supporting parents in understanding the importance of high-quality parent–child sex communication (Flores & Barroso, 2017), especially considering that most parents desire to be their child's primary sex educator (Lagus et al., 2011). Policy makers could consider a parent-based approach (Jaccard et al., 2002), which utilizes evidence-based interventions aimed at reducing parental barriers to communication, such as lack of sexual health knowledge (Crichton et al., 2012), and lack of communication skills (Malacane & Beckmeyer, 2016). Successful interventions include a focus on strengthening the parent–child relationship (Wilson et al., 2010a), sexuality education aimed at parents (Sutton et al., 2014), improved parental communication tools (Sevilla et al., 2016), as well as opportunities to practice communicating (Kamala et al., 2017) in order to target parents' attitudes and confidence (Newby et al., 2011). Interventions could take place using methods such as media campaigns and written materials (Malacane & Beckmeyer, 2016), healthcare setting interventions (e.g., nurses talking to parents and adolescents together Teitelman et al., 2008), or in-person workshops (Klein et al., 2005). Providing comprehensive sex education (Eisenberg et al., 2008) alongside parent interventions aimed at improving parent–child sex communication (Jaccard et al., 2002) would allow parents to transmit their family values about sexuality, while ensuring that adolescents receive accurate sexual health information. This dual approach could prove to be synergistic for policy makers and advocates who hope to promote adolescent sexual health (Prendergast et al., 2019).

Future Directions and Conclusions

The current chapter has provided an overview of research on parenting adolescents, and it is

clear that this field is rich with research suggesting that the importance of the parent–child relationship continues, despite transitions both adolescents and parents experience during this time period. One focus of the current volume is on policy applications of research, and while space did not allow for every policy application, it is clear that basic parenting research has important implications for practical issues involving parents and adolescent children. Future research should continue to explore ways in which basic parenting research can inform parents, educators, and policy makers regarding changes that may be especially beneficial to teenagers. For example, additional areas of research could include the role of parents in promoting adolescents’ political and civic engagement and policy surrounding the age at which teens can vote (Hart & Atkins, 2011), the role that parents play in policies surrounding adolescent work (Steinberg & Dornbusch, 1991), and the role of parents in the promotion of adolescent sleep (Meijer et al., 2016), especially as it relates to school start times.

Taken together, there is plenty of room for continued research exploring the role of parenting on adolescent development. Research that continues to improve measurements of the nuances of parenting is increasing our understanding of the multidimensionality of parenting during this developmental time period. The most consistent finding in our review of research – cutting across studies of parental control, behavioral monitoring, media monitoring, peer management, parent–child sex communication, and policy research – was the importance of the parent–adolescent relationship in enhancing the effects of positive parenting and buffering the effects of negative parenting. A review of research did not reveal consistent differences in parenting as a function of gender or ethnicity/culture, but the field is clearly in need of additional research

exploring the parent–child relationship in diverse populations. This becomes especially essential when considering policy applications of parenting, as low-income, single parents are typically at the greatest disadvantage because of additional strain that may contribute to less than optimal parenting. This not only has direct implications for child outcomes but also cascading implications in areas such as a teenager’s interaction with the juvenile justice system, involvement in driver’s education, and parent–child sex education. It was clear that in all of these areas, and this was only a sampling, adolescents whose parents are not optimally involved are at a clear disadvantage, necessitating prevention and interventions that directly target at-risk teens and parents. In sum, contrary to stereotypes in popular media and television, research suggests that parents continue to be a meaningful and impactful influence in the lives of their adolescent children in ways that highlight a myriad of practical implications for those involved in improving the lives of young people around the world.

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12 Parenting during Emerging Adulthood

Larry J. Nelson

As the average age of marriage has risen (approximately age 30 for women and 32 for men, across developed countries; OECD, 2019) the number of jobs available to those without higher education has decreased, and the cost of housing has increased, many young people are single, living at home, and financially dependent on parents well into their twenties (Srinivas, 2019; Vespa, 2017). As a result, there is a need for many parents to remain engaged in the parenting process longer than in the past. Indeed, today, parents are instrumental to the well-being of young people as they make their way through the third decade of life (Nelson & Padilla-Walker, 2013). Hence, the purpose of this chapter is to identify aspects of the parent–child relationship that are tied to flourishing and floundering, respectively, during emerging adulthood (ages 18–29) including the ways in which culture, parenting, and developmental issues (e.g., strivings for autonomy) may intersect to influence (mal)adjustment. Furthermore, the chapter will provide a lens (i.e., transfer of resources) to help parents think about the ways they may assist their emerging-adult children develop capacities needed for adulthood.

Theoretical and Developmental Foundation for Examining Parenting in Emerging Adulthood

Separation-individuation theory posits that adolescence is marked by the psychological

shift away from dependence on parents towards a sense of self that is separate and distinct from parents (Blos, 1979). However, research suggests (a) that emerging adulthood, rather than adolescence, is the time during which the process of individuation reaches fruition (Tanner, 2006), and (b) that separation from parents does not mean an end to the parent–child relationship (see Soenens et al., 2019). Indeed, it is during emerging adulthood that “the shift in power, agency, responsibility, and dependence between emerging adults and their social contexts –primarily experienced by emerging adults as a period during which parent regulation is replaced by self-regulation” (Tanner, 2006, p. 27). Through this lens we see that emerging adulthood is when the process of separation and individuation reaches its peak and includes not a complete separation from parents but, instead, a renegotiation of the parent–child relationship in which emerging adults come to see themselves as separate from but still connected to their parents (Tanner, 2006). Thus, the parent–child relationship continues to play a very important role in the process of healthy development in the third decade of life.

There are a number of developmental and contextual factors that, together, may account for why the process of separation and individuation now takes longer than in the past and, therefore, requires parental involvement further into the twenties. First, we know that brain development is not finished by the end of adolescence. Changes in the brain (i.e.,

synaptic pruning, myelination of the prefrontal cortex, and changes in the limbic system) begin in early adolescence but do not end with a fully developed brain until well into the twenties (Steinberg, 2008; Taber-Thomas & Perez-Edgar, 2016). Until the maturation of the prefrontal cortex is complete, young people often still struggle with the abilities to calculate risk, delay gratification, control emotions, and think rationally and beyond the moment, because they tend to be easily distracted, often overgeneralize, and powerfully experience emotion, passion, and pleasure (e.g., Steinberg, 2008; Taber-Thomas & Perez-Edgar, 2016).

Now, there is no evidence that the age at which brain development is complete has changed over time but the context within which that maturation takes place has changed for many (albeit not all) young people, and it is because of that change that parents may be more important than ever before in the years immediately following adolescence. Specifically, in many Western, industrialized nations, young people do not take on adult roles (e.g., marriage, parenthood, careers) as early as they once did (significant exceptions will be discussed) resulting in less structure in the lives of young people.

Throughout childhood and adolescence, individuals experience a fair amount of structure as parents, teachers, and other adults (e.g., coaches, employers) establish boundaries for when things should occur (e.g., bedtime), rules for what behavior is expected (e.g., in the classroom), and discipline for when those expectations are not met. There is little, if any, time throughout the day that children and adolescents are not within the protective care afforded them by the structure of their lives. This structure is meant to protect children who may not yet have developed the cognitive, linguistic, social, or emotional maturity (i.e., competence) to exercise

complete autonomy. As those skills mature though, children and adolescents are given more autonomy commensurate with that maturity to make decisions for themselves.

In the past, structure continued to play a protective role for the not-yet-fully-developed brain as young people frequently married, started careers, or entered the military (i.e., there was some accountability for where one should be, what one should be doing, etc.). Today, a defining feature of emerging adulthood in most industrialized nations is the almost complete lack of structure afforded to young people which means that, for many, it is the first time in their lives that they have almost complete autonomy on how to spend their time and who (if anybody) to spend it with. Thus, despite a developing brain that has significant limitations well into the twenties, the third decade of life becomes a time during which young people have more control than ever before over what they do with their lives.

The combination of very little structure and a brain that is still developing may explain why so many young people flounder during the early portion of emerging adulthood (18–25). In regard to externalizing problems, evidence points to the twenties as being higher than any other point in the life course for alcohol and drug use (Andrews & Westling, 2016), driving while drunk (Centers for Disease Control and Prevention, 2019), risky sexual practices (Lam & Lefkowitz, 2013), and reckless behaviors that are potentially life threatening to oneself or others (see Park et al., 2014). Furthermore, it is the period of time with the highest risk of being a victim of a violent crime (e.g., U.S. Department of Justice, 2019), or of perpetuating a crime (see Craig & Piquero, 2016). Concerning internalizing problems, it is a period with high rates of self-harm, disordered eating, anxiety, depression, and challenges in many other aspects of mental health (World Health Organization, 2016). Taken together,

these externalizing and internalizing challenges exhibited in the twenties may explain why the leading causes of death in industrialized nations between ages 18–30 are accidents and suicide (e.g., World Health Organization, 2016).

In painting what may be considered a rather bleak picture of emerging adulthood, my intent is not to suggest that there are not positive outcomes associated with increased autonomy and prolonged brain development. In fact, there is evidence that the underdeveloped brain opens young people up to being influenced in many ways for the better (see Steinberg, 2014) and that many, in turn, use the autonomy afforded them in emerging adulthood to engage in tremendously positive behaviors (e.g., service, environmental activism, civic engagement, humanitarian efforts) (see Padilla-Walker & Nelson, 2017). My intent is simply to demonstrate that with the combination of biological (e.g., underdeveloped brain) and contextual features (lack of structure) it should be clear why parents may continue to play a vital role in the lives of young people. A strong parent–child relationship may become one of the most important forms of structure that young people have because without it many tend to see emerging adulthood as being filled with instability (Schnyders & Lane, 2018). Hence, with this theoretical and developmental foundation for why parents still matter, the following sections will focus on the specific ways in which parents can contribute to flourishing or floundering in emerging adulthood.

The Role of Parents in Emerging Adulthood

As noted, a close relationship with parents appears to be an important factor in flourishing in emerging adulthood (e.g., Nelson & Padilla-Walker, 2013). Hence, it is important

to better understand factors that may characterize and predict a close (or not) relationship between parents and their emerging-adult children and, also, be linked to flourishing and floundering, respectively.

Flourishing. Studies examining the quality of the parent–child relationship (including attachment) in emerging adulthood and the specific features of the relationship (e.g., companionship, intimacy, aid, relative power, conflict) have found it to be positively related to self-worth/self-esteem (Clifford & Nelson, 2019; Laible et al., 2004), prosocial behavior (Barry et al., 2008), well-being when making the university-to-work transition (Buhl, 2007), happiness (Demir, 2010), perceived personal efficacy (Leondari & Kiosseoglou, 2002), identity development (Kaniūšonytė & Žukauskienė, 2018), romantic relationship quality (Karre, 2015), and intimate relationship satisfaction (Cruz, 2014). Furthermore, a high-quality relationship protects against negative outcomes such as risky drinking (Serido et al., 2014) and nonmedical prescription opioid use (Serido et al., 2014).

High-quality parent–child relationships also appear to be important because they promote other positive aspects of the relationship that benefit young people. For example, research shows that a high-quality relationship with mothers leads to child disclosure of information which, in turn, protects against externalizing behaviors such as drinking, drug use, and risky sexual behaviors (e.g., Urry et al., 2011). This evidence suggests that a strong parent–child relationship creates the climate in which healthy relational processes (e.g., communication) can occur which, in turn, promote positive outcomes in emerging adulthood.

Because of the significant role that the parent–child relationship plays, it is important to examine factors that may promote strong relationships and lead to indices of flourishing directly. Specifically, parental behaviors that

provide support (acceptance, warmth, involvement, and nurturance) and foster autonomy have been linked to both a positive parent–child relationship and to indices of flourishing. For example, parental social support is associated with less self-criticism and higher levels of goal attainment (Dickson & Shulman, 2016), lower levels of risky sexual behavior (Simons et al., 2013), less depression and loneliness (Mounts, 2004), and better academic and social adjustment (Smojver-Ažić et al., 2015). Similarly, paternal acceptance is linked with psychological adjustment (Filus & Roszak, 2014) and lower levels of anxiety (Reitman & Asseff, 2010). Maternal warmth and acceptance is associated with lower levels of drug use (Luk et al., 2015), higher levels of self-regulation and social competence (Moilanen & Manuel, 2017), and higher academic achievement via perceived teacher social support (de la Iglesia et al., 2014). Parental autonomy support is associated with better well-being (Ratelle et al., 2013), and more prosocial behaviors (Padilla-Walker et al., 2018). Finally, in work examining parenting styles that reflect warmth and support (i.e., authoritative parenting), results reveal the same pattern of outcomes. Authoritative parenting by mothers and fathers is associated with greater motivation for learning (Alt, 2015), better adjustment to university, lower levels of depression and impulsiveness (Patock-Peckham et al., 2011), and lower levels of internalizing and externalizing problems (McKinney et al., 2016; Steele & McKinney, 2019).

It should be underscored here that providing support *to an* emerging adult is very different than *doing for* grown children what they developmentally should be doing for themselves (i.e., helicopter parenting, Nelson et al., 2019). The distinction between supporting children and doing things for children is shown in a study that examined “intense levels of support” from parents by examining how many times a

week parents provided their grown children (mean age = 23.82) emotional support, practical support, socializing, advice, financial support, and listening to them talk about daily events. Note that none of this support involved doing things for their children what they should be doing for themselves. The grown children who received frequent support reported better psychological adjustment and higher life satisfaction than those who did not (Fingerman et al., 2012). This is an example of parents providing support, even at high levels, with positive results. It did not reflect doing things for their children which, as will be outlined, is certainly not linked to flourishing.

Taken together, a strong parent–child relationship (fostered by parental warmth and support) provides a level of structure and security during a time of life that is marked by its lack of stability and structure (Nelson, 2020). Furthermore, support should not be confused with directing young people’s choices or doing for them what they developmentally should now be attempting to do for themselves. The negative outcomes of doing the latter will be addressed next.

Floundering

Although there is nothing developmentally inappropriate with parents providing support, parental behaviors that limit autonomy are problematic. Specifically, *behavioral control* (aimed at limiting children’s behavioral autonomy), *psychological control* (aimed at limiting children’s psychological and emotional autonomy), and *helicopter parenting* (parents doing for their children what they developmentally should be doing for themselves) have been found to be distinguishable as related but unique forms of control in emerging adulthood (e.g., Padilla-Walker & Nelson, 2012) and to be linked to a variety of mostly negative outcomes.

Although emerging adults should have more autonomy over their choices, many parents still attempt to exercise *behavioral control* by setting rules about how children spend their money, trying to control what classes they take, or controlling what they do on nights and weekends. Not surprisingly, behavioral control is linked to indices of floundering in emerging adulthood. For example, parental behavioral control has been positively related to lower emotional control, aggression, risk/externalizing behaviors (e.g., stealing, cheating, drug use), and negatively to sympathy, self-regulation, prosocial behavior, and a moral ethic of autonomy (Manzeske & Stright, 2009; Nelson et al., 2019). Furthermore, parents who use high levels of punishment and hostility have emerging-adult children with higher levels of depression, anxiety, impulsivity, and lower levels of social competence, self-worth, and kindness (Nelson et al., 2011). Finally, parental use of physical force (e.g., “hit you with a fist or kicked you hard”) is related to emerging adults’ alcohol problems directly and indirectly via increased psychological problems (depressive and anti-social problems) (Pollard & McKinney, 2019). In sum, parental behavioral control is a significant risk factor in development during emerging adulthood.

Psychological control is manifested through strategies of manipulating the relationship, such as inducing guilt if the child does not do what parents want, ignoring the child if their behavior is disagreeable (e.g., love withdrawal), and trying to change the child’s thoughts or emotions (e.g., invalidating feelings) (Barber, 1996). It may not be surprising that coercive attempts to manipulate how children feel would be related to problems of an internalizing nature including depression and anxiety (e.g., Reitman & Asseff, 2010), difficulty coping with interpersonal stress (Abaied & Emond, 2013), and lower levels of emotion

regulation (Manzeske & Stright, 2009; Nelson et al., 2019).

The indices of floundering linked to psychological control are not limited to internalizing problems though. Psychological control is also associated with externalizing behaviors (e.g., drug and alcohol use, risky sexual behaviors, stealing) (Nelson et al., 2019; Urry et al., 2011), problems in romantic relationships (more conflict, Karre, 2015), fewer positive behaviors (e.g., sympathy, benevolence, prosocial behaviors) (Nelson et al., 2019) and, longitudinally, with lower levels of identity commitment (Luyckx et al., 2007). Furthermore, domains of education and work appear to be impacted negatively, as shown in a longitudinal study that revealed father psychological control to be associated with lower levels of education completed, lower employment status, more employment stress/worry, and, for males, lower perceived educational abilities, while maternal psychological control was linked to lower perceived educational abilities (for sons) and more occupational problems (Desjardins & Leadbeater, 2017). Taken together, limiting young people’s strivings for autonomy via psychologically and emotionally manipulative means appears to harm numerous important areas of emerging-adult development.

As noted, *helicopter parenting* is conceptualized as parents doing for their children what they developmentally are able to do for themselves (Padilla-Walker & Nelson, 2012). Although extreme examples of helicopter parenting often capture media headlines (e.g., the U.S. college admission bribery scandal, www.nytimes.com/news-event/college-admissions-scandal), research shows that helicopter parenting is not employed that often in emerging adulthood (e.g., Padilla-Walker et al., 2019) and mainly only in the first years of emerging adulthood (ages 18–19) (Nelson et al., 2020). Of those few who use it, mothers tend to use it a little more than fathers, as

shown in a study that found that the use of helicopter parenting declined from ages 19 to 21 but at age 19 mothers used helicopter parenting 9 percent more than fathers (Padilla-Walker et al., 2019). In this study, it was also found that when fathers do use helicopter parenting (again, rather infrequently), it tends to be with sons more than daughters and possibly in response to their sons' behaviors (e.g., video game use).

In emphasizing that the use of helicopter parenting is not widespread, I do not mean to suggest it is not linked to indices of floundering. Helicopter parenting is negatively associated with indices of adjustment in college, such as school engagement (Padilla-Walker & Nelson, 2012), school functioning (Luebbe et al., 2018), coping skills (Odenweller et al., 2014; Segrin et al., 2013), decision-making skills (Luebbe et al., 2018), and school burnout via lower self-control (Love et al., 2020). Helicopter parenting is also associated with internalizing problems including anxiety, depression, and emotional dysregulation (e.g., Cook, 2020; Luebbe et al., 2018; Rousseau & Scharf, 2015; Reed et al., 2016), poorer coping skills (Segrin et al., 2013), lower self-efficacy (Bradley-Geist & Olson-Buchanan, 2014), and lower well-being (via a reduced sense of autonomy and competence) (Schiffrin et al., 2019). Beyond educational and internalizing problems, helicopter parenting has also been linked to externalizing and relationship problems such as substance use problems (Cook, 2020), higher levels of narcissism (Segrin et al., 2013), negative marital attitudes (Willoughby et al., 2015), and lower relationship competence (Cook, 2020). Finally, helicopter parenting is negatively related to internal locus of control, prosocial behaviors, moral reasoning, and empathy (Kwon et al., 2016; McGinley, 2018; Nelson et al., 2019).

It should be noted, though, that while the overall picture suggests that helicopter

parenting is a risk factor for emerging adults' floundering, there are nuances to this general picture. Specifically, helicopter parenting that occurs in conjunction with parental warmth appears to have fewer negative outcomes than when lacking warmth (Nelson et al., 2015) or when combined with psychological control (Padilla-Walker et al., 2019). In other words, the evidence points to the negative correlates of helicopter parenting generally but a closer look shows parental warmth may protect young people against some of the negative effects of helicopter parenting while psychological control may exacerbate them.

Taken together, there is now a rather substantial amount of work from which it is possible to start making some points about the role of parents in the lives of emerging adults who flounder. It is rather clear that parenting is problematic when it infringes upon the developmental strivings for autonomy and when it hinders the formation of a strong parent-child relationship which, in turn, leaves emerging adults without a source of structure during a time of instability.

Cultural Contexts for the Interaction of Parenting and Development

It is important to note, though, that the developmental process of individuation occurs within contexts and it is, therefore, necessary to consider the ways in which these contexts (i.e., culture) may shape the role of parents in their children's individuation process. Up to this point, my intent has been to demonstrate broadly the links between parenting and flourishing and floundering in emerging adulthood. In general, the evidence suggests that culture does not often alter the importance of positive parenting. In fact, many of the studies reported previously, showing links between child flourishing and parental warmth, support, and autonomy granting, came from a variety of

countries and cultural contexts, including Israel (Dickson & Shulman, 2016), Croatia (Smojver-Ažić et al., 2014), Argentina (de la Iglesia et al., 2014), Poland (Filus & Roszak, 2014), Turkey (Moilanen & Manuel, 2017), and Mozambique (Cruz, 2014), to name just a few. Thus, across most cultures, positive parenting matters.

However, culture may affect what specific aspects of positive parenting are valued, how they are expressed (i.e., whether support looks differently across cultures), or the role that parents are expected to take in the individuation process such that there may be features of a culture that are conducive to the developmental process of individuation and others that may hinder the process. In other words, it is important to consider the ways in which culture (i.e., values, beliefs, behaviors), parenting (e.g., support, control), and development (e.g., strivings for autonomy) may intersect in a variety of ways to influence flourishing or floundering.

For example, in many Asian cultures, the notion of filial piety remains a strong backdrop for the parent–child relationship. Filial piety is a central tenet of Confucianism, which prescribes respect, love, reverence, and obedience toward one's parents and elders (Confucius, 2013, 1.2). These prescriptions have the potential to facilitate a strong parent–child relationship that may provide structure for Asian emerging adults. However, China has undergone a dramatic shift in its exposure to and relationship with Western cultures and ideals. As such, Western individualistic values and ideologies (e.g., liberty and individual freedom) have been introduced into the country and have been increasingly accepted by many Chinese people, especially in the younger generation (e.g., Chen & Wang, 2010). Thus, what it means to become an adult to young people may differ (sometimes rather drastically) from what it

means to parents. In a study examining the criteria that Chinese young people and their parents deem necessary for adulthood, emerging adults endorsed criteria reflecting independence (e.g., independent decision-making, financial independence) to a much greater extent than did their parents (Nelson et al., 2013). Hence, it appears that emerging adulthood in China might be a time during which a self-versus-parent conflict may exist due to cultural values.

Indeed, the potential conflict can be seen in a study of 612 Chinese college students in which it was found that filial piety moderated the association between parenting and self-esteem with parental psychological control (as perceived by the child) predicting lower self-esteem when emerging-adult children felt highly devoted to parents (i.e., high filial piety), but had no influence on self-esteem when they did not (Jorgensen et al., 2017). These results demonstrate how the potential exists in Confucian cultures for conflict between developmental strivings of the self and a cultural emphasis on devotion to parents and their desires, especially when parents attempt to control their emerging-adult children. In other words, if parents are attempting to direct young people's decisions, children may be conflicted between their sense of duty to parents and their desire to become independent decision makers, resulting in resentment and self-doubt.

Similar to the notion of filial piety is the concept of *familismo*, or familism, which refers to individuals' identification with and attachment to family, including a sense of responsibility, loyalty, support, and solidarity among family members (Sabogal et al., 1987). Familism is a central cultural value in Latino families and has repeatedly been linked to positive outcomes throughout childhood and adolescence (see Stein et al., 2014 for a review) and in emerging adulthood (Kline et al., 2016).

Although a sense of familism typically leads to positive outcomes, when Latino adolescents and their parents do not agree on autonomy expectations, there is increased risk for parent–child conflict and greater psychopathology (e.g., Bámaca-Colbert et al., 2012). This raises the possibility that as children get older there may be the potential for conflict if parental behavior appears to conflict with developmental strivings for autonomy, or if parenting practices do not coincide with the values of *familismo* (i.e., if parenting is not loving and supportive) (Shigeto et al., 2019). Indeed, taken together, findings support the notion that family relationships are important for Latinx emerging adults while also showing the complex associations between cultural values, parental behaviors, and young people’s developmental strivings for autonomy.

The results also point to the need to examine how differences can exist even between cultures that are characterized as more group/family oriented. For example, in a study of Mexican American college students, it was found that high familism values buffered the risk of paternal psychological control in that paternal psychological control was negatively associated with self-esteem when college students reported low familism values but not in the group that reported high familism values (Kline et al. 2016). These findings raise the question why it was among the group with *low* levels of devotion to family in Mexican American young people (familism) that psychological control was negatively linked to self-esteem while, in a similar study in China, discussed previously (Jorgensen et al., 2017), it was in the group that had a *high* level of devotion to parents (filial piety) that psychological control was negatively related to self-esteem. There may be very subtle but important distinctions between the related but distinct constructs of filial piety and familism.

Specifically, studies repeatedly emphasize that familism includes a desire to maintain strong family ties wherein it is expected that the family will be the primary source of instrumental and emotional support, and will promote feelings of loyalty and commitment to the family over individual needs and desires (e.g., Halgunseth et al., 2006). This seems to convey a bi-directional process between parents and children that is underscored by mutual support. Filial piety, on the other hand, seems to reflect a process that is more uni-directional in nature. Again, filial piety is the prescription that children show respect, reverence, and obedience toward one’s parents and elders (Analects 1.2, trans. 2003). There may be a difference between emphasizing obligation to all members of the family (of which children are a part) and emphasizing obligation to parents only. A focus on mutual support within the family versus expectation to honor parents may result in differences in how emerging adults perceive the actions of their parents. For example, in the context of a supportive family (high familism), emerging adults may see their parents’ controlling behaviors as attempts to advise, teach, and help them rather than hurt or control them (Halgunseth et al., 2006; Kline et al., 2016). Obviously, more work is needed to examine how cultural values that foster commitment towards one’s family over one’s own desires intersect with developmental self-strivings, but these extant studies underscore that even subtle differences in cultural values can influence factors in the parent–child relationship that can, in turn, influence flourishing or floundering in emerging adulthood.

Finally, the ways in which cultural values and developmental strivings intersect in unique ways can be seen in Italy, where it is quite normative for emerging adults to live in “hotel families” (Sartori, 2007). This is a term given to the family home where Italian young

people often live throughout much of their twenties and experience a great deal of freedom with few expectations to contribute to household finances or responsibilities (Carrà et al., 2014; Sartori, 2007). However, the role of parents in Italy goes well beyond just providing living accommodations to their children during emerging adulthood. In most studies examining the criteria that emerging adults endorse as requisite for adulthood, the criterion “not deeply tied to parents emotionally” tends to be rated quite high regardless of whether the study is conducted in countries that reflect more individualistic values or collectivistic values (Nelson & Luster, 2015). In Italy, though, the criterion “not deeply tied to parents emotionally” ranks as the least important criterion for adulthood (Crocetti & Tagliabue, 2016). Thus, it appears that having and maintaining a strong connection with parents during emerging adulthood may be rather normative, even important, in Italy. Indeed, it may represent a culture in which the process of becoming an adult is not about separating oneself *from* the family but rather a culture in which the process of individuation occurs *within* the context of the family in that young people may “modify their relationship with parents, without perceiving their emotional bond with them as detrimental for their autonomy and adult status” (Crocetti & Tagliabue, 2016, p. 46).

However, even within this culture that emphasizes the importance of family relationships, recent work in Italy underscores that there is a difference between parental support and parental control. Specifically, emerging adults whose parents are high in psychological control and low in support report lower levels of self-acceptance, positive relations, autonomy, environmental mastery, purpose in life, and personal growth than those whose parents are high in support and low in psychological control (Liga et al., 2020). Thus, even in a

culture that emphasizes the familial context, flourishing is compromised when parental control is high and support is low.

These examples are not meant to be a comprehensive overview of all the ways in which parents in every culture may vary in the role they play in the lives of their emerging-adult children. Instead, they are meant to highlight how culture, parenting, and development may intersect in a variety of ways to influence flourishing or floundering. In general, the findings suggest that when there is a good “fit” between cultural values, parenting practices (e.g., support, warmth), and developmental processes (e.g., strivings for autonomy), then positive outcomes emerge but when there is a clash between cultural values (e.g., filial piety), parenting practices (e.g., control), and developmental strivings (e.g., desire for independence), then young people may be at risk for floundering.

Transfer of Resources – A Lens for Practical Advice for Parents

In this final task of this chapter, I would like to provide a lens for how parents can organize, think about, and apply the research presented thus far as they make practical decisions about their parenting. It is beyond the scope of one chapter to provide direction for every area of importance for emerging adults to flourish but it is certainly possible to begin to think in a very practical way about the role of parents in the lives of their emerging-adult children. To do that, it is important at this point to provide a rather targeted definition of “flourishing.” Because the aspect of flourishing that many parents want to see accomplished is young people’s preparation for and reaching adulthood (Nelson et al., 2007), in this final section of the chapter flourishing is specifically defined as the achievement of capacities requisite for adulthood. But what are those capacities?

There has been a substantial amount of work done examining what emerging adults and their parents feel is needed to be considered an adult. There is overwhelming evidence (see Nelson & Luster, 2015, for a review) that young people around the world tend not to see discrete events, or role transitions, (e.g., completing one's education, buying a home, or marrying, having a child) as indicating the beginning of

adulthood but, instead, the criteria that are repeatedly endorsed by young people and their parents (Nelson et al., 2007) as requisite markers of adulthood reflect *relational maturity, independence, family capacities, and norm compliance* (see Table 12.1 for a description of these markers of adulthood). As a result, it is important to understand how parents can assist in the development of these capacities.

Table 12.1. *Capacities that parents and emerging adults see as needed to be considered an adult* and which parents can target as they transfer resources*

Relational maturity

Accept responsibility for the consequences of your actions
 Establish a relationship with parents as an equal adult
 Learn always to have good control of your emotions
 Become less self-oriented, develop greater consideration for others

Independence

Financially independent of parents
 No longer living in parents' household
 Decide on personal beliefs and values independently of parents or other influences

Norm compliance

Avoid becoming drunk
 Avoid drunk driving
 Avoid illegal drugs
 Have no more than one sexual partner
 Avoid committing petty crimes like vandalism and shoplifting
 Drive safely and close to the speed limit
 Avoid use of profanity/vulgar language
 Use contraception if sexually active and not trying to conceive a child

Family capacities

Become capable of supporting a family financially
 Become capable of caring for children
 Become capable of running a household
 Become capable of keeping family physically safe

Role transitions

Finish education
 Married/partnered
 Have at least one child
 Settle into a long-term career
 Purchase a house or have a stable living environment

*See Nelson & Luster, 2015 for a review

The notion of intergenerational exchange has been employed to capture any transfer of material goods, money, or emotional support between family members of different generations to one who needs help (e.g., Eggebeen, 1992). In other words, parents need to consider what resources, or assets, they are transferring to their children that, as emerging adults, they can draw upon in order to develop these essential capacities of adulthood. For example, it appears that parents who are warm, supportive, and have a strong relationship with their children are transferring resources that are leading to progress in *independence* (e.g., perceived personal efficacy, Leondari & Kiosseoglou (2002); identity development, Kaniušonytė & Žukauskienė (2018); *interdependence*, e.g., better social adjustment, Smojver-Ažić et al. (2014); higher levels of social competence, Moilanen & Manuel (2017); *family capacities*, e.g., positive views of marriage, Willoughby et al. (2015); *norm compliance*, e.g., less risky substance use and fewer risky sexual behaviors, Cerda et al. (2014), Luk et al. (2015); *role transitions*, e.g., higher academic adjustment and achievement, de la Iglesia et al. (2014); and well-being in the university-to-work transition, Buhl (2007)).

This is not to say that warmth and support are the resources being conveyed but rather that they appear to be the means by which assets are being transferred, as their presence increases the likelihood of them being received by the child (e.g., Barry et al., 2008). Thus, the extant research is underscoring that positive parenting is associated with flourishing because it contributes to the acquisition of attributes needed to be an adult. Again, it is beyond the scope of the chapter to outline all of the specific resources that might be transferred, but I would like to provide a few specific examples to help parents think about how to be intentional in their parenting by

providing assets that young people can use to develop capacities of adulthood.

Resources that parents might transfer include knowledge, skills, values, time, and material assets. An example of the type of skills, knowledge, and values that can be conveyed for a young person's benefit is shown in the areas of romantic relationships and sex. Emerging adults report having communicated with their parents about morals, sexual health, the importance of sex occurring in the context of a relationship, and pregnancy (e.g., Lefkowitz & Espinosa-Hernandez, 2007), and this type of communication leads to healthier sexual practices in emerging adulthood (Riggio et al., 2013). The results are very similar in regard to the transfer of knowledge, skills, and values in the area of finances. Parents' socialization (modeling and direct teaching of sound financial behavior) leads to emerging adults' financial knowledge (e.g., understanding concepts such as compounding interest) and healthy financial attitudes (e.g., their views about performing positive financial behaviors), resulting in positive financial behaviors (e.g., spending within a budget) (Shim et al., 2010). In yet another domain, it has been found that when parents talk to emerging adults about various aspects of alcohol use (e.g., "the negative social experiences that could happen because of drinking" and "how people think they are sober when in fact they are not"), young people are more likely to use protective behavioral strategies which are behaviors that individuals can engage in before, during, or after drinking in order to limit negative alcohol-related consequences (e.g., drink water while drinking alcohol, stop drinking at a predetermined time, avoid drinking games, use a designated driver) (Litt et al., 2020).

It should be noted that the transfer of knowledge, skills, and values begins well before emerging adulthood but the effects thereof are to be seen extensively in emerging adulthood.

In sum, these examples underscore that as parents provide their children with resources (e.g., knowledge, skills, and values), especially in the context of positive parenting and a strong parent–child relationship (e.g., Simons et al., 2013), their children are more likely to experience indices of flourishing, generally (e.g., fewer risky behaviors, healthier relationships, social competence), and to achieve attributes of adulthood, specifically (e.g., norm compliance, self-regulation, financial independence, being able to care for a family).

Another example of the type of resources that parents might transfer is in the realm of financial and material assets. It is interesting how many parents do not see providing financial resources as supporting the development of independence but rather view it as hindering independence. Because it takes longer to become financially independent due to the fact that a living wage is increasingly dependent on post-secondary education, many young people may suffer without some material support. For example, Bell et al. (2005) explored levels of income adequacy (“sufficient wage and salary income to live on their own at-or-above the national poverty threshold in a given year,” p. 36), and using a multi-national dataset, found that from 1985 until 2000, many emerging adults experienced a decline in income adequacy. This might explain why a transfer of financial resources during this time appears to be linked to flourishing, and the lack thereof to floundering (e.g., Kendrig et al., 2014).

This is an appropriate place to reiterate the need to distinguish between (a) providing support (i.e., transferring resources) to young people, and (b) controlling their choices or doing everything for them. In a study of U.S. college students, it was found that parents who paid for everything (tuition, books, housing, daily expenses, and recreation) had emerging adults who were floundering (e.g., they had more risk behaviors, lower levels of

occupational identity) more than those who received less (but, in several areas, still some) parental financial support (Padilla-Walker et al., 2012). Thus, there is a big difference between providing support (e.g., transferring resources) and doing everything for the child. This is because providing for all of a young person’s financial needs does not promote the development of adult capacities because young people are not starting to do anything for themselves. However, it is important to reiterate that providing little or no material assistance likewise hinders the development of adult capacities (Schoeni & Ross, 2005; Kendrig et al., 2014). Thus, again, doing everything for an emerging adult hinders growth but transferring resources (providing support) facilitates it.

And material or tangible assistance does not refer solely to financial assistance. In examining the decline in income stability experienced by emerging adults, Bell and colleagues (2005) found that emerging adults who continued to live with their parents or were claimed as part of their parents’ household sustained smaller losses in income adequacy than those who lived more independently. This is important because bigger losses in income adequacy often delays education for those emerging adults who live independently because they are forced to either drop out of school or work full-time to pay for their education, which then delays their obtaining employment that is sufficient for the establishment of economic independence (e.g., Schoeni & Ross, 2005). In sum, although it may seem counterintuitive that parental financial assistance would lead to financial independence, research suggests that helping children to complete their education via the transfer of tangible resources (e.g., place to live, financial assistance) results in faster pathways to financial independence.

So what can parents do? With these examples of transfer of resources, parents can begin to

ask themselves questions to guide their own decision-making and behaviors. For example, parents may ask: What resources do I have that I might convey to my children to help them more readily and successfully acquire attributes of adulthood? Am I withholding an asset that might assist my child? By withholding a resource might I be setting my child up for failure? Do I have unrealistic expectations for which attributes of adulthood my children can acquire without resources? Am I withholding the potential for growth by doing everything for my children? Am I transferring resources in some domains but withholding it in others? Am I attempting to transfer resources but is the way I am doing it (harshly, psychologically manipulative, guilt inducing, etc.) hindering the likelihood that my child will accept them?

Parents may also wonder where the line is between transferring resources and doing “too much” for their children? Remember that the definition of helicopter parenting is “doing for children what they developmentally can do for themselves” (Nelson et al., 2019). This is a good guiding principle for parents. For example, providing advice, encouragement, and a listening ear does not involve doing anything *for* a child. Having a child seek help (notice who is instigating the transfer of knowledge) in discussing the pros and cons of choosing a particular major in school is different from parents deciding for them, or coercing them into a major (i.e., “I will only help pay for your tuition if you major in what I tell you to”), or calling a university’s academic advisement office to gather information about majors for the child.

Another guiding principle is whether providing a resource is facilitating or hindering the development of capacities of adulthood. For example, one young person may use the opportunity to live at home as a chance to play video games all day while another may use it

as the opportunity to take more classes in school in order to finish sooner (by not having to work to pay rent). Using this example we can see that living at home in itself is neither good nor bad but the reason (i.e., fostering or delaying capacities for adulthood) behind this practice is what determines its value. So, parents can make choices about how and whether to transfer resources by asking whether it would facilitate a young person developing capacities of adulthood.

In summary, these are the key points I am attempting to help parents think about by focusing on the transfer of resources. First, young people flourish when they have a high-quality relationship with their parents in which resources are conveyed. Second, resources take numerous forms (e.g., skills, knowledge, values, time, material assets). Third, the goal of transferring resources is to help facilitate the development of capacities for adulthood, with the reminder that there is not just one aspect of adulthood (e.g., financial independence) that parents can help facilitate but many (independence, interdependence, family capacities, norm compliance and role transitions). Fourth, doing things for children that they developmentally can do on their own does not transfer any resources. Fifth, simply telling young people what to do, or coercing emerging adults to do something, is not transferring resources. In fact, sixth, the way in which parents interact with their children can either increase or decrease the likelihood that their children will accept the resources being offered and use them to develop capacities of adulthood.

The final concern to be addressed, then, is what happens when circumstances prevent the transfer of resources? Parents may be experiencing challenges in their own lives that hinder the transfer of resources or that adversely affect the parent–child relationship. For example, parental maladaptive cognitions,

mental health issues (e.g., depression, anxiety), and other indices of psychopathology have all been linked to children experiencing indices of maladjustment (e.g., internalizing problems, externalizing problems) in emerging adulthood (e.g., McKinney & Franz, 2019; Szkody & McKinney, 2020). It may be that parents experiencing these types of challenges may find it difficult to transfer resources.

It may also be difficult to transfer resources if assets are lacking in a family to begin with. As noted previously, Schoeni and Ross (2005) found that, on average, parents provided roughly \$38,000 in material assistance – housing, food, educational expenses, or direct cash assistance – throughout the transition to adulthood (from age 18 to age 34). Unfortunately, there were large differences in financial support across income groups with emerging adults from families in the lower-income groups (bottom two income quartiles) receiving markedly less financial support (approximately \$25,000 on average) during the years 18 to 34 while those in the top quartile received \$70,965 of economic assistance from their families. And, it should be noted that this distinction in financial resources transferred does not appear to reflect in any way a difference in desire to support children as there was no difference across family income levels in the amount of support that was given in the form of time.

More recent work paints a similar picture, that there is a wide disparity in families' ability to transfer financial resources and this discrepancy affects the likelihood of young people making a successful transition to adulthood. Kendrig and colleagues (2014) found that emerging adults who grew up in poverty were (a) in adolescence, more likely to help and support their parents and siblings every day, be worried about money, and pay family bills, and, (b) in emerging adulthood, less likely to have finished high school, to ever enroll in

college, or to receive at least two financial transfers from parents or relatives, and more likely to become a young parent. These findings were particularly strong for Black young people, and those individuals who as adolescents lived in family situations that did not include both of their biological or adopted parents (e.g., biological mother not present).

In sum, emerging adults from low-income families not only have fewer financial resources transferred to them, they are actually transferring resources back to their family starting in adolescence. This reflects a process referred to as “childhood adultification” (Burton, 2007) and is problematic because taking on adult roles (e.g., paying bills, caring for children) before one is ready, or equipped to do so, has been linked to numerous indices of floundering (e.g., Gennetian et al., 2004). Thus, the absence of financial resources to transfer to children becomes a significant impediment to flourishing in emerging adulthood and this burden falls disproportionately upon Black families (e.g., Kendrig et al., 2014), immigrant families (see Bornstein & Cote, 2019), families with an incarcerated parent (Siennick, 2016), and families led by divorced or remarried parents (López Turley & Desmond, 2011). Hence, there is a glaring need to address systemic factors (e.g., racism, poorly funded schools, policy brutality, unfair arrest, prosecutorial, and incarceration practices, lack of access to health care, cost of college education) that continue to harm and prevent growth in these at-risk families and their emerging-adult children.

Conclusion

The research presented in this chapter underscores the point that parents still matter in the lives of their children during emerging adulthood. During a time that is full of instability for many, parents have the opportunity to

provide structure via the support they provide to, the strong relationship they form with, and the resources that they transfer to their children. Warmth, support, and communication mark some of the forms of positive parenting in emerging adulthood while control, criticism, and conflict harm the parent–child relationship and are linked to young people’s floundering. The process of recentering the parent–child relationship during emerging adulthood occurs in contexts, though, so the chapter underscored how the intersection of culture, parenting, and development can either “fit” to foster positive outcomes, or conflict in a way that might cause conflict in the parent–child relationship and struggles for emerging adults’ development. Finally, in this chapter, I have attempted to provide a framework for parents to think about how to parent during this time. Specifically, research shows that parents who transfer resources (e.g., skills, knowledge, material assets) to their children help facilitate the development of capacities needed for adulthood. In sum, the process of becoming an adult is taking longer than ever before and is fraught with opportunities to flourish or flounder along the way. Fortunately, young people do not have to undertake this journey on their own if their parents will continue to play their important, albeit changing, role in the lives of their children.

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Part III

Parental Factors That Impact Parenting

13 Mental Health and Parenting

Sandra T. Azar and Michelle F. Wright

History portrays parents with mental health (MH) problems as automatically placing children at risk. Indeed, psychological disorders were considered legal grounds for removal of children, even citing “predictive determinancy” to justify removal during pregnancy (Field & Sanchez, 1999). This led to parents’ rights being ignored (Benjet et al., 2003) and failures to provide intervention and legal representation (*Rocking the Cradle Report*, 2012), reducing parents’ seeking treatment as it could increase potential discrimination in custody determinations.

In this chapter, we review work linking child risk to parental psychopathology (e.g., schizophrenia, affective disorder, and substance use) and other disorders (e.g., post-traumatic stress disorder, anxiety disorders, attention deficit hyperactivity disorder (ADHD)). Framed using a family model of normative family behavior and accomplishment of tasks involved (i.e., the McMasters Model, (Epstein et al., 1982)), we draw conclusions regarding overlap across diagnostic categories in the mechanisms for child risk. Impairments associated with MH problems that interfere with caregiving have been studied. We consider functional parenting impairments with the most evidence (e.g., negative affect, disengagement, poor synchrony, poor affective recognition skills, poor attention). Our social cognitive and neurocognitive perspective of parenting risk (Azar & Weinzierl, 2005) provides transdiagnostic mechanisms that may explain the origins of parenting risk and

starting points for interventions. Innovative approaches, such as using technology to address injury risk, and examples of prevention efforts to prevent intergenerational transmission are highlighted. Finally, we discuss improvements needed to answer policy questions about parenting (e.g., custody determinations, termination of parental rights) and services needed to support optimal caregiving, drawing on the *Rocking the Cradle Report* (2012).

Epidemiology: How Common is MI in Parents?

During the past year, 18 percent of all parents in the United States have had some type of mental illness (MI) and about 4 percent had serious mental illness (SMI; Stambaugh et al., 2017). Estimates from Canada and the UK are similar, suggesting that between 12.1 percent and 25 percent of children are exposed to maternal MI (Abel et al., 2019). The prevalence for children from 0 to 3 years old is 7 percent and increases to nearly 11 percent by 15–17 years old (Pierce et al., unpublished data, 2020). These numbers vary based on diagnosis (e.g., one in five children live with a parent with severe depression (Child Trends, 2017)). In the United States, MI is more common among mothers than fathers and least common among Asians compared with other races (Stambaugh, et al., 2017). A Canadian study (Abel et al., 2019) found that approximately 78 percent of such parents report receiving

no mental healthcare, suggesting identification issues when addressing risk.

Core Tasks in Parenting where Impairments May Lead to Risks

Parental MH problems might disrupt crucial parenting tasks. These domains include: (1) *basic tasks* that involve fundamental needs (e.g., food, shelter), (2) *developmental tasks* that involve adjusting and fostering the developmental needs of family members; and (3) *hazardous tasks* that involve handling emergency situations (e.g., illness, injuries) (Epstein et al., 1982). As each domain is considered, keep in mind that society sets standards around adequate parenting and has designated legal actions when parents fail to meet them (Azar et al., 2004; IMNRC, 2015). A neglectful household, for example, includes disengagement from or indifference toward the child, a lack of monitoring, bizarre behavior, apathy, immaturity, rigidity, stereotyped views of the child (e.g., the child is a “burden”), poor planning, unresponsiveness to the child’s emotional distress, and minimization of the child’s needs (IMNRC, 2015). These are associated with parental MI and may lead to child risk. We argue for an understanding of mediating mechanisms to address the needs of parents with MI.

Basic Tasks

Families provide food and shelter, and these require income. MI is linked to problems in these domains. Difficulties in basic tasks bode poorly for child outcomes. Nutritional deficiencies, especially in young children, can adversely impact brain development (Benton, 2008), resulting in cognitive, behavioral, and emotional problems (Lozoff et al., 2000). Food insecurity is associated with child externalizing and internalizing problems (e.g.,

Slopen et al., 2010). Maternal depression is linked to food insecurity. It is unclear whether the lack of nutrition plays a causal role in depression or depression plays a causal role in food insecurity. Increases in emotional distress and stress may disrupt the parent–child relationship, decreasing parenting quality. Parents may attend less to children’s needs and their cognitive and social growth, which may explain behavioral problems (Alpert et al., 2000).

For parents living with SMI, financial security may be difficult to achieve or sustain. Luciano, Nicholson, and Meara (2014) found low employment rates for parents with SMI (38 percent full-time and 17 percent part-time among mothers; 60 percent full-time and 9 percent part-time among fathers) compared to parents with no MI (50 percent full-time and 19 percent part-time among mothers; 85 percent full-time and 5 percent part-time among fathers). Figures from advocacy groups are somewhat higher (e.g., National Alliance on Mental Illness, 2014). Mothers and fathers with MI were twice as likely to fall below the U.S. census poverty threshold than those without MI.

It is not clear how much employment and income decrements are a consequence of MI or result from the stigma. Data link unemployment and the severity of post-traumatic stress disorder (PTSD) symptoms (Smith et al., 2005). MI is associated with low self-esteem, apathy, negative cognitions, trouble making decisions, and social disengagement, all of which can make employment more tenuous. Further, people with MI are perceived as not ambitious, motivated, intelligent, and as less able to handle stress. They are viewed as less employable, which jeopardizes their available income (National Alliance on Mental Illness, 2014). Brouwers (2020) outlined four areas of stigma that hinder employment prospects of people with MI: (1) employers hold negative

attitudes toward people with MI, decreasing the chances of being hired or supported; (2) the disclosure and non-disclosure of MI can lead to job loss; (3) anticipated discrimination, self-stigma, and the “Why Try” effect can decrease motivation to keep or find employment, resulting in unemployment; and (4) stigma is a barrier to seeking healthcare, leading to untreated and worsened health conditions and adverse occupational outcomes (e.g. missing work, sick leave).

Housing insecurity refers to a range of unstable living conditions, from inability to pay rent to homelessness. Homelessness, for example, is a common occurrence for mothers with MI (e.g., lifetime rates of homeless with depression range from 40 percent to 85 percent; Bassuk et al., 2020) and anxiety disorders, PTSD, and substance use issues can also co-occur, often untreated. This can cause stress to the parent and the child. MI increases adversity exposure and inhibits the accomplishment of survival tasks (Ranning et al., 2020), with high mobility being problematic (e.g., Lee et al., 2020). Neighborhood-concentrated disadvantage (e.g., high rates of unemployment) and residential instability (e.g., neighborhood residential changes) are linked to physical qualities (e.g., home safety) and cognitive stimulation (e.g., provision of learning materials) of homes (May et al., 2018). Both may reflect economic constraints and stigma associated with where parents with MH can rent or own housing (Corrigan, 2004).

Households touched by severe MI are described as disorganized, inconsistent in daily activities (e.g., forgotten hygiene routines), and having poor provision of basic needs. Other issues arise for children with special medical needs (e.g., special diets) and provisions needed for school attendance. Food insecurity is linked to poor parental mental health, psychological distress, and diagnoses of depression and anxiety. Some evidence

exists for causal chains. For example, deterioration of maternal mental health is associated with food insecurity and food insecurity precedes depression outcomes (Loopstra, 2018). As will be discussed later, the social cognitive and neurocognitive impairments that are associated with various diagnoses (Azar et al., 2016) may also account for inconsistency of daily childcare.

Social resources can often compensate for or precipitate parents' incapacities through the stress that is generated by interpersonal relationships. Indeed, social support mediates associations between qualities of neighborhoods and parenting quality (May et al., 2018). Parents with MI may be limited in social skills and just have a smaller social network (e.g., fewer family members or more limited ability to access assistance from others). The efforts expended by the occurrence of MI may also drain extended family and friends of instrumental and emotional support. Child protective services utilize family as surrogates and giving them supplemental economic support and service. Such placements with relatives may reduce risk and give respite, and may be more familiar to the child (see Azar & Hill, 2006), affording continuity crucial for development and an alternative to foster care.

Interpersonal stressors relevant to survival are heightened in parents with MI, such as domestic violence in the family. This exposure, especially during pregnancy, can disrupt caregiving and is linked to offspring's development (Toso et al., 2020). Heighten cortisol due to stress influencing the fetus has been posited. Overall, greater exposure to adversity in parents with MI may jeopardize survival needs and explain negative child outcomes. This is substantiated by a study with 2 million participants using a Danish registry where it was concluded that higher morbidity and mortality in the offspring of SMI-afflicted parents was linked to higher exposure to psychosocial

stressors and not necessarily specific diagnoses (Ranning et al., 2020).

Developmental Tasks

The research on MI and child outcomes has focused on impaired capacities to foster children's development. Psychopathology involves emotional dysregulation, an underlying mechanism that cuts across a myriad of disorders (Aldao et al., 2010). For example, Lavi and colleagues (2019) identified that parental neglect risk (measured by later Child Prevention Services (CPS) involvement) was associated with lower emotional control, coping strategies, and emotion regulation skills; each are broadband transdiagnostic risks. Although disorder-specific failings of the parents are reported in studies, it is the adversity exposure associated with SMI that creates risk and consequences (e.g., inability to meet developmental needs). MH disorders share genetic components that need to be considered in poor child outcomes, but identification may not be as helpful in crafting interventions as first thought. Moreover, children with genetic vulnerability may require more optimal parenting than children without this, and they may develop problems even when parenting is adequate. This review emphasizes common mechanisms for parent and child outcomes, focusing on malleable psychosocial factors. We first review parenting risks across MH disorders, all of which may have common genetic roots, and then move to social cognitive and neurocognitive mechanisms that may account for these risks.

Before we do this, we alert the reader to important methodological issues within existing studies that make it difficult to draw strong conclusions. Many reviews and studies we cite do not consider the contextual issues raised earlier (e.g., do not control for income, housing instability). Second, the measurement

of psychopathology is global. Issues included whether the work used valid structured diagnostic interviews or symptom ratings on a screening instrument. If diagnosis was used in sampling, studies did not consider changes in diagnostic criteria over time and/or across different locales. Also, it is not clear whether the history of hospital stays versus community samples as a source of samples was considered. Whether the data were collected at the first episode or after multiple episodes was not typically noted. The level of effect multiple absences from the home may have different effects on child outcome.

Most studies focused on maternal MI, with only a small number including data on fathers. In one study, MI in fathers was not associated with child behavioral and emotional problems (Kahn et al., 2004), although other studies have found associations between fathers' MH and internalizing and externalizing problems in infancy and childhood (Ramchandani & Psychogios, 2009). A lack of research on fathers with MI and conflicting research results limits our understanding of how the mental health of both parents affects child outcomes and hinders our ability to develop interventions.

Child gender is not routinely considered in outcomes. The processes of intergenerational transmission are different for male and female offspring (Bridgett et al., 2015). The role of demographics (socioeconomic status (SES), marital status) and context of the data collected also matter (e.g., availability of resources for treatment). Future work needs to consider such dimensions and with more depth.

Alcoholism and substance abuse: Parents with substance use issues often engage in parent-child role reversals, assuming that children can take care of themselves and the parent as well (Gallant, et al., 1998); they are less responsive and more rejecting of their child than non-substance-using parents

(Hans, et al., 1999). Most common are studies during infancy and toddlerhood. Mothers who are substance abusers show impaired maternal affective involvement at three months that negatively influences dyadic affective behavior at 12 months compared to non-substance-using mothers (Sigveland, Olafsen, & Moe, 2013). At later ages, children report less parental monitoring, higher child violence exposure, harsh discipline, and emotional maltreatment that influence the intergenerational transmission of substance use. A recent 18-month study found parent-reported emotional maltreatment was the best predictor of child internalized and externalized behavior, controlling for child age, gender, poverty, and prior CPS involvement (Seay, 2020). Thus, emotional maltreatment may be a target for preventive interventions.

Anxiety disorders and PTSD. Anxiety disorders are linked to parenting problems (Whipple & Webster-Stratton, 1991). Amos et al. (2011) argued that the child's behaviors and needs can trigger the mother's fear schema from previous trauma, leading mothers to avoid anything associated with their child. In a cross-sectional study, Epkins and Harper (2016) reported that parents' anxiety increased parenting problems. PTSD is linked to less secure infant attachment (Bosquet et al., 2014), negative parenting practices (Hershkowitz et al., 2017), and inconsistent discipline and poor supervision (Gewirtz et al., 2010). Symptoms decrease monitoring behavior, lead to inaccurate developmental expectations for their child, and increase rigid or chaotic reactions (Kritikos et al., 2019; Saltzman et al., 2011). All contribute to poor outcomes.

ADHD. In a meta-analysis, parental ADHD was linked to harsh parenting, lax parenting, and less positive parenting (Park, et al. 2017). Marital relationships and social support networks are reduced as resources for

parenting. Finally, intergenerational transmission contributes to childcare burdens (i.e., parental inattentiveness and impulsivity possibly combine with child's ADHD to increase parenting burden).

Depression. Depressive symptoms are the most studied with respect to impairments in parent-child interaction (Rishel, 2012), with early mother-child interactions playing a central role (Cummings & Davies, 1994). Depressed mothers show significant impairments in their emotional involvement, communication, and affection, and increased hostility and resentment, all of which impact child outcomes. They speak less, express fewer emotions, show more covert and overt hostility, and are less involved and more intrusive than comparison mothers (Radke-Yarrow et al., 1992). When mothers are emotionally unavailable, infants exhibit behavioral disorganization, avoidance, and lack positive affect, which negatively effects maternal behavior (Tronick & Weinberg, 1997) and mutual regulation of the dyad. Intrusive mothers have hostile affect and disrupt the infant's activity, leading to anger, turning away from their mothers to limit intrusiveness, and internalization of angry and protective styles of coping. Withdrawn mothers are disengaged, unresponsive, affectively flat, and do little to support infants' activity. Infants are unable to self-regulate this negative state, and develop passivity, withdrawal, and problematic regulatory behaviors (e.g., sucking on thumb) (Hart et al., 1998). Although there are fewer studies with older children and adolescents, findings from a meta-analysis suggest that maternal depression also is linked to negative behavior and disengagement in mothers of school-age children and adolescents (Lovejoy, Graczyk, O'Hare, & Neuman, 2000). Similar findings were included in a more recent review by Goodman (2020). Depressed mother's difficulties with engagement also

constrain involvement with interventions (LaForett & Mendez, 2010)

Schizophrenia/psychotic disorders. Mothers with schizophrenia are less responsive to their infants and have poor emotion recognition skills (Healy et al., 2016). Their negative processing bias may also be translated to offspring.

Hazardous Tasks

MI in parents is linked to concerns regarding physical safety, including failures in healthcare and injury risk. A sample of such work is presented below.

Health care during pregnancy. MI is linked with difficulties complying with health regimens during pregnancy and there are negative consequences for newborns. Depression, for example, is associated with inadequate prenatal care, poor nutrition, preterm birth, pre-eclampsia, spontaneous abortion, and substance abuse. Mannisto et al. (2016) found that maternal psychiatric disorders, especially comorbid psychiatric conditions (e.g., anxiety, bipolar disorder, depression with anxiety disorder) were associated with prematurity. Preterm births are associated with a myriad of poor child outcomes, poor nutrition and lower birth weights, which are associated with SES, rather than the presence of symptoms. Pregnancy itself precipitates symptoms of MI (e.g., postpartum depression; O'Hara, 2009). Less is known about the treatments for maternal MI and adverse pregnancy outcomes, although concerns regarding medication use exist (e.g., effects of antipsychotics during pregnancy) (Galbally et al., 2014).

Children's ongoing healthcare. Oversights in routine medical care have been found with maternal MI. For example, in a five-year longitudinal UK study, mothers with MI were less likely to receive preventative health vaccinations (e.g., Diphtheria) (Osam et al., 2020).

Children of mothers with depression and psychotic illnesses were 14 percent less likely and children of mothers with alcohol/substance misuse disorders were 50 percent less likely to receive necessary vaccinations at two years compared to those of unaffected women.

Other data exist for over-use of medical care. One study found children of depressed parents were 41 percent more likely to visit the emergency room, 47 percent more likely to be admitted to the hospital, and 67 percent more likely to have clinic visits than those without depressed parents (Dreyer et al., 2018). The studies do not provide explanations. It could mean that poorer childcare requires emergency room (ER) visits. This may result from social- and neurocognitive impairments seen in MI (e.g., low maternal self-efficacy).

Injury risk. Poor monitoring of child safety is an ongoing concern. Parental MI is associated with increased risk of child injuries, particularly during the first year. Adjusted rate ratios ranged from 1.28 (1.24 to 1.32) for fall injuries to 3.54 (2.28 to 5.48) for violence-related injuries among children aged 0–1 in a national study in Sweden (Nerviana et al., 2020). In our own work, we have found links between maladaptive injury risk beliefs and child neglect (Azar, et al., 2016). Social cognitive and neurocognitive problems (e.g., attention) play a role in these associations (Morrongiello & Corbett, 2008). As discussed later, impairments are associated with MI.

Special childcare needs. The ability to care for children with special needs may be of particular concern. Care must be at a higher level with these children. MI may interfere with a parent's ability or motivation to follow treatment accommodations (e.g., sending the child to school without proper medications or assistive devices) (Azar & Hetzel-Riggins, 2021). Special needs require a parent with good

organizational capacities, memory, and attentional capacities.

Environmental crises. When risks increase during social crises (e.g., hurricanes, epidemics), parents with MI may struggle to adjust their responses. For example, the recent COVID-19 crisis uncovered the vulnerability of those living in poverty common among MI parents, and ones with SMI may be hardest hit. Residential instability found among those with MI may add to the COVID crisis (National Low-Income Housing Coalition, 2020). Residential instability and homelessness increase the risk of infection and make it harder to treat those who are infected. Individuals with SMI who are employed may have challenges taking time off work and may lack insurance to cover treatment. Similarly, small social networks may limit support from friends and family should individuals with SMI become ill. These factors may elevate infection rates and worsen prognoses of adults and their children. The influence of parental psychopathology has been studied in relation to natural disasters. In a study of displaced families 5–7 months post-Hurricane Katrina, global maternal psychological distress and PTSD moderated the relation between child hurricane exposure and child internalizing and externalizing symptoms (Spell et al., 2008). Such events tax coping capacities (e.g., problem solving) and resources.

Symptoms that increase child physical risk. MI may play a causal role in the hazards the family is exposed to. Higher rates of violence in people with SMI, particularly in those untreated, than in individuals without MI have been reported. In a study in the UK, an estimated 14 percent of patients with first-episode psychosis were violent within 12 months (Senior et al., 2020), and a study in Sweden found a 20 percent increased risk of repeat violent offending in people with diagnosed

schizophrenia-spectrum disorders released from prison (Chang et al., 2015).

Transdiagnostic Mechanisms of Risk

Failures in parenting adequately hint at mediating mechanisms that cross diagnostic categories. Broadband impairments in cognitive and affective functioning are linked to parenting risk. We favor a social information processing (SIP) approach to discussing such transdiagnostic mechanisms. We devote our discussion to this approach, which we outline below.

SIP views of parenting risk posit factors that are relevant to parents with MH problems (Azar & Twentyman 1986; Azar & Weinzierl, 2005; Crandall et al., 2015; Crittenden, 1993; Milner, 2003). Social cognitive and neurocognitive factors are linked to parenting and general social functioning problems. For example, cognitive impairments are a feature of psychotic disorders and have negative consequences for daily functioning (Fett et al., 2011). In the SIP model that has guided our parenting work (see Figure 13.1), three elements are a source of parenting risk and also occur in psychological disorders. These include (a) rigid and inflexible role schema, (b) executive function (EF) capacities, and (c) appraisals, which are the products of the first two. Problems in these factors create inappropriate and less responsive parenting behavior and are detrimental to children's outcomes (e.g., lack of adequate attention to supervision, safety, and hygiene) (Azar, & Weinzierl, 2005). Problems are associated with many diagnostic categories.

Schema (Expectations/Standards). Schemas are like filters, determining what information about social interactions, and the contexts in which they occur, enters the cognitive system. Parental expectations for children reflect schemas of children, including their role in relation to others (e.g., parents expect a child as young

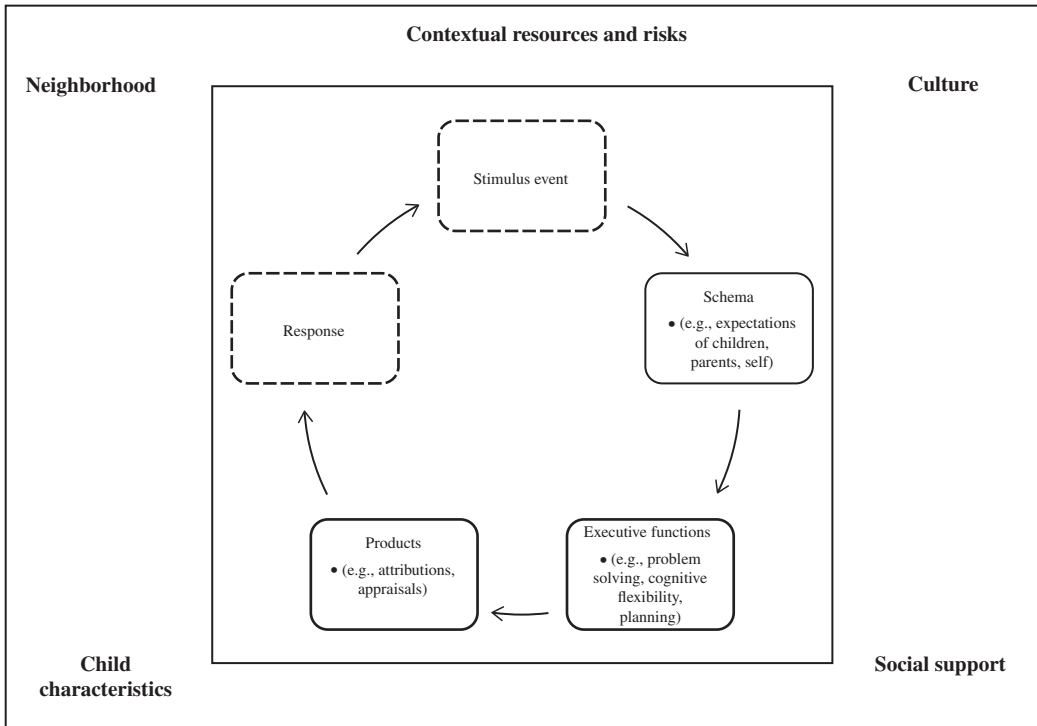


Figure 13.1 Social informational processing (SIP) model

as 18 months old to comfort them when they are sad) and what *should* happen in everyday situations (e.g., usually a four-year-old can choose the right clothing for the weather). Overly simplistic, rigid, or developmentally inappropriate child-related schema, or ones that fail to differentiate the internal construction of children from that of adults, may create parenting risk. Parents with SMI may fail to actively attend to environmental cues and changes in their children's developmental capacities (Azar et al., 2016). Maltreating mothers show higher levels of unrealistic expectations of children and these are linked to dysfunctional responses to children (Azar et al., 1984, 2012). We found such expectations are associated with maladaptive injury prevention beliefs (Azar et al., 2016).

Unrealistic expectations are linked to the quality of adult relationships in those with

MH problems (Beck et al., 1979; deJong & Daniels, 2020; Ellis, 1962). Individuals have assumptions for how family members and other social partners "should" behave and unrealistic expectations are linked to interpersonal disengagement and active relational disruptions (e.g., poor communication patterns) (Eidelson & Epstein, 1982). Wright and Roloff (2015) found that unrealistic expectations regarding mind-reading (i.e., that a romantic partner should know their partner's thoughts and feelings without the partner communicating them) were associated with problematic couple interactions among young adults, including increased conflict and withdrawal during conflict. Similarly, Hall (2014) reported that less flexible expectations regarding friendships were positively associated with withdrawal and avoidance during conflict and less friendship satisfaction. Thus,

unrealistic expectations may affect parenting quality by reducing support parents receive from other adults.

Executive functions (EF) and Problem Solving (PS). Impairments in EF capacities and PS are common in individuals with schizophrenia, depression, ADHD, and substance use disorders. EF is the foundation of higher-order cognitive skills and involves the ability to perceive situations in complex and flexible ways to identify problems, self-regulate, prioritize, enact responses and evaluate outcomes, and adjust behavior when responses are not effective. EF capacities allow for trial and error learning as one interacts socially, and they shift responses as children's developmental capacities change or are affected by environmental factors or physical states (e.g., illnesses). Adapting behavior in response to changes is crucial to child safety and supervision (e.g., installing barriers with greater child mobility), and are key to parents accomplishing childcare (Azar et al., 2012; Azar et al., 2016; Deater-Deckard, 2014).

Problem-solving deficits are linked to impairments in social functioning crucial to other functions (e.g., attaining and sustaining employment) and are deficient in various clinical samples (Nezu, 2004). In the SIP model these deficits lead to judgment errors by parents (e.g., poor problem identification and alternative solution generation; Hansen et al., 1989). Cognitive inflexibility and less cognitive complexity are two EF deficits that might contribute to impaired parenting and increase child risk (Azar & Weinzierl, 2005). Flexible behavior depends on our ability to cope with distracting stimuli that can interfere with the attainment of goals. Home settings and child caregiving often involve coping with distracting stimuli. Emotional distracters interfere with ongoing cognitive processes and involve the interaction between a dorsal neural system associated with "cold"

executive processing and a ventral system associated with "hot" emotional processing (Zelazo, 2020).

Parents need to approach parenting with complex decision-making skills, attending to detailed cues at multiple levels, selecting the most useful ones, and incorporating adequate levels of information before reaching a decision. Parents with less complex decision-making strategies may generate fewer responses and/or make inappropriate choices (e.g., fail to notice their child is feverish before sending them to school). Indeed, a less conscientious style is related to inadequate supervision and more child injuries among a non-clinical sample of mothers (Morrongiello & House, 2004). Mothers who have maltreated children exhibit poorer cognitive flexibility than non-maltreating mothers (Azar et al., 2012; Fontaine & Nolin, 2012) and poorer problem solving in caretaking situations (Azar et al., 1984, 2012; Hansen et al., 1989).

Mothers with CPS histories show poorer performance on neuropsychological tests that measure working memory, attention, cognitive inflexibility, planning, and problem solving (Azar et al., 2012; Azar et al., 2016). Slow processing speed may diminish responsiveness in threat situations. These problems are linked to more dyssynchronous mother-child interactions (child ages 3–5) (Azar, McGuier, & Britt, in submission). Teen mothers perform more poorly than adults on tasks of cognitive flexibility and are less sensitive in their infant interactions (Chico et al., 2014). Parenting at a younger age is linked to neglect. Parents with poor inhibition may have difficulty sustaining attention to children when faced with competing stimuli, and poor cognitive flexibility may hinder shifts in responses as children develop (Johnston et al., 2012).

Appraisals. Appraisals are the products of the first two model elements. When expectancies are violated, an appraisal of this

discrepancy must be generated, influencing thoughts, emotions, and responses. When children fail to meet their expectancies, parents must seek an explanation (an attribution). EF deficits may impede modifying expectations in response to child cues, and maladaptive appraisals are likely when discrepancies occur. Parenting can be frustrating and unrewarding. This may lead to helplessness and withdrawal from the functions of their parental role, especially when they blame their own skills. To protect the self, they may also see the child as purposefully withholding appropriate responses to frustrate them and may make negative attributions to normal child behavior (e.g., infant crying) and respond aversively or avoid caretaking. One study found negative child attributions in neglectful mothers (Larrance & Twentyman, 1983), and in another study we have shown that inaccurate emotion appraisals in auditory and sensory modalities are associated with dyssynchronous interactions between mothers and their preschool children (McGuier, Azar, & Britt, in submission).

As the SIP model posits, these three SIP elements in the parenting domain are associated with maladaptive parenting (e.g., poorer physical childcare and supervision, coercive responses, less use of explanation, and more desynchrony in parent-child interactions; Azar, et al., 1988; Azar et al., 2012; Azar & Weinzierl, 2005). These cognitive problems are associated with affective dysregulation in parenting, further increasing child and social problems.

Problems in all three areas are common in psychopathology. For example, depression is associated with negative appraisal biases. Schizophrenia, depression, and PTSD are associated with appraisal difficulties and mislabeling of affective and behavioral social cues. Thus, they provide an understanding of mechanisms that link MI and child outcomes.

Implications for Interventions

This cognitive work has implications for intervention at multiple ecological levels, from decreasing stigma through legal mandates to building supports in employment to direct parenting interventions. Those in leadership positions can influence policy and procedures for parents with MI, promote collaborative case management with governmental and non-profit organizations and remove cross-system barriers. Efforts to change biases in service providers and employment have not yet been evaluated. Suggestions include value assessment and discussions, such as exposing staff to role models and ongoing supervision and training (McDowell et al., 2003). Cognitive re-evaluation strategies might reduce the impact of stereotypes on professional responses (e.g., people with mental illness are dangerous) (Azar, 2000). Managers must promote openness and communication regarding values for the workplace, which has shown to produce organizational change regarding diversity (Nybell & Gray, 2004), and counter negative automatized responses to MI in parents and the belief that their children have problems too (e.g., Dowdney & Skuse, 1993). Reductions in stereotypes by increasing intergroup contact has empirical support (Pettigrew & Tropp, 2006). However, the role model and contact approaches failed to show differences in one recent study (Mittal et al., 2020).

In constructing interventions, points of entry are key. Parents may not seek or resist intervention, fearing the involvement of child protection, or may be reluctant to acknowledge the effects of their illness on their children and depress engagement. Children presenting for mental health services have a high rate of parental MH problems and their parents were more amenable to services when the child is the identified patient (Campbell

et al., 2020). More work has been given to some disorders (e.g., depression) than others. Interventions that target parenting in people with psychosis are rare, despite the extensive parental needs (see reviews: Gearing et al. 2012). Treatment for some disorders involves mental health courts that apply a team approach, which integrates multiple disciplines as intervention agents and involve longer-term commitment with a reduction in punitive responses for relapses (e.g., infant mental health courts, Stacks, Baron, & Wong, 2019).

Interventions have occurred in the three domains covered in this chapter: *survival issues*, promotion of *capacities to support developmental needs regarding the child (parenting skills)*, and the handling of *hazardous tasks* in caring for children (e.g., reducing risks for child household injuries). Interventions addressing social cognitive and neurocognitive foundations that need specific attention. We offer some examples of how these issues are being addressed in the field. We will highlight prevention efforts on intergenerational risks where offspring of MI parents are the target of intervention.

Survival. Both societal and person-based supports are needed. Employment obstacles must be addressed. Societal-level efforts are partially addressed in the Americans with Disabilities Act (ADA, 1990), which bars discrimination based on mental disorder and requires reasonable accommodations. It protects parents with MI from discriminatory practices in employment but is rarely applied to child custody cases. Professional caregivers (i.e., coaches or service providers for parents with MI), employers, and parents themselves need to be knowledgeable of ADA. Reports, such as the *Rocking the Cradle* report (2012), delineate important provisions where better implementation of ADA's tenets is needed. Formal organizational change is required which reduces pessimism and rejection regarding individuals with MI (e.g., labeled

as "therapeutic disdain" and also found in employment) (Bender, 1993).

To fully implement ADA, human capacity building is needed (Azar & Read, 2005; Azar et al., 2009; Azar et al., 2012). Equipping employers with skill sets and hiring new staff with training on mental health issues are also needed. Human capacity building involves increasing community service providers' capacities to serve families of MI with earmarked funding. Reducing automatic beliefs in incapacities of individuals with MI histories is important. Adaptations in workplace procedures and practices are necessary too. The CDC's Workplace Health Resource Center (see CDC, 2018) aims to bring awareness to mental health related issues in the workplace and provides examples of programs, including assessing employees' attitudes and well-being with anonymous surveys about mental health, outlining strategies to reduce stigma, and normalizing discussion of mental health. For example, workplaces might employ a clinical psychologist to provide support for professional and personal growth opportunities through employee goal setting, one-on-one coaching, and development sessions. Other suggestions comprise providing resources to employees, including disseminating mental health information through infographics, parent support groups, weekly table tents, and educational and storytelling videos. The CDC describes economic supports, such as income supplements, improving paid-time-off policies, providing free mental health services, and improving insurance coverage of mental and behavioral health (CDC, 2018).

Promoting parenting capacities (developmental focus). Evidenced-based behavioral and cognitive behavioral parenting approaches that address parenting problems in parents with MI (e.g., Parent Child Interaction Therapy (Timmer et al., 2011), Incredible Years (IY) (Seabra-Santos et al., 2016)) and

fine-tune parent–infant interaction (e.g., attachment approaches) enhance parenting (see Azar & Maggi, 2016). Although these parenting treatments reduce hostile interactions, and increase positive interactions, parenting practices, and emotional availability, they might require some adaptations. How interventions are conducted needs to be considered by therapists for adaptation. For example, as noted in depression earlier, MI may reduce engagement or increase cognitive obstacles to learning new parenting strategies. Slow cognitive processing, a characteristic of many disorders, may require breaking material into smaller bits and addressing appraisal biases. Parent–child interaction therapy is successful with depressed mothers (Timmer et al., 2011). Depressed mothers reported higher levels of child behavior problems than did non-depressed mothers and after treatment there were decreases in child behavior problems. However, depressed mothers more often withdrew from therapy before completing treatment. Thus, improving engagement is required, such as increasing praise as new skills are achieved. Other engagement strategies such as motivational interviewing, patient education, self-management supports, personalized care planning, shared decision-making, and family or peer supports have also been described as useful (Meneara, et al., 2020; Sampson et al., 2013).

Cognitive impairment of parents with MI has been targeted through cognitive remediation therapies (e.g., vocational rehabilitation) (Wexler & Bell, 2005). Changing social cognitive capacities such as problem-solving training, stress management strategies, and cognitive restructuring, are important for improving parenting (Azar, 1997; Azar & Wolfe, 2006). Behavioral training to improve cognitive processes (attention, memory, executive function, social cognitive, metacognition) involves anger management, self-control training, and stress management, and improvements in these skills

reduce child maltreatment. Although effects are small, treatment with a therapist and psychosocial rehabilitation produce larger effects.

Cognitive compensatory approaches are delivered as stand-alone treatments and as part of cognitive remediation and/or psychosocial interventions, with improved community functioning as a target (Allot et al, 2020). Compensatory techniques aim to compensate or circumvent cognitive deficit, with reliance on intact cognitive strategies and supports for working *around* cognitive deficits.

Addressing healthcare, injury prevention, and nutrition. Traditional parenting skills are targeted in behavioral treatment packages, as are life skills for MI populations. Some work has integrated medical care professionals (e.g., Mental Health Nurse Practitioners) into community healthcare to improve health outcomes (e.g., detection of cardiovascular disease risk, improvement of metabolic functioning) of parents with SMH (Furness, et al., 2020). This study highlights the need for healthcare professionals in preventive interventions with young parents (e.g., the Nurse Family Partnership). Nurses show better effects than social workers and paraprofessionals (Korftnacher, O'Brien, Hiatt, & Olds, 1999).

Some small-scale studies employ behavior strategies to reduce family functioning risks (e.g., meal planning, financial planning, home safety). These efforts, along with parenting interventions, may be enhanced by integrating cognitive strategies (e.g., reattribution training, cognitive rehabilitation to improve EF) and adapted services to address parents' learning capacities and ability to work cooperatively with professionals (see Azar et al., 2010). Integration of these factors into existing intervention and prevention efforts may reduce recidivism in families involved with the child welfare system. For example, executive functioning deficits are linked to incompetence on tasks like shopping (e.g., shopping for healthy

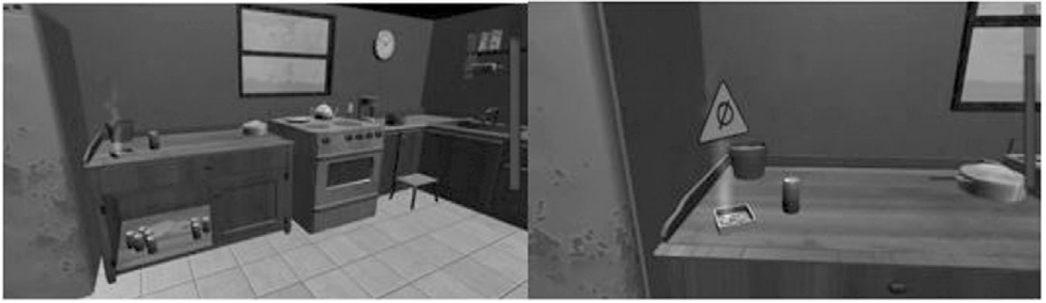


Figure 13.2 Sample of screens in Home Safety Hero game showing household room Azar et al., (2018)

food) among adults with schizophrenia (Laroi et al., 2009). Cognitive capacities may also be involved in financial planning (Willner et al., 2010) and attention should be given here as well.

Difficulties with organization and planning around food preparation, health, and home safety can be addressed. To promote understanding, acquisition, and retention of skills, skills should be broken into simple, manageable steps, and include multiple modalities, such as pictorial prompts and auditory stimuli (Azar et al., 2012a; Feldman, 1994; Wilson et al., 2013). Training in decision-making around medical care (e.g., calling the doctor or not) is used in Safe Care (Chaffin et al., 2012). Parents learn about the need for medical attention through pictures of children's faces with varying spots depicting injury or illness. Pictorial checklists of everyday tasks are posted on refrigerators to remind parents of steps being trained.

It is important that materials are sensitive to literacy difficulties and learning styles of parents with MI. Parents are able to use self-teaching materials, including visual checklists with detailed instructions and audio recordings when they are unable to read (Feldman, 2004). Information should be clear and logical to facilitate better retention. The Connecticut Parents with Cognitive Limitations Work Group (2002) provides tips for writing for these parents.

Improving injury prevention capacities should be addressed. Injury prevention work often has high attrition rates. Tailored approaches have shown some effectiveness. A preliminary effort using gaming technology to keep parents engaged and improve home safety and the reduction of risks has been developed (Azar et al., 2018). The pace of learning is slow to circumvent slow processing speeds found in parents with MI. The computer game, Home Safety Hero, targets home safety through training parents on the identification of home hazards, expanding repertoires of resolutions, and practicing under distraction (See Figure 13.2) (Azar et al., 2018). A recent small randomized control trial was carried out with teen mothers who show many of the cognitive problems seen in parents with MI. Mothers showed faster reaction times for identifying risks in a virtual household and performed better than a comparison group with less exposure to the game when faced with distraction. The game addresses engagement and integrates multiple trials to promote retention.

Preventing the Intergenerational Transmission of MI

The heritability of SMI has directed attention toward the offspring of MI parents. The cognitive style of the MI parent may intertwine

with their children's interpretation of the world and can be targeted in preventative interventions. One of the earliest evidence-based interventions for these children is *Family Talk*; it is used by various practitioners (e.g., pediatricians) and is designed to provide information about mood disorders, improve family communication about depression, and promote positive interactions and resilience (Beardslee et al., 2003). The program decreases depression in youth and has been disseminated in other countries (Finland and Costa Rica).

Cognitive behavioral (CBT) prevention programs teach cognitive strategies to offspring, such as challenging automatic negative thoughts. One study randomly assigned adolescents with subclinical levels of depression to usual care or usual care plus 15 sessions of CBT (Clarke et al., 2001). Teens learned cognitive restructuring techniques to challenge negative thoughts regarding having a depressed parent. The group receiving CBT had a lower incidence of depression after 15 months. Interpersonal skills (IP) approaches also prevent depression among adolescents. CBT and IP showed better outcomes compared to a no-treatment control group (Horowitz et al., 2007).

Some efforts are directed at the intergenerational transmission of other disorders. One such program (e.g., Families Overcoming Risk and Building Opportunities for Well Being or FORBOW) focuses on the antecedents of developing psychosis (e.g., sleep problems, cognitive delay), targets various ages, and involves training in well-being skills (Uher et al., 2014). Assessment occurs yearly to examine the short- and long-term efficacy for preventing the onset of SMI. A variety of psychosocial risk factors exist in the literature but more work is needed to derail the development of MI, utilizing what is known about antecedents of SMI.

Policy Issues

Legal issues in practice with parents with MI need discussion. In the past, certain conditions (mental illness, substance abuse, etc.) warranted the removal of children and the termination of parental rights (TPR) (Grisso, 2003). For TPR, the evidence must be "clear and convincing" and the effects of continuing the parent-child relationship must be worse than the consequences of TPR (Azar et al., 1998; Benjet et al., 2003). TPR is more common among Black, indigenous, and people of color (BIPOC) parents (e.g., 3.0 percent for Native American and 1.5 percent for African American children) (Wildeman et al., 2020). Discrimination within family law systems faces parents with MI, often resulting in the loss of custody. Three problem areas in the legal system were noted for parents with MI, including little understanding of MI and the need for assistance beyond litigation, including access to additional legal services (Powell et al., 2020). Legal education needs to address these issues. Current models of parenting are inadequate in TPR evaluations and are often based on research with specific groups, leading to bias (Azar et al., 1998; Azar & Goff, 2008). Data linking parents' skills with their capacity to parent within the contexts available to them are needed. Such a prescriptive approach helps with risk judgments, services, and/or decision making (Azar et al., 1998; Azar & Goff, 2008). The SIP framework suggests domains for assessment protocols. Future work needs to operationalize the constructs noted in validated and normal evaluation approaches.

Conclusion

Clearly, evidence of parental MI problems has been linked to negative child outcomes, with genetic transmission being one obvious

connection. However, in this chapter we argue for a more contextualized view of risk. As often discussed in the disability literature generally, we argue for viewing such parents' capacity to parent as disabled only to the extent to which the context in which they live is "disabling". This view argues for the nature of parents' psychosocial contexts being considered. This fosters the development of a system of supports that best fits parents' needs and those of their children. Our model of risk draws attention to social support and assistance with the economic stresses to which they are subjected (e.g., employment and housing biases, and crises in the community). It also highlights cognitive cross-diagnostic symptom patterns that need to be considered in interventions. Finally, two generation models of intervention that involve targeting transactions between mothers and children must also be studied to achieve better outcomes. The authors recognize these recommendations are a "tall order" when it comes to existing systems of care, but hold this as an aspirational set of recommendations which fits with the available science they have reviewed.

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14 Ethnicity and Race as a Context for Parenting: An Examination of Academic Socialization, Cultural Socialization, and Teachings about Discrimination

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Introduction

Over the past several decades, the ethnic-racial landscape of the United States has become more diverse than it has ever been in our nation's history. Twenty years ago, at the turn of the twenty-first century, 75 percent of the U.S. population was classified as non-Hispanic White, whereas roughly 12 percent each were classified as Black or Hispanic, and just under 4 percent were classified as Asian (U.S. Census Bureau, 2001). Recent census data indicate that, as of 2019, the proportion of the U.S. population that was non-Hispanic White has dropped to 60 percent, whereas the proportion of Blacks, Hispanics, and Asians has increased to 14 percent, 18 percent, and 6 percent, respectively (U.S. Census Bureau, 2019). Among children under 18, these trends have been even more pronounced. In 2018, only 50 percent of children 18 years or younger were non-Hispanic White, whereas 5 percent, 14 percent, and 25 percent were Asian, Black, or Hispanic, respectively (The Annie E. Casey Foundation, 2020b). These unprecedented demographic shifts have important implications for understanding family socialization processes both because parents themselves now come from more diverse backgrounds, and because the schools, neighborhoods, and other contexts within which their children learn are also more diverse.

On average, families of different ethnic-racial backgrounds navigate different family, economic, and social conditions. This diversity in social conditions can influence parents' beliefs about their roles as parents and how they enact them. For example, while the majority of non-Hispanic White, Asian, and Hispanic mothers of children ages 0–18 were married with a spouse present in 2019 (80 percent, 91 percent, and 71 percent respectively), only 52.6 percent of their African American counterparts were married with a spouse present (U.S. Census Bureau, 2020). Asian (29 percent), Hispanic (27 percent) and African American (26 percent) children are far more likely than are their White counterparts (16 percent) to live in multi-generational households (Pew Research Center, 2018). These trends in marital status and household composition have implications for the stresses faced and supports available to parents. For example, parents in two-parent and multi-generational households benefit from decreased financial and child care stress but also have decreased control over child-rearing decisions and practices (Generations United, 2011).

In addition to having different social backgrounds, parents of varied ethnic-racial backgrounds are rearing their children in vastly different economic environments, which can also influence their parenting practices.

The median annual family income of households was over \$90,000 for White (\$91,000) and Asian (\$106,700) children but under \$50,000 per year in the households of Hispanic (\$49,000), African American (\$41,000) and Native American (\$45,000) children (The Annie E. Casey Foundation, 2020a). Moreover, over one-third of African American children are at or below the federal poverty line, compared to 26 percent of Hispanic children and 11 percent of their White and Asian counterparts. African American and Hispanic parents are twice as likely as their White and Asian counterparts to rear their children in neighborhoods that they perceive to be unsafe, and are five times as likely as their White and Asian counterparts to rear children in high-poverty neighborhoods (The Annie E. Casey Foundation, 2020a).

Despite considerable ethnic-racial diversity in today's families, all parents share universal goals for their children. These goals include ensuring children's physical and mental safety, and well-being, and promoting successful progressions through key developmental tasks (e.g., walking, reading, friendship formation, transition into puberty). Not surprisingly, though, *how* and *how well* parents from varied ethnic-racial backgrounds are able to accomplish these goals is closely tied to their particular social, economic, and cultural contexts. For instance, in addition to the tasks that all parents manage, ethnic-racial minority parents in the United States are also tasked with navigating numerous discriminatory situations (economic, political, institutional, interpersonal) at their child's school (Hill, 2011), in their neighborhoods (Russell et al., 2018), workplaces (Triana, Jayasinghe, & Pieper, 2015), and service establishments (Brewster, Lynn, & Cocroft, 2014). These additional issues complicate parenting and, in our view, constitute a supplemental tax that ethnic-racial parents carry in raising their children.

In this chapter, our overarching goal is to highlight the current state of research knowledge regarding parenting processes in ethnic-racial minority parents. We focus on three key domains of parenting that are applicable to both childhood and adolescence: (1) promoting academic achievement and success in school, (2) promoting positive ethnic-racial identities, and (3) promoting awareness of and skills for coping with discrimination. We examine parenting in these domains because school success, positive identity development, and resistance in the face of discrimination are key protective factors in the lives of ethnic-racial minority youth.

Our chapter is embedded in the recognition that all parenting occurs in and is shaped by broader ecological contexts (Bronfenbrenner & Morris, 2006). Our conceptualization of the intersection between these ecological contexts is shown in Figure 14.1. The most distal context is a universal one, as we mentioned above: All parents strive to insure the mental and physical safety of their children and to shepherd them through key developmental tasks. Existing within this most distal context is a more local national context, which is comparable to Bronfenbrenner and Morris' conceptualization of the macrosystem. The macrosystem encompasses the overarching beliefs and values of the culture or cultures within which the developing individual exists. It provides an overarching blueprint for the ways in which other systems function. For example, in the United States, the messages that permeate all other systems include an emphasis on individual effort, educational success, and achievement. Ethnic-racial minority parents are also part of local cultural communities that shape their parenting values, beliefs, and goals, all of which are distinct in some ways from those of the dominant culture. Some of these values, beliefs, and goals are "culturally promotive," that is, they emanate

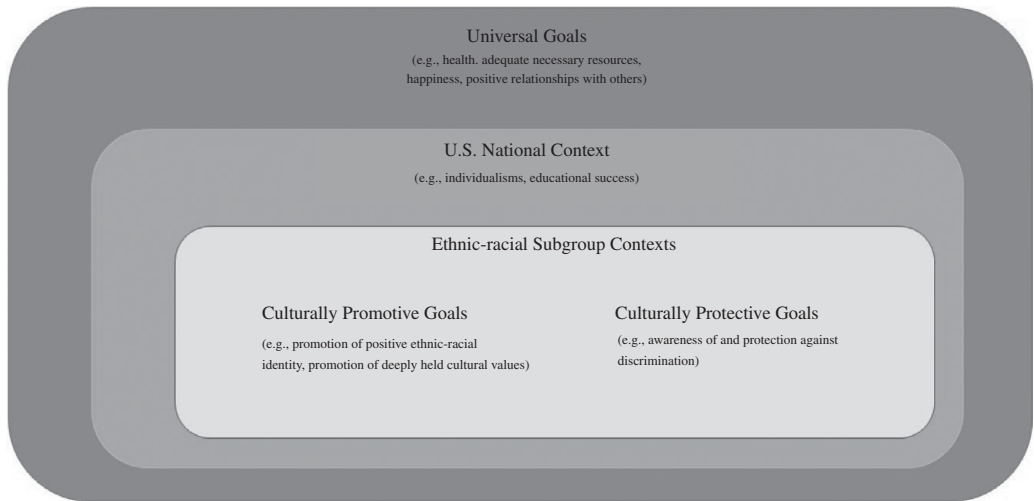


Figure 14.1 Heuristic model for examining ethnicity and race as a context for parenting

from parents' desires to promote their child's positive views of and connections with their culture of origin. Other parental values, beliefs, and goals are "culturally protective": they emanate from parents' efforts to maintain their child's health and well-being despite "threats" from the dominant culture. Academic socialization, cultural socialization, and teaching about discrimination reflect the later three levels of the ecological environment. Promoting school success is a goal that most American parents share, whereas promoting a positive ethnic-racial identity is one that is prompted by parents' desire for their children to maintain ties to their family roots. Teaching about discrimination, on the other hand, is a protective strategy that emanates from families' position in society relative to the dominant culture.

In discussing each, we first outline the importance of these domains of development for children's and adolescents' positive functioning. Then, we describe the strategies ethnic-racial minority parents employ to address them. Due to the fact that African Americans, Hispanics, and Asians constitute the largest ethnic-racial minority groups in the United States, we focus on these three

groups rather than attempting to describe issues within a diverse array of other ethnic-racial minority groups that have arrived in the United States more recently (e.g., East Indian, Arab, African/Caribbean immigrants). These three groups are also of theoretical interest to us due to patterns of commonality and distinction between them. African American and Hispanic parents are more socioeconomically similar to each other than they are to U.S. Whites and Asians and have reported common experiences with harmful stereotypes, negative media representation, discrimination, and marginalization in the United States (Way et al., 2013). However, unlike African Americans, a group with a distinctive history of racial oppression in the United States, many Hispanic and Asian groups are more recent immigrants to the United States and share processes related to acculturation and language barriers. Thus, examining common and unique processes across these groups may provide insight into the interplay between cultural and social position variables vis-à-vis varying facets of parenting.

Prior to pursuing our main focus, several caveats are in order. First, it seems important

to clarify terminology. Although the terms “ethnicity” and “race” refer to distinct phenomena, there is an emerging consensus within the literature on ethnic and racial minority families that “ethnicity” and “race” are not meaningfully distinguishable in the context of individuals’ psychological experiences (Hughes et al., 2006; Quintana, 2008). Thus, we use the hybrid term ethnicity-race in referring to groups and to dynamics such as identity development or discrimination. Second, we focus our discussion on groups as defined by panethnic categories, using the term “African American” to refer to literature on Black parents born in the United States, Hispanic to refer to immigrant and U.S.-born parents originating from Latin American countries, and Asian to refer broadly to U.S.-born or immigrant parents from East Asian countries. However, where studies have specified a national origin group (e.g., Korean versus Asian), we use the same terminology that the original author(s) used. We recognize that our reliance on panethnic categories is somewhat unsatisfying, as these categories fail to adequately capture the nuances of these groups’ experiences. For example, the practices and experiences of Cuban versus Dominican families, both of whom are categorized as Hispanic, or of Chinese versus Hmong families, both of whom are categorized as Asian, may be quite distinct. However, within the social sciences we have more empirically-based knowledge about parenting within these larger panethnic categories, although researchers are now beginning to address ethnic-racial differences within panethnic groups (Ceballo, Jocson, & Alers-Rojas, 2017). Last, we do not delve into the vast within-group variability that exists within panethnic categories according to SES, geographical area, parental education, etc., but instead focus on providing a broad-brush view of current scholarship regarding academic

socialization, identity socialization, and teachings about discrimination. With this in mind, we turn now to exploring these three issues.

Promoting Academic Achievement

Academic engagement and performance in school are key markers of adaptation and adjustment for children and adolescents in the United States, providing a foundation for a successful transition to adulthood (Wigfield et al., 2007). Unfortunately, beginning at school entry, American children’s academic performance varies considerably across ethnic-racial groups. African American and Hispanic children, on average, enter preschool with literacy, numeracy, and self-regulation skills that lag behind those of White and Asian children (Reardon & Portilla, 2016). These “racial achievement gaps,” which persist over the course of schooling, are evident on multiple indicators (e.g., grades, standardized test scores, grade retention rates, high school graduation rates, college entrance exams, college attendance rates, etc.), and are especially evident at key transition periods including transitions into kindergarten, middle school, and high school (Sutton, Langenkamp et al., 2018; Vanlaar et al., 2014).

The school contexts that ethnically-racially diverse youth encounter also vary widely. Compared to White and Asian students, African American and Hispanic students are more likely to attend under-resourced schools with less qualified teachers, even when controlling for students’ individual socioeconomic status (Diamond & Huguley, 2011; Hanushek & Rivkin, 2009). Furthermore, even in racially-integrated and/or affluent schools, Black and Hispanic students are less likely than are their White or Asian counterparts to be in advanced courses despite possessing the necessary qualifications (Lewis & Diamond, 2015). They are also more likely to encounter

unsupportive and unwelcoming classroom and school environments (Soumah & Hoover, 2013), and in varied state-wide school climate surveys have been found to report lower feelings of safety and less support from teachers than do their White and Asian counterparts *who attend the same schools* (Shukla, Konold, & Cornell, 2016; Voight, Hanson, O'Malley, & Adekanye, 2015). In addition, they face lower expectations for achievement from their teachers (Ferguson, 2007; McKown & Weinstein, 2008), harsher and more frequent disciplinary consequences (Gilliam et al., 2016; Spilt & Hughes, 2015), and racial discrimination from teachers and peers in schools (Ceballo, et al., 2017; Hughes et al., 2017). Ethnic-racial minority students with immigrant parents also face the additional challenge of navigating their language acquisition alongside their parents' limited familiarity with and lack of comfort in the U.S. school system (Garcia Coll et al., 2012). With this as background, it is critical for scholars to understand the ways in which ethnic-racial minority parents navigate issues related to children's achievement and school experiences.

Being engaged and involved in children's and adolescents' academic and school experiences, known as parent involvement, is one of the key ways that parents of all racial-ethnic backgrounds support their children's success in school. In the scholarly literature, parent involvement is commonly defined as parents' interactions with their children and/or school that promote academic success (Hill & Tyson, 2009). In most frameworks, parental involvement is conceived as a multi-faceted concept comprised of the transmission of educational goals, values, and aspirations (termed academic socialization), involvement at school (e.g., visiting the school, volunteering at school, contact with teachers and/or counselors), and involvement at home (e.g., structuring time for study, assistance with homework, taking

children to a library or museum) (Epstein & Sanders, 2002; Hill & Craft, 2003). Accordingly, in this section, we review what is known about how Black, Hispanic, and Asian parents engage with these types of involvement, the differential effects involvement has on their children's academic adjustment, and challenges that ethnic-racial minority families encounter in these processes. The different patterns of involvement and their differential relationships to children's and adolescents' well-being underscore the importance of recognizing that there are both universal and culturally specific components of parenting.

Academic socialization: educational expectations and aspirations. Holding high performance and attainment expectations for students' course grades, and high aspirations for their school completion and college attendance is among the most important ways that parents support their children's academic success. Multiple meta-analyses have demonstrated that parents' higher performance expectations and educational aspirations are positively associated with children's academic engagement and performance (e.g., Hill & Tyson, 2009; Jeynes, 2005; 2007; Wilder, 2014). Holding high educational expectations and aspirations increases parents' involvement behaviors and communicates to youth that educational success is an important norm within the family (Wilder, 2014).

Despite existing narratives that African American and Hispanic parents hold low educational expectations for their children and place little emphasis on the importance of education (Hill, 2011; Ceballo, Joeson, & Alers-Rojas, 2017), empirical data suggest that African American, Hispanic, and Asian parents all, on average, prioritize education and have high educational expectations and aspirations for their children's educational futures. Several studies using large and nationally representative U.S. samples have found that when academic

proficiency, school type, and family and personal background variables are controlled, all non-White groups in the United States have higher educational aspirations for their children compared to White Americans (Cheng & Starks, 2002; Mau & Bikos, 2000). Some studies have found that Asian parents hold higher academic expectations than all other groups (Yamamoto & Holloway, 2010). In one study that used data from the 1990 National Education Longitudinal Study (NELS), Asian (70 percent), African American (68 percent), and Hispanic (61 percent) parents were all more likely than were White parents (58.9 percent) to expect their children to complete a college degree or more (Cheng & Starks, 2022). In our work, Black, Asian, Hispanic and White parents all held high standards for their child's school-related behavior (e.g., bringing materials to class, completing homework, getting good grades) and performance (e.g., not failing a class in a major subject, finishing high school), although Hispanic parents held slightly lower short-term expectations compared to parents from other ethnic-racial groups, according to adolescent reports. For example, although 80 percent of African American, 88 percent of Asian, and 83 percent of White youth indicated that their parents would be "very upset" if they received a grade of C in a major subject, 74 percent of Hispanic youth indicated this expectation (Hughes, McFadden, Harding, Tubbs, & Sengal, 2013).

Although most parents hold high academic expectations and aspirations for their children, there is some evidence that such expectations function differently across ethnic-racial groups. In a review of the literature on academic expectations, Yamamoto and Holloway (2010) concluded that parental expectations are strongly related to student performance among White families but less so among African American, Hispanic, or Asian families. In considering mediating mechanisms,

the authors speculated that mothers from these ethnic-racial minority groups may be less effective than White mothers in communicating expectations in ways that influence children's own expectations or academic efficacy, that high expectations may be less likely to translate into increased school involvement for ethnic-racial minority parents than for White parents, and that high parental expectations may be less likely to shape teachers' expectations in ways that enhance teachers' interactions with students. Thus, for the most part, specificity within ethnic-racial groups in the structural and cultural context in which parental attitudes take form means that parental attitudes about school have differential roles across groups.

Parental practices: home and school involvement. Most studies have found that the large majority of parents employ practices to engage with their children's education and school success both at home and at school, regardless of their ethnic-racial background. However, there is substantial variation across ethnic-racial groups in how parents are behaviorally involved. According to national data, the average early educational environments of Black and Hispanic families have fewer learning materials, less language stimulation, and less frequent outside activities, as assessed by traditional measures, compared to the average educational environments in White and Asian households (Iruka, Dotterer, & Pungello, 2014). Even among low-income families, African American and Hispanic (English and non-English speaking) mothers are less likely than White mothers to endorse mainstream ideas about child development, believe that daily reading and language use with young children is important, and read to their children on a daily basis (Keels, 2009). More recent work with the ECLS-B indicated that controlling for family structure, day care quality, and immigrant status attenuated the

Black/Latino–White cognitive skill gap at the 10-month assessment and that factors such as maternal warmth, maternal reading, and maternal story telling substantially reduced, but did not eliminate, the gap at 24 months (Gibbs & Downey, 2020).

Extant literature suggests that ethnic-racial differences are also embedded in cultural belief systems as well as resources. For instance, the concept of *educación* in some Latino cultures places emphasis on parents' roles as teachers of moral development and goodness more so than on their roles as academic teachers. Also, low-income Latino parents often expose their children to rich oral discourse that includes *consejos* (advice), and family stories marked by *dichos* or traditional sayings (Cortez, 2008; Delgado-Gaitán, 1994; Espinoza-Herald, 2007; Sánchez et al., 2010; Melzi, Schick, & Scarola, 2018). This discourse helps parents and children bond, and encourages children to make independent decisions and think critically, which are both integral to preschoolers' development and continued academic success (Melzi et al., 2018).

Unfortunately, most comparative studies fail to control for maternal education or family incomes and, thus, it is difficult to disentangle socioeconomic versus ethnic-racial differences in children's early home environments. Moreover, scholars have not fully disentangled differences in early child rearing environments that are due to economic and social context, from those that are due to indigenous belief systems that differ across ethnic-racial groups. Indeed, sociologist Annette Lareau, writing about child rearing in European American and African American families, coined the terms "concerted cultivation" and "natural growth" to describe social class differences in how families approach child rearing, with poor and lower-class families allowing for more unstructured time and social interactions (natural growth) and middle-/upper-class families

operating with highly scheduled routines and activities and deliberate socialization for how to interact with adults and mainstream institutions (Lareau, 1999). In Lareau's work, Black middle-class families' approaches to child-rearing were more similar to those of White middle-class families than to those of Black poor and lower-class families.. For example, there are very few studies of early child rearing environments of the 15 percent of African American families (Black Demographics, 2020) or the 20 percent of Hispanic families (Statista, 2020) who sit squarely in the upper-middle class, earning over \$100,000 per year. Thus, attributing ethnic-racial group differences in early child rearing to ethnicity-race seems unwarranted.

Interestingly, the early child rearing environments of Chinese families in the United States and China are quite distinct from those of African American and Hispanic or even White families, regardless of maternal education or social class. In analyses of the ECLS-B, learning materials and language stimulation in the homes of Asian preschoolers were comparable to those of Black and Hispanic households (and lower than those in White households) but Asian preschoolers had significantly higher literacy and numeracy skills compared to any other group. What explains this counterintuitive pattern? Chao and colleagues argued that Chinese parents are guided by the notion of "Guan" (training children), a philosophy based on Confucian principles regarding how children are best nurtured (Chao et al., 2009). Compared to their White counterparts, Chinese parents of infants and toddlers are more likely to implement rigid schedules that include formal learning time, hands-on guidance of children's learning, and practice drills for skill development (Luo, Tamis-LeMonda, & Song, 2013). Compared to White and African American parents in the United States, Chinese parents in the

United States and China are more directive in their teaching, focusing more on the accuracy of children's answers and children's mistakes than on their successes (Pomerantz et al., 2014). Parmar, Harkness and Super (2008), in comparing how Asian immigrant and White parents were involved in their 3–6-year-old children's daily activities, observed that during play activities with children, Asian parents spent far more time than did parents from other groups on pre-academic activities such as learning letters and numbers, playing math games and working with the computer. Lower maternal educational and economic resources do not appear to dampen the emphasis on skills training that Chinese immigrant parents engage in, as high SES and low SES immigrant mothers place similar emphasis on cognitive engagement and enrichment activities (Yamamoto & Li, 2016). In an expansion of Chao et al.'s (2009) argument that Chinese mothers' behaviors are rooted in specific worldviews and belief systems, Chinese mothers are significantly more likely than are American (White or African American) mothers to endorse ideas that children's learning reflects children's morality and that parents' support of children's learning reflects parents' love and duty (Ng, Pomerantz, & Lam, 2013). They are also more likely to tie their children's performance to their own self-worth (Ng, Pomerantz, & Deng, 2014).

Once children enter formal schooling, ethnic-racial minority groups in the United States also support their children's education in a variety of ways, some of which are recognized and valued within the American mainstream and some of which are less widely recognized. In some large national data sets, African American mothers' levels of home involvement and school involvement have been comparable to those reported by White mothers (James, Rudy, & Dotterer, 2019). In other studies, African American mothers are

heavily involved in home-based activities but report lower school-based involvement compared to White mothers (Lee & Bowen, 2006; Hill, 2011; Sibley & Dearing, 2014; Wang, Hill & Hofkens, 2014; Wang & Sheikh-Khalil, 2014). Similar patterns have been found when comparing Hispanic mothers to White or Asian mothers (Lee & Bowen, 2006; Niemeyer, Wong & Westerhaus, 2008; Pong, Hao, & Gardner, 2002). In one study, first-, second-, and third-generation Hispanic mothers were less likely to be formally involved in school through participation in PTA, for example, but equally likely to have discussions about school compared to third- + generation White mothers (Pong et al., 2002). Interestingly, second-generation Hispanic mothers were significantly more likely to report PTA involvement than were first-generation mothers and third-generation Hispanic mothers, underscoring the importance of considering time in the United States when making these sorts of group comparisons.

Despite their children's high achievement, Asian American parents have been found to have lower involvement in school and at home relative to parents from other ethnic-racial groups once children begin formal schooling (Chao et al., 2009). For example, Asian parents reported lower school-based (e.g., attending a PTO meeting, volunteering in school) and home-based (visiting a library, attending a play, concert, or live show) involvement compared to their European American and African American counterparts in the High School Longitudinal Survey. Hughes et al. (2013) also found that Chinese American mothers of middle school youth and their adolescent children both reported lower involvement from mothers compared to White, African American and Hispanic (Puerto Rican and Dominican) mothers. Parents from many Asian cultures tend to endorse a clear separation of parent and

teacher responsibilities, and recognize teachers as the educational authority once their children enter school. Asian American parents who hold these values may consider it inappropriate or disrespectful to participate in the school environment, where they believe the teacher has primary responsibility for children's learning. Similar attitudes of respect have been reported by Hispanic parents (Mendez, Westerber, & Thibault, 2013; Wong & Hughes, 2006). Thus, parents' views of parental involvement may differ based on their cultural values and attitudes.

Although parents' views of parental involvement differ according to their ethnic-racial background, existing conceptualizations of the process are mainly based on the experiences of White parents in the United States. Recently, however, scholars have identified manifestations of parent involvement that are unique to ethnic-racial minority parents. In our work, for example, we found that cultural socialization and racial literacy training were important components of African American caregivers' efforts to "set the stage" for their adolescent boys' academic success (Francis, Hughes, Watford & Way, 2021). Ceballo et al. (2017) argued that in poor immigrant Hispanic families, involvement and commitment to children's education includes excusing children from work and chores so they can focus on schoolwork, limiting disruptions in small living spaces so that children can study, teaching life lessons about work available to unskilled and poorly educated people, and motivating children to do well by making sacrifices and modeling hard work. In a study of Hispanic ninth graders, Ceballo and colleagues found that gift/sacrifice – making sacrifices and modeling hard work – predicted Hispanic students' achievement values and school effort over and above measures of conventionally acknowledged facets of parent involvement (Ceballo et al., 2014).

Schools also contribute to the observed patterns of involvement among diverse ethnic-racial minority parents in a number of ways. First, as of 2015, over 80 percent of teachers in public elementary and secondary schools were White and female (NCES, 2019). Lack of ethnic-racial diversity among teachers means ethnic-racial minority students have less access to educators who "look like" and represent their families and children. Second, school teachers, administrators, and other staff respond most favorably to the cultural and social capital they share with White middle-class families (Lee and Bowen, 2006). White parents' capacity to be involved at school in ways and at times that school personnel feel necessary provides them with access to valued resources that ethnic-racial minority parents are in less of a position to develop. Thus, interactions between ethnic-racial minority parents and school staff are often stressful, combative, and unsatisfying (Hill, 2011). Studies have documented that teachers and educators tend to believe that lower-income, ethnic-racial minority, and immigrant parents are not involved in and do not value their children's education (Hill, 2011; Ceballo et al., 2017; Wilkerson & Kim, 2010) and they rate their relationships with White parents more positively than they rate their relationships with ethnic-racial minority parents (Hughes, Gleason & Zhang, 2005). Using national data, Cherng (2016) found that, first, controlling for a host of demographics alongside children's classroom performance and behavior, math teachers were more likely to contact Black and Latino parents than White parents about their children's disruptive behaviors. Second, teachers were also less likely to contact any of the ethnic-racial minority parents with news of their children's accomplishments (Cherng, 2016). Third, ethnic-racial minority parents are well aware that teachers value certain

types of behaviors, attitudes, and styles of communicating over others. Hill (2011) reviewed evidence that African American parents often feel that teachers are condescending and insensitive. Ceballo et al. (2017) described similar feelings of exclusion and disrespect from teachers among low-income Latino families, whose involvement difficulties may be compounded by language barriers and lack of familiarity with the U.S. school system. Teachers have also been found to have difficulty accepting Chinese parents' child-rearing perspectives. They resist Chinese immigrant parents' insistence that their children use English rather than their native language (Hu, Torr, & Whiteman, 2014), and their emphasis on correctness and continual improvement rather than on the joyful process of learning (Ran, 2001).

In sum, the vast majority of parents in the U.S. value education and seek to support their children's educational performance and school success. However, parents instantiate involvement practices differently across ethnic-racial groups, with African American and Hispanic parents placing less emphasis on early traditionally assessed indicators, literacy, and numeracy activities compared to U.S. White and Asian parents and more emphasis on oral story-telling, singing, expressive movement, and such. Once children are of school age, African American and Hispanic parents report being far more involved at home than at school. Asian parents are intensely involved in hands-on educational training with young children but have low involvement at home and at school relative to other groups once children are older. Traditionally, although school staff, especially teachers, primarily see and value school-based involvement, they often do not operate in ways that engage ethnic-racial minority parents. Thus, the domain of children's academic success is one in which parents pursue the same set of values but with different

beliefs and practices, due to their differing cultural values and contextual constraints.

Promoting Positive Ethnic-Racial Identities

Developing a positive ethnic-racial identity is a fundamental task for ethnic-racial minority children and adolescents in the United States. Broadly, ethnic-racial identity is the part of one's social identity that is based on membership in one or more ethnic-racial groups (Verkuyten, 2016). Numerous studies over the past three decades have focused on understanding the nature, consequences, and correlates of multiple facets of ethnic-racial identity. Scholars have sought to articulate the content, meaning, and significance of ethnic-racial identity (Sellers et al., 1998) as well as the course of ethnic-racial identity development (Phinney, 1990; Galliher, Rivas-Drake, & Dubow, 2017). Identity content includes positive feelings about one's group (private regard), awareness of others view of one's group (public regard), the significance of ethnicity-race to one's overall self-definition (centrality), and the substantive meaning of one's ethnicity-race (ideology) (Sellers et al., 1998). Ethnic-racial identity development involves movement from a status in which the meaning of ethnicity-race is unexplored and diffusion to one in which it is highly valued (Phinney, 1990). Between childhood and adolescence, most youth develop a positive ethnic-racial identity that includes favorable group attitudes, feelings of group belonging, and an increasingly clear perspective on the role of ethnicity-race in their self-definition (Vedder & van Geel, 2017). Further, youth who report more positive ethnic-racial identities also report holding more favorable outcomes across multiple domains, including better social, psychological, academic, behavioral, and relational adjustment (see Rivas-Drake, et al., 2014 for a review).

Parents play a key role in fostering children's and adolescent's positive ethnic-racial identities. Beginning at a very young age, many ethnic-racial minority parents engage in deliberate and intentional efforts to bolster children's feelings of attachment to their ethnic-racial group, to promote their internalization of group-based values and beliefs, and to ground them deeply in knowledge about their group's history and heritage. Scholars have termed this type of socialization *cultural socialization* (Hughes, et al., 2006). Cultural socialization is one of four types of ethnic-racial socialization that are commonly examined in existing studies (Hughes et al., 2006). A plethora of empirical studies have documented that children of parents who engage in more frequent cultural socialization have more favorable identity outcomes, including earlier and more frequent identity exploration, higher ethnic affirmation and private regard, greater identity centrality and more identity commitment. In the empirical literature, associations between cultural socialization and ethnic-racial identity are quite robust, having been documented in studies of African American (e.g., Hughes et al., 2009; Seaton et al., 2012), Hispanic (e.g., Christophe et al., 2020), Asian (Atkin & Yoo, 2020; Woo et al., 2020) and White (Hughes et al., 2009) families. Relationships between cultural socialization and components of ethnic-racial identity have also been documented across multiple developmental stages, from age five (Derlan Umana-Taylor, Updegraff, & Jahromi, 2017) through emerging adulthood (Nelson, Syed, Tran, Hu, & Lee, 2018), in studies based on adolescent (Hughes, Witherspoon et al., 2009) and parent (Hughes, Hagelskamp et al., 2009) reports of cultural socialization, and in cross-sectional as well as longitudinal studies (Derlan et al., 2017). In a recent meta-analysis, the effect size for relationships between cultural socialization and ethnic

identity (defined globally) was moderate to large, according to commonly used metrics (Huguley et al., 2019). Scholars have emphasized that parents' cultural socialization messages operate by focusing children's attention on their ethnic-racial group membership (Coard et al., 2004), by shaping the ways in which youth cognitively organize information about ethnicity-race (Whaley, 2003) and by providing concrete information that shapes how children perceive, interpret, or interact with others (Stevenson, 1995).

The study of parents' cultural socialization has its foundation in scholars' efforts to understand how ethnic-racial minority and immigrant parents transmit their cultural history, values and traditions to their children and, simultaneously, ensure that children acquire social skills or other characteristics necessary to function effectively in society. Thus, in contrast to the literature on academic socialization among ethnic-racial minority parents (which has focused largely on how their practices differ from those of White parents), literature on cultural socialization stems from efforts to articulate the unique child rearing demands that ethnic-racial minority parents encounter. For instance, in an early conceptualization, Boykin and Tom's (1985) described the triple quandary that African American parents face in raising their children: (1) balancing the tasks of promoting cultural pride, (2) preparing children for life in mainstream society, and (3) preparing children to deal with racism and discrimination. Knight et al. (1993) focused on how Mexican parents promote a sense of cultural connectedness in their children and Ou and McAdoo (1987) similarly, focused on how Chinese American parents promote a sense of cultural connectedness in their children. This literature on ethnic differences in cultural socialization has grown exponentially over the past 25 years. In this section, we give a broad-brush overview of what is known about

cultural socialization among African American, Hispanic, and Asian families. We begin by describing our own work that has sought to describe nuances in the manifestation of cultural socialization across multiple ethnic-racial groups. Then, we discuss how the large majority of ethnic-racial minority parents engage in cultural socialization. Finally, we describe studies that have sought to understand sources of variation in parents' cultural socialization practices.

The nature of cultural socialization.

Although cultural socialization has been defined as "parental practices that teach children about cultural heritage, ancestry, and history; that maintain and promote cultural customs and traditions; and that instill cultural, racial, and ethnic pride" (Hughes & Chen, 1999, p. 474), studies that have described the varying expressions of cultural socialization suggest that this broad concept is multifaceted and manifests in distinct ways across different ethnic-racial groups, due to the fact that ethnic-racial minority groups differ in their social positioning in the United States, and in the values and beliefs they hold close. In an early focus group study, African American parents described numerous strategies they used to connect their 4–12-year-old children with their African American heritage. These strategies included exposing children frequently to African American books, magazines, dolls, and documentaries; trips to African American museums and historical sites; discussions with children about slavery, reconstruction, Jim Crow, and the Civil Rights movement; and emphasizing the beauty of Black phenotypic characteristics such as dark skin and natural hair (Hughes and Dumont, 1993). In a later study in which we interviewed an ethnically-racially diverse sample of early adolescents and their primary caregiver, this particular manifestation of ethnic-racial socialization was much more

characteristic of African American parents than of Hispanic or Asian parents (Hughes, et al., 2008), with foundations in African American parents' awareness of the low social status of African Americans in the dominant culture, the invisibility of the contributions of African Americans in mainstream outlets, and parents' expectations that their children would likely encounter prejudice and discrimination in external settings. African American parents' cultural socialization occurred both reactively – following children's encounters with or indirect exposure to toxic stimuli – and proactively as parents anticipated racialized dynamics in the settings children navigate. Thus, African American parents' cultural socialization, whether reactive or proactive, was rooted largely in efforts to arm children with a strong sense of racial pride and cultural heritage, particularly in the context of negative messages from the larger society.

The cultural socialization practices of Hispanic and Asian parents appear, thus far, to be more rooted in goals for cultural knowledge and cultural values than in goals for arming youth to resist negative societal messages, as for African American parents. For example, Ayon, Ojeda, & Ruano (2018) found that Mexican immigrant parents' cultural socialization included exposure to food, religious practices, celebrations and community events, travel to their country of origin, use of Spanish language, family photos, and web resources. Similarly, in Hughes et al. (2008), Dominican, Puerto Rican, and Chinese parents often alluded to special events, including family gatherings, birthday parties, religious or holiday celebrations, attending festivals (e.g., Chinese New Year, Puerto Rican or Dominican Day parade), school cultural history projects, and visits back to the homeland as cultural socialization tools. Many of these practices were embedded in the family traditions inherent in being an

ethnic group member. Umaña-Taylor and Fine (2004) term this form of cultural socialization, *covert ethnic socialization*, a term that includes everyday food preparation and meals, native language use, music, dancing, conversation, media use, and the like.

Cultural socialization can also include the transmission of specific core cultural values. For instance, in Stevenson's conceptualization of proactive racial socialization, reliance on extended family and spiritual/religious coping were described as aspects of cultural socialization among African American families (Stevenson, 1994). Among Latino families, studies have examined respect (*respeto*) and familism (*familismo*) (Knight et al., 2011) while studies of both Asian and Latino families have examined the transmission and functions of a sense of family obligation (Fuligni, Yip, & Tseng, 2002; Tsai et al., 2015). In our work, Chinese parents have reported socializing their children both through communicating cultural norms such as frugality, discipline, hard work, and wholesome eating, and emphasizing the development of strong English language skills and assimilation into mainstream ways (Hughes, et al., 2008).

Regardless of its various manifestations, the overwhelming majority of ethnic-racial minority parents engage in cultural socialization with their children. In our research with different samples in diverse settings, including northeastern and mid-western urban areas and northeastern suburban towns, over 90 percent of African American, Dominican, and Puerto Rican and 64 percent of Mexican parents of children aged 6–17 reported having engaged in cultural socialization within the past 12 months (Hughes & Chen, 1999). Among studies that have compared cultural socialization across ethnic-racial minority groups, some have documented group differences in the frequency of socialization whereas other studies have not found such differences.

For instance, African American parents reported more cultural socialization compared to Hispanic parents in several investigations (e.g., Else-Quest & Morse, 2015; French, Coleman, & DiLorenzo, 2013; Hughes, 2003). In a study conducted in New York and Chicago, however, African American, Dominican, and Puerto Rican parents did not differ reliably in the likelihood of reporting cultural socialization (Hughes & Chen, 1999). Asian parents' reports of cultural socialization have also been found to be similar to those of African American parents, according to parents or youths' reports in some studies (Else-Quest & Morse, 2015; French et al., 2013). In our work in New York City, however, Chinese American early adolescents and their parents have reported less frequent cultural socialization, compared to their African American, Dominican, and Puerto Rican counterparts (Harding, Hughes, & Way, 2017; Hughes et al., 2008; Rivas-Drake, et al., 2009). It may be that the local context is quite important in determining whether there are ethnic-racial differences in cultural socialization.

Perhaps not surprisingly, cultural socialization processes are more prevalent and more of a priority for ethnic-racial minority parents as compared to White parents (Harding et al., 2017; Hughes et al., 2008). In all studies that have included White parents, they have been found to engage in less-frequent cultural socialization compared to Asian and Hispanic families (Huynh & Fuligni, 2008; Kiang & Fuligni, 2009) as well African American families (Hughes, Witherspoon et al., 2009; Rivas-Drake, Hughes, & Way, 2009). In one study (Harding et al., 2017), parents of early adolescents from diverse ethnic-racial backgrounds completed a card sorting task in which they rank-ordered their goals across four domains (academics, peers, proper demeanor, ethnicity-race) from most to

least important. Then, they ranked the importance of contrasting goals within each domain (e.g., in the academic domain, they ranked performance versus learning goals). In examining the overall importance of domains, African American mothers gave a significantly higher ranking to ethnicity-race as a domain of importance compared to White mothers. Dominican and Chinese mothers did not differ reliably from either group. Even more interesting, within the ethnicity-race domain, African American and Latino (Puerto Rican and Dominican) mothers were significantly more likely to choose cultural socialization as being more important than promoting egalitarian views whereas the reverse was true for White mothers as well as for Chinese mothers. In fact, over half of the White parents in the sample ranked the domain of teaching children about race as least important and, within the domain, they were 15 times less likely than African Americans and 11 times less likely than Hispanics to choose cultural socialization as a goal over promoting egalitarian or color-blind views.

Cultural versus contextual predictors of parents' cultural socialization. Considering how cultural socialization functions in promoting positive ethnic-racial identities among ethnic-racial minority youth, questions about what factors prompt caregivers to engage in cultural socialization are both theoretically and practically significant. Theoretically, comparing the frequency of cultural socialization across ethnic-racial minority groups may provide information about the desirability and feasibility of assimilation into the dominant U.S. culture at a group level. Practically, examining the frequency of cultural socialization across groups can suggest potential levers for shifting parents' cultural socialization behaviors. Not surprisingly, given that parents' experiences in their families of origin shape many aspects of their subsequent parenting

(Spilman et al., 2013), one factor that has been associated with parents' more frequent cultural socialization is having been reared in a context in which they received more cultural socialization (Hughes & Chen, 1997; Kurtz-Costes et al., 2019; White-Johnson, Ford, & Sellers, 2010; Williams et al., 2019). This finding has been documented not only in retrospective correlational studies (e.g., Hughes & Chen, 1997) but in studies that use independent reports from family members as well. Among African American families, parents' memories of having received cultural pride socialization predicted their children's reports of received pride socialization (Kurtz-Costes, 2019). Among Mexican families, grandmothers' reports of their cultural socialization of mothers predicted mothers' reports of their socialization of children (Williams et al., 2019). Thus, cultural socialization appears to be intergenerationally transmitted.

Parents who hold a strong, positive ethnic-racial identity also report providing more frequent cultural socialization to their children. Identity has emerged as a particularly prominent predictor of cultural socialization for Hispanic families. For example, Mexican American mothers and mothers who more strongly endorsed traditional Mexican values reported more frequent cultural socialization, which predicted adolescents' ethnic identity two years later (Knight, et al., 2011). Mexican American mothers of adolescent girls who initially reported more positive feelings about their ethnic-racial group, and mothers of boys who initially reported higher involvement in Mexican culture and higher ethnic-racial centrality, reported more cultural socialization one year later (Derlan et al. 2017; Derlan et al., 2018). Interestingly, in one study, ethnic-racial identity was associated with cultural socialization among Dominican and Puerto Rican parents but not among African American parents: African American parents

engaged in frequent cultural socialization irrespective of their own ethnic-racial identity. Hughes (2003) suggested that African American cultural socialization is, in part, driven by parents' awareness of the ubiquity and persistence of negative group images that their children are likely to encounter, as we discuss next.

Promoting Awareness of and Skills for Coping with Discrimination

The National Academy of Sciences defines ethnic-racial discrimination as differential treatment on the basis of ethnicity-race or on the basis of inadequately justified factors other than ethnicity-race that disadvantage an ethnic-racial group (National Research Council, 2004). The majority of adolescents, especially ethnic-racial minority adolescents, report having been exposed to ethnic-racial discrimination (Hughes et al., 2017). In some cases, it is overt and explicit; in other cases, it is subtle. It occurs across multiple settings including school, neighborhood, and extracurricular settings (Flanagan et al., 2009) and from various sources including peers, familiar adults, and unfamiliar adults (Benner & Graham, 2013; Del Toro, Hughes, & Way, 2020). A recent national survey found that 81 percent of Black, 71 percent of Hispanic, 75 percent of Asian, and 45 percent of White youth aged 15–24 reported having either witnessed or experienced discrimination within the past year (Jones et al., 2018). Discrimination is a negative, stressful, and toxic experience for youth, and experiencing it more frequently has been associated with a host of adverse outcomes including poorer psychological, health, and academic adjustment outcomes (Hughes et al., 2016).

Ethnic-racial minority parents are acutely aware of the existence of ethnic-racial discrimination and the likelihood that their children may encounter it (Hannah-Jones, 2016). Some parents take this awareness as a cue to prepare

their children to correctly interpret and handle experiences of discrimination, including various micro-aggressions. Other parents hope and pray that their children will not encounter discrimination, and only have conversations with them about it if and when they learn that such discrimination has occurred. Thus, teachings about discrimination occurs both proactively, in anticipation of experiences that parents expect their adolescents to have, and reactively, following instances of discrimination that require parental guidance or intervention (Hughes & Chen, 1999; Hughes et al., 2008; Stevenson, Reed, & Bodison, 1996). However, in many families, including ethnic-racial minority families, teaching about discrimination does not occur at all. Studies of the prevalence of such teachings have yielded a wide range of results, depending on the sample and how questions have been asked. Among a sample of about 500 Black families in Prince Georges County Maryland, 47 percent of parents reported having conversations with their children about the likelihood of discrimination (Peck, Brodish, Malanchuk, Banerjee, & Eccles, 2014). Similarly, in a sample of 444 Chinese American families, 51 percent of mothers and 56 percent of fathers reported that they sometimes or often talked to their children about how being Chinese meant they had to do better in school than other people in order to achieve comparable success in the future (Benner & Kim, 2009). Although the majority of Puerto Rican, Dominican, African American, and Chinese parents in our various research projects have also reported some discussions with their children about discrimination (Hughes & Chen, 1999), parents are much more likely to report transmitting cultural socialization messages, as described above. In addition, in the few comparative studies of ethnic-racial socialization, African American parents have been much more likely to report teaching about

discrimination compared to parents from other ethnic-racial minority groups (Else-Quest & Morse, 2015; Hughes, 2003; 2008; Rivas-Drake et al., 2009) and compared to White parents (Hughes et al., 2009).

In contrast to parents' cultural socialization practices, literature on parents' teachings about discrimination is complicated. For one, whereas the idea that ethnic-racial pride and knowledge about one's own history and heritage is beneficial for children, the basic question about whether parents should or should not incorporate discussions about discrimination into their overarching socialization practices is hotly contested. Opponents argue that talking to children about possible ethnic-racial discrimination calls attention to differences that are unimportant and leads to racial antagonism, divisiveness, and raced-based stereotyping, whereas proponents argue that explicitly and intentionally preparing children of color to navigate existing racial realities is the only way to parent responsibly (Hughes et al., 2016). In addition, research studies do not provide clear answers as to whether preparing ethnic-racial minority youth for experiences with discrimination is beneficial for them, with studies finding positive, negative, and no relationships between the prevalence and/or frequency of such parental discussions and a range of psychosocial, health, and relational adjustment outcomes (Hughes et al., 2006; Umana-Taylor & Hill, 2020). Finally, having discussions about ethnic-racial discrimination with children is uncomfortable and difficult for parents. In our studies, African American, Hispanic, Asian, and White parents were much more likely to indicate that they believed it was important to teach children about the realities of discrimination than they were to actually do so in practice (Hughes, et al., 2008). Thus, teaching about discrimination in ways that are helpful to children is neither straightforward nor easy.

Parents' attitudes and comfort levels are not the only factors that play into the effectiveness

of parents' teachings about discrimination. Children bring their own attitudes and dispositions to the socialization process. For one, children and adolescents rarely disclose their experiences of discrimination with their parents either because they expect negative parental reactions or because children want to protect their parents from worrying about them (Juan & Syed, 2014). In our qualitative research, we have found that many parents whose early adolescents reported experiencing and/or witnessing ethnic-racial discrimination believe that their children know nothing about such discrimination or are too young to have conversations about it (Hughes, et al., 2008). Moreover, when these conversations do occur, children and adolescents can miss, misinterpret, or ignore them. In several studies using both interview and survey-based methods, we have found that the correspondence between messages about discrimination that parents believe they send and those that their children report receiving is quite low. In one study of an ethnically-racially diverse sample of second and fourth grade children, fewer than half of the parent-child dyads agreed on the frequency of conversations about discrimination. In the majority of cases where there was disagreement, children reported hearing more messages about discrimination than the parent reported sending (Hughes et al., 2006). In a separate study of diverse adolescents, mothers' reports of discussions about discrimination with adolescents were unrelated to adolescents' own reports, especially for girls (Hughes et al., 2009).

Due to the documented difficulty parents have in talking about discrimination with children, as well as the mixed findings regarding the benefits and challenges of this sort of socialization, several projects have implemented intervention methods to empirically evaluate the effects of programs aimed at increasing parents skills for and effectiveness in discussing racial intolerance and injustice

with their children. A sampling of these programs are shown in Table 14.1. For example, researchers at the University of Pennsylvania developed the Engaging, Managing, and Bonding through Race (EMBRace) Intervention, which is a culturally based therapeutic intervention for Black youth aged 10–14 and their caregiver that focuses on enhancing African American youth and families' capacity to confront racial stress and trauma together (Anderson, McKenny & Stevenson, 2019). The New York City-based Center for Racial Justice in Education (CRJE) developed a training program entitled "Talking about race: Raising children for racial justice." The program is customized for parents, guardians, and caregivers to explore strategies for addressing race and racism with their children. Participants learn how children see race, gain skills for approaching age-appropriate conversations with a racial equity lens, and practice strategies to promote positive racial identity development in children. "One Talk a Time", developed by a multi-ethnic team of scholars at the University of North Carolina, Greensboro, is a collaborative intervention program that provides support for Latinx, Asian, African American and Black youth and their families to have conversations about race and ethnicity. The program uses a "Ready, Set, Talk" training module that is intended to develop parents skills through videos and prompts. The results of these intervention trials will provide necessary information about whether providing parents with skills to engage in conversations with their children about race relations and racial realities has consequences for youths' identity development and well-being.

In sum, then, although helping children successfully navigate ethnic-racial discrimination is a necessary area of socialization, it is also a very sensitive area that parents manage in distinct and highly personal ways. Although most studies that compare the frequency of such messages across ethnic-racial groups have

found that African American parents are more likely than Hispanic, Asian, or White parents to engage in such discussions, there is little consensus in the literature about whether such discussions are helpful for youth. Empirical relations between having more frequent conversations and negative adjustment outcomes suggests that placing too much emphasis on the possibility of experiencing discrimination may undermine youths' sense of agency and efficacy.

Summary and Conclusion

The increasing ethnic-racial diversity of the U.S. population means that a complete understanding of parenting processes among American parents requires that we consider ways in which parents from diverse ethnic-racial groups are similar and unique to each other. The concepts deemed important to study and the ways researchers understand them can no longer be predicated only on the experiences of a single group. In this chapter, we reviewed the current state of research knowledge regarding parenting among African American, Hispanic, and Asian parents in three key domains of development: (1) promoting academic achievement and success in school, (2) promoting positive ethnic-racial identities, and (3) promoting awareness of and skills for coping with discrimination. Not only are these three domains critically important for healthy development among ethnic-racial minority youth, but they represent differing ecological contexts reflecting larger macrostructural values that are common to all parents (e.g., educational success in the United States), goals that are promotive within a subcultural context in which children's retention of cultural values and pride in their group are important to parents, and goals that are protective within a subcultural context, stemming from parents awareness that children need to navigate a dominant culture that can be oppressive. Thus, we drew

Table 14.1. *Examples of parenting interventions focused on cultural socialization and teaching about discrimination*

Intervention	Description	Reference
Engaging, Managing, and Bonding through Race (EMBRace) Intervention	5-session, culturally-based therapeutic intervention for Black youth aged 10–14 and at least one caregiver that empowers African American youth and families to confront racial stress and trauma together. Provides clinical strategies and psychoeducation about racial socialization in order to enhance coping mechanisms. “Provides tools to: • Engage in bi-directional conversations on race based on psychoeducation from the RS literature • Manage RST through healthy approach coping strategies • Foster bonding between parent and child through parent–child communication and strengthening of parent–child relationship.”	https://www.recastingrace.com/research/embrace
Parent Village at PRIDE (Positive Racial Identity Development in Early Education) at PittEducation	6-week program for parents and caregivers of Black children blending “child development information with Africana history and culture by using interactive activities such as drumming and dancing. The program equips participants with the tools and knowledge necessary to have conversations and engage in activities with their children about African people, Black identity, and feeling proud of who they are.”	www.racepride.pitt.edu/parent-village/
Fathers and Sons	A program for African American fathers and their 8–12 year old sons who are not living in the same home. Programs center around 3 primary themes: effective communication; cultural awareness; and skill building. While the primary goal is to improve father–son relationships, the focus on cultural awareness relates to racial socialization.	http://prc.sph.umich.edu/projects/fathers-and-sons/

The Strong African American Families (SAAF) Program

“... a 7 session program designed for youth aged 10–14 and their caregivers. The goal of SAAF is to build on the strengths of African American families and support parents and youth during the transition from early adolescence to the teen years with a specific emphasis on helping young people avoid risky and dangerous behaviors (e.g., substance use).” A caregiver/family topic includes “Encouraging Racial Pride.”

<https://cfr.uga.edu/saaf-programs/saaf/>

Pact Family Camp

“... a weeklong summer retreat where adopted children of color of all ages—and their families—share experiences and build community while learning from experts and each other.”

www.pactadopt.org/resources/racial-identity.html

Topics relevant to racial socialization:

- Adopted children of color get support making sense of their identity and building strong self-esteem.
- The children are surrounded by role models – other people of color who are adopted or staunch adoption allies.
- All camp counselors and most featured speakers are people of color with a personal connection to adoption.

National Museum of African American History and Culture – online resources talking about race for parents or caregivers

“Each of us has a racial identity, and racial beliefs affect all of us. In America, there is still much pain, confusion, fear, and anger around racial identity; it is no wonder that families struggle with what to tell their children. How do we teach such challenging subject matter? We are all the same in our humanity, yet we can be very different when it comes to our racial identity. How can we support our children’s pride in their racial heritage, yet discourage feelings of racial superiority or inferiority? This site is here to help you answer these challenging questions and more.”

<https://nmaahc.si.edu/learn/talking-about-race/audiences/parent-caregiver>

Table 14.1. (cont.)

Intervention	Description	Reference
Center for Racial Justice in Education: Talking about race: raising children for racial justice	“This version of Talking about Race is customized for parents, guardians, and caregivers to explore strategies for addressing race and racism with their children. Participants learn how children see race, gain skills for approaching age-appropriate conversations with a racial equity lens, and practice strategies to promote positive racial identity development in children.”	https://centerracialjustice.org/trainings/
The Naja Program	Rites of passage intervention over four months and an overnight retreat for 10–12-year-old AA girls. • creative dance • other art media. Topics included: • Health and nutrition • Etiquette • Female sexual development.	Belgrave, Chase, Vaughn, Gray, Addison, & Cherry, 2000
Imani Rites of Passage Program (IROP)	Youth and family rites of passage program for Black boys aged 10–15 and their families. 15 two-hour weekly sessions that included lectures and workshops that centered around identity development. Activities included: • role plays • cultural field trips • writing exercises. The intervention also included a monthly parenting program geared towards helping parents racially socialize their children.	Whaley & McQueen, 2004

MAAT Afrocentric Adolescent and Family Rites of Passage Program	The intervention included three components: • activities to promote family enhancement • family counseling • an after-school program. All components incorporated and emphasized African American culture and heritage.	Harvey and Hill, 2004
Aban Aya Youth Project	16–21 lessons per year in school. Topics covered: • self-esteem • stress management, • interpersonal relationships, • peer pressure, • decision-making and problem-solving, • conflict resolution. A parental support program was also included simultaneously with the school-based intervention. Interventions included the Nguzo Saba principles, story-telling and proverbs, and African and African American history and literature	Flay, 2004; Liu and Flay, 2009
One Talk a Time	An online resource to provide support for Latinx, Asian, African American and Black youth and their families to have conversations about race and ethnicity through videos and prompts • empower parents with have courageous conversations • provides tools to support parents in having conversations about race with kids.	www.caminoslab.org/research

attention to elements in parenting that parents from varying ethnic-racial groups respond to in different ways based on their cultural values, and the constraints and supports in their local contexts.

As we described, promoting academic success is a universal goal shared by American parents but addressed in very different ways across groups. Some of these ways are driven by differing cultural belief systems, while others are driven by economic or logistical constraints on parents' ability to be involved in their children's schooling, and others by the nature of interactions between parents and school staff. Unlike promoting school success, promoting children's positive ethnic-racial identity is a goal that is shared more so among ethnic-racial minority parents than among American-born White parents. It emerges from intra-group processes and parents' desire for their children to retain their cultural lineage and is implemented in distinct ways across ethnic-racial groups, since cultural traditions, history, and heritage are distinct across groups as well. Finally, preparing youth to navigate situations of discrimination is a parenting process that is less widely shared across ethnic-racial minority groups, with African American parents, who have a long history of oppression and discrimination in the United States, placing a higher value on this aspect of socialization and engaging in it more frequently. Scholarly literature is less clear in this area than in others on ways that parents have such discussions productively.

In closing, it seems important to acknowledge that parenting in these domains of development likely operates synergistically. In our own work, we have found that the combination of higher parent involvement in children's schooling, embedding children deeply in their cultural history and heritage, and effectively preparing them to recognize and navigate potentially discriminatory events is associated with better outcomes among youth

overall (Francis et al., 2021; McGill, Hughes, Alicea & Way, 2012). In future work on parenting, researchers will need to explore how these key areas operate in concert across a range of ethnic-racial minority groups.

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15 The Role of Fathers in Caregiving

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Throughout most of human history, mothers' role in the family has been to provide emotional support and care for their children whereas fathers' role has been mostly instrumental, that is, to provide financial support for their children with limited involvement in day-to-day parenting (Cabrera et al., 2000). This dichotomous view minimizes the financial contribution mothers make to their families and the emotional support that fathers provide for their children. It is clear that this restrictive and reductionist view of parents has outlived its purpose. Over the past few decades, dramatic sociocultural and economic shifts in families, such as increased out-of-wedlock births and increased women participation in the labor force, have led to significant changes in both women's and men's roles in the family (U.S. Bureau of Labor Statistics, 2019).

Beginning in the 1960s, trend data show that fathers have been increasingly engaged in the direct care of their children (Bianchi et al., 2012). National data from 2011 to 2013 show that the majority of resident fathers with children under age five spent their time in caregiving (e.g., diapering/toileting), reading, and other activities, such as eating dinner together and playing, at least once a week with their child (Astone et al., 2016). And as women's participation in the labor force increased, so did fathers' day-to-day responsibility for the care of their young children. In 2016, fathers made up 17 percent

of all parents who stayed at home, representing an increase in stay-home fathers from 4 percent in 1989 to 7 percent in 2016 (Livingston, 2018). The time fathers spend with children under 6 years is divided among various activities, with most of the time spent in playing and physical care and less time spent in reading and educational activities (U.S. Bureau of Labor Statistics, 2020).

These statistics suggest that the emotional versus instrumental dichotomy of roles assigned to fathers and mothers is no longer an accurate portrayal of contemporary parenting in the United States. The erosion of traditional gender roles about who is in charge of caregiving and breadwinning raises the need to further understand how fathers are involved in childrearing and how their caregiving is related to children's developmental trajectories.

In this chapter, we synthesize the empirical evidence mostly from the United States on fathers' caregiving roles and impacts on children across developmental domains and from early childhood to adolescence. To the degree possible, we also include studies conducted in other countries to illustrate cross-cultural patterns of father caregiving. We also discuss the theoretical and methodological approaches used to conceptualize and assess fathers' contributions to children's development. Lastly, we draw implications from existing findings to guide policies and programs targeting families and children.

Fathers in the United States: Demographic Considerations

Most fathers in the United States live with their children under 18 years (U.S. Census Bureau Current Population Survey, 2020). Even though the proportion of children under age 18 living in two-parent families has been decreasing since the 1960s, in 2020 the majority of children lived in two-parent households with their biological/adoptive married parents (U.S. Census Bureau Current Population Survey, 2020). Of the proportion of children living in single-parent households in 2020, most (82 percent) lived with their mothers and only a small proportion (18 percent) lived with their fathers (U.S. Census Bureau Current Population Survey, 2020).

Approximately, 25 percent of American children do not live with their fathers (U.S. Census Bureau Current Population Survey, 2020). In 2020, 79 percent of non-Hispanic White children lived in married or cohabiting two-parent (biological or adoptive) families compared to 68 percent of Hispanic and 41 percent of Black children (U.S. Census Bureau Current Population Survey, 2020).

It is important to underscore that father non-residency does not necessarily mean non-involvement. The formation and dissolution of Black and Latino families is historically rooted so that the adaptation of what it means to be a family is different for other groups (McAdoo, 1991; McAdoo & McAdoo, 1997; Manning & Brown, 2006). Marriage is considered the most stable form of family because it provides children with stability and economic security, which, in turn, ensure optimal wellbeing for children. But marriage is most beneficial for White, not for Latino or Black families (Manning & Brown, 2006). Married White families tend to be of higher educational levels than cohabiting White parents, but among Latinos and Black families, educational levels

are similar for married and cohabiting couples, making marriage less economically beneficial for Latino and Black children.

Theoretical Approaches

In contrast to research on the influence of mothers on children that has been dominated by attachment theory, research on fathers' influence on children has been either atheoretical, that is not guided by any particular theory, or driven by a variety of content-specific micro-theories (Cabrera, Fitzgerald, et al., 2007; Cabrera, Fitzgerald, et al., 2014). The relative lack of attention to theoretical framing in understanding father effects on children's development reflects, in part, the multidisciplinary nature of the research on fathers as well as the "catching up" to research on mothers approach taken especially by scholars examining the impact of parenting on children's development (Schoppe-Sullivan & Fagan, 2020). Theories and conceptual models are important because they enable researchers to delineate how patterns of parent-child interactions and caregiving influence the development of children's language, cognitive, and social-emotional skills (Grusec, 2017). The most commonly used theoretical frameworks to study father caregiving are the bioecological model and family systems theories (Bronfenbrenner & Morris, 2006; Cox & Paley, 1997). Other models that consider the multiple contextual influences on parent-child interactions include parenting process models and models that derive from attachment theory (Belsky, 1984; Cabrera et al., 2007, 2014; Paquette, 2004). In addition to these models, parenting research, more broadly, has been framed by principles or core ideas that include global versus specific parenting behaviors, domains of parental engagement, and the timing of early experiences, including parenting behaviors (e.g.,

Baumrind, 1967; Bornstein, 2006; Sameroff, 2010; Wachs & Chan, 1986).

Overall, father–child interactions, and caregiving more generally, have been studied using multiple theoretical perspectives that share one common principle: fathers impact their children’s development in the context of a network of relationships embedded in particular socio-cultural contexts through direct and indirect effects on children (Cabrera, Fitzgerald, et al., 2014; Coley, 2003; Suizzo et al., 2017; Verhoeven et al., 2012). These relationships are dynamic, complex, and change over time.

Methodological Approaches

The study of how fathers influence children’s development has also used a variety of methodological approaches and measures, including quantitative (e.g., survey) and qualitative (e.g., focus groups, semi-structured interviews) methodologies. Methodological and conceptual challenges in conducting research on fathers include the *identification of fathers* (i.e., who is the father of the focal child), *recruitment and retention of father* participants into research studies and programs (Mitchell et al., 2007), and the lack of consensus on theoretical framing about *how fathers should be involved with their children* to foster their development. The various ways to define father involvement (e.g., instrumental versus emotional support) have resulted in a variety of survey instruments that makes comparison of findings across studies arduous, at best.

Early studies of the impact of fathering behaviors on children’s development relied on mothers as proxy respondents for fathers. In contrast, more recent studies have relied on fathers themselves to tell their own stories (Cabrera, Fitzgerald, et al., 2007). The methodological shift from maternal perceptions of father involvement to fathers’ reports of their own involvement has been the single most

important innovation on how researchers collect data on fathers (Cabrera et al., 2002). Unfortunately, it is not the norm. Maternal perceptions of father involvement are still prevalent even though they only provide information about maternal perceptions of fathers rather than about father caregiving. It is important that fathers’ own voices and views about their parenting behaviors and beliefs be included in studies of fathers and parenting. If this is not possible, it behooves researchers to clearly use “parenting” only when both parents are included and “maternal parenting” when only mothers are included.

A common approach to studying fathers is the use of small-scale studies, which are typically based on samples of convenience that may underrepresent low-income and ethnic and racial minority fathers, and mostly include maternal perceptions of fathers’ behaviors. On the other hand, the availability of nationally representative data sets has been an invaluable resource to the field because they enable researchers to conduct studies that generalize to the larger population. These studies are especially helpful to policymakers interested in promoting father involvement in a way that is beneficial for children. Although there is some overlap, large-scale studies in the United States can be categorized according to two foci, those that focus on *becoming a dad* (e.g., the National Survey of Family Growth) and those that focus on *being a dad* (e.g., the Early Childhood Longitudinal Study–Birth Cohort) (Cabrera et al., 2004).

Correlates of Father Caregiving

One of the most salient findings in the fatherhood literature is the heterogeneity in how fathers are involved in their children’s lives (Cabrera, Hofferth, et al., 2011). Some fathers are highly involved and invested in the day-to-day caregiving of their children whereas others

are less so. A considerable body of research has found that fathers' caregiving varies across a number of individuals, family, and community and cultural dimensions. These factors influence fathers' involvement and parenting behaviors as hypothesized by the parenting process model (Belsky, 1984) and the heuristic model of the dynamics of paternal influence on children (Cabrera, Fitzgerald, et al., 2014; Cabrera, Fitzgerald, et al., 2007). Here, we highlight research on the most commonly investigated covariates of father involvement: sociodemographic (e.g., family structure, education, income), psychological (e.g., depression), and neurobiological (e.g., levels of testosterone) characteristics.

Multiple studies have shown that fathers' residential status and education are significantly related to fathers' expenditures of time and money on their children. Married fathers and those with a high school diploma are more frequently engaged in literacy activities with their toddlers than unmarried fathers and those without a high school diploma (Cabrera, Hofferth, et al., 2011; Cabrera et al., 2008; Leavell et al., 2012). A small-scale study of low-income African American and Latino fathers found that fathers with less than a high school education were more likely to talk less often to their children and use lower-quality language input during their interactions with children (Malin et al., 2012).

Although fathers' resources, such as education and family structure, are significant contributors of father involvement, there are other characteristics that trump parents' resources. Controlling for fathers' education, mental health, and family structure, a study with a national sample of ethnic minority fathers found that African American and Latino fathers reported more frequent engagement in caregiving and physical play activities than White fathers (Cabrera, Hofferth, et al., 2011). These findings suggest that fathers'

behaviors might be rooted in cultural expectations and beliefs about their role in children's lives. Although studies about how culturally specific values and beliefs motivate fathers to be involved is quite scarce (e.g., Aldoney & Cabrera, 2016), there is empirical evidence that supports this hypothesis. Leavell and colleagues (2012) found that compared to White fathers of sons, African American and Latino fathers of sons reported the highest levels of engagement in caregiving (e.g., put child to bed), play (e.g., play outside), and visiting activities (e.g., visit relatives). Similar patterns of involvement favoring ethnic minority fathers has been reported with nonresident fathers (Cabrera et al., 2008).

Parents' mental health, especially during the early years, is a significant predictor of children's maladjustment because depressed parents are less likely to be responsive and attentive to their children's needs during this critical time of development (Nuttall et al., 2019; Paulson et al., 2011). Little is known, however, about the effects of paternal depression on children's development (Wilson & Durbin, 2010). A meta-analysis with 28 empirical studies showed that paternal depression had significant but small negative effects on parenting, with depressed fathers demonstrating decreased positive and increased negative parenting behaviors (Wilson & Durbin, 2010). Other studies, including one that used national data, found that fathers across race and ethnic groups who reported fewer depressive symptoms participated more frequently in learning activities (e.g., reading, playing, teaching) with their children at 24 months (Baker, 2014; Nuttall et al., 2019).

Over the last couple of decades, advancements in understanding the biological process of human behavior have resulted in significant insights into the neurobiology of fathers' caregiving behaviors. Most of this work suffers from the inevitable methodological constraints

common to that type of research such as using samples of convenience and cross-sectional designs. Hence, generalizability of these findings is limited. Nevertheless, there are important insights from this work that could help us better understand how to promote desirable father caregiving behaviors among fathers (Abraham & Feldman, 2018).

A significant insight from studies on hormones such as oxytocin (OT), critical for lactation and maternal care, and vasopressin (AVP), which men experience more strongly because it interacts with the male sex hormone testosterone (T), suggests that they are important for caregiving behaviors in mothers and fathers. High levels of OT and AVP are found to be reciprocally associated with more affectionate contact and more stimulatory contact with infants, respectively (Apter-Levi et al., 2014). Studies have shown that both mothers and fathers of newborns have similar and consistently high levels of OT, but the increased levels of OT elicit different parenting behaviors from mothers than from fathers (Gordon et al., 2010). Mothers are more likely to engage in affectionate contact, mutual gazing, and vocalizations with infants, whereas fathers tend to perform positive arousal and stimulatory contact with infants, such as moving the infant around the room (Gordon et al., 2010).

Levels of T have also been implicated in parenting behaviors. A study with a small sample of couples found that maternal T levels were neither stable nor predictive of mother–infant synchrony (i.e., affectionate responses to infants' social signals). In contrast, paternal T was stable across the first six months after the birth of their first child and fathers with higher levels of T displayed lower levels of father–infant synchrony (Gordon et al., 2017). Similarly, a study of Filipino fathers found that fathers who reported three hours or more of daily childcare had lower

T compared to fathers who were not involved in care (Gettler et al., 2011).

Overall, the emerging empirical evidence on fathers' and mothers' hormonal levels is mostly descriptive, indicating that caregiving hormones affect both mothers' and fathers' parenting behaviors but in distinct ways: OT and AVP promote warm and sensitive contact with the child and T is negatively associated with caregiving. These caregiving hormones shape how parents structure their social interactions with children.

Fathers' Caregiving and Children's Development

Decades of developmental science research have shown that parenting, broadly construed, is unequivocally the most significant influence on children's development, in particular in the early years (Bernier et al., 2010). In this section, we review studies examining the role of fathers' caregiving for children's cognitive, language, academic, and social-emotional skills from early childhood to adolescence. Given space considerations, we do not provide a systematic review of the literature, but rather focus on a select set of notable studies published over the last two decades and synthesize the most recent, emerging literature on how fathers' caregiving matters for child development.

Father Caregiving and Children's Cognitive and Language Skills

Cognitive Development

Specific parenting behaviors, such as reading or telling stories, as well as the quality of interactions between parents and children, that is how responsive parents are to the child's signals in a developmentally appropriate and prompt manner, are significant contributors

to children's cognitive development (Bruner & Bornstein, 1989). Contrary to popular belief, fathers are as sensitive and responsive to their children's needs as are mothers and they influence children's development in similar ways (Cabrera, Shannon, et al., 2007). During play interactions with children, fathers who are observed to be sensitive and provide high levels of cognitive stimulation (e.g., teaching or scaffolding the child's current abilities) to their toddlers have children with better cognitive scores on the Bayley Scales of Infant Development (e.g., problem-solving, early number concepts) at age three than fathers who are less sensitive and stimulating (Cabrera, Shannon, et al., 2007; Easterbrooks & Goldberg, 1984; Ryan et al., 2006; Tamis-LeMonda et al., 2004). A small-scale study of families in the United Kingdom, controlling for parents' education, socioeconomic status (SES), and family income, found similar associations between father sensitivity during parent-child play interactions at 12 months and toddlers' cognitive skills (assessed using the Bayley) at 18 months (Malmberg et al., 2016).

Fathers' caregiving is also associated with children's executive function (EF) skills, although the evidence is quite limited. The neglect of studies in this area is notable because EF skills are foundational to the development of other key cognitive skills, such as theory of mind, are malleable, and have significant implications for children's academic achievement (Carlson et al., 2004; Clark et al., 2010). Children's EF consists of three main components: working memory (holding and manipulating information in one's mind), cognitive flexibility (switching between rules and adapting thinking to different situations), and inhibitory control (stopping impulsive behaviors; for review see Carlson et al., 2013). The empirical evidence thus far shows that among low-income White

and Black parents, paternal sensitivity at 24 months, but not earlier, is related to better EF scores at age three (Towe-Goodman et al., 2014). Other studies with 30-month-old children have shown that these associations were significant only for Black fathers but not Latino fathers (Owen et al., 2013). A potential explanation is that due to high levels of employment, Latino fathers have less contact with their children compared to Black fathers, who are more likely to be unemployed but have more time to spend with children (Owen et al., 2013). The role that ethnic background plays in the development of these cognitive processes in children is really in its infancy. But these findings have implications for identifying cultural-specific parenting practices – based on parents' beliefs and norms about appropriate ways to foster and promote developmental skills – that differentially influence children's skills. In other words, strategies used by middle-class White families to promote children's development are not necessarily the only strategies or the “correct” strategies to achieve this goal. This is an area ripe for future investigation.

In contrast to how positive parenting practices are related to children's cognitive development, the way in which negative parenting undermines development, in particular among fathers, has been relatively neglected. A study with White middle-class families found that more paternal control (e.g., providing excessive help), but not autonomy-supporting behaviors (e.g., encouraging child to participate), at age three was concurrently and significantly negatively associated with lower EF skills at age three (Meuwissen & Carlson, 2015). Studies of paternal spanking are virtually nonexistent. A study with mothers and fathers found that only mothers' spanking (not fathers') had a significant association with receptive vocabulary and externalizing behaviors (MacKenzie et al., 2013). And a study of

Dutch families found that self-reported paternal harsh parenting (e.g., screaming angrily at child) at age three was related to weaker maternal-reported metacognitive (i.e., planning) and inhibitory self-control skills of children at age four (Lucassen et al., 2015).

Language Development

Language scholars frame their work within a sociocultural framework that children learn language in the context of rich and reciprocal social interactions with significant others (Vygotsky, 1978). Parents influence early language development through linguistic interactions with their children during quotidian activities, such as playing, and through engaging in literacy activities (e.g., reading, singing, telling stories). To assess the frequency and quality of literacy activities, studies use both maternal reports and observational data coded from parent–child interactions across the early years in childhood (from toddlerhood to kindergarten) (Pancsofar & Vernon-Feagans 2010; Malin et al., 2014; Reynolds et al., 2019). The frequency with which parents read to their children is a hallmark of most educational interventions in the United States (Xie et al., 2018). Thus, the finding that mothers read more often to their children than fathers is of consequence (Cabrera et al., 2020; Duursma, 2016). Studies that include mothers and fathers confirm that mothers read more often than fathers (Baker, 2014; Baker, 2013; Cabrera et al., 2020; Duursma, 2014). But these studies also find that fathers' reading frequency at 24 months, rather than mother's reading frequency, is more important for children's language skills (Cabrera et al., 2020; Duursma, 2014). The one exception is a study conducted with Finnish children – although both mothers' and fathers' literacy activities were associated with children's vocabulary,

mothers' literacy activities showed stronger associations (Lyytinen et al., 1998). These findings might reflect cohort effects as well as demographic differences and differential cross-cultural investments in early childhood.

The finding that among parents in the United States fathers' frequency of reading matters more than mothers' for children's language skills might be explained by findings that fathers use higher-quality linguistic inputs with their children than mothers. That is, fathers use what researchers have called “metalingual talk,” such as asking children to label objects in the book, recasting what children are trying to say (e.g., “no, it's not a cat, it's a dog”), and using more *wh*-questions (e.g., “where is the cat?”), which are more conversationally challenging for children and consequently a better predictor of language skills (Malin, Cabrera, & Rowe, 2014; Varghese & Wachen, 2016). Several studies have shown that during interactions with toddlers or during shared book reading, fathers tend to produce more *wh*-questions than mothers (Leech et al., 2023; Malin et al., 2014; Rowe et al., 2017).

Another line of inquiry examines whether different types of home activities elicit different types of linguistic input. When researchers compare language input during book reading versus toy play with toddlers, they find that African American and Latino father–child dyads produce significantly more word types, questions, and labels per minute during book reading than during toy play. However, fathers' mean length of utterance tends to be significantly greater during toy play (Salo et al., 2016). It is unclear, however, whether these differential inputs are related to language skills. Cabrera and colleagues (2017) found that fathers' and mothers' playfulness (i.e., levels of creativity, imagination, humor, and/or curiosity during play) in toddlerhood was

associated with children's vocabulary skills at prekindergarten. These findings open the door for future studies on whether some contexts are more likely to prime parents to talk more to their children than others, which would be informative for parenting programs to include in their curricula.

Father Caregiving and Children's Social-Emotional Development

A critical milestone of early childhood is the development of social and emotional skills that enable children to interact with others in meaningful ways. The development of these skills during the early years is significant because they set off developmental cascades that affect other skills across domains of development (Masten & Cicchetti, 2010). Children learn to be socially competent, that is, to communicate, regulate their emotions, take turns, and learn how to get along with others, when they experience consistent and positive social interactions with their caregivers starting from birth (Zeman et al., 2006; Cabrera, 2012). Parents' "serve and return" interactions are reciprocal and essential to their children's emotional experiences by providing stimulation through touch, vocalizations, and facial expressions during infancy and by engaging in sensitive, reciprocal social exchanges as children get older (Papoušek & Papoušek, 2002). At first, fathers and mothers are external regulators of infants' emotions and arousal, but as children get older, intrinsic and extrinsic processes come together to help the child monitor, evaluate, and modify emotional reactions to achieve their goals (Eisenberg & Morris, 2002; Eisenberg & Spinrad, 2004; Thompson, 1994).

Social-Emotional Skills During Early Childhood

Studies examining the role of fathers in the early development of social-emotional skills

have focused on fathers' parenting practices and the quality of their relationship with their children. As has been demonstrated with mothers, the quality of the interactions between fathers and children is robustly related to children's social adaptation (Keown, 2012; Lunkenheimer et al., 2020; Rispoli et al., 2013). In samples of low-income families, fathers who are attuned to their children's cues, try to understand what their children are communicating, and respond in sensitive ways, have children who are observed to be more sociable and emotionally regulated (Cabrera et al., 2007). Other observational studies with low-income families find that during play interactions with their toddlers, fathers who use regulatory behavior strategies (i.e., physical support and modeling) and regulatory language (e.g., direct commands) at 24 months had children with greater sustained attention and emotion regulation, respectively, at prekindergarten (Malin, Cabrera, Karberg, et al., 2014). In other words, higher quality father-child interactions and relationships are predictive of children's positive socioemotional adaptation.

Mothers and fathers' parenting strategies change over time according to their children's developmental age. As children get older and develop better language skills, fathers increasingly rely on verbal strategies to help children regulate their emotions. In a study of White, educated mothers and fathers and their three-year-olds, fathers who talked more about negative emotions and provided more causal explanations about desires had children who were better able to attribute mental states to others and to themselves (i.e., theory of mind) at age three (LaBounty et al., 2008). The attribution of mental states (e.g., "you're sad") enables children to regulate their emotions and adapt their behaviors accordingly. Similar findings have been reported with White rural families: fathers who reported

being more positive about their life and used more positive parenting practices (e.g., responsive to child's verbalizations) at ages three and four had children with higher parent-reported social competence (e.g., "helps with everyday tasks") at age five (Jeon & Neppl, 2019).

As in other areas of research, there is a dearth of research that considers how parents' beliefs, values, and norms about childrearing vary within and across cultural groups in ways that influence their social-emotional socialization practices. Studies that have compared mothers to fathers find parent gender differences in the way parents socialize their children to manage their emotions. In a cross-sectional study of Chinese families in China, Chinese fathers reported providing more controlling responses (i.e., punish and minimize) to their preschoolers' negative emotions, as compared to Chinese mothers. This was significantly related to preschoolers' internalizing and externalizing behaviors (Yu et al., 2015). Studies with White middle-class families in the United States show that during a discussion of emotions while reading with their three-year-old children, mothers used significantly more "think" and "emotion" words than did fathers (LaBounty et al., 2008). Similar findings have been reported with ethnically diverse families. In a study of middle-class, White and African American families, fathers engaged in less emotion coaching behaviors (e.g., labeling emotions, teaching children about emotions) and showed less positive affect than mothers (Gerhardt et al., 2020). Yet, fathers who used more emotion-coaching and showed more positive affect had children with more positive emotional expression concurrently a year later compared to fathers who used less (Gerhardt et al., 2020).

School-Age Children

The bulk of studies on how fathers' caregiving influences the development of social skills

among school-age children (ages 6 to 12) has focused less on the paternal behaviors and father-child relationship qualities that promote social competence (e.g., social self-efficacy) and prosocial behaviors (e.g., positive peer relationships), and instead has focused more on the aspects of fathering that explain children's behavioral problems. Based on attachment theory, several studies have shown that securely attached school-age children tend to have an increased sense of social self-efficacy (e.g., beliefs in one's competency relative to interpersonal functioning), which is associated with more positive friendships with peers (Coleman, 2003; Diener et al., 2008). Conversely, infants who are insecurely attached with both parents (assessed via the Strange Situation) are more likely to have higher levels of externalizing behaviors at ages six (teacher-reported) and eight (self-reported), as compared to children with a secure attachment to at least one parent (Kochanska & Kim, 2013).

Studies have also shown that certain types of fathering behaviors are more likely to promote children's social adaptation than others. Fathers who use an emotionally open communication style (e.g., emotion-focused reactions to children) during observed interactions with their children are more likely to have children with more constructive coping styles (e.g., greater reliance on cognitive decision making), as compared to fathers who have a less open style of communication. Mode of communication is concurrently linked to greater social competence with peers and fewer internalizing and externalizing symptoms (Gentzler et al., 2005). In contrast, fathers who are overcontrolling (e.g., excessive regulation of children's activities) have children with increased anxiety in elementary school, which in turn predicts anxiety symptoms during adolescence (Verhoeven et al., 2012). These associations also tend to be

bidirectional. Fathers' sensitivity at grade 1 was negatively associated with externalizing behaviors at grades 3 and 5, which in turn, predicted to increased externalizing behaviors in boys at age 15 (Zvara et al., 2018).

Because a quarter of children in the United States do not live with their fathers, it is important to understand under what conditions nonresident father involvement matters for children's wellbeing. Due to methodological issues, assessing the quality and quantity of nonresident father involvement is challenging and is operationalized differently, making comparisons across studies difficult at best (Cabrera et al., 2000). A series of meta-analyses found that the quantity (e.g., time spent with child) and quality (e.g., sensitive and warm interaction with child) of nonresident father involvement with their children was positively associated with children's social-emotional well-being (e.g., level of anxiety) and behavioral adjustment (e.g., externalizing and internalizing behaviors) (Adamsons & Johnson, 2013; Amato & Gilbreth, 1999; Coates & Phares, 2019). Additionally, fifth grade children who reported feeling close to their fathers during middle childhood were rated by their mothers as having fewer behavior problems and more positive peer relationships (Cabrera, Cook, et al., 2011). Other studies have assessed the impact of father absence on children's social development. A large study of low-income, nonresidential fathers found that "father absence" was negatively related to children's behaviors, such that experiencing paternal first-time incarceration between the ages of one and nine increased children's self-reported antisocial behaviors (e.g., internalizing, externalizing, delinquency) and led to a one- to two-month schooling setback (Haskins, 2015). However, paternal incarceration was not related to children's reports of prosocial skills (Haskins, 2015). Father absence during the first years of life that is followed by father presence

creates instability in children's lives, which negatively affects their social adaptation (Karberg & Cabrera, 2017). For instance, family instability during ages three to five, assessed as mothers' report of changes in residential romantic partners, was negatively related to co-parenting support, which was predictive of children's externalizing behaviors at age five (Karberg & Cabrera, 2017).

Adolescents

The empirical evidence on how fathers' caregiving is related to adolescents' social adaptation is quite sparse; most research has almost exclusively focused on the role of mothers (Flouri, 2008). The broad literature on the influences of parenting on adolescents' social adaptation is guided by Baumrind's parenting model that authoritative parenting is best for children's development (1967). Several specific dimensions of authoritative parenting are thought to be key for healthy adolescent psychological adjustment: parental acceptance or warmth, behavioral supervision, and autonomy granting (Steinberg, 1990; Steinberg et al., 1989). This model is based on parenting practices of middle-class parents in California and hence its applicability to other populations is limited.

The importance of fathers in the lives of adolescents has been underestimated and understudied; consequently the empirical evidence is spotty (Flouri, 2008). Fathers develop healthy and nurturing relationships with their adolescents when they spend time together in various activities, which affords adolescents the opportunity to interact and strengthen the bond with their fathers. There is empirical evidence that nonresident father involvement trumps residency as a predictor of children's social adaptation or at least acts as a protective factor. In a study of African American, urban, teenage boys, adolescents who reported

spending time, feeling emotionally supported and viewed their father as a role model, also self-reported fewer internalizing symptoms, greater psychological wellbeing, less delinquency and drug use, and lower rates of school drop-out (Zimmerman et al., 1995). Similarly, adolescents who did not live with their biological fathers were rated as having fewer behavior problems than their peers only when their fathers were highly involved (e.g., internalizing and externalizing) (Carlson, 2006).

Studies in the United States find that the quality of the father–adolescent relationship, including emotional closeness, supportiveness, nurturance, and parenting style, is also significantly related to adolescents' first delinquency, substance use, behavior problems, and emotional health (Bronte-Tinkew et al., 2006; Coley, 2003; Suizzo et al., 2017). Similar findings were reported in studies of Dutch and British families. A study of adolescents living in the Netherlands found that across four ethnic groups (i.e., Dutch, Moroccan, Turkish, Surinamese), adolescents who reported having a negative relationship (e.g., conflict) with mothers and fathers also reported increased aggression and delinquency and lower self-esteem (Wissink et al., 2006). In a cross-sectional study, British fathers who reported being highly involved (e.g., gave praise) with their adolescents also reported that their adolescents had fewer total difficulties and were less hyperactive and exhibited more prosocial behaviors (Flouri, 2008).

Whether patterns of father relationships are the same across adolescents of diverse ethnic and cultural backgrounds is not yet settled, as there are comparatively fewer studies on this population. A study of African American families using interview and questionnaire data found that daughters who self-reported feeling angry with and alienated from their fathers reported higher levels of depressive symptoms and school behavior problems only when their

fathers had low levels of contact (Coley, 2003). Similarly, in a sample of Mexican American families, fathers' perception of neighborhood danger was related to greater levels of fathers' harsh parenting only for fathers who had higher levels of familism. The strong value that family comes first resulted in more harsh parenting for youth, which was related to more youth internalizing problems when youth lived in highly disadvantaged neighborhoods (White & Roosa, 2012). These findings center fathers as important contributors to adolescents' wellbeing and highlight the way in which contextual fathers shape parenting behaviors.

Mechanisms of Father Caregiving

The specific mechanisms (e.g. direct, moderation, or mediation pathways) by which father caregiving matters for children's development are not well understood. Framed within a family systems theory, it is expected that fathers contribute *indirectly* to their children's outcomes, through their effects on other aspects of the family system, such as interparental relationships (Cabrera et al., 2017; Lewin et al., 2015; Majdandžić et al., 2018; Ryan et al., 2006). Studies have found that fathers who are engaged and involved with their children have stronger relationships with their partners, which, in turn, are associated with better cognitive outcomes for children (Cabrera, Fagan, et al., 2011; Varga et al., 2017).

Other studies have shown that the impact of father engagement operates through its impact on earlier skills, which then support the development of later skills. This developmental cascade phenomenon is central to developmental theories that development is cumulative, depending at each period of development on skills that developed in previous periods. Several studies have shown that father engagement during toddlerhood is associated with stronger expressive vocabulary at 24 months,

which, in turn, is associated with better language skills at 36 months (Rowe et al., 2017). High quality of fathers' reading to toddlers has been found to increase children's interest in reading at 24 months, which in turn is related to better language skills at prekindergarten (Malin, Cabrera, & Rowe, 2014). Similar findings have been reported with adolescents. A study of sixth graders from low-income, ethnic minority families found that fathers' warmth affected their adolescents' beliefs about themselves (i.e., feeling more positive about themselves) and their determination to persist in school tasks, which in turn influenced their academic achievement (Suizzo et al., 2017).

A central core of developmental theory is that human development is shaped through interactions with proximal and distal factors in their environment. That is, the association between proximal processes, such as fathers' caregiving, and children's outcomes is moderated (i.e., cumulative or compensatory effects) by other proximal or distal aspects of their environment (Lewin et al., 2015; Majdandžić et al., 2018; Ryan et al., 2006; Martin et al., 2010). Studies have shown that children with both a supportive mother and a supportive father (joint effects) had significantly better cognitive performance than toddlers with two unsupportive or just one supportive parent (Ryan et al., 2006). However, a study using a national data set did not find evidence of joint effects of mothers' and fathers' cognitive stimulation (e.g., reading to child, telling stories) at 24 months on children's reading or math skills at 48 months (Cabrera et al., 2020). In a study of two-parent Dutch families, mothers or fathers' low levels of challenging parenting behaviors (e.g., tickling, chasing, teasing the child) were associated with reduced anxiety only when the other parent exhibited high levels of challenging behaviors (Majdandžić et al., 2018). Similarly, a study of African American mothers found that the negative

effects of maternal depression on infants' distress was reduced only when fathers were reported to be highly involved with their infants (Lewin et al., 2015). The protective effect of fathers has also been observed with adolescents. Having a positive relationship with their fathers protects adolescents from the negative effect that fathers' authoritarian parenting style (i.e., controlling and not warm) can have on adolescents' delinquent behaviors (Bronte-Tinkew et al., 2006).

In summary, the empirical evidence on how father caregiving promotes children's development across domains and developmental periods has increased exponentially over the last decade or so, yet it is still comparatively a fraction of the research on mothers' influence on children. Moreover, most of the research on father effects on children's development is, with some exceptions, based on small samples of convenience; is correlational and cross-sectional; does not examine the ways in which cultural context might play a role; is mostly based on parental reports, with few observational studies especially of fathers and adolescents; and, does not systematically examine the mechanisms through which fathers matter for children. These limitations and data constraints make it difficult to draw causal conclusions, discern directionality of findings, and generalize to other groups.

Despite shortcomings in research on fathers, the descriptive data we reviewed point to several conclusions. First, both fathers and mothers meet their children's basic needs of being loved, emotionally supported, and protected from harm, but there are notable differences in *how* mothers communicate and interact with their children. Second, fathers who are sensitive and cognitively stimulating have children with better cognitive and socio-emotional outcomes, such as working memory, problem-solving, and EF skills, than parents who are not as sensitive. And this is

true across cultural groups. Third, there are differences in terms of the quantity and quality of how fathers interact with their children. Although fathers engage in literacy activities, such as reading, less often than mothers, the linguistic input they provide through reading or linguistic exchanges is of much higher quality than mothers' and therefore more predictive of children's language skills. Fourth, fathers' contributions to children's development seem to be mostly through direct effects, over and above maternal contributions. Fifth, fathers, as mothers, engage in socio-emotional socialization practices that help children develop the emotional and social skills they need to form and maintain meaningful relationships with others.

Implications of Research for Policy and Programs of Parenting

The extensive research on how fathers' caregiving influences children's development has significant implications for research and policy and programs.

At the *research level*, we make several recommendations to advance the field:

- Studies need to be grounded in theoretical frameworks to better conceptualize the role of fathers in caregiving and the mechanisms through which father caregiving influences child development.
- Data should be collected directly from fathers through surveys and/or observations instead of being based on mothers' report of fathers' engagement with children.
- It is imperative to examine father caregiving in an ecological context that includes significant relationships with partners, children, and others.
- We need to move beyond descriptive, correlational studies to conduct studies that can facilitate the examination of the mechanisms

through which father caregiving influences children's outcomes.

- We need to consider placing fathers in a cultural context to understand the unique strengths and challenges of childrearing associated with culturally specific norms, beliefs, and practices. This includes ethnicity and race but goes beyond it by focusing on aspects of fathers' sociocultural contexts that are strongly related to how they parent.
- Father involvement is predicated by multiple factors, among them, depressive symptoms. It is important that we recognize that fathers' mental health is a serious threat to fathers and their families.
- We need better understanding of the specificity that links quality father-child interactions, especially during the critical early years of development, and children's cognitive and socio-emotional skills across developmental stages.
- Longitudinal designs including economically and culturally diverse families would strengthen the current knowledge on father caregiving and its impact on children's development over time.

At the *policy and program* levels, the empirical findings make a strong case for including fathers into parenting programs that aim to increase parental investments in learning activities and in developing higher-quality (e.g., responsive, warm, and cognitively stimulating) interactions with children. Given the influence fathers have on children's development, it is no longer cost-effective or efficacious to design parenting programs only for mothers. Providing support for mothers and fathers to build and maintain healthy relationships with their partners should be a central goal of programs that aim to improve the lives of children; this is equally important for resident and nonresident fathers.

Indeed, fatherhood programs in recent decades have focused on supporting fathers in their role as caregivers and working together with mothers as co-parents. The Healthy Marriage and Relationship Education (HMRE) and Responsible Fatherhood (RF) programs aim to strengthen couples' relationships and help fathers overcome barriers to become more engaged in their children's lives (Alamillo et al., 2020; Patnaik & Avellar, 2020). These programs differ substantially from the traditional child support programs that enforce fathers' financial contributions to the child (Osborne, 2020). However, fathers often face substantial barriers that prevent them from participating in those programs. Fathers' inability to participate in programs reflects work schedules and non-resident status, traditional beliefs about gender roles such that men should be breadwinners and women should be caregivers, and providers' ability to adapt the services to fathers' needs (McBride et al., 2017).

Policies that facilitate prenatal involvement and postnatal involvement are critically important to signal to fathers that society at large values their parenting roles. Prenatal involvement has long-term effects on father involvement in their children's lives (Cabrera et al., 2008). Paternal leave policies enable fathers to become more engaged with their children, especially soon after birth. Findings from a study that used a national data set across Australia, Denmark, the United Kingdom, and the United States showed that fathers who took leave, especially two weeks or more around childbirth, were more likely to carry out childcare-related activities (e.g., feeding, changing diapers, and reading) when children were under age three and that children with highly involved fathers performed better in cognitive assessments at later years (Huerta et al., 2013). These findings offer more evidence that parental leave for fathers and

mothers is a pathway to child wellbeing. Moving forward, it may be beneficial for programs to allocate more efforts in supporting nonresident or noncustodial fathers by establishing the minimum amount of time they are required to spend with their child and by making broader programs (e.g., health care, parenting education, housing vouchers) more accessible to those fathers (Osborne, 2020). These policies align with the research showing that more-engaged fathers tend to have children with better outcomes, and that fathers with more economic resources and knowledge of child development are more likely to have quality interactions with their child, which in turn would promote children's development.

In summary, programs and policies interested in promoting school readiness and academic achievement among children should target both mothers and fathers and start as early as possible in children's lives. Programs and policies should also invest in parents' education to encourage them to interact with their children in ways that support early cognitive and social development.

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16 Understanding Parenting through a Family Systems Lens

Carolyn S. Henry

A challenge in traditional research and parenting prevention and intervention is the limitation of focusing only on specific parent–child dyads, often mother–child relationships. This approach is focused primarily on how parenting techniques relate to child outcomes. Valuable insights continue to be gained from this line of inquiry, yet most parenting scholars and practitioners recognize that parent–child relationships are bidirectional such that children and parents influence each other. As general systems theory was increasingly applied to families by the mid-twentieth century, approaches to scholarship and practice on parenting increasingly recognized that parenting is intertwined with a complex set of family relational patterns that emerge through interaction. Later, several systemically-oriented theories of parenting began to emerge, incorporating principles of general systems theory. The goal of the present chapter is to provide an overview of family systems and applications in parenting, including how interrelated family adaptive systems at multiple family system levels can be used to conceptualize parenting. Potential applications of using a family systems lens to view parenting across the family life cycle are discussed, including diversity in family forms, and for families living with adversity illustrated through the example of poverty.

Family Systems Perspectives on Parenting

Substantial valuable parenting research and practice is based on theories that coalesce around the *families as contexts lens* focusing on one or more parent figures interacting with at least one child within contexts such as families, schools, neighborhoods, or cultures. The defining feature of the families as context approach is that the focus is on an individual, usually a child, and families are one of many contexts of development. In contrast, the focus in this chapter is on the *families as systems lens*, where parenting is part of larger family relational systems involving individual family members and family subsystems (e.g., parent–child, couple, sibling, intergenerational) (Kerig, 2019). The premise of this approach is that complex family relational systems emerge and navigate change as needs at multiple family system levels evolve over time. Family relational systems are embedded in broader ecosystems spanning the molar to the societal levels that may influence or be influenced by families.

The families as system lens encompasses scholarship in multiple fields relevant to parenting, including classic scholarship such as the work by Cowan and Cowan (2002), Cox and Paley (2003), Lewis and Fehring (1998), Minuchin (1985), and more recent

work (Fiese, Jones, & Saltzman, 2019; Kerig, 2019). The current chapter focuses on using the families as systems lens to guide parenting research and practice.

Families as Systems

Eloquently described by Broderick (1993), families are “open, ongoing, goal-seeking, self-regulating social systems that share features of all such systems” (p. 37). This next section defines the fundamental family system features as guided by general systems theory (von Bertalanffy, 1968) such as relationship processes, the circular rather than linear nature of systems, holism, hierarchy, family life cycles, and perturbations.

Fundamental features of family systems

Relationship processes. Family systems involve a complex web of family relational processes (Minuchin, 1974) that arise through interaction over time involving family dynamics such as connectedness, authority and influence, a family worldview and identity, and addressing basic needs. Relational patterns arise as one or more family subsystems (e.g., parent–child) interact and interface with other family subsystems (e.g., couple) that coalesce around important areas of family life. Key relational processes include circular relational patterns, interdependence/mutual influence, holism, hierarchy, and change and evolution over time.

Circular relational patterns. Understanding relational processes requires moving beyond a focus on cause and effect. Parenting research, for example, supports parenting styles and techniques characterized by responsive parenting combined with firm control being associated with positive child outcomes (see Bush & Peterson, 2013). Yet, a disproportionate emphasis on parent effects on children

obscures the bidirectional nature of parent–child socialization whereby parents and children mutually socialize each other. And, parent–child dynamics are intertwined with the relational processes among co-parents, siblings, and other parts of family systems; intersecting with culture, ecosystems, relationship history, and other factors. Thus, viewing parenting through a systems lens involves moving beyond cause and effect (or linear thinking) to understand the interwoven relational patterns in circular thinking such that the focus is on the dynamics of interaction between one or more people.

Interdependence/mutual influence. Complex relational patterns in families are such that change in one part of a family system reverberates through the entire system. For example, a parent who has a difficult day at work may be more irritable and less patient with other family members who in turn respond less favorably than usual to the parent. At least temporarily, family interactions may become less supportive or nurturing than usual and flow across multiple family relationships beyond the specific parent and child to sibling or couple relationships. Also, a warm greeting and expression of fondness by family members might improve the parent’s mood, enhancing family interactions that day, recharging the parent for the next day at work. This example was punctuated to assume the artificial starting place of the parents’ challenging day at work. *Punctuation* means acknowledging that any part of an interaction one examines was preceded by other interactions and is embedded in ongoing system dynamics (Broderick, 1993). Simply put, each part of a family system mutually influences other parts and it is challenging to completely capture the full scope of mutual influence in interactions between parents and children as they interface with other subsystems (co-parent, sibling), overall family system and family-ecosystem interfaces.

Holism. A fundamental concept in systems perspectives is *holism* or the idea that systems involve relationships that are more than adding together the qualities of individuals or subsystems (von Bertalanffy, 1968). Thus, each part of a family (e.g., family member or subsystem) is best understood as part of the system. To understand parent–child relations requires consideration that interactions are part of broader family dynamics within ecosystems. For example, parental supportive behaviors toward children occur within the parental triangle of mother and father (or other primary parent figures) and child (Fivaz-Depeursinge, & Corboz-Warnery, 1999), other subsystems (e.g., sibling) (Feinberg, Solmeyer, & McHale, 2012), and overall family systems (Henry et al., 2006), and the ecosystem (Bush & Peterson, 2013) (see Figure 16.1).

Holism is illustrated in co-parenting as more than one adult sharing responsibility for a child, regardless of whether they are currently a couple or share a residence (McHale, 2011). Effective co-parenting goes beyond the couple subsystem, involving co-parents' solidarity with each other, mutual involvement with a child, or unresolved dissonance and interfaces with the quality of parent–child interactions (McHale, 2011). These interactions interface with a broader set of ongoing relational patterns in the family system and extended family or kin systems, and occur within specific communities and cultures.

Hierarchy. A hierarchy emerges, reflecting the internal structure of a family system. In two-parent families, adult partners often serve as an executive system guiding the overall system and other subsystems (e.g., parent–child, sibling, grandparent–child) toward common family goals (Minuchin, 1974). Minuchin saw married couple families with children as functioning best with a unified and supportive executive system. Yet, many

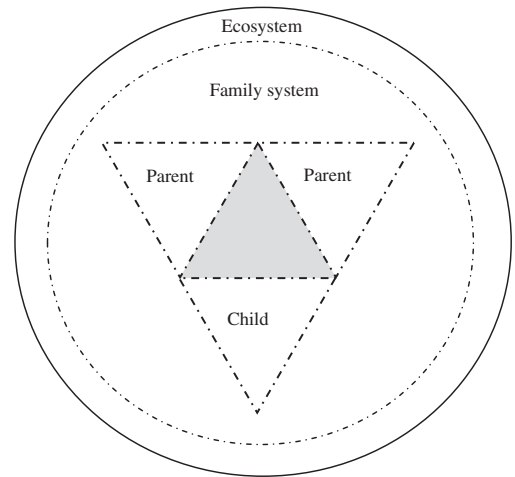


Figure 16.1 Parent–parent–child triangle within a family system and ecosystem

Notes: The grey area represents relational patterns in the parent–child triangle. Using a systems lens, multiple systems levels are considered in relation to parenting: each individual, four dyadic subsystems (coparent dyad, first parent–child dyad, second parent–child dyad, and parent–child triangle), the overall family system, and interfaces of each with aspects of the ecosystem (e.g., culture, school, workplace(s), neighborhood, health care, social services). If additional children, kin networks (voluntary and involuntary) are part of a family system, the relational patterns increase in complexity. The parent–parent–child triangle was inspired by the conceptualization of the mother–father–child triangle by Fivaz-Depeursinge and Corboz-Warnery (1999) and centered in the overall family system and ecosystem.

children are part of binuclear family systems, spending time in two parents' households, where co-parents are not or no longer romantic partners (Ahrons, 1979). In this family form, parents may seek to maintain a hierarchy of the parental subsystem, despite other differences between parents, with a common focus on child well-being. In multi-generational households, the executive system

may cross generational lines such as when a parent and grandparent serve as “co-parents.” Thus, co-parenting varies, in part, by family form (or composition), the degree of mutual recognition between parenting figures of their mutual rights as a parent figure, and the establishment of a cooperative rather than competitive parenting subsystem (Ahrons, 1979). A hierarchy between children may occur when siblings’ ages are distant, rather than having a sibling relationship that serves as the first peer-like relationship.

Family relational patterns include overall family system qualities such as family cohesion or family adaptability that combine to regulate day-to-day family life. Also, families and their members interact with others in neighborhoods, schools, workplaces, healthcare systems, and other parts of ecosystems. Thus, hierarchies may arise at multiple levels for family systems including individual family members, family subsystems (e.g., coparents, sibling, parent-child), and family-ecosystem interfaces.

Change and evolution over time. As living systems, families engage in on-going relational processes seeking a balance between stability and change (von Bertalanffy, 1968). In the same way that human bodies change and evolve across the life cycle, family systems experience changes from within and through interactions with the ecosystem over time (Broderick, 1993). As detailed later, family stress response systems are activated to regulate the balance between stability and change as families experience change or disruptions in the rhythms of family life. In turn, feedback loops occur to either restore or modify the ongoing steady state of the system. **Perturbations** are sufficiently strong disturbances to family systems that, at least temporarily, disrupt the steady state of family dynamics (Kunnen & van Greet, 2011). In the next section, common forms of perturbations are addressed.

Family Systems and Perturbations

As family systems progress over time, they experience two fundamental forms of perturbations: (a) family developmental changes or (b) other perturbations. Family systems progress through family life cycles such that new challenges tend to arise as perturbations in family relational patterns (for a review, see Martin, 2018). Early family life cycle work focused on 20th century two-parent married couple families in the U.S. based on changes in human development, couple and parent-child subsystems, and existing social norms. Later, family life cycles were developed to address diversity of family life cycles based on family forms, cultures, and socioeconomic contexts, (McGoldrick et al., 2016). The next section addresses life cycle changes in family systems with children.

Systems perspectives emphasize the importance of balancing stability and change in relational patterns over time. Family transitions into new life cycle stages are central to parenting family systems encounter change from within the system based on social norms and culture (Martin, 2018). Family systems tend to experience new challenges associated with somewhat predictable family transitions such as forming a couple relationship, marriage, the birth of a child, a child starting school, a child leaving home, or retirement. For example, an adult child establishing a home away from the family home might prompt changes in the nature of contact between the adult child and parents, couples might reassess priorities in their relationship and family systems, household tasks might change or be completed by different family members.

Relational processes in boundaries, rules, laws, rituals, or roles within and between family subsystems occur and may be modified as families go through life cycle changes. For

example, the transition to parenthood often involves changes in couple subsystems, managing additional family tasks, and changes in relationships with extended family members. In turn, boundaries tend change to allow for an additional family member, and modifications may be needed in rules, laws, and rituals. Ecosystem changes also occur involving schools, work, healthcare or other entities as families address current family needs and changes such as the addition of vulnerable new family member(s).

Specific stressor events are a second type of perturbation that creates temporary, and sometimes lasting, disruptions in family relational patterns (Hill, 1958). Family daily hassles may occur in areas such as family relationships, social networks and commitments, healthcare, home management, work- or school-related stressors, family finances, and entertainment (Almeida, 2005; Rollins, Garrison, & Pierce, 2002). Daily hassles, depending upon their severity, may prompt disruptions in routines, finances, health and well-being, or plans for the future or other areas of family life (Almeida, 2005). For example, a child waking up sick may disrupt the regular routines of a parent taking the child to school and continuing on to work, since plans to care for the child's health and safety may take priority that day.

Ongoing family vulnerabilities such as abilities and disabilities, poverty, discrimination (e.g., racism, sexism, classism); or aspects of ongoing family interaction that increase stress (e.g., family secrets, tension among family members, ineffective communication or problem solving) (McGoldrick et al., 2016) also involve perturbations. In the above example of a child waking up sick one morning and disrupting the family routines, at least for the day, the parent may be the only source of income for the family or work in a low-paying insecure job (Smith, 2020). Thus, the intersection of daily

hassle and ongoing stressors may increase the risk to needed income or may mean losing the job by not being present.

Stressors beyond daily hassles are common across families' life cycles, including historic events (e.g., a pandemic, economic recession, war, policy changes due to political changes) or unpredictable events (e.g., untimely death, natural disasters, trauma, unemployment) (McGoldrick et al., 2016). Stressors vary in several ways, e.g. ambiguity versus clarity, since sometimes families lack clarity such as if a parent is missing in action during military deployment (Boss, 2002). Family stressors can be desirable (e.g., a desired job change or a pregnancy) or lack free choice (e.g., being adopted or laid-off) (Boss, 2002).

In sum, family perturbations arise from a variety of internal and external sources, may be acute, chronic or cumulative, and may or may not have a clear resolution over time. In turn, changes in ongoing relational patterns may occur in multiple aspects of family functioning represented through family adaptive systems (FAS) (Harrist, Henry, Liu, & Morris; 2019; Henry et al., 2015). The next section presents how multiple FAS arise as part of family relational systems and can be used to examine the delicate balance between stability and change, as families navigate family developmental transitions or other perturbations.

Family Adaptive Systems and Parenting

A promising way to understand parenting through a systems lens involves multiple FAS representing domains of family functioning. FAS are analogous to interrelated human adaptive systems such as the circulatory system, respiratory system, or digestive system, and they sustain functioning over time (Henry et al., 2015). Masten (2001) pioneered

the application of human adaptive systems as the “ordinary magic” of child resilience or competence despite adversity. Strengths in human adaptive systems (e.g., health, information processing, parenting) that arise and develop over the lifespan often explain the difference in children showing resilience. The idea of human adaptive systems has been extended to identify FAS critical areas of family functioning (Harrist et al., 2019; Henry et al., 2015).

FAS are complex relational patterns that arise and regulate key areas of family functioning such as emotion, control, meaning, maintenance, or stress response (Henry et al., 2015). When perturbations arise in family systems, multiple FAS are likely to be involved, often at multiple system levels, in restoring the prior steady state or in adapting to a new steady state in relational patterns. Table 16.1 provides examples of how FAS address key family functions in key areas of family life.

FAS function well when families have the awareness of and abilities to address the inevitable perturbations that arise in family systems, ranging from daily hassles to developmental changes to crises and trauma. Figure 16.2 shows how family systems concepts can be understood through FAS and, in turn, how these ideas translate to parenting systems. The next section provides an overview of family adaptive systems and how they are useful in understanding parenting through a systems lens.

Family Emotion Systems and Parenting

Family emotion systems emerge and function to reflect and guide interaction patterns that maintain and regulate the balance of connectedness versus separateness in family systems. Family emotional systems play a key role in parenting in multiple ways. At the individual level, emotional development and regulation

are critical to human development across the life cycle. In parent–child subsystems the ability of parents to connect with their children includes demonstrating their own emotional well-being to their children, engaging in emotion coaching, and regulating their own emotions; these are all assets in socializing children toward emotional competence (Morris et al., 2007, 2017). Prior conceptualizations of parenting focused on connectedness include attachment, bonding, responsiveness, warmth, support, and emotion regulation.

Likewise, the nature of emotional connectedness in co-parenting and/or couple relationships is critical in the family system in which parenting occurs. For example, in couple relationships established before co-parenting, the relationship expands to include co-parenting. Positive connectedness in the couple relationship before entering parenting is likely to facilitate a smooth transition to parenting. Further, this is more likely to occur when the family control system (described below) is also functioning well. In family systems, aspects of connectedness such as cohesion (emotional bonding) and effective communication involve relational patterns that open the door for effective couple, co-parental, parent–child, sibling, or other family subsystems and emotional health among individual family members.

Parenting involves an emotional component in the interface between families and parts of their ecosystems. The importance of family–school connectedness is well-established in facilitating positive child developmental outcomes (e.g., LaForett & Mendez, 2010). When a child’s chronic health challenge requires close connections with medical providers, family-centered care may not only foster positive child outcomes, but contribute to creating a new steady state in the family system (Nabors et al., 2019). Establishing trust among medical providers, schools, and family systems, for example, may be central to

Table 16.1. *Family adaptive systems (FAS) and parenting across family system levels*

Family adaptive systems		Family system levels			
		<i>Individuals</i>	<i>Subsystems</i>	<i>Family system</i>	<i>Family–ecosystem fit</i>
	Function	Child, mother, father	Mother–child, father–child, co-parent, couple, sibling	Complex web of family relational patterns	Interface of family system with broader systems
Basic FAS					
<i>Emotion system</i>	Emotions and connections in family and beyond family;	Emotion regulation, empathy	Bonding, attachment, responsiveness, warmth, emotion coaching, support, encouragement, cooperation	Cohesion, communication and conflict resolution, emotional expressiveness, sequences/cycles	Emotional connectedness of family and members to the ecosystem (e.g., work, school, community)
<i>Control system</i>	Authority, organization, empowerment	Self-efficacy, autonomy, problem solving ability	Demandingness and guidance, sibling rivalry, competition	Power, authority, roles, rules, flexibility, routines, boundaries	Boundaries outside family, social norms, access to services
<i>Meaning system</i>	Subjective perspectives on self, family, culture, and community	Individual situational meaning, identity, individual goals	Shared identity, child-centered vs parent-centered parenting, sibling, goals of parenting	Family situational meaning, family identity, family worldview, family goals, coherence & hardiness, rituals	Cultural, community, and societal goals and values
<i>Maintenance system</i>	Meet basic needs of the family and family members	Health, safety, and economic needs such as food, clothing, and shelter	Provide or secure care for vulnerable members	A structure for meeting needs of family and protecting vulnerable members	Interface systems and resources (e.g., health, education, social, education, economic)
Meta-level FAS					
<i>Stress response system</i>	Regulates the balance of stability and change in the individual, subsystem, overall family system, and family–ecosystem fit consistent with development and ability to meeting needs; activates as change occurs and regulates through feedback loops to restore balance by dampening or amplifying change				

Notes: The basic FAS develop and regulate specific areas of family functioning whereas of the meta-level stress response system regulates family dynamics when stress occurs. Adapted from Henry et al. (2015).

Family systems	Family adaptive systems	Parenting
• Relational processes • Circular rather than linear • Equifinality • Multifinality • Interdependence • Holism • Hierarchy • Perturbations • Feedback loops • Positive feedback • Amplifies change • Morphogenesis • Negative feedback • Dampens change • Morphostasis	• Basic relational processes • Emotion system • Self and connection • Control system • Roles, authority, decisions • Meaning system • Worldview, identity, situational meanings • Maintenance system • Meet needs--food, clothing, shelter, health, health & safety, economic • Meta-level • Stress response system	• Parenting processes • Connectedness, responsiveness, emotion regulation/co-regulation • Demandingness, rules, laws, routines, rituals, boundaries • Goals, values, and perceptions of situations • Providing/teaching how meet basic needs and protecting, consistent with child development and ability • Feedback loops in the stress response system • Processes of balancing stability and change as perturbations arise (<i>daily hassles, developmental changes, unexpected situations</i>).

Family systems, family adaptive systems, and parenting involve multiple interrelated family system levels: individual members, subsystems such as parent–child, sibling, coparent), overall family system, and family-ecosystem interface.

Figure 16.2 Family systems, family adaptive systems, and parenting

effectively addressing the physical, socio-emotional, and educational needs of a hospitalized child (American Academy of Pediatrics Council Committee on Hospital Care and Child Life Council, 2014).

Family Control Systems and Parenting

Though interaction, family systems develop family control systems, or complex patterns of interaction that emerge as a structure of authority and responsibility. There tends to be a hierarchy in the control system regarding the relative influence among systems or subsystems in families, as well as influence by broader ecosystems (Minuchin, 1974). In a two-parent family where the couple relationship precedes the children in the family, the couple subsystem tends to serve as the executive system for families (Minuchin, 1974). In families where the child is part of two parent's

separate households (or a binuclear family system) the executive system may continue to be the parents or each parent's family might have an executive system for their specific household (Ahrns, 1979). The control system involves interrelated family rules, rituals, and roles at multiple system levels such as individuals, family subsystems, overall family systems, and family-ecosystem interfaces (Henry et al., 2015). Through interaction, family systems develop relational patterns involving control (or power) that regulate the flexibility and stability in ongoing family dynamics over time.

Boundaries and Triangulation

Family *boundaries* are features of families defining who participates in them and how they participate (Minuchin, 1974). Within families, internal boundaries arise and

regulate how and when family members interact with each other. External boundaries arise and regulate how family systems and their members participate with others outside the family, such as extended family members, neighborhoods, schools, or healthcare providers. Boundaries may be clear, unclear, or diffuse among family members as part of the steady state of family dynamics. Boundaries may be modified as perturbations arise across the family life cycle and as the needs or ecosystems of families and their members evolve. *Triangulation* is an example of where a family member who is frustrated with another brings in a third person, either inside or outside of the family instead of addressing the original concern (Bowen, 1978). A child frustrated with a parent may attempt to triangulate another parent (or other family member) against the parent rather than directly address the issue. Although this may temporarily bring a sense of control to the child, the initial issue between the first parent and the child remains unresolved, since the child crossed traditional boundaries between co-parents and a children.

Rules, Laws, Routines, Rituals, and Roles

Family boundaries are often evident through family rules and laws that develop and are modified over time (Broderick, 1993; Minuchin, 1974). *Family rules* are unspoken, yet shared, expectations within family systems that regulate emotion, control (e.g., one family member has priority to sit in their chair), management of family resources, or family beliefs (Broderick, 1993). Similar to rules, *family laws* are shared expectations among family members that are spoken, such as a parent who asks a child “what do we do with dirty clothes” and the child picks up the clothes they just took off and carries them to the laundry basket.

Through interaction, families’ interwoven set of rules and laws emerge as routines that guide day-to-day family life (McCubbin & McCubbin, 1988). *Family routines* provide regularity in family life such that discussion is not necessary each time regular interaction patterns occur. For example, parents may use a bath, putting on pajamas, or reading to children as routines to move young children from the active part of the day into sleeping. Family routines emerge as families repeat patterns with inherent rules and laws until they become a natural part of “how we do things in our family.” Family routines are adapted as family needs change over time and are present across family forms.

Some routines become *family rituals* as regularly occurring events take on symbolic meaning (Fiese et al., 2002), for example, going to visit extended family for a specific holiday may reflect honoring older family members. Rituals play a key role in maintaining control systems by formalizing routines through the intersection of family control and meaning systems. *Family roles* refer to family expectations for specific family members or family subsystems relating to specific aspects of family life (Bush & Peterson, 2013). Family roles such as provider, nurturer, or family entertainer are intertwined with family rules; laws emerge and may evolve and change over time. For example, which parent is responsible for which roles is intrinsically grounded in family meaning (or belief) systems, often reflecting societal expectations for parents.

Family Meaning Systems and Parenting

Understanding parenting through a systems lens involves the subjective meanings family members subscribe to before, during, and after specific parent–child interactions. Family meaning systems emerge, reflecting, guiding, and regulating interactions through family

beliefs, values, identity, and perceptions of situations (Henry et al., 2015). Family meaning occurs at three levels of abstraction: situational meaning, identity, and worldview (Patterson & Garwick, 1994). As families face perturbations, not only do individual family members subjectively perceive the situation. Often, through communication, family systems gradually share *situational meanings* based on a collective sense of whether there is a balance between the family demands and capabilities of the situation. This occurs in concert with *family identity*, or the overall view of who is in a family, their relationships with each other, and interfaces with the broader ecosystem. Family meaning is manifested through “we” statements about the family as a whole, such as “we are,” “we believe,” “we include.” Family identity may include related or voluntary kin who are regularly involved in ongoing family functioning (Sanner, Ganong, & Coleman, 2021).

Family situational meaning and identity occur in concert with the *family worldview* or a shared sense of purpose, relation to the larger scope of life, optimism based in reality, making the most of the current situation, and a sense internal versus external control (Patterson & Garwick, 1994). Families hold worldviews about issues such as meaning making processes, hope, or the ability to connect experiences with a bigger picture of life (Walsh, 1998). Culture is often integrated into the family worldview and situational meanings about interfaces with the broader ecosystem. Thus, family cultural identity is understood within the family in relation to extended family or kin networks, or in relation to current or historical cultural contexts.

Family meaning is often manifested through the concept of *punctuation*, or the idea that since there is no beginning or endpoint in systemic relationships, there is not an objective frame from which to view episodes in family

interaction. For example, two siblings may tell a parent the other sibling started a conflict, illustrating how each child punctuates or takes a mental snapshot using an arbitrary starting point in the series of exchanges between the two. Both siblings may believe their view to be accurate based on where in the series of exchanges as they punctuated it. Often, developing shared meaning and viewing events and relationships occurs in a broader relational context (Patterson & Garwick, 1994). For example, parenting research shows a positive association between parenting style and child social competence, suggesting a possible causative link between parenting styles and child outcomes. It is equally possible that specific child behaviors elicit particular parenting techniques. Thus, it is important to recognize that parenting styles or techniques occur as part of sequences of interactions embedded in larger relational patterns reflecting parental values.

Parenting styles involve at least three FAS: the meaning system through which parents’ values (based on family worldview and identity) define and regulate whether parenting is best parent-centered (authoritarian), child-centered (permissive), or an integration of parent- and child-centered (authoritative) (see Baumrind, 2013). Reflecting the family’s worldview, an authoritarian parent may prioritize child compliance with parental demands (control system) and with minimal responsiveness to the child. In a child-centered family system, a permissive parent might emphasize child autonomy (control system) to the neglect of parental guidance either by being indulgent or neglectful (emotion systems). In a family with a parent- and child-centered worldview, an authoritative style might involve parent-child dynamics with balance of parental demandingness and child autonomy (control system), and with responsiveness to the child.

Family Maintenance Systems and Parenting

A fundamental function of families is to provide and protect family members in accordance with societal and family expectations. Over time and through interaction, family maintenance systems emerge and regulate key family tasks associated with meeting basic needs, maintaining and adapting family structure, health and safety, and resources (Henry et al., 2015). Meeting family needs includes adequate food, nutrition, clothing, and shelter for family systems. Family structure includes (a) family form (or composition), describing who family members see as part of their family and (b) family organization or roles and responsibilities of specific family members or subsystems. Family health and safety includes promoting family well-being, protecting vulnerable family members against danger, access to physical and mental health services, and acquiring and managing economic resources. In families with children, societal expectations tend to place the majority of responsibility of providing for and protecting family members on adult family members. Thus, developing and maintaining family maintenance systems tends to be an intrinsically tied to parenting. Notably, family composition and the family cycle stage can be critical elements in determining family maintenance needs.

Providing for and protecting vulnerable family members varies according to family worldviews, family identity, race/ethnicity, poverty, or societal norms relating to group discrimination (e.g., racism, ableism). For example, parents in ecosystems that tolerate or promote racism toward members of the family, or experience poverty, may utilize parenting styles designed to protect children against potential discrimination or adverse effects of living in neighborhoods with high poverty rates (James et al., 2018). Parents

who experience discrimination may seek to protect their children by instilling them with racial/ethnic identity and pride as well as readiness to respond to discrimination (see also Chapter 14 on ethnic-racial socialization).

Family Stress Response Systems and Parenting

The family stress response system is a meta-level family adaptive system that emerges and functions to regulate the processes of other family adaptive systems. In the human body, the stress response system activates a physiological response, often described as fight, flight, or freeze response that activates resistance to change in cardiovascular, respiratory, gastrointestinal, musculoskeletal, endocrine or other adaptive systems. Physical responses to stress may include changes in heart rate, blood pressure, blood sugar, perspiration, respiration, muscle tightening, or other changes. Over time, the body generally returns to the normal state. Yet continued exposure to stress may bring about ongoing vulnerabilities (e.g., a compromised immune system) (Ha & Granger, 2016). Similarly, family systems initially experience disruptions in one or more FAS when perturbations occur and are then followed by either a return to earlier functioning or sustained changes.

Multiple FAS are often involved in resisting change at one or more family system levels. For example, a child that wakes up sick may not go to school; a parent stays home from work to care for the child and reschedules appointments until another day; the parent may schedule an appointment for the child with a medical professional that is only available when other children in the family are coming home from school; an older sibling may be asked to be responsible for younger siblings. Multiple FAS (e.g., emotion, control, meaning, and maintenance) may be involved

in attempts to manage the illness, as the parent fears losing clients or income while attending to the child or the spread of the illness to other family members. Short-term change is all that is necessary if the illness is resolved in a few days and the parent and child are both able to return to their normal activities and routines. If the child's illness is later discovered to be an chronic illness, such as diabetes, greater change is needed as the family learns to manage the illness through changes in the meaning and routines of healthy eating, exercise, blood sugar testing, and insulin treatments. In this case, long-term change in the family maintenance and meaning systems are likely necessary as the family finds manageable ways to modify their ongoing relational patterns to address the diabetes.

When a perturbation occurs in one or more FAS, the family stress response system activates. Like the human body responds to fight an infection, the family stress response system may initially resist a change, regardless of the source of change and whether it is desired or not. Like the human body initially resisting or embracing change, over time the family returns to prior functioning or makes changes in one or more FAS to re-stabilize the family system. If a perturbation requires a modification in one or more FAS to restore effective functioning, and the family continues to resist making modifications, they are likely to continue to struggle until they change the patterns that are no longer working.

Through the basic FAS, ongoing relational patterns interface with key areas of functioning including, but not limited to, emotion, control, meaning, and maintenance to create a *steady state* (or *equilibrium*) in the ongoing family dynamics. Family systems allow for a range of deviation from their steady state before they perceive that *disequilibrium* occurs as family members perceive, at least temporarily, that family demands exceed their capabilities

(Patterson, 2002). In turn, the stress response system activates the feedback loops.

Feedback loops are activated as perturbations disrupt the steady state, resulting either in resisting the change and returning to the prior steady state or amplifying the change to incorporate it into a modified steady state that addresses the change (von Bertalanffy, 1968). *Negative feedback* occurs as family systems resist change and seek to return to their original interaction patterns. *Positive feedback* occurs as families embrace change with adaptive modifications in one or more FAS. As a negative feedback loop develops, *morphostasis* occurs as family systems return to and maintain the family interaction patterns. A positive feedback loop yields *morphogenesis* as the family system recalibrates their interaction patterns in the relevant FAS to create a new balance.

Borrowing from the study of child resilience (Masten & Cicchetti, 2010), the concept of cascades illustrates how multiple FAS may be involved in restoring balance or allowing continued disruption in family functioning. Cascades are the cumulative consequences of family life cycle and other perturbations involving transitions and co-actions between and across multiple family adaptive systems over time (Henry et al., 2015). For example, as a child progresses into pre-adolescence or early adolescence, family boundaries change as the family system changes from engaging with the elementary school to the middle school. At the same time, the steady state of the family system may change based on the pre-adolescent's cognitive or biological developmental changes. Correspondingly, the pre-adolescent may increase their engagement with peers, creating a perturbation in ongoing patterns in family emotion and control systems to accommodate increased desires for greater autonomy from the family and greater connectedness with peers. Meanwhile, an older

sibling may become licensed to drive a car, creating both an opportunity for more flexibility in drivers and possibly introducing another perturbation in an increased demand by family drivers for the existing cars.

Managing the multiple changes in the family emotion, control, meaning, and maintenance systems is necessary to restore balance to the family. In turn, biological change may result in a need to change how the family regulates emotions (emotion system), manages pre-adolescents' greater desire for autonomy (control system), redefines relationships with peers (meaning system), or manages transportation needs (maintenance system).

A related systems concept is the *principle of equifinality*, which posits that different starting points can have the same outcomes (von Bertalanffy, 1968). For example, forms of parental control that are generally effective in specific cultural groups might not be as effective for families in the broader ecosystem. Parenting research generally shows that forms of control such as punitiveness or psychological control tend to be associated with negative child outcomes (Bush & Peterson, 2013). Yet, in Latinx cultures the value of *familismo*, or loyalty to the extended family over individual needs, appears to create a different family climate such that parental punitiveness or psychological control may not be negatively associated with psychological adjustment if used in concert with overall family cohesion or parental emotional support (Bámaca-Colbert et al., 2018). These findings are consistent with earlier research showing that no-nonsense parenting involving high expectations for compliance, acceptance, and moderately harsh parenting can be effective in Mexican American (White et al., 2013) and rural single-mother African American (Brody & Flor, 1998) families.

The *principle of multifinality* posits that the same starting point can yield different

outcomes (von Bertalanffy, 1968). For example, a parenting technique that is effective with one child may not have the same result in another child in the same family that has a different temperament, yielding variation in social competence, academic achievement, psychological adjustment, or other aspects of life even when the parent uses the same parenting style with both children. Also, although some aspects of effective parenting are generalizable to multiple family forms, it is also possible that effective parenting may vary by family form or other factors.

Parenting and Diversity

This section provides illustrations of family diversity and parenting; parenting by fathers as well as mothers, variation in family forms, and families living with sustained poverty.

Fathers and Mothers as Parents

Although it may seem unusual to include parenting by fathers and mothers as a form of diversity, a disproportionate emphasis on mothers as parents can obscure fathers as parents. In the past quarter of a century, notable advances in conceptualizing fathers as parents occurred, rather than an overreliance on the scholarship on mothers as parents (Schoppe-Sullivan & Fagan, 2020). Systems perspectives support conceptualizing co-parents as engaging in parenting functions rather than emphasizing distinctions by sex. This allows recognition of multiple parent-child subsystems as part of larger relational systems to be considered in ways that allow parenting researchers and practitioners to move beyond stereotyped views of mothers as nurturers and fathers as providers and protectors.

Another promising approach is to recognize that parent-child relationships often include at

least two parent figures. For example, Fivaz-Depeursinge and Corboz-Warnery (1999) conceptualized the mother–father–infant triangle as a subsystem. A systems lens provides for consideration of parenting as involving each family member, subsystem, overall family system, and the family-ecosystem without making gender-specific generalizations about parenting. Systems perspectives have the benefit of applying systems principles to understand effective co-parenting by focusing on relational patterns rather than gender of parents or family forms (McHale & Sirotkin, 2019; Prendergast & MacPhee, 2018). Instead, the focus is on two distinct parent–child subsystems: sibling subsystems and co-parent subsystems that, in two-parent families, overlap with couple subsystems. In the next section, two family forms, adoptive and sexual- and gender-minority families, are provided to illustrate the application of systems concepts to parenting across family forms.

Family Form

Substantial variation exists in the living arrangements of children in the U.S. and yields multiple family forms in which parenting occurs for children (at 0–17 years of age) (Federal Interagency Forum on Child and Family Statistics, 2019). In 2018, 69 percent of children lived with two biological or adoptive parents (65 percent married, 4 percent cohabiting); of these, 92 percent lived with biological or adoptive parents and 8 percent lived with a step-parent. Of the 26 percent of children living with a single parent (22 percent mothers, 4 percent fathers), a cohabiting partner of the parent was present in 29 percent of single-father families and 11 percent of single-mother families. Of the 4 percent of children not living with a parent, 54 percent lived with grandparents, 21 percent with other relatives, and 25 percent with non-relatives (foster

parents or other). Statistics are less accessible on the prevalence of children living in binuclear families where two biological or adoptive parents co-parent while living in separate households. Next, adoptive family systems, and sexual- and gender-minority families are briefly addressed.

Adoptive Family Systems

Adoption typically requires biological parent(s) to relinquish parental rights and responsibilities or that a child has no living parents. Adoptive family systems include one or more non-biological children where parents have legal rights and responsibilities for the child(ren) (Wiley, 2017). Adoptive parenting is a transition in family form, activating transition into a family life cycle of the adoptive family network directly or indirectly, and involving both the birth and adoptive families rather than simply moving a child from one family to another (Grotevant & Lo, 2017). Thus, adoption initiates a lifelong process rather than occurring as an event. The initial transition into parenting in adoptive families is often followed by other transitions such as deciding how to discuss the adoption with the child, supporting the child's curiosity about the birth family, addressing adoption-related loss, developing positive views of the child's origins allowing for a positive sense of self, and possibly the child's desire or efforts to connect with the birth family (Brodzinsky, 1987). Adoptive family life cycles may vary based on the age and history of the child, the presence of non-adoptive children, societal acceptance of adoption as a family form (or variations such as transracial adoptions or adoptions of children with special needs).

Best practice in adoptive parenting includes attending to both pre- and post-adoption family networks. Child pre-adoption experiences may include institutionalization or

trauma. Child post-adoption ecosystems may include contact with the birth parent(s) or their relatives; differences between adoptive parent–child race, ethnicity, or national origin; or being raised by opposite-sex or same-sex parents in a variety of family forms (Grotevant & Lo, 2017). Variation occurs in whether birth parents are involved with children, potentially creating boundary ambiguity about multiple parent figures in the family and their participation in the family (Boss, 2002). When biological parents are not part of a child's life, ambiguous loss may arise, as the child's birth parents are psychologically present and physically absent.

Transitions into parenthood in adoptive family systems may occur after a range of experiences such as infertility, delaying parenthood until older age, addressing other kin network needs, or a variety of other reasons (Wiley, 2007). Adoptive siblings may be present in the household or live elsewhere. Members of the adoptive kin network, both from the birth and adoptive families, may hold beliefs that are similar to or different from the adoptive parents. Prior foster families may be part of kin networks, since 25 percent of children leaving the U.S. foster care system in 2018 were adopted (Children's Bureau/ACYF/ACF/HH, 2020).

Family adaptive systems are integral to positive adaptation in adoptive families. For example, providing a safe and secure home (maintenance system), developing a sense of identity in the adoptive family and establishing emotional bonds (meaning system), and realigning the control system may be necessary. Research shows that open communication about adoption tends to be associated with more positive adoptive family functioning (Brodzinsky, 2006). Next, family systems in sexual- or gender-minority parent families are examined as a family form.

Sexual- and Gender-Minority Family Systems

A systems lens is useful in considering sexual- and gender-minority (SGM) families. With legal changes in the accessibility of marriage and adoption for SGM individuals in more locations, there can be great variation in the social and legal ecosystems of specific SGM families. Thus, the interfaces of families and their ecosystems are particularly salient for SGM families. Further, parents often go through a processes of accepting their SGM identity and disclosing this to their families of origin. This may occur before or after parenting begins. Also, given the diversity in SGM family systems, it is critical to consider existing research and needed research on parenting in families with at least one parent who identifies as LGBTQAI+ (lesbian, gay, bisexual, transgender, queer/questioning, asexual, intersex, and other sexual- and gender-identities) (Reczek, 2020). The majority of studies on parenting in SGM families are on gay and lesbian families, with less known about other SGM families. Among same-sex couples, 40 percent of Black, 28 percent of Latino, and 16 percent of white individuals are parents (Gates, 2015).

Two key parenting themes are evident in recent research on SGM families: (a) parenthood and parenting experiences in context and (b) the effects on children. The pathways to parenthood can be complex given the variation in legal constraints, community norms, potential social stigma, cost of reproductive technology, and access to adoption (Baumle & Compton, 2015). Specific pathways may include parenthood via a previous heterosexual relationship, surrogacy, reproductive technology, or adoption. The transition to parenthood may occur before or after the self-acceptance and identification of oneself as LGBTQAI+ (Reczek, 2020). Parenting research tends to focus on the types of activity

shared with children, self-care activities, social stigma, and social support. In general, child behavioral, emotional, and health outcomes in SGM families do not differ from children in cisgender heterosexual families, with most differences explained by factors such as SES or family transitions (Reczek, 2020).

Parenting and Poverty: Adversity and Resilience

Systems theories are central to individual and family adversity and resilience and address both relational and broader ecosystem dynamics (Henry et al., 2015; Masten, 2018). Two common types of adversity across the family life cycle: (a) *status risk* or ongoing challenges such as discrimination based on race/ethnicity, chronic physical or mental health issues, and poverty and (b) *specific stressors* or events that are perturbations that activate the family stress response systems (Henry et al., 2015; Rutter, 1987). By viewing parenting through a family systems lens, potential risks and protections associated with adversity become evident. Although parenting can intersect with a variety of forms of adversity, this chapter focuses on the specific adversity of poverty.

A systems lens is promising for considering adversity and resilience associated with parenting in families living with poverty (Teti, 2018). Poverty is a status risk which continues over a period of time or which indefinitely involves multiple specific stressors (or perturbations). Children living in poverty face heightened risk for adverse childhood experiences (ACEs). Children experiencing ACEs are at increased risk for residential mobility, witnessing or experiencing violence, unwanted pregnancies, premature death, and other developmental and health challenges (Hays-Grudo & Morris, 2020). Parents living in poverty often experience stress when trying to raise their children, maintain couple relationships, and

address the challenges of having experienced ACEs during their own childhoods. Siblings may be expected to place family responsibilities over future goals and engage in parenting. Families living with poverty find maintaining food and clothing, having a safe and secure shelter, and securing healthcare to be recurring stressors. The lack of economic resources makes it difficult to have regularity in routines associated with the family maintenance system, such as providing for family system needs. Thus, interfaces with ecosystems, whether extended kin networks or formal services, can be fundamental to maintaining family well-being.

Families struggling for survival may face challenges in maintaining routines in emotional connectedness. Parental control techniques for children may focus more on authority and safety than promoting progress toward developmental milestones. The sense of family identity may be challenged as family members face difficulty fulfilling their own and societal expectations for parenting. Family adaptive systems at multiple system levels can offer protection against risks for families who can effectively mobilize their strengths to address economic difficulties. For example, support from extended kin networks may assist when household items need to be fixed or children need care when parents are not available due to work or other challenges. Schools may provide meals for children to foster physical well-being. Teachers may provide guidance and support to foster social and academic competence. Religious organizations may provide a sense of meaning and connectedness or maintenance-related supports. Other health and social services in the community can provide needed protection to families. A promising strategy for supporting effective parenting for families living with poverty is prevention and intervention, focusing on helping families mobilize or develop the skills needed to meet basic needs.

Implications for Service Providers and Policy Makers

Through a systems lens, overall family systems and parenting are both parts of family relational patterns in key areas of functioning such as emotion, control, meaning, and maintenance. Researchers and professionals engaged in developing policies or providing services for children and parents should take into account multiple functions societies expect parents to provide and the how the FAS can be points or vulnerability or strengths that become risks or offer protection. Perpetrations occur through family life cycle transitions or other stressors. Using a systemic lens holds the promise of moving beyond one-size-fits-all policies, including policies that claim to be race neutral and to address parenting issues within ecosystems. Based on these ideas, key principles to guide implications and research follow.

1. *Families are best viewed as a whole with complex, intertwined relational patterns.* Parenting occurs in concert with the experiences of individual family members, other subsystems, overall family systems, and family–ecosystem interfaces.
2. *Definitions of “family” may vary based on family form, race/ethnicity, national origin, family worldview, and family identity.* In turn, co-parenting may be quite different for families in cultures where extended family or kin networks (relatives or others considered “family”) engage in parenting than for those in cultures where mothers and fathers are primary parents.
3. *Race/ethnicity are integral to family meaning systems.* The family worldview and identity often reflects shared family views on race/ethnicity. Yet, there can be gaps among family members or subsystems in terms of orientation toward the historic culture of families and current contexts (Bámaca-Colbert et al., 2019).
4. *Co-parenting may involve opposite- or same-sex partners currently in a couple relationship, former partners, step-parents, (residential or non-residential), birth parents and foster or adoptive parents, a grandparent and a parent, grandparents, or other family members.* Depending on family form and worldview, co-parenting relationships may involve current couples, former couples, step-parents and parents, parents and siblings, parents and grandparents, birth parents and adoptive parents, legal parents and foster parents, or other extended family relationships. From a systems perspective, effective co-parenting is evident as families effectively manage key aspects of family systems functioning evidenced in family adaptive systems. Thus, the demographic qualities of co-parents are less important than the ability to effectively navigate everyday family life and perturbations that arise over time.
5. *In families with two parents (e.g., father and mother) involved in the lives of their children, the involvement of both parents with the children is important.* Researchers are increasingly studying fathers as parents as well as mothers as parents, such that ways mothers and fathers tend to parent similarly or differently are important to understanding family dynamics (Schoppe-Sullivan & Fagan, 2020). Parent–child triads in research may provide different insights and opportunities for prevention and intervention relying only on specific dyads such as father–child or mother–child.
6. *Binuclear family systems are a distinct family form in which co-parenting occurs.* Depending upon the nature of the relationship between parents in binuclear family systems, the ability to cooperatively promoting the child may require setting

boundaries on the adult relationships to focus only on parenting.

7. *Within family systems, cascades of risk or protection can occur.* These arise across (a) domains of family functioning such as emotion, control, meaning, and maintenance or (b) multiple family system levels. Thus, child behavior problems may be evidence of cascading challenges in the basic FAS or signify disequilibrium in the broader family system or subsystems (e.g., couple subsystems). Cascades are promising for intervention in parenting (Doty, Davis, & Ardititi, 2017).
8. As the science of bio-physiological aspects of family life continues to progress, the integration of theory and research findings with systems perspectives has notable potential to better account for the family system relational patterns as well as individual responses to changes.

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Part IV

Child Factors that Impact Parenting
